Tangent-Space Methods for Nonlinear Control and Biomechanics

Tangent-Space Methods for Nonlinear Dynamics and Control

A unified treatment of how exact local linearity, contraction metrics, and optimal control share a common geometric foundation.

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*"Every nonlinear system is locally linear—not approximately, but exactly.
The art of control is navigating between these exact local pictures."*

— The Tangent Hyperplane Manifesto

Preface

This book grew from a simple observation that proved surprisingly difficult to articulate cleanly: *superposition is exact in tangent spaces*. Not approximately true. Not "good enough for small perturbations." Exact—by the definition of what a tangent space is.

That observation is not new mathematics. Calculus established it centuries ago. What is new is the insistence that control theory, dynamics, and engineering practice should fully internalize what calculus already proved. When we linearize a nonlinear system, we are not making an approximation. We are computing the exact infinitesimal structure of the dynamics at a point. The approximation enters only when we step away from that point—and even then, the error is not sloppiness but geometry: the curvature of the manifold.

This book develops that perspective systematically. We show that contraction theory, trajectory optimization (LQR, DDP, MPC), and the decomposition of motion into drift and control all rest on the same geometric foundation: *tangent-space dynamics with a metric*.

The treatment is aimed at graduate students and practicing engineers who have encountered linearization, Lyapunov stability, and optimal control as separate topics and want to see them unified. We assume familiarity with linear algebra, multivariable calculus, and ordinary differential equations. Differential geometry is introduced as needed, always paired with physical intuition.

Every chapter follows a consistent structure: a qualitative framing of why the material matters, a plain-language explanation, a geometric intuition box, and then the formal development. This is deliberate. The mathematics must be rigorous, but the reader should always know *why* a calculation is being done before seeing *how*.

A word on notation: we are deliberately explicit. Every matrix dimension, every smoothness assumption, every domain restriction is stated. This is not pedantry—it is the scaffolding that prevents the subtle errors that plague "hand-wavy" treatments of linearization and superposition.

February 2026

Contents

Nomenclature

General Conventions

- Scalars are lowercase or Greek letters (e.g., m, t, θ).
- Vectors are bold lowercase (e.g., $\mathbf{x}, \mathbf{u}, \mathbf{q}$).
- Matrices are bold uppercase (e.g., $\mathbf{A}, \mathbf{B}, \mathbf{M}, \mathbf{J}$).
- Expected values and sets are denoted by blackboard bold or script (e.g., $\mathbb{E}, \mathcal{S}$).

State and Kinematics

$\mathbf{q} \in \mathbb{R}^n$ Joint configuration vector (generalized coordinates).

$\dot{\mathbf{q}} \in \mathbb{R}^n$ Joint velocity vector (generalized velocities).

$\ddot{\mathbf{q}} \in \mathbb{R}^n$ Joint acceleration vector.

$\mathbf{x} \in \mathbb{R}^{nx}$ System state vector; commonly $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$ for mechanical systems.

$\mathbf{u} \in \mathbb{R}^m$ Control input vector (e.g., joint torques, muscle activations).

$\boldsymbol{\tau} \in \mathbb{R}^n$ Generalized forces/torques acting on the joints.

$t \in \mathbb{R}$ Time.

Inertia and Dynamics

$\mathbf{M}(\mathbf{q})$ Mass (inertia) matrix, symmetric positive definite.

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ Coriolis and centrifugal force matrix.

$\mathbf{g}(\mathbf{q})$ Gravity vector.

$\mathbf{J}(\mathbf{q})$ Jacobian matrix relating joint velocities to task-space velocities ($\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$).

Control and Optimization

$\mathbf{A}_k = \frac{\partial f}{\partial \mathbf{x}}$ Local Jacobian matrix of the state dynamics.

$\mathbf{B}_k = \frac{\partial f}{\partial \mathbf{u}}$ Local Jacobian matrix of the control input.

$V(\mathbf{x})$ **or** $\mathcal{V}$ Value function or Lyapunov function.

$\mathbf{Q}$ State penalty matrix in LQR/DDP.

$\mathbf{R}$ Control penalty matrix in LQR/DDP.

$\mathbf{K}$ Feedback gain matrix ($\delta\mathbf{u} = -\mathbf{K}\delta\mathbf{x}$).

$\boldsymbol{\gamma}$ A trajectory or flow, mapping time and initial conditions to states.

Affine Control and Counterfactuals

$f(\mathbf{x})$ Drift vector field: complete autonomous evolution of the declared effective plant when $\mathbf{u} = \mathbf{0}$.

$G(\mathbf{x})$ Input (control) matrix field mapping control inputs to state derivatives.

$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$ Control-affine system equations of motion.

$\mathbf{x}^{\mathrm{ZTCF}}(t)$ Forward Zero-Torque Counterfactual trajectory under $\mathbf{u}(t) \equiv \mathbf{0}$.

$\mathbf{a}_{\mathrm{ZVCF}}(\mathbf{q}, \mathbf{z})$ Instantaneous Zero-Velocity Counterfactual acceleration with velocity and declared control set to zero.

$\mathrm{DCR}_{W,\mathcal{U}}(\mathbf{x})$ Drift-Control Ratio in a declared acceleration or task-projected space: $\frac{\|W\mathbf{a}_{\mathrm{drift}}\|_2}{\sup_{\mathbf{u}\in\mathcal{U}} \|WB_a\mathbf{u}\|_2+\varepsilon}$.

Biomechanics and Motor Control

$a_i \in [0, 1]$ Muscle activation level.

ℓ_i^M Muscle fiber length.

v_i^M Muscle fiber contraction velocity.

ℓ_i^{MT} Musculotendon unit length.

$\mathbf{F}_{\mathbf{muscle}}$ Force generated by muscles.

$\mathbf{W}$ Muscle synergy matrix (weights).

$\mathbf{c}(t)$ Muscle synergy activation vector over time.

Geometry and Manifolds

$SE(3)$ Special Euclidean group of rigid body motions.

$SO(3)$ Special Orthogonal group of 3D rotations.

$\mathfrak{se}(3)$ Lie algebra of $SE(3)$, space of spatial twists (velocities).

$\mathcal{M}$ A smooth manifold.

$T_{\mathbf{x}}\mathcal{M}$ Tangent space at point $\mathbf{x}$.

Chapter 1

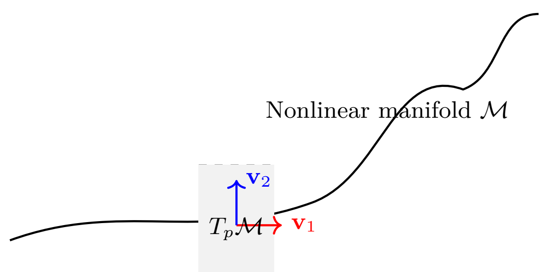
Foundations: Tangent Spaces and Exactness

In Plain Language 1.1. Zooming in on the Map

Every curved surface, no matter how complex, has a flat patch touching it at each point. On that flat patch, the familiar rules of addition and scaling work perfectly. This chapter establishes that these flat patches—tangent spaces—are not approximations of the surface. They *are* the exact infinitesimal structure of the surface, by the very definition of what "smooth" means.

1.1 Motivation: Why Geometry Matters for Control

Nonlinear dynamics are often described as systems in which superposition does not apply. Students learn early that while linear systems permit the combination of solutions—if $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ are solutions, then so is $\alpha\boldsymbol{x}_1(t) + \beta\boldsymbol{x}_2(t)$—nonlinear systems deny this convenience. The narrative typically stops there, leaving the impression that nonlinearity and superposition are fundamentally incompatible.



The key insight of this chapter is that while trajectories cannot be superposed on the manifold itself, *infinitesimal perturbations* live in the tangent space $T_p\mathcal{M}$, where superposition holds exactly.

That impression is incomplete. It is true that *trajectories* in a nonlinear system do not generally superpose. But at each point in state space, there exists a tangent space that is a genuine vector space, and within that vector space, superposition holds exactly. Not approximately. Not "for small enough perturbations." *Exactly*, by the axioms of linear algebra.

The purpose of this book is to take that geometric fact seriously and trace its consequences through contraction theory, optimal control, and the practical decomposition of complex motions into interpretable components.

Geometric Intuition

A trajectory is a curve on a state-space manifold. At each point on that curve, there is a tangent plane capturing first-order motion of nearby trajectories. All local stability and local optimality calculations happen in these moving tangent planes. A metric is a local ruler; changing the metric changes which perturbation directions count as "large" or "small."

Remark 1.1 (Lie Brackets and Nonlinear Coupling). While vector fields are added and scaled linearly in the tangent space, their composition via the flow map is nonlinear. The *Lie bracket* $[\,\mathbf{f}, \mathbf{g}\,]$ of two vector fields measures this nonlinear coupling: it quantifies how the order of application matters. Formally, for vector fields $\mathbf{f}$ and $\mathbf{g}$ on a manifold $\mathcal{M}$,

$$[\,\mathbf{f}, \mathbf{g}\,] = \frac{\partial \mathbf{g}}{\partial \boldsymbol{x}}\mathbf{f} - \frac{\partial \mathbf{f}}{\partial \boldsymbol{x}}\mathbf{g}. \tag{1.1}$$

As Agrachev and Sachkov detail in [AS04, Ch. 1–2], the Lie bracket is the fundamental tool for understanding how infinitesimal perturbations in different directions interact. In control theory, brackets $[\,\mathbf{g}_i, \mathbf{g}_j\,]$ between input vector fields characterize what motions can be generated through iterated applications of controls—the foundation of controllability analysis via the Lie algebra rank condition, discussed in Chapter 3.

1.2 Historical Context: From Newton to Modern Differential Geometry

To appreciate the force of the tangent-space framework, it helps to understand how it emerged from the practical concerns of mechanics and the rigorous formalization of analysis.

1.2.1 Newton, Leibniz, and the Birth of the Infinitesimal

The modern concept of tangent space originates in the Newton–Leibniz calculus of the late 17th century [GPS02, Arn89]. Newton's method of "fluxions" (rates of change) and Leibniz's differential notation $\mathrm{d}x$ both captured an intuitive idea: at each point on a smooth curve, there is a direction of motion, a "tangent line." For a curve $y = f(x)$ in the plane, the tangent line at $(x_0, f(x_0))$ has slope $f'(x_0)$ and can be written as

$$y - f(x_0) = f'(x_0)(x - x_0). \tag{1.2}$$

This is purely one-dimensional geometry. But the insight—that the tangent line is the "best linear approximation" to the curve at that point—proved far more general than the original calculus allowed.

1.2.2 Cauchy, Weierstrass, and the Rigorous Foundation

By the 19th century, Cauchy and especially Weierstrass placed the derivative on a rigorous footing [GPS02]. The limit definition

$$f'(x_0) := \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \tag{1.3}$$

made the tangent line precise: it is the unique linear function whose error vanishes faster than the perturbation itself.

The power of this formulation is that it is *coordinate-free*: it depends only on the behavior of f in a neighborhood of x_0, not on the choice of coordinates. A function is differentiable at x_0 if and only if the above limit exists, regardless of whether we write the function as $y = f(x)$ or in any other smooth parameterization.

1.2.3 Riemann's Vision of Intrinsic Geometry

Bernhard Riemann's 1854 paper *Über die Hypothesen, welche der Geometrie zu Grunde liegen* (On the Hypotheses Which Lie at the Foundations of Geometry) was the pivotal moment [dC92]. Riemann observed that just as Newton and Leibniz could construct a local approximation (the tangent line) at each point of a curve, one could construct a local vector space (the tangent space) at each point of any smooth manifold, even if the manifold itself were curved, high-dimensional, or not embedded in Euclidean space.

Riemann's innovation—and its modern development by Élie Cartan, Hermann Weyl, and others [dC92, Arn89]—fundamentally separated two ideas:

1. The *manifold itself*: a geometric object that may be curved.
2. The *tangent space at a point*: a vector space, always flat and linear.

This separation is conceptually important. Curvature is not a failure of linearity; it is the nonlinearity of how tangent spaces connect as you move through the manifold.

1.2.4 From Approximation to Exact Structure

For most of the 20th century, even when differential geometry was fully developed [AM78, Arn89], the relationship between a nonlinear dynamical system and its tangent-space (linearized) dynamics remained framed as an *approximation* [Kha02]. Textbooks would write:

> To study the qualitative behavior of a nonlinear system $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ near an equilibrium $\bar{\boldsymbol{x}}$, we linearize: $\dot{\delta\boldsymbol{x}} = \mathbf{A}\,\delta\boldsymbol{x}$ where $\mathbf{A} = \frac{\partial f}{\partial \boldsymbol{x}}|_{\bar{\boldsymbol{x}}}$. This linear system is "close" to the original system for small perturbations.

The language—"linearize," "approximate," "close to"—suggests that the tangent space is a crutch, a convenience for analysis. But this framing obscures the true mathematical situation.

1.2.5 The Conceptual Shift

The recognition that linearization is *exact* at the infinitesimal level, not an approximation, came from careful attention to the definition of the Fréchet derivative. If we define smoothness via the condition

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}) = f(\bar{\boldsymbol{x}}) + \mathbf{A}\,\delta\boldsymbol{x} + o(\|\delta\boldsymbol{x}\|), \tag{1.4}$$

then there is *exactly one* matrix $\mathbf{A}$ satisfying this property (at the infinitesimal scale). The matrix $\mathbf{A}$ is not an approximation; it is the *definition* of what smoothness means at that point.

This modern viewpoint—which permeates contemporary differential geometry [dC92], control theory [Kha02, Sas99], and optimization (especially gradient-based learning)—is the foundation of this book. We do not linearize for convenience and accept error. We recognize that linearization *is* the exact infinitesimal description, and we use this exactness as a structural principle for analyzing and controlling nonlinear systems.

1.3 Smooth Dynamical Systems

Consider a dynamical system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}), \quad \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{u} \in \mathbb{R}^m, \tag{1.5}$$

where $\boldsymbol{x}$ is the state vector and $\boldsymbol{u}$ is the control input. We impose the following standing assumption throughout this book.

Assumption 1.2 (Smoothness)**.** The vector field $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is continuously differentiable (C^1) in both arguments. Where higher-order residual bounds are needed, we assume C^2 smoothness.

This assumption is minimal yet physically reasonable. Virtually all systems derived from classical mechanics—Newtonian, Lagrangian, or Hamiltonian—produce smooth vector fields. The C^1 requirement is the bare mathematical condition for derivatives to exist.

Common Misconception

Systems with impacts, Coulomb friction, switching controllers, or hard state constraints violate C^1 smoothness. Extensions to such hybrid systems exist (saltation matrices, Filippov solutions, complementarity formulations) but lie outside the scope of this treatment. We address smooth systems exclusively and note hybrid extensions where relevant.

1.4 The Fréchet Derivative: Linearization as Exact Structure

1.4.1 Definition and Uniqueness

At a fixed operating point $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}})$, the **Jacobian** of the vector field with respect to the state is the matrix

$$\mathbf{A} := \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}})} \in \mathbb{R}^{n \times n}, \tag{1.6}$$

defined as the unique linear map satisfying the Fréchet derivative condition:

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}, \bar{\boldsymbol{u}}) = f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) + \mathbf{A}\,\delta\boldsymbol{x} + o(\|\delta\boldsymbol{x}\|). \tag{1.7}$$

The notation $o(\|\delta\boldsymbol{x}\|)$ denotes a remainder that vanishes faster than $\|\delta\boldsymbol{x}\|$ as $\|\delta\boldsymbol{x}\| \to 0$:

$$\lim_{\|\delta\boldsymbol{x}\|\to 0} \frac{\|f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}, \bar{\boldsymbol{u}}) - f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) - \mathbf{A}\delta\boldsymbol{x}\|}{\|\delta\boldsymbol{x}\|} = 0. \tag{1.8}$$

1.4.2 Proof of Uniqueness

Proposition 1.3 (Uniqueness of the Fréchet Derivative)**.** *If a linear map* $\mathbf{A}$ *satisfies Eq.* (1.7) *at* $\bar{\boldsymbol{x}}$*, then* $\mathbf{A}$ *is unique.*

Proof. Suppose two matrices $\mathbf{A}$ and $\mathbf{A}'$ both satisfy

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}, \bar{\boldsymbol{u}}) = f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) + \mathbf{A}\,\delta\boldsymbol{x} + o(\|\delta\boldsymbol{x}\|), \tag{1.9}$$

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}, \bar{\boldsymbol{u}}) = f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) + \mathbf{A}'\,\delta\boldsymbol{x} + o(\|\delta\boldsymbol{x}\|). \tag{1.10}$$

Subtracting these equations gives

$$(\mathbf{A} - \mathbf{A}')\,\delta\boldsymbol{x} = o(\|\delta\boldsymbol{x}\|). \tag{1.11}$$

For any nonzero vector $\boldsymbol{v} \in \mathbb{R}^n$, set $\delta\boldsymbol{x} = t\boldsymbol{v}$ where $t > 0$ is small. Then

$$(\mathbf{A} - \mathbf{A}')\boldsymbol{v} = \frac{o(t\,\|\boldsymbol{v}\|)}{t} = t \cdot \frac{o(t\,\|\boldsymbol{v}\|)}{t \cdot \|\boldsymbol{v}\|} \cdot \|\boldsymbol{v}\|\,. \tag{1.12}$$

As $t \to 0^+$, the term $\frac{o(t\|\boldsymbol{v}\|)}{t\cdot\|\boldsymbol{v}\|} \to 0$ by definition of $o(\cdot)$. Therefore, $(\mathbf{A} - \mathbf{A}')\boldsymbol{v} = 0$ for all $\boldsymbol{v}$, implying $\mathbf{A} = \mathbf{A}'$. □

This uniqueness is the cornerstone of the book's philosophy. There is exactly one candidate for "the infinitesimal structure" of f at $\bar{\boldsymbol{x}}$, and it is $\mathbf{A}$.

Principle 1: The Derivative is Exact The linear map $\mathbf{A}$ is not an approximation to f. It *is* the exact infinitesimal structure of f at $\bar{\boldsymbol{x}}$. There is no "better" first-order description—$\mathbf{A}$ is unique by the definition of the Fréchet derivative.

This distinction is the conceptual foundation of the entire book. The standard narrative—"we linearize for convenience and accept some error"—conflates two distinct operations:

1. **Computing the derivative** (exact, by definition);
2. **Using the linear model for finite perturbations** (introduces residuals proportional to curvature).

1.4.3 Relationship to the Gâteaux Derivative

The Fréchet derivative is stronger than the more permissive Gâteaux derivative. The **Gâteaux derivative** of f at $\bar{\boldsymbol{x}}$ in direction $\boldsymbol{v}$ is

$$D_{\boldsymbol{v}} f(\bar{\boldsymbol{x}}) := \lim_{t\to 0} \frac{f(\bar{\boldsymbol{x}} + t\boldsymbol{v}) - f(\bar{\boldsymbol{x}})}{t}, \tag{1.13}$$

provided the limit exists. If f is Fréchet-differentiable at $\bar{\boldsymbol{x}}$ with derivative $\mathbf{A}$, then the Gâteaux derivative in any direction $\boldsymbol{v}$ exists and equals $\mathbf{A}\boldsymbol{v}$.

However, the converse is false: a function can have all Gâteaux derivatives in all directions (even linearly) without being Fréchet-differentiable. The Fréchet condition is uniform in all directions; the remainder $o(\|\delta\boldsymbol{x}\|)$ must be small relative to $\|\delta\boldsymbol{x}\|$ for *all* perturbations $\delta\boldsymbol{x}$, not just along specific rays.

For the purposes of this book, we always assume Fréchet differentiability (part of the C^1 assumption), which ensures the robust infinitesimal linearization on which all subsequent theory depends.

1.4.4 Worked Example: The Pendulum Jacobian

Consider a simple pendulum with friction and torque input:

$$\dot{\theta} = \omega, \quad \dot{\omega} = -g\sin(\theta) - b\omega + u, \tag{1.14}$$

where θ is angle, ω is angular velocity, g is gravitational acceleration, b is friction, and u is applied torque. The state is $\boldsymbol{x} = (\theta, \omega)$ and control is $\boldsymbol{u} = u$.

The vector field is

$$f(\boldsymbol{x}, u) = \begin{pmatrix} \omega \\ -g\sin\theta - b\omega + u \end{pmatrix}. \tag{1.15}$$

At an equilibrium $(\bar{\theta}, \bar{\omega}) = (\bar{\theta}, 0)$ (the pendulum at rest), with input $\bar{u} = g\sin\bar{\theta}$ (input compensating gravity), the Jacobians are:

$$\mathbf{A} = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{(\bar{\theta},0)} = \begin{pmatrix} 0 & 1 \\ -g\cos\bar{\theta} & -b \end{pmatrix}, \tag{1.16}$$

$$\mathbf{B} = \left.\frac{\partial f}{\partial u}\right|_{(\bar{\theta},0)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.17}$$

Now, consider a perturbation $\delta\boldsymbol{x} = (\delta\theta, \delta\omega)$ from the equilibrium. The Fréchet property says

$$f(\bar{\theta} + \delta\theta, 0 + \delta\omega, \bar{u}) \approx f(\bar{\theta}, 0, \bar{u}) + \mathbf{A}\,\delta\boldsymbol{x} \tag{1.18}$$

for small $\delta\boldsymbol{x}$. Let us verify this numerically.

Numerical Verification

Take $g = 10\,\mathrm{m/s}^2$, $b = 0.1\,\mathrm{s}^{-1}$, and $\bar{\theta} = \pi/6$ (30 degrees). Then $\cos(\pi/6) = \sqrt{3}/2 \approx 0.866$, so

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -10 \times 0.866 & -0.1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -8.66 & -0.1 \end{pmatrix}. \tag{1.19}$$

Consider perturbations of increasing size. At equilibrium, $f(\bar{\theta}, 0, \bar{u}) = (0, 0)$.

For $\delta\boldsymbol{x} = (0.01, 0)$ (small angle perturbation):

$$\text{True:} \quad f(\bar{\theta} + 0.01, 0, \bar{u}) = \begin{pmatrix} 0 \\ -g(\sin(\pi/6 + 0.01) - \sin(\pi/6)) \end{pmatrix} \tag{1.20}$$

$$= \begin{pmatrix} 0 \\ -10(0.5087 - 0.5) \end{pmatrix} = \begin{pmatrix} 0 \\ -0.0867 \end{pmatrix}, \tag{1.21}$$

$$\text{Linear:} \quad \mathbf{A}\,\delta\boldsymbol{x} = \begin{pmatrix} 0 & 1 \\ -8.66 & -0.1 \end{pmatrix} \begin{pmatrix} 0.01 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -0.0866 \end{pmatrix}, \tag{1.22}$$

$$\text{Residual:} \quad r(\delta\boldsymbol{x}) = \begin{pmatrix} 0 \\ -0.0001 \end{pmatrix}, \quad \|r(\delta\boldsymbol{x})\| \approx 4.3 \times 10^{-4}, \quad \frac{\|r(\delta\boldsymbol{x})\|}{\|\delta\boldsymbol{x}\|} = 0.043. \tag{1.23}$$

For $\delta\boldsymbol{x} = (0.001, 0)$ (one-tenth the size):

$$\text{Residual:} \quad \|r(\delta\boldsymbol{x})\| \approx 4.3 \times 10^{-6}, \quad \frac{\|r(\delta\boldsymbol{x})\|}{\|\delta\boldsymbol{x}\|} = 0.0043. \tag{1.24}$$

For $\delta\boldsymbol{x} = (0.0001, 0)$:

$$\text{Residual:} \quad \|r(\delta\boldsymbol{x})\| \approx 4.3 \times 10^{-8}, \quad \frac{\|r(\delta\boldsymbol{x})\|}{\|\delta\boldsymbol{x}\|} = 0.00043. \tag{1.25}$$

The ratio $\|r(\delta\boldsymbol{x})\| \,/\, \|\delta\boldsymbol{x}\|$ decreases approximately as $\|\delta\boldsymbol{x}\|$ (i.e., $o(\|\delta\boldsymbol{x}\|)$ behavior), confirming the Fréchet property. For a mixed perturbation like $\delta\boldsymbol{x} = (0.01, 0.01)$, we would see this same scaling: the residual is dominated by the quadratic term (curvature) in the sin function, which is $O(\|\delta\boldsymbol{x}\|^2)$.

1.4.5 The Input Jacobian

Similarly, the Jacobian with respect to the input is

$$\mathbf{B} := \left.\frac{\partial f}{\partial \boldsymbol{u}}\right|_{(\bar{\boldsymbol{x}},\bar{\boldsymbol{u}})} \in \mathbb{R}^{n\times m}, \tag{1.26}$$

satisfying

$$f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}} + \delta\boldsymbol{u}) = f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) + \mathbf{B}\,\delta\boldsymbol{u} + o(\|\delta\boldsymbol{u}\|). \tag{1.27}$$

Together, the pair $(\mathbf{A}, \mathbf{B})$ encodes the complete first-order sensitivity of the dynamics at $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}})$. These are the objects that every subsequent chapter will build upon.

1.5 Tangent Spaces as Vector Spaces

1.5.1 Abstract Definition

Definition 1.4 (Tangent Space)**.** Let $\mathcal{M}$ be a smooth manifold and $p \in \mathcal{M}$. The **tangent space** $T_p\mathcal{M}$ is the vector space of all tangent vectors at p—equivalently, the space of all velocity vectors of smooth curves passing through p.

More formally, two smooth curves $\gamma_1, \gamma_2 : (-\epsilon, \epsilon) \to \mathcal{M}$ with $\gamma_1(0) = \gamma_2(0) = p$ are said to be *equivalent at order one* if they have the same velocity vector at $t = 0$:

$$\dot{\gamma}_1(0) = \dot{\gamma}_2(0). \tag{1.28}$$

The tangent space $T_p\mathcal{M}$ is the set of equivalence classes of such curves, where the equivalence relation is having the same velocity. Vector addition and scalar multiplication are defined by concatenating velocities: if $[\gamma_1]$ and $[\gamma_2]$ are two tangent vectors (equivalence classes), then $[\gamma_1] + [\gamma_2]$ is the equivalence class of the curve that travels at velocity $\dot{\gamma}_1(0) + \dot{\gamma}_2(0)$. This makes $T_p\mathcal{M}$ a vector space.

For systems evolving in $\mathbb{R}^n$, the tangent space at any point is isomorphic to $\mathbb{R}^n$ itself. The distinction becomes non-trivial on curved manifolds (e.g., SO(3) for rotational dynamics), where the tangent space is a genuinely different object from the manifold. More fundamentally, the control-affine structure defines a *distribution*—a linear subspace of the tangent bundle whose closure under Lie brackets determines what states are reachable by the control system (see Agrachev and Sachkov [AS04, Ch. 3]). manifold.

1.5.2 Isomorphism With Directional Derivatives

There is another perspective on tangent vectors: as directional derivatives of functions on the manifold. A tangent vector $v_p \in T_p\mathcal{M}$ can be viewed as a linear functional on the space of smooth functions $f : \mathcal{M} \to \mathbb{R}$, defined by

$$v_p(f) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\gamma(t)), \tag{1.29}$$

where γ is any curve representing v_p. This definition is independent of the choice of curve (by equivalence), and it gives a derivation: a linear map that satisfies the Leibniz rule $v_p(fg) = f(p)v_p(g) + g(p)v_p(f)$.

Conversely, every derivation at p arises from a unique tangent vector. This isomorphism between geometric tangent vectors (equivalence classes of curves) and algebraic derivations is fundamental to modern differential geometry and will underlie our treatment of variational equations in Chapter 2.

1.5.3 Why Tangent Spaces Are the Natural Home of Superposition

Principle 2: Superposition Lives in Tangent Spaces Vector addition $\delta\boldsymbol{x}_1 + \delta\boldsymbol{x}_2$ is only defined in vector spaces. Tangent spaces are vector spaces *by construction*. Therefore, superposition of infinitesimal perturbations is exact—not because the nonlinear system is "approximately linear," but because tangent spaces are *inherently* linear.

When one says "superposition fails in nonlinear systems," the precise meaning is: *you cannot add trajectories in state space*. But you *can* add infinitesimal variations in the tangent space. This is not an approximation—it is the definition of the tangent space as a vector space.

1.5.4 Coordinate Representation

In local coordinates $\boldsymbol{x} = (x_1, \ldots, x_n)$ on a chart of $\mathcal{M}$, a tangent vector $\delta\boldsymbol{x} \in T_{\bar{\boldsymbol{x}}}\mathcal{M}$ is represented by its components $(\delta x_1, \ldots, \delta x_n) \in \mathbb{R}^n$. The Jacobian $\mathbf{A}$ then acts as a linear map $\mathbf{A} : T_{\bar{\boldsymbol{x}}}\mathcal{M} \to T_{\bar{\boldsymbol{x}}}\mathcal{M}$, sending one tangent vector to another.

Under a smooth coordinate change $\boldsymbol{z} = \phi(\boldsymbol{x})$ with Jacobian $\mathbf{T} = \partial\phi/\partial\boldsymbol{x}$, tangent vectors transform as

$$\delta\boldsymbol{z} = \mathbf{T}\,\delta\boldsymbol{x}, \tag{1.30}$$

and the Jacobian transforms as

$$\mathbf{A}_z = \mathbf{T}\,\mathbf{A}\,\mathbf{T}^{-1} + \dot{\mathbf{T}}\,\mathbf{T}^{-1}, \tag{1.31}$$

so the extra $\dot{\mathbf{T}}\mathbf{T}^{-1}$ term must be retained whenever the coordinate chart varies along the trajectory. For a time-independent coordinate change, $\dot{\mathbf{T}} = 0$ and the transformation is a similarity transform, so the eigenvalues of $\mathbf{A}$ are unchanged. For a time-dependent coordinate change, the instantaneous eigenvalues of $\mathbf{A}_z$ need not match those of $\mathbf{A}$; invariant statements must instead be made about the underlying geometric dynamics, such as stability conclusions, Lyapunov exponents, or contraction rates.

1.6 Residuals: Geometry, Not Error

When we use the linear model $\mathbf{A}\delta\boldsymbol{x}$ to predict the evolution of a *finite* perturbation, a residual appears:

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}, \bar{\boldsymbol{u}}) - f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) - \mathbf{A}\delta\boldsymbol{x} = r(\delta\boldsymbol{x}), \tag{1.32}$$

where $\|r(\delta\boldsymbol{x})\| = o(\|\delta\boldsymbol{x}\|)$ by the Fréchet property. If f is C^2, a sharper bound holds via the mean-value form of Taylor's theorem:

$$\|r(\delta\boldsymbol{x})\| \le \frac{1}{2} \sup_{\xi \in [\bar{\boldsymbol{x}},\, \bar{\boldsymbol{x}} + \delta\boldsymbol{x}]} \left\| \frac{\partial^2 f}{\partial \boldsymbol{x}^2}(\xi, \bar{\boldsymbol{u}}) \right\| \cdot \|\delta\boldsymbol{x}\|^2 . \tag{1.33}$$

Principle 3: Residuals Are Geometry, Not Error The residual $r(\delta\boldsymbol{x})$ measures how far the true dynamics deviate from the tangent-space prediction. This deviation comes from *manifold curvature* (second-order geometry), not from "our approximation being bad." A large residual at a point means the manifold is sharply curved there—it is a geometric signal, not a failure of the method.

1.6.1 Residual Scaling: A Concrete Example

To illustrate the $O(\|\delta\boldsymbol{x}\|^2)$ scaling of the residual, consider the one-dimensional function

$$f(x) = \sin(x). \tag{1.34}$$

At $\bar{x} = 0$, the derivative is $f'(0) = \cos(0) = 1$. The linear approximation is $\hat{f}(\delta x) = \delta x$. The residual is

$$r(\delta x) = \sin(\delta x) - \delta x. \tag{1.35}$$

Table 1.1: Residual scaling for $f(x) = \sin(x)$ at $\bar{x} = 0$ with linear approximation $\hat{f} = x$. Because $f''(0) = -\sin(0) = 0$, the residual is $O(\delta x^3)$, not $O(\delta x^2)$.

δx	$\sin(\delta x)$	Linear: δx	Residual r	$r/(\delta x)^2$	$r/(\delta x)^3$
0.1	0.0998334	0.1	-1.666×10^{-4}	-0.01666	-0.1666
0.01	0.0099998	0.01	-1.667×10^{-7}	-0.001667	-0.1667
0.001	0.001000	0.001	-1.667×10^{-10}	-0.000167	-0.1667
0.0001	0.000100	0.0001	-1.667×10^{-13}	-0.0000167	-0.1667

Common Misconception

This example is deliberately instructive. The function $\sin(x)$ has $f''(0) = -\sin(0) = 0$, so the second-order residual bound $\|r(\delta\boldsymbol{x})\| \leq \frac{1}{2}C_H \|\delta\boldsymbol{x}\|^2$ (Eq. (1.33)) still holds—it is simply a loose bound in this case. The *actual* residual is $r(\delta x) = -(\delta x)^3/6 + O(\delta x^5)$ from the Taylor series $\sin(\delta x) = \delta x - (\delta x)^3/6 + \cdots$.

The column $r/(\delta x)^3 \to -1/6 \approx -0.1667$ confirms cubic scaling. The column $r/(\delta x)^2 \to 0$ shows that the $O(\delta x^2)$ bound is satisfied but not tight. This illustrates a general principle: *the Fréchet property guarantees $o(\|\delta\boldsymbol{x}\|)$ residuals; the actual rate depends on the local curvature tensor.* When curvature vanishes (flat directions), the residual shrinks even faster than the bound suggests.

For a function with nonzero second derivative—such as $f(x) = \cos(x)$ at $\bar{x} = 0$, where $f''(0) = -1$—the residual *is* quadratic: $r(\delta x) = -(\delta x)^2/2 + O(\delta x^4)$, and the ratio $r/(\delta x)^2 \to -1/2$ exactly. The general bound (1.33) is tight whenever $f''(\bar{x}) \neq 0$.

The practical lesson: the residual quantifies curvature. Zero residual at second order means the manifold is locally "flatter than expected"—which happens precisely at inflection points and other geometrically special configurations.

1.6.2 Two Perspectives on Linearization

Table 1.2: Two Perspectives on the Linearization Residual.

	Standard View	**Tangent-Space View**
Residual is...	Approximation error	Manifold curvature
Strategy	Minimize error	Measure and exploit curvature
Linearization is...	"Good enough"	Exact at the infinitesimal limit
Goal	Better approximation	Navigate the tangent bundle

In the standard view, we apologize for the residual: "Yes, the linear model is wrong for finite perturbations, but for small perturbations it is good enough." In the tangent-space view, the residual is information: it tells us about the curvature of the state-space manifold, and this information is essential for understanding transport and nonlinear phenomena.

1.7 Manifold Examples: From Theory to Application

The abstract theory of tangent spaces becomes concrete and powerful when applied to specific manifolds that arise in dynamics and control. Three examples are essential.

1.7.1 SO(3): Rotations and Attitude Dynamics

The special orthogonal group SO(3) is the set of all 3×3 orthogonal matrices with determinant $+1$:

$$\mathrm{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3\times 3} : \mathbf{R}\top\mathbf{R} = \mathbf{I},\, \det(\mathbf{R}) = 1\}. \tag{1.36}$$

SO(3) is a 3-dimensional Lie group: it is a smooth manifold that is also a group under matrix multiplication. At the identity, the tangent space is the space of skew-symmetric matrices, called the Lie algebra $\mathfrak{so}(3)$:

$$T_{\mathbf{I}}\mathrm{SO}(3) = \mathfrak{so}(3) = \{\mathbf{S} \in \mathbb{R}^{3\times 3} : \mathbf{S} + \mathbf{S}\top = 0\}. \tag{1.37}$$

At an arbitrary rotation $\mathbf{R}$, the tangent space is the left translation of that Lie algebra:

$$T_{\mathbf{R}}\mathrm{SO}(3) = \{\mathbf{RS} : \mathbf{S} \in \mathfrak{so}(3)\} = \mathbf{R}\mathfrak{so}(3). \tag{1.38}$$

This space is 3-dimensional, but its elements are generally not themselves skew-symmetric unless $\mathbf{R} = \mathbf{I}$. The Lie algebra structure—defined through the commutator bracket $[\mathbf{S}_1, \mathbf{S}_2] = \mathbf{S}_1\mathbf{S}_2 - \mathbf{S}_2\mathbf{S}_1$—is exactly the Lie bracket of vector fields on the manifold SO(3). This bridge between algebraic (Lie algebra) and geometric (Lie bracket) structures is central to Agrachev and Sachkov's treatment of geometric control on Lie groups [AS04, Ch. 2].

Jacobian on SO(3)

Consider a rigid body with rotation matrix $\mathbf{R} \in \mathrm{SO}(3)$ and angular velocity $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$. The kinematic equation is

$$\dot{\mathbf{R}} = \mathbf{R}\,[\boldsymbol{\omega}]_\times, \tag{1.39}$$

where $[\boldsymbol{\omega}]_\times$ is the skew-symmetric matrix representation:

$$[\boldsymbol{\omega}]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}. \tag{1.40}$$

The Jacobian of this system with respect to a perturbation in angular velocity is

$$\frac{\partial \dot{\mathbf{R}}}{\partial \boldsymbol{\omega}} = \mathbf{R}\,\frac{\partial [\boldsymbol{\omega}]_\times}{\partial \boldsymbol{\omega}}. \tag{1.41}$$

This reveals a key point: perturbations to the angular velocity propagate through the system as skew-symmetric matrices. More precisely, the tangent space at the identity $T_I\mathrm{SO}(3) = \mathfrak{so}(3)$ consists of skew-symmetric matrices. The tangent space at an arbitrary element $R \in \mathrm{SO}(3)$ is $T_R\mathrm{SO}(3) = \{R \cdot S : S \in \mathfrak{so}(3)\}$. The linear perturbation dynamics lives naturally in $\mathfrak{so}(3)$ when expressed in body-frame coordinates (i.e., left-trivialized).

Why SO(3) Matters

For quadcopters, satellites, and any system with rotational degrees of freedom, using Euler angles or quaternions often introduces artificial singularities or redundancy. By working directly with SO(3) and its tangent space, we eliminate these complications and gain access to the rich structure of Lie groups and Lie algebras, which provide exact (non-approximate) transport laws through the state space.

1.7.2 SE(3): Rigid Body Transformations

The special Euclidean group SE(3) describes rigid body motions in 3D: rotations and translations. It is the semidirect product $\mathrm{SO}(3) \ltimes \mathbb{R}^3$, represented as 4×4 homogeneous transformation matrices:

$$\mathbf{T} = \begin{pmatrix} \mathbf{R} & \boldsymbol{p} \\ 0 & 1 \end{pmatrix}, \quad \mathbf{R} \in \mathrm{SO}(3), \quad \boldsymbol{p} \in \mathbb{R}^3. \tag{1.42}$$

The tangent space to SE(3) at any $\mathbf{T}$ is the space of Plücker coordinates or twist vectors, represented as 4×4 matrices of the form

$$\begin{pmatrix} [\boldsymbol{\omega}]_\times & \boldsymbol{v} \\ 0 & 0 \end{pmatrix}, \tag{1.43}$$

where $[\boldsymbol{\omega}]_\times \in \mathfrak{so}(3)$ and $\boldsymbol{v} \in \mathbb{R}^3$ is a linear velocity. This is the Lie algebra $\mathfrak{se}(3)$, a 6-dimensional space.

Kinematics and Dynamics

For a rigid body in space with configuration $\mathbf{T}(t) \in \mathrm{SE}(3)$ and twist (body velocity) $\boldsymbol{\xi} = (\boldsymbol{v}, \boldsymbol{\omega})$, the kinematic equation is

$$\dot{\mathbf{T}} = \mathbf{T} \begin{pmatrix} [\boldsymbol{\omega}]_\times & \boldsymbol{v} \\ 0 & 0 \end{pmatrix}. \tag{1.44}$$

Again, the dynamics live in the tangent space (the Lie algebra), and perturbations evolve linearly within that space—an expression of Axiom 2. The nonlinearity re-enters through the composition of infinitesimal steps, as formalized in Axiom 4 (transport).

1.7.3 The Double Pendulum Configuration Manifold

A double pendulum consists of two rigid links connected by hinges, with angles θ_1 and θ_2. The configuration space is naturally a torus:

$$\mathcal{M} = T^2 = S^1 \times S^1, \tag{1.45}$$

where each S^1 is a circle (angle wraparound).

Coordinates and Tangent Space

Coordinates on T^2 are $(\theta_1, \theta_2) \in [0, 2\pi) \times [0, 2\pi)$ with wraparound identification. The tangent space $T_{(\theta_1,\theta_2)}(T^2)$ is a 2-dimensional space representing the angular velocities $(\dot{\theta}_1, \dot{\theta}_2)$.

The state is $\boldsymbol{x} = (\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) \in T^2 \times \mathbb{R}^2$. The dynamics come from the Lagrangian or energy conservation, with gravity and friction. For example:

$$\ddot{\theta}_1 = g(\sin\theta_1 + \frac{1}{2}\sin(\theta_1 + \theta_2)) + (\text{friction, control}), \tag{1.46}$$

and similarly for θ_2.

Nonlinear Effects and Coupling

The double pendulum exhibits rich nonlinear phenomena:

- **Tangent-space superposition.** Small perturbations around a nominal trajectory add linearly in the tangent space.
- **Residual geometry.** The residual $r(\delta\boldsymbol{x})$ from Eq. (1.32) is large when the nominal trajectory passes through regions of high kinetic-energy curvature or steep potential gradients.
- **Transport and coupling.** As the system evolves, the principal directions of stability rotate through the torus. The state-transition matrix (Section 1.8) encodes this rotation exactly.

The torus structure is essential: wraparound in angles is handled correctly only by recognizing that the manifold is curved (topologically), not flat. Standard treatments that treat θ_i as real variables miss this structure.

1.8 Nonlinearity Re-Enters Through Transport

1.8.1 Transport as Curved Geometry

Principle 4: Transport Between Tangent Spaces Moving along a trajectory from $\bar{\boldsymbol{x}}(t_0)$ to $\bar{\boldsymbol{x}}(t_1)$ means moving between tangent spaces $T_{\bar{\boldsymbol{x}}(t_0)}\mathcal{M}$ and $T_{\bar{\boldsymbol{x}}(t_1)}\mathcal{M}$. The connection between these spaces—parallel transport, Christoffel symbols in Riemannian geometry—encodes the nonlinearity. Optimal control (DDP, iLQR) is therefore *exact local optimization + careful transport*, not "global optimization with approximations."

The relationship between successive tangent spaces is governed by the *state transition matrix* $\Phi(t_1, t_0)$, which we develop fully in Chapter 2. For now, we sketch the idea.

1.8.2 State Transition Matrix and Variational Equations

Consider the system $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ and a reference trajectory $\bar{\boldsymbol{x}}(t)$. A nearby trajectory with initial condition $\bar{\boldsymbol{x}}(t_0) + \delta\boldsymbol{x}(t_0)$ follows

$$\boldsymbol{x}(t) = \bar{\boldsymbol{x}}(t) + \delta\boldsymbol{x}(t) + o(\|\delta\boldsymbol{x}(t_0)\|), \tag{1.47}$$

where $\delta\boldsymbol{x}(t)$ satisfies the variational equation:

$$\dot{\delta\boldsymbol{x}}(t) = \mathbf{A}(t)\,\delta\boldsymbol{x}(t), \quad \mathbf{A}(t) := \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\bar{\boldsymbol{x}}(t)}. \tag{1.48}$$

The unique solution with initial condition $\delta\boldsymbol{x}(t_0)$ is

$$\delta\boldsymbol{x}(t) = \Phi(t, t_0)\,\delta\boldsymbol{x}(t_0), \tag{1.49}$$

where $\Phi(t, t_0)$ is the state transition matrix (also called fundamental matrix).

1.8.3 Christoffel Symbols and Parallel Transport (Intuitive Picture)

In Riemannian geometry on a curved manifold, "parallel transport" means moving a tangent vector along a curve while keeping it "as parallel as possible" (minimizing rotation relative to the manifold curvature). This is formalized by the Christoffel symbols Γ^k_{ij}, which measure how basis vectors change as you move along the manifold.

For our nonlinear dynamics, an analogous concept applies: as the reference trajectory $\bar{\boldsymbol{x}}(t)$ evolves, the linearized dynamics at successive points are not simply comparable; they must be "transported" via the state transition matrix $\Phi(t, t_1)$. The transport is exact (no approximation), but it encodes the manifold's curvature.

1.8.4 Connection to Differential Dynamic Programming

In Differential Dynamic Programming (DDP) and iLQR (discussed in later chapters), the algorithm repeatedly:

1. Linearize the dynamics along a nominal trajectory, computing $\mathbf{A}(t), \mathbf{B}(t)$.
2. Solve a linear-quadratic problem in the tangent spaces, obtaining optimal perturbations $\delta\boldsymbol{x}(t), \delta\boldsymbol{u}(t)$.
3. Transport these perturbations back to the full manifold via $\Phi(t, t_0)$, updating the nominal trajectory.
4. Repeat.

Each iteration solves an *exact* local problem (Axiom 1) within the tangent spaces, then accounts for curvature via *exact* transport (Axiom 4). The algorithm does not approximate; it refines the nominal trajectory by exploiting the local structure at each point.

1.9 Scope of Exactness: When the Tangent-Space Picture Breaks Down

The tangent-space framework is powerful and exact for smooth systems. But it has boundaries. Understanding these boundaries is essential for knowing when the theory applies and when generalizations are needed.

1.9.1 Discontinuities and Piecewise-Smooth Systems

The C^1 smoothness assumption (Assumption 1.2) is critical. If the vector field f or its Jacobian $\partial f/\partial \boldsymbol{x}$ jumps or is undefined at a point, the Fréchet derivative fails to exist. Common examples:

Coulomb Friction. The friction force has a discontinuous dependence on velocity:

$$f_{\text{friction}} = \begin{cases} -\mu_s |N| \cdot \operatorname{sign}(v), & v = 0 \text{ (static)} \\ -\mu_k |N| \cdot \operatorname{sign}(v), & v \neq 0 \text{ (kinetic)} \end{cases} \tag{1.50}$$

The sign function is discontinuous; hence $\partial f/\partial v$ does not exist.

Switching Controllers. A controller that switches between $u = u_1$ and $u = u_2$ based on crossing a threshold introduces a discontinuity in f or $\partial f/\partial u$.

Hard Constraints. A state constraint $x_i \leq x_{\max}$ enforced by a spring-like potential $V = \frac{1}{2}k(x_i - x_{\max})^2$ near the boundary is okay (smooth), but a hard wall (infinite force at the boundary) is not.

For such systems, the Fréchet derivative and Axiom 1 do not apply. Instead, one must use generalized notions: Filippov solutions, saltation matrices (for impacts), or differential inclusions. These are beyond the scope of this book but represent important extensions.

1.9.2 Chattering and Sliding Modes

In optimal control with control constraints (e.g., $|u| \leq u_{\max}$), the optimal control can be *bang-bang*: switching infinitely often between $u = u_{\max}$ and $u = -u_{\max}$. On a time interval where infinitely many switches occur (sliding mode), the averaged effect of the rapid switching can be captured by a generalized velocity constraint, but the instantaneous dynamics are discontinuous and do not satisfy C^1 smoothness.

The tangent-space framework applies to the intervals between switches and to the averaged dynamics, but not at the switching surfaces themselves.

1.9.3 Zeno Behavior and Accumulation of Events

In some hybrid systems, an infinite number of discrete events (impacts, mode switches) can accumulate in finite time, a phenomenon called Zeno behavior. At the accumulation point, the system's state may be well-defined (by continuity), but the dynamics are singular and do not admit a classical smooth vector field.

1.9.4 Singular Configurations on Manifolds

On manifolds like SO(3) or SE(3), certain configurations are singular: the exponential map or logarithmic map may not be invertible, or the intrinsic metric may degenerate. For example, the quaternion representation of rotations has a singularity at the antipodal point: two different quaternions represent the same rotation. When the system passes through such a point, local coordinates become singular, and care must be taken to handle the transition.

The tangent-space theory remains valid (the Lie group structure is smooth everywhere), but global coordinate charts cannot cover the entire manifold without singularities (a topological consequence of the Hairy Ball Theorem for S^2 and its generalizations).

Table 1.3: Scope of Tangent-Space Exactness and When Extensions Are Needed.

Feature	**Smooth Dynamics**	**Extension Needed**
Discontinuous f or $\partial f/\partial \boldsymbol{x}$	Not applicable	Filippov, differential inclusions
Switching controllers	OK between switches	Saltation matrices at switch
Bang-bang control	Averaged dynamics okay	Singular control on sliding surface
Impacts/collisions	Not applicable	Saltation matrix, impact law
Zeno behavior	Not applicable	Hybrid system theory
Singular configurations	Handled locally	Global charts may be singular

1.9.5 Summary: Boundaries of the Theory

Throughout this book, we work exclusively with smooth systems and note extensions where relevant. The exactness of the tangent-space picture is one of its greatest strengths, and understanding its boundaries is essential for applying it responsibly.

1.10 The Five Axioms: A Summary

The conceptual framework of this book rests on five axioms, each a consequence of standard differential geometry but rarely stated explicitly in control texts.

1. **Tangent spaces are exact, not approximate.** The derivative is the unique first-order structure at a point.

2. **Superposition lives in tangent spaces.** Tangent spaces are vector spaces; perturbation addition is well-defined and exact.

3. **Residuals are geometry, not error.** Deviations from linearity measure curvature, not modeling failure.

4. **Nonlinearity re-enters through transport.** Moving between tangent spaces requires navigating the manifold's curvature.

5. **Integration preserves exactness.** Accumulating infinitesimal linear contributions is itself exact; residuals arise from curvature, not from integration.

These axioms inform every derivation in the chapters that follow. When we write a Jacobian, we are not approximating. When we solve a Riccati equation in tangent coordinates, we are solving an exact local problem. When we iterate (as in DDP), we are re-centering the exact local analysis at a new base point. The "error" is always transport cost—the price of moving through curved space.

1.11 Scope and Limitations

The mathematical framework developed here applies to smooth (C^1) nonlinear systems, including mechanical systems (Newtonian, Lagrangian, Hamiltonian), robotic manipulators, vehicles, aerospace systems, and chemical process control.

The framework *does not* directly apply to discontinuous systems, impact dynamics, Coulomb friction, or hybrid switching systems. For these, generalizations exist—saltation matrices for impacts, differential inclusions for set-valued dynamics, complementarity systems for contact—and we note these extensions where appropriate.

1.11.1 When to Use This Framework

This textbook is most applicable when:

- The system dynamics arise from conservation laws (energy, momentum) or classical mechanics.
- The control inputs are continuous and bounded (not discontinuous switching).
- The state evolves on a smooth manifold (including $\mathbb{R}^n$, SO(3), SE(3), and their products).
- Local behavior around trajectories is of primary interest (not global existence or long-time stability in highly chaotic regimes).
- Numerical precision and computational efficiency are important, justifying exploitation of local structure.

1.11.2 Important Distinction: Linearization vs. Linear Approximation

Remark 1.5. Throughout this book, we use "linearization" to mean "computing the Jacobian (Fréchet derivative)"—an exact operation. We reserve "linear approximation" for the distinct act of using the Jacobian to predict finite perturbations, which introduces the residual of Eq. (1.32). Conflating these two operations is the source of most conceptual confusion about linearization in nonlinear systems.

A linearization is always exact. A linear approximation introduces error (residual). By keeping these concepts distinct, we unlock the structural power of tangent spaces and avoid the psychological barrier of thinking that everything about perturbations is approximate.

1.11.3 Looking Ahead

Chapter 2 develops the theory of perturbation evolution: how infinitesimal variations propagate along a trajectory, quantifying the sensitivity of orbits to initial conditions. The state transition matrix and Lyapunov exponents emerge as central objects.

Chapter 5 introduces contractivity and metric tensors, translating the intuition that "some directions are more stable than others" into a precise geometric framework.

Chapter 6 applies these ideas to optimal control, showing how DDP and iLQR are exact local algorithms operating in tangent spaces, with control constraints and terminal costs handled via their residual geometry.

The journey from foundations (this chapter) to applications (later chapters) is one of increasing structure and pragmatism: we leverage the exactness of the tangent space to build algorithms that are both conceptually clear and computationally efficient.

1.12 Preview: Why These Ideas Matter for the Golf Swing

Before proceeding to the formal theory, it is worth previewing how the foundations of this chapter connect to the book's primary application domain: the biomechanics of the golf swing. This preview is deliberately non-technical; the full mathematical treatment appears in Chapter 9.

In Plain Language 1.2. The Golfer's Frame of Reference

A golfer's swing is a chain of rotating body segments—torso, arms, wrists, club—that accelerates a clubhead to high speed and launches a ball. The physics is ferociously nonlinear: centrifugal forces grow as the square of rotational speed, the club shaft bends and rebounds elastically, and the mass distribution changes as the golfer's posture evolves. Yet at each instant, the tangent-space picture gives us an exact local model of how small changes propagate.

1.12.1 The Golfer's Tangent Space

At any instant during the downswing, the golfer's state is a point on a high-dimensional manifold: joint angles, angular velocities, shaft bending amplitudes, and their rates. The tangent space at this point is the space of all infinitesimal variations—a slightly different wrist angle, a marginally faster shoulder rotation, a small change in shaft flex.

By Axiom 1, the Jacobian at this point is the *exact* description of how these small variations evolve over the next instant. By Axiom 2, these variations add as vectors. A small wrist perturbation and a small shoulder perturbation evolve independently at first order, even though the full nonlinear swing couples them through centrifugal and Coriolis forces.

1.12.2 Curvature as a Diagnostic

The residual—the gap between tangent-space prediction and actual evolution—measures the "curvature" of the swing manifold. During the early backswing, velocities are low and the dynamics are nearly linear (small residual). During the late downswing, when the clubhead is moving at high speed, centrifugal forces dominate and the residual becomes large. This is not a failure of the theory; it is a geometric signal that the system is passing through a high-curvature region.

Practically, this means:

- Controllers (biological or engineered) that linearize once and hold the gain constant work well in the backswing but deteriorate in the downswing.
- Iterative methods (DDP, iLQR) that re-linearize at each time step naturally adapt to the changing curvature.
- The transition from "control-dominated" to "drift-dominated" dynamics (Chapter 8) is precisely the transition from small to large residuals.

1.12.3 Transport and the Kinematic Chain

Axiom 4 (transport) is especially vivid in the golf swing. As the club sweeps through a large arc, the tangent space rotates and stretches. The state transition matrix $\Phi(t_1, t_0)$ maps perturbations at one phase of the swing to perturbations at another. A small error in wrist lag at the top of the backswing may amplify into a large clubface angle error at impact—or it may be "washed out" by the dynamics. The eigenvalues of Φ tell us which scenario applies.

This is the deep connection between the abstract mathematics of this chapter and the practical question every golfer cares about: *which aspects of technique are forgiving, and which are critical*? Tangent-space analysis provides a principled, quantitative answer.

Exercises

1. **Fréchet derivative computation.** Compute the Fréchet derivative of $f(x, y) = (x^2y,\ \sin(xy))$ at the point $(1, \pi)$. Verify that the residual $\|f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}) - f(\bar{\boldsymbol{x}}) - Df(\bar{\boldsymbol{x}})\delta\boldsymbol{x}\| = O(\|\delta\boldsymbol{x}\|^2)$ by evaluating at $\delta\boldsymbol{x} = (0.1, 0.1)$, $(0.01, 0.01)$, and $(0.001, 0.001)$.

2. **Residual scaling table.** Construct a residual scaling table (analogous to Table 1.1) for $f(x) = e^x$ at $\bar{x} = 0$. Compute $r/(\delta x)^2$ for $\delta x = 0.1, 0.01, 0.001, 0.0001$. What value does the ratio converge to? Relate this to the Taylor series of e^x.

3. **Tangent space of SO(3).** The rotation group SO(3) consists of 3×3 orthogonal matrices with determinant 1. Using the constraint $R^TR = I$, show that the tangent space at $R = I$ consists of skew-symmetric matrices. What is the dimension of this tangent space?

4. **Tangent space of a torus.** The torus $\mathbb{T}^2$ is parameterized by two angles $(\theta_1, \theta_2) \in [0, 2\pi)^2$. Describe the tangent space at an arbitrary point. Is it isomorphic to $\mathbb{R}^2$? Explain why the tangent space is "flat" even though the torus is curved.

5. **When linearization is not an approximation.** Consider the system $\dot{x} = ax + bx^3$ for constants a, b. Compute the linearization at $\bar{x} = 0$. In what sense is this linearization "exact"? In what sense is it "approximate"? Use the language of Axiom 1 (tangent-space exactness) to articulate the distinction.

6. **Curvature and residual.** For $f(x) = \tan(x)$ at $\bar{x} = 0$, compute $f''(0)$ and predict the leading-order residual behavior. Construct a scaling table to verify. Compare with the $\sin(x)$ example in the text.

7. **Connection (Axiom 3).** For a 2D system $\dot{x}_1 = x_2$, $\dot{x}_2 = -\sin(x_1)$ (simple pendulum), compute the Jacobian $A(\bar{\boldsymbol{x}})$ at three different equilibria: $\bar{\boldsymbol{x}} = (0, 0)$, $(\pi, 0)$, and $(0, 1)$. How do the eigenvalues change? Interpret the change geometrically.

8. **Programming exercise.** Implement a function that, given $f : \mathbb{R}^n \to \mathbb{R}^n$ and a point $\bar{\boldsymbol{x}}$, computes the Fréchet derivative numerically using finite differences. Test it on the pendulum system from Exercise 7. Compare with the analytical Jacobian.

Chapter 2

Variational Dynamics and the Moving Frame

In Plain Language 2.1. Linearizing the Curve

Even though a nonlinear system curves and twists through state space, tiny errors around a reference trajectory evolve by a linear rule that we can compute and exploit. This chapter derives that rule—the variational equation—from first principles, constructs its exact solution via the Peano-Baker series, and shows how it provides a "moving local coordinate system" that travels along with the nominal motion. We examine rigorous proofs of its fundamental properties, connect it to sensitivity analysis and optimal control, and confront the numerical challenges in computing it accurately.

2.1 From Global Nonlinearity to Local Linearity

2.1.1 Setup: Nominal and Perturbed Trajectories

The variational equation is one of the most important tools in dynamical systems theory [Kha02, Sas99]. It provides an exact linear characterization of infinitesimal perturbations around a nominal trajectory and is fundamental to stability analysis, sensitivity computation, and optimal control [PBGM62, SL91].

Given the control-affine dynamics

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\,\boldsymbol{u}, \quad \boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{u} \in \mathbb{R}^m, \tag{2.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}^n$ and $G : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are smooth, suppose we have a *nominal trajectory* $\bar{\boldsymbol{x}}(t)$ generated by a nominal input $\bar{\boldsymbol{u}}(t)$:

$$\dot{\bar{\boldsymbol{x}}}(t) = f\big(\bar{\boldsymbol{x}}(t)\big) + G\big(\bar{\boldsymbol{x}}(t)\big)\,\bar{\boldsymbol{u}}(t). \tag{2.2}$$

Consider a nearby trajectory $\boldsymbol{x}(t) = \bar{\boldsymbol{x}}(t) + \delta\boldsymbol{x}(t)$ under a perturbed input $\boldsymbol{u}(t) = \bar{\boldsymbol{u}}(t) + \delta\boldsymbol{u}(t)$ with $\|\delta\boldsymbol{x}\|$ small. Then $\boldsymbol{x}$ satisfies

$$\dot{\bar{\boldsymbol{x}}} + \dot{\delta\boldsymbol{x}} = f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}) + G(\bar{\boldsymbol{x}} + \delta\boldsymbol{x})(\bar{\boldsymbol{u}} + \delta\boldsymbol{u}). \tag{2.3}$$

2.1.2 Derivation of the Variational Equation From First Principles

We expand the right-hand side of (2.3) in a Taylor series about $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}})$.

Expansion of $f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x})$

Using the multivariate Taylor theorem,

$$f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}) = f(\bar{\boldsymbol{x}}) + \frac{\partial f}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} \delta\boldsymbol{x} + \frac{1}{2}\int_0^1 (1-\tau)\frac{\partial^2 f}{\partial \boldsymbol{x}^2}\bigg|_{\bar{\boldsymbol{x}}+\tau\delta\boldsymbol{x}} (\delta\boldsymbol{x}, \delta\boldsymbol{x})\,\mathrm{d}\tau. \tag{2.4}$$

The first derivative is the Jacobian $J_f(\bar{\boldsymbol{x}}) := \partial f/\partial \boldsymbol{x}|_{\bar{\boldsymbol{x}}}$, which is an $n \times n$ matrix. The second derivative term represents the Hessian applied to the perturbation: for each state component i, we have

$$\left[\frac{\partial^2 f_i}{\partial \boldsymbol{x}^2}(\delta\boldsymbol{x}, \delta\boldsymbol{x})\right]_i = \sum_{j,k} \frac{\partial^2 f_i}{\partial x_j \partial x_k}\delta\boldsymbol{x}_j\,\delta\boldsymbol{x}_k.$$

The integral bound on the second derivative is $O(\|\delta\boldsymbol{x}\|^2)$. Let us denote

$$R_f(\delta\boldsymbol{x}) := \frac{1}{2}\int_0^1 (1-\tau)\frac{\partial^2 f}{\partial \boldsymbol{x}^2}\bigg|_{\bar{\boldsymbol{x}}+\tau\delta\boldsymbol{x}} (\delta\boldsymbol{x}, \delta\boldsymbol{x})\,\mathrm{d}\tau.$$

Then

$$f(\bar{\boldsymbol{x}}+\delta\boldsymbol{x}) = f(\bar{\boldsymbol{x}}) + J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x} + R_f(\delta\boldsymbol{x}), \quad \|R_f(\delta\boldsymbol{x})\| = O(\|\delta\boldsymbol{x}\|^2). \tag{2.5}$$

Expansion of $G(\bar{\boldsymbol{x}}+\delta\boldsymbol{x})(\bar{\boldsymbol{u}}+\delta\boldsymbol{u})$

Similarly, expand $G(\bar{\boldsymbol{x}}+\delta\boldsymbol{x})$ as

$$G(\bar{\boldsymbol{x}}+\delta\boldsymbol{x}) = G(\bar{\boldsymbol{x}}) + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} \delta\boldsymbol{x} + O(\|\delta\boldsymbol{x}\|^2). \tag{2.6}$$

Thus

$$G(\bar{\boldsymbol{x}}+\delta\boldsymbol{x})(\bar{\boldsymbol{u}}+\delta\boldsymbol{u}) = \left[G(\bar{\boldsymbol{x}}) + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} \delta\boldsymbol{x} + O(\|\delta\boldsymbol{x}\|^2)\right](\bar{\boldsymbol{u}}+\delta\boldsymbol{u}) \tag{2.7}$$

$$= G(\bar{\boldsymbol{x}})\bar{\boldsymbol{u}} + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u} + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} (\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}}) + O(\|\delta\boldsymbol{x}\|^2) + O(\|\delta\boldsymbol{x}\|\|\delta\boldsymbol{u}\|). \tag{2.8}$$

Here the term $\frac{\partial G}{\partial \boldsymbol{x}}(\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}})$ involves the directional derivative of G with respect to $\boldsymbol{x}$ applied to $\delta\boldsymbol{x}$, scaled by $\bar{\boldsymbol{u}}$. In index form, the i-th component is $\sum_{j,k}\frac{\partial G_{ij}}{\partial x_k}\delta\boldsymbol{x}_k\,\bar{u}_j$.

Combining Terms

Substituting (2.5) and (2.8) into (2.3):

$$\dot{\bar{\boldsymbol{x}}} + \dot{\delta\boldsymbol{x}} = f(\bar{\boldsymbol{x}}) + J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x} + R_f(\delta\boldsymbol{x}) \tag{2.9}$$

$$+ G(\bar{\boldsymbol{x}})\bar{\boldsymbol{u}} + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u} + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} (\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}}) + O(\|\delta\boldsymbol{x}\|^2). \tag{2.10}$$

Subtracting the nominal equation (2.2):

$$\dot{\delta\boldsymbol{x}} = J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x} + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} (\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}}) + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u} + R_f(\delta\boldsymbol{x}) + O(\|\delta\boldsymbol{x}\|^2). \tag{2.11}$$

Dropping Higher-Order Terms

To obtain the linearized (first-order) equation, we retain only terms linear in $\delta\boldsymbol{x}$ and $\delta\boldsymbol{u}$ and drop all $O(\|\delta\boldsymbol{x}\|^2)$ terms:

$$\dot{\delta\boldsymbol{x}} = J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x} + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}} (\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}}) + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u}. \tag{2.12}$$

The $R_f(\delta\boldsymbol{x})$ and $O(\|\delta\boldsymbol{x}\|^2)$ terms represent curvature of the vector field. They vanish in the limit $\|\delta\boldsymbol{x}\|\to 0$, making them negligible for infinitesimal perturbations. Note that the term $\frac{\partial G}{\partial \boldsymbol{x}}(\delta\boldsymbol{x}\otimes\bar{\boldsymbol{u}})$ is linear in $\delta\boldsymbol{x}$ (when $\bar{\boldsymbol{u}}$ is fixed along the nominal trajectory), so it is retained.

Compact Matrix Form

Define

$$\mathbf{A}(t) := \frac{\partial}{\partial \boldsymbol{x}}\big[f(\boldsymbol{x}) + G(\boldsymbol{x})\bar{\boldsymbol{u}}(t)\big]\bigg|_{\boldsymbol{x}=\bar{\boldsymbol{x}}(t)}, \qquad \mathbf{B}(t) := G\big(\bar{\boldsymbol{x}}(t)\big). \tag{2.13}$$

The matrix $\mathbf{A}(t)$ is the Jacobian of the vector field $f + G\bar{\boldsymbol{u}}$ along the nominal trajectory. Explicitly,

$$\mathbf{A}(t) = J_f(\bar{\boldsymbol{x}}(t)) + \frac{\partial G}{\partial \boldsymbol{x}}\bigg|_{\bar{\boldsymbol{x}}(t)} (\mathbf{1}\otimes\bar{\boldsymbol{u}}(t))^T, \tag{2.14}$$

where the second term arises from the control input contribution to the state sensitivity. Then (2.12) becomes:

$$\boxed{\dot{\delta\boldsymbol{x}}(t) = \mathbf{A}(t)\,\delta\boldsymbol{x}(t) + \mathbf{B}(t)\,\delta\boldsymbol{u}(t),} \tag{2.15}$$

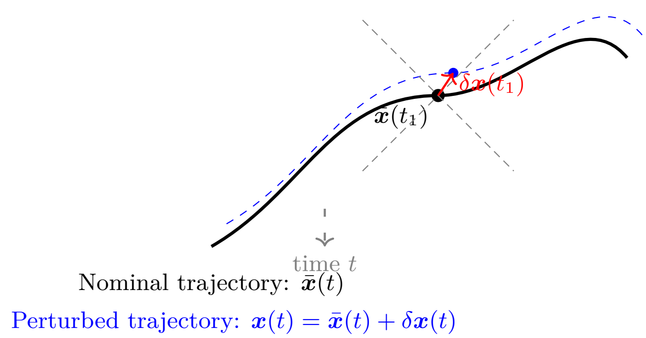
This is the **variational equation**: a linear, time-varying ODE that governs the evolution of infinitesimal perturbations.

Design Implication

The error incurred by dropping the $O(\|\delta\boldsymbol{x}\|^2)$ terms in the derivation is the residual. For initial conditions with $\|\delta\boldsymbol{x}(t_0)\| \ll 1$, this residual remains small (growing only quadratically with $\|\delta\boldsymbol{x}\|$) over a finite time interval. This is the precise justification for using linearization to study local stability and design feedback control.

Geometric Intuition

The variational equation (2.15) is a *linear, time-varying* system that lives in the tangent space along $\bar{\boldsymbol{x}}(t)$. It describes how infinitesimal perturbations propagate. The matrices $\mathbf{A}(t)$ and $\mathbf{B}(t)$ change as the nominal trajectory evolves, reflecting the fact that the tangent space itself moves with the base point. Notice that $\mathbf{A}$ depends on the nominal input $\bar{\boldsymbol{u}}(t)$: different reference inputs generate different tangent-space dynamics.



The key insight is that the perturbation $\delta\boldsymbol{x}(t)$ evolves linearly in the tangent space along the nominal trajectory, governed by the variational equation (2.15). This linear evolution is exact at the infinitesimal level.

2.2 The State Transition Matrix

The state transition matrix (STM) is the fundamental solution object for linear time-varying systems. It is covered extensively in classical control texts [Kha02, Sas99] and forms the basis for sensitivity analysis and optimal control computations.

2.2.1 Definition and Fundamental ODE

When $\delta\boldsymbol{u} = 0$, the variational equation reduces to the homogeneous system

$$\dot{\delta\boldsymbol{x}} = \mathbf{A}(t)\,\delta\boldsymbol{x}. \tag{2.16}$$

The solution is characterized by the **state transition matrix** (STM) $\Phi(t, t_0)$, defined as the unique $n \times n$ matrix satisfying

$$\boxed{\dot{\Phi}(t, t_0) = \mathbf{A}(t)\,\Phi(t, t_0), \quad \Phi(t_0, t_0) = \mathbf{I}_n.} \tag{2.17}$$

Once $\Phi(t, t_0)$ is known, the solution to (2.16) with initial condition $\delta\boldsymbol{x}(t_0)$ is

$$\delta\boldsymbol{x}(t) = \Phi(t, t_0)\,\delta\boldsymbol{x}(t_0). \tag{2.18}$$

Remark 2.1. The STM is not a solution in the usual sense; it is a fundamental solution matrix. Each column of $\Phi(t, t_0)$ is a solution to the homogeneous equation with a unit initial condition. Specifically, if $\delta\boldsymbol{x}_i(t_0) = \mathbf{e}_i$ (the i-th unit vector), then $\delta\boldsymbol{x}_i(t) = \Phi(t, t_0)\mathbf{e}_i$ is the i-th column of $\Phi(t, t_0)$.

2.2.2 Properties and Rigorous Proofs

The state transition matrix inherits fundamental properties from the structure of linear ODEs:

Theorem 2.2 (Semigroup Property)**.** *For any $t_0 \le t_1 \le t_2$,*

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\,\Phi(t_1, t_0). \tag{2.19}$$

Proof. For any initial condition $\delta\boldsymbol{x}_0 \in \mathbb{R}^n$, define $\delta\boldsymbol{x}(t) := \Phi(t, t_0)\delta\boldsymbol{x}_0$, which is the unique solution to $\delta\dot{\boldsymbol{x}} = \mathbf{A}(t)\delta\boldsymbol{x}$ with $\delta\boldsymbol{x}(t_0) = \delta\boldsymbol{x}_0$.

At time t_1, we have $\delta\boldsymbol{x}(t_1) = \Phi(t_1, t_0)\delta\boldsymbol{x}_0$. The solution on $[t_1, t_2]$ with initial condition $\delta\boldsymbol{x}(t_1)$ is, by definition, $\delta\boldsymbol{x}(t) = \Phi(t, t_1)\delta\boldsymbol{x}(t_1) = \Phi(t, t_1)\Phi(t_1, t_0)\delta\boldsymbol{x}_0$.

Evaluating at $t = t_2$:

$$\delta\boldsymbol{x}(t_2) = \Phi(t_2, t_1)\,\Phi(t_1, t_0)\,\delta\boldsymbol{x}_0.$$

But also $\delta\boldsymbol{x}(t_2) = \Phi(t_2, t_0)\delta\boldsymbol{x}_0$ by the definition of $\Phi(t_2, t_0)$. Since $\delta\boldsymbol{x}_0$ was arbitrary, $\Phi(t_2, t_0) = \Phi(t_2, t_1)\,\Phi(t_1, t_0)$. □

Theorem 2.3 (Invertibility)**.** *The state transition matrix is invertible for all t, t_0, and*

$$\Phi(t, t_0)^{-1} = \Phi(t_0, t). \tag{2.20}$$

Proof. From the semigroup property,

$$\Phi(t, t_0)\,\Phi(t_0, t) = \Phi(t, t) = \mathbf{I}_n.$$

Thus $\Phi(t_0, t)$ is indeed the right inverse. Similarly, from $\Phi(t_0, t)\,\Phi(t, t_0) = \Phi(t_0, t_0) = \mathbf{I}_n$, it is also the left inverse. Hence $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$. □

Theorem 2.4 (Liouville's Formula)**.** *The determinant of the state transition matrix satisfies*

$$\det \Phi(t, t_0) = \exp\left(\int_{t_0}^{t} \operatorname{tr} \mathbf{A}(s)\,\mathrm{d}s\right). \tag{2.21}$$

Proof. We use Jacobi's formula:

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(\Phi(t, t_0)) = \det(\Phi)\ \operatorname{tr}\left(\Phi^{-1}\,\frac{\mathrm{d}\Phi}{\mathrm{d}t}\right). \tag{2.22}$$

Since $\dot{\Phi} = \mathbf{A}\,\Phi$, this gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(\Phi) = \det(\Phi)\ \operatorname{tr}(\Phi^{-1}\,\mathbf{A}\,\Phi) \tag{2.23}$$
$$= \det(\Phi)\ \operatorname{tr}(\mathbf{A}), \tag{2.24}$$

where the last step uses the similarity invariance of the trace: $\operatorname{tr}(\Phi^{-1}\mathbf{A}\Phi) = \operatorname{tr}(\mathbf{A})$.

Integrating from t_0 to t, with initial condition $\det(\Phi(t_0, t_0)) = \det(\mathbf{I}_n) = 1$:

$$\det(\Phi(t, t_0)) = \exp\left(\int_{t_0}^{t} \operatorname{tr}(\mathbf{A}(s))\,\mathrm{d}s\right). \tag{2.25}$$

□

Principle 5: Integration Preserves Exactness The state transition matrix $\Phi(t, t_0)$ accumulates *exact* infinitesimal linear contributions along the trajectory. Each infinitesimal step $\mathrm{d}(\delta \boldsymbol{x}) = \mathbf{A}(t)\delta \boldsymbol{x}\, \mathrm{d}t$ is exact in its tangent space. The integral

$$\delta \boldsymbol{x}(t_1) = \Phi(t_1, t_0)\delta \boldsymbol{x}(t_0) + \int_{t_0}^{t_1} \Phi(t_1, s)\mathbf{B}(s)\delta \boldsymbol{u}(s)\, \mathrm{d}s$$

is the exact accumulation of these exact local effects. Residuals arise from curvature (when we compare the true finite-displacement trajectory to the linear prediction), not from integration itself.

2.3 The Peano-Baker Series and Exact Solution

While the ODE (2.17) defines the STM implicitly, we can construct an explicit series solution called the Peano-Baker series [Mag54] (also known as the Dyson series in physics). This series has deep connections to Lie group theory and geometric analysis [dC92, AM78].

2.3.1 Derivation of the Series

Suppose we have a formal integral representation of (2.17):

$$\Phi(t, t_0) = \mathbf{I} + \int_{t_0}^{t} \mathbf{A}(s_1)\, \Phi(s_1, t_0)\, \mathrm{d}s_1. \tag{2.26}$$

Verify: differentiating both sides with respect to t gives $\dot{\Phi} = \mathbf{A}(t)\Phi(t, t_0)$, and at $t = t_0$, $\Phi(t_0, t_0) = \mathbf{I}$. This integral equation is equivalent to the ODE.

Now, substitute the integral equation into itself iteratively:

$$\begin{aligned}
\Phi(t, t_0) &= \mathbf{I} + \int_{t_0}^{t} \mathbf{A}(s_1) \left[\mathbf{I} + \int_{t_0}^{s_1} \mathbf{A}(s_2)\Phi(s_2, t_0)\, \mathrm{d}s_2\right] \mathrm{d}s_1 && (2.27)\\
&= \mathbf{I} + \int_{t_0}^{t} \mathbf{A}(s_1)\, \mathrm{d}s_1 && (2.28)\\
&\quad + \int_{t_0}^{t} \mathbf{A}(s_1) \int_{t_0}^{s_1} \mathbf{A}(s_2)\Phi(s_2, t_0)\, \mathrm{d}s_2\, \mathrm{d}s_1. && (2.29)
\end{aligned}$$

Continuing this recursion, we obtain the formal series:

$$\boxed{\Phi(t, t_0) = \sum_{k=0}^{\infty} \Phi_k(t, t_0),} \tag{2.30}$$

where

$$\begin{aligned}
\Phi_0(t, t_0) &:= \mathbf{I}, && (2.31)\\
\Phi_1(t, t_0) &:= \int_{t_0}^{t} \mathbf{A}(s_1)\, \mathrm{d}s_1, && (2.32)\\
\Phi_2(t, t_0) &:= \int_{t_0}^{t} \int_{t_0}^{s_1} \mathbf{A}(s_1)\mathbf{A}(s_2)\, \mathrm{d}s_2\, \mathrm{d}s_1, && (2.33)\\
\Phi_k(t, t_0) &:= \int_{t_0}^{t} \int_{t_0}^{s_1} \cdots \int_{t_0}^{s_{k-1}} \mathbf{A}(s_1)\mathbf{A}(s_2)\cdots\mathbf{A}(s_k)\, \mathrm{d}s_k \cdots \mathrm{d}s_1. && (2.34)
\end{aligned}$$

This is the **Peano-Baker series**: an infinite product series where Φ_k represents the cumulative effect of k applications of $\mathbf{A}$, with all possible orderings integrated over the time interval.

2.3.2 Convergence

For time-varying linear systems with bounded coefficients, the series converges uniformly:

Proposition 2.5 (Peano-Baker Convergence)**.** *Let $\mathbf{A}(t)$ be piecewise continuous with $\|\mathbf{A}(t)\| \leq M$ for all $t \in [t_0, T]$. Then the Peano-Baker series converges absolutely and uniformly to the unique solution of (2.17).*

Proof. We bound the series term by term. For the k-th term:

$$\|\Phi_k(t, t_0)\| \leq \int_{t_0}^{t} \int_{t_0}^{s_1} \cdots \int_{t_0}^{s_{k-1}} \|\mathbf{A}(s_1)\| \cdots \|\mathbf{A}(s_k)\| \, \mathrm{d}s_k \cdots \mathrm{d}s_1 \tag{2.35}$$

$$\leq M^k \int_{t_0}^{t} \int_{t_0}^{s_1} \cdots \int_{t_0}^{s_{k-1}} \mathrm{d}s_k \cdots \mathrm{d}s_1 \tag{2.36}$$

$$= M^k \frac{(t - t_0)^k}{k!}. \tag{2.37}$$

Thus the series is bounded by the exponential series:

$$\sum_{k=0}^{\infty} \|\Phi_k(t, t_0)\| \leq \sum_{k=0}^{\infty} \frac{M^k (t - t_0)^k}{k!} = e^{M(t-t_0)},$$

which is finite and independent of the ordering in $\mathbf{A}$. Hence the series converges uniformly on any finite interval $[t_0, T]$, and the sum satisfies the ODE (2.17). □

Remark 2.6. The Peano-Baker series highlights a crucial feature: even though $[\mathbf{A}(t_1), \mathbf{A}(t_2)] \neq 0$ in general (the matrices do not commute), the series still provides the exact solution. The summation over all orderings effectively accounts for non-commutativity. This is in stark contrast to the *Magnus expansion* from perturbation theory, which organizes the series differently to emphasize commutators.

2.4 Numerical Computation of the State Transition Matrix

For practical implementation, we rarely sum the Peano-Baker series explicitly. Instead, we integrate the matrix ODE (2.17) numerically.

2.4.1 Direct Integration of the Matrix ODE

The most straightforward approach is to view $\Phi(t, t_0)$ as an $n \times n$ matrix and integrate

$$\dot{\Phi} = \mathbf{A}(t)\, \Phi, \quad \Phi(t_0) = \mathbf{I}_n, \tag{2.38}$$

using a standard ODE solver. This requires integrating n^2 coupled scalar ODEs (the entries of Φ).

Design Implication

At each time step, we compute the Jacobian $\mathbf{A}(t)$ by evaluating the partial derivatives of the nominal dynamics at $(\bar{\boldsymbol{x}}(t), \bar{\boldsymbol{u}}(t))$. This requires either automatic differentiation, manual Jacobian formulas, or finite-difference approximations of the derivatives. In production code, automatic differentiation is often most reliable and accurate.

2.4.2 The Matrix Exponential and Padé Approximation

For time-invariant systems (constant $\mathbf{A}$), the STM has a closed form:

$$\Phi(t, t_0) = \exp\big[\mathbf{A}(t - t_0)\big]. \tag{2.39}$$

The matrix exponential $\exp(\mathbf{A}\tau) = \sum_{k=0}^{\infty} \frac{(\mathbf{A}\tau)^k}{k!}$ is a fundamental object in control theory. Computing it accurately is essential but non-trivial.

Padé Approximation

The Padé rational approximation $[\ell/m]$ of $\exp(x)$ is the ratio of two polynomials that matches the Taylor series up to order $\ell + m$:

$$[\ell/m]_{\exp}(x) = \frac{N_\ell(x)}{D_m(x)}, \quad N_\ell, D_m \text{ are polynomials of degree } \ell, m. \tag{2.40}$$

For the matrix exponential, we compute $N_\ell(\mathbf{A}\tau)$ and $D_m(\mathbf{A}\tau)$ as matrix polynomials and then solve the linear system

$$D_m(\mathbf{A}\tau)\,\Phi(t, t_0) = N_\ell(\mathbf{A}\tau), \tag{2.41}$$

which gives Φ with spectral accuracy.

A common choice is the [7/8] Padé approximation with scaling and squaring:

1. Scale: Find the smallest integer q such that $\|\mathbf{A}\tau/2^q\| < \rho$ for some threshold ρ (typically 0.5).
2. Compute: Apply the [7/8] Padé approximant to $\mathbf{A}\tau/2^q$.
3. Square: Repeatedly square the result q times to recover $\exp(\mathbf{A}\tau)$.

This approach achieves unit roundoff accuracy in double precision with minimal computational cost.

2.4.3 Computational Cost

For an $n \times n$ STM:

- **Direct integration:** Each step of an RK4 integrator performs ~ 4 matrix multiplications (one per stage), each costing $O(n^3)$. For N steps: $O(4Nn^3)$ flops.
- **Matrix exponential (scaling & squaring):** Computing $N_\ell(\mathbf{A})$ costs $O(n^3)$; solving the linear system costs $O(n^3)$ (via LU factorization). Squaring q times adds $qO(n^3)$. Total per call: $O((q+2)n^3)$ flops.
- **Evaluation cost per time step:** Regardless of method, if we evaluate Φ at N time points, the dominant cost is $O(Nn^3)$.

For large n (e.g., $n = 100$), the $O(n^3)$ factor is significant. Specialized structure in $\mathbf{A}$ (sparse, low-rank, block-diagonal) can be exploited to reduce cost. For general dense problems, direct integration with adaptive step-size control is often most efficient in practice.

Common Misconception

Computing Φ for stiff systems (those with widely separated time scales) is notoriously ill-conditioned. We discuss remedies in Section 2.10.

2.5 Variation of Constants Formula

When both initial perturbations $\delta\boldsymbol{x}(t_0) \neq 0$ and control perturbations $\delta\boldsymbol{u}(t) \neq 0$ are present, the complete solution of the variational equation is given by the **variation of constants** (also called Duhamel formula):

$$\boxed{\delta\boldsymbol{x}(t) = \Phi(t, t_0)\,\delta\boldsymbol{x}(t_0) + \int_{t_0}^{t} \Phi(t, s)\,\mathbf{B}(s)\,\delta\boldsymbol{u}(s)\,\mathrm{d}s.} \tag{2.42}$$

2.5.1 Derivation

Consider the non-homogeneous equation

$$\dot{\delta \boldsymbol{x}} = \mathbf{A}(t)\delta \boldsymbol{x} + \mathbf{B}(t)\delta \boldsymbol{u}(t). \tag{2.43}$$

We seek a particular solution using the method of variation of parameters. Assume

$$\delta \boldsymbol{x}(t) = \Phi(t, t_0)\, \mathbf{c}(t), \tag{2.44}$$

where $\mathbf{c}(t)$ is a time-varying coefficient vector to be determined. Then

$$\dot{\delta \boldsymbol{x}} = \dot{\Phi}(t, t_0)\, \mathbf{c} + \Phi(t, t_0)\, \dot{\mathbf{c}} = \mathbf{A}(t)\Phi(t, t_0)\, \mathbf{c} + \Phi(t, t_0)\, \dot{\mathbf{c}}. \tag{2.45}$$

Matching with the desired dynamics:

$$\mathbf{A}(t)\Phi(t, t_0)\, \mathbf{c} + \Phi(t, t_0)\, \dot{\mathbf{c}} = \mathbf{A}(t)\Phi(t, t_0)\, \mathbf{c} + \mathbf{B}(t)\delta \boldsymbol{u}. \tag{2.46}$$

This simplifies to

$$\Phi(t, t_0)\, \dot{\mathbf{c}} = \mathbf{B}(t)\delta \boldsymbol{u}, \quad \Rightarrow \quad \dot{\mathbf{c}} = \Phi(t_0, t)\mathbf{B}(t)\delta \boldsymbol{u}, \tag{2.47}$$

using $\Phi(t_0, t) = \Phi(t, t_0)^{-1}$. Integrating from t_0 to t:

$$\mathbf{c}(t) = \mathbf{c}(t_0) + \int_{t_0}^{t} \Phi(t_0, s)\mathbf{B}(s)\delta \boldsymbol{u}(s)\, \mathrm{d}s. \tag{2.48}$$

With the initial condition $\delta \boldsymbol{x}(t_0) = \Phi(t_0, t_0)\mathbf{c}(t_0) = \mathbf{c}(t_0)$, we have

$$\delta \boldsymbol{x}(t) = \Phi(t, t_0) \left[\delta \boldsymbol{x}(t_0) + \int_{t_0}^{t} \Phi(t_0, s)\mathbf{B}(s)\delta \boldsymbol{u}(s)\, \mathrm{d}s \right]. \tag{2.49}$$

Using the semigroup property $\Phi(t, t_0)\Phi(t_0, s) = \Phi(t, s)$:

$$\delta \boldsymbol{x}(t) = \Phi(t, t_0)\delta \boldsymbol{x}(t_0) + \int_{t_0}^{t} \Phi(t, s)\mathbf{B}(s)\delta \boldsymbol{u}(s)\, \mathrm{d}s, \tag{2.50}$$

which is the Duhamel formula.

2.5.2 Physical Interpretation

The Duhamel formula decomposes the state perturbation into two contributions:

1. **Homogeneous term** $\Phi(t, t_0)\delta \boldsymbol{x}(t_0)$: The initial perturbation $\delta \boldsymbol{x}(t_0)$ is propagated forward via the STM. This is the response to initial conditions alone.
2. **Forced term** $\int_{t_0}^{t} \Phi(t, s)\mathbf{B}(s)\delta \boldsymbol{u}(s)\, \mathrm{d}s$: At each time s, a control perturbation $\delta \boldsymbol{u}(s)$ is applied. It enters through the actuator channels $\mathbf{B}(s)$, and its effect is then propagated forward from time s to time t by the STM $\Phi(t, s)$. The integral sums the contributions of all past control inputs, weighted by their propagation matrices.

Design Implication

The Duhamel formula is the engine behind sensitivity analysis, adjoint methods, and trajectory optimization. It tells us how each input perturbation at time s contributes to the final state perturbation at time t, weighted by the transition matrix $\Phi(t, s)$. This is the precise sense in which input effects “add up”—they superpose linearly in the tangent space. For control design, this formula shows which times and which channels have the most leverage: early times (with large $\Phi(t, s)$) and actuator-rich directions (large column spaces of $\mathbf{B}(s)$) are most influential.

2.6 Discrete-Time Variational Dynamics

For computational implementation, we discretize the continuous-time dynamics. Given a sampling period h and zero-order-hold (ZOH) input, the discrete-time flow map is

$$\boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k), \tag{2.51}$$

where f_d is the time-h solution operator (flow map) of the continuous-time system (2.1). Formally,

$$f_d(\boldsymbol{x}_k, \boldsymbol{u}_k) := \Phi_c\big(t_k + h, t_k; \boldsymbol{u}_k\big), \tag{2.52}$$

where Φ_c denotes the flow of the continuous-time system with zero-order-held input $\boldsymbol{u}(t) = \boldsymbol{u}_k$ for $t \in [t_k, t_k{+}h)$.

2.6.1 Linearization and Discrete Jacobians

Linearizing the discrete flow map around the nominal sequence $(\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{u}}_k)$:

$$\delta\boldsymbol{x}_{k+1} = \mathbf{A}_k\, \delta\boldsymbol{x}_k + \mathbf{B}_k\, \delta\boldsymbol{u}_k, \tag{2.53}$$

where

$$\mathbf{A}_k = \frac{\partial f_d}{\partial \boldsymbol{x}}\bigg|_{(\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{u}}_k)}, \qquad \mathbf{B}_k = \frac{\partial f_d}{\partial \boldsymbol{u}}\bigg|_{(\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{u}}_k)}. \tag{2.54}$$

2.6.2 Computing Discrete Jacobians via Matrix Exponential

For a control-affine continuous system, the discrete flow under ZOH control satisfies

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \int_{t_k}^{t_k+h} \Phi_c(t_k + h, \tau)\, G(\bar{\boldsymbol{x}}(\tau))\, \bar{\boldsymbol{u}}_k\, \mathrm{d}\tau, \tag{2.55}$$

where $\Phi_c(t_k + h, \tau)$ is the state transition matrix of the linearized system along the nominal trajectory.

At the nominal trajectory point $\bar{\boldsymbol{x}}_k$, assuming the input is held constant at $\bar{\boldsymbol{u}}_k$ over the interval:

$$\bar{\boldsymbol{x}}_{k+1} = \bar{\boldsymbol{x}}_k + \int_{t_k}^{t_k+h} \Phi_c\big(t_k + h, \tau; \bar{\boldsymbol{u}}_k\big) G(\bar{\boldsymbol{x}}(\tau))\bar{\boldsymbol{u}}_k\, \mathrm{d}\tau. \tag{2.56}$$

For the Jacobians:

$$\mathbf{A}_k = \Phi_c(t_k + h, t_k; \bar{\boldsymbol{u}}_k)\bigg|_{\bar{\boldsymbol{x}}_k}, \tag{2.57}$$

which is the STM over one sampling interval.

$$\mathbf{B}_k = \int_{t_k}^{t_k+h} \Phi_c(t_k + h, \tau; \bar{\boldsymbol{u}}_k)\, G(\bar{\boldsymbol{x}}(\tau))\, \mathrm{d}\tau, \tag{2.58}$$

which is an integral of the STM weighted by the input matrix.

For time-invariant systems with constant A and B:

$$\mathbf{A}_k = \exp(Ah), \quad \mathbf{B}_k = \left(\int_0^h \exp(As)\, \mathrm{d}s\right) B = A^{-1}\left[\exp(Ah) - \mathbf{I}\right] B. \tag{2.59}$$

2.6.3 Discrete State Transition Matrix

The discrete state transition matrix from stage k to stage $\ell > k$ is the product of individual Jacobians:

$$\Phi_{d,\ell,k} = \mathbf{A}_{\ell-1}\, \mathbf{A}_{\ell-2} \cdots \mathbf{A}_k, \quad \ell > k, \qquad \Phi_{d,k,k} = \mathbf{I}_n. \tag{2.60}$$

This satisfies a discrete semigroup property: $\Phi_{d,\ell,k} = \Phi_{d,\ell,j}\Phi_{d,j,k}$ for $k \le j \le \ell$.

Remark 2.7. Unlike the continuous STM, the discrete STM is not a fundamental solution to a matrix ODE; it is a pure product of matrices. For numerical work, computing $\Phi_{d,\ell,k}$ by successive multiplication is standard, but for large $\ell - k$, this accumulates rounding errors and can become numerically unstable (Section 2.10).

2.7 A Worked Example: Linearized Pendulum

To concretize the theory, we compute the variational equation and STM explicitly for a simple pendulum linearized about the downward equilibrium.

2.7.1 Pendulum Dynamics

Consider a simple pendulum with mass m, length ℓ, and a motor torque τ applied at the pivot:

$$\ddot{\theta} = -\frac{g}{\ell}\sin\theta + \frac{\tau}{m\ell^2}. \tag{2.61}$$

Writing this as a first-order system with $x_1 = \theta$ and $x_2 = \dot{\theta}$:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -\frac{g}{\ell}\sin x_1 + \frac{u}{m\ell^2} \end{pmatrix}, \tag{2.62}$$

where $u = \tau$ is the control input.

2.7.2 Nominal Trajectory

Suppose the nominal trajectory is the equilibrium at the downward position with zero control:

$$\bar{\boldsymbol{x}}(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \bar{u}(t) = 0. \tag{2.63}$$

Verify: $\dot{\bar{x}}_1 = 0$, $\dot{\bar{x}}_2 = -\frac{g}{\ell}\sin(0) + 0 = 0$. This is indeed a fixed point.

2.7.3 Jacobians

At the point $(\bar{\boldsymbol{x}}, \bar{u}) = (0, 0)$:

$$\mathbf{A} = \frac{\partial}{\partial \boldsymbol{x}} \begin{pmatrix} x_2 \\ -\frac{g}{\ell}\sin x_1 \end{pmatrix}\Bigg|_{(0,0)} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{\ell}\cos(0) & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{\ell} & 0 \end{pmatrix}. \tag{2.64}$$

$$\mathbf{B} = \frac{\partial}{\partial u} \begin{pmatrix} x_2 \\ -\frac{g}{\ell}\sin x_1 + \frac{u}{m\ell^2} \end{pmatrix}\Bigg|_{(0,0)} = \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix}. \tag{2.65}$$

The variational equation is thus

$$\begin{pmatrix} \delta\dot{x}_1 \\ \delta\dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{\ell} & 0 \end{pmatrix} \begin{pmatrix} \delta x_1 \\ \delta x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix} \delta u. \tag{2.66}$$

2.7.4 Eigenvalues and State Transition Matrix

For the time-invariant case, we compute the eigenvalues of $\mathbf{A}$:

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \det \begin{pmatrix} \lambda & -1 \\ \frac{g}{\ell} & \lambda \end{pmatrix} = \lambda^2 + \frac{g}{\ell} = 0. \tag{2.67}$$

Thus $\lambda = \pm i\sqrt{g/\ell}$, which are purely imaginary. Let $\omega_0 := \sqrt{g/\ell}$ (the natural frequency). The matrix exponential can be computed using the formula for a 2D system:

$$\exp(\mathbf{A}t) = \cos(\omega_0 t)\mathbf{I} + \frac{\sin(\omega_0 t)}{\omega_0}\mathbf{A} = \begin{pmatrix} \cos(\omega_0 t) & \frac{\sin(\omega_0 t)}{\omega_0} \\ -\omega_0 \sin(\omega_0 t) & \cos(\omega_0 t) \end{pmatrix}. \tag{2.68}$$

Verify: $\frac{\mathrm{d}}{\mathrm{d}t}\Phi(t, 0) = \begin{pmatrix} -\omega_0 \sin(\omega_0 t) & \cos(\omega_0 t) \\ -\omega_0^2 \cos(\omega_0 t) & -\omega_0 \sin(\omega_0 t) \end{pmatrix}$. Comparing with $\mathbf{A}\Phi(t, 0)$:

$$\begin{pmatrix} 0 & 1 \\ -\frac{g}{\ell} & 0 \end{pmatrix} \begin{pmatrix} \cos(\omega_0 t) & \frac{\sin(\omega_0 t)}{\omega_0} \\ -\omega_0 \sin(\omega_0 t) & \cos(\omega_0 t) \end{pmatrix} = \begin{pmatrix} -\omega_0 \sin(\omega_0 t) & \cos(\omega_0 t) \\ -\omega_0^2 \cos(\omega_0 t) & -\omega_0 \sin(\omega_0 t) \end{pmatrix}. \tag{2.69}$$

Good (noting $-\omega_0^2 = -g/\ell$).

2.7.5 Numerical Example

For concreteness, let $\ell = 1$ m and $g = 10$ m/s^2 (rounded for clarity; other chapters use $g = 9.81$ m/s^2), so $\omega_0 = \sqrt{10} \approx 3.162$ rad/s. The natural period is $T = 2\pi/\omega_0 \approx 1.987$ s.

At time $t = 0.5$ s:

$$\omega_0 t = \sqrt{10} \times 0.5 \approx 1.581 \text{ rad}, \tag{2.70}$$

$$\cos(\omega_0 t) \approx \cos(1.581) \approx -0.0108, \tag{2.71}$$

$$\sin(\omega_0 t) \approx \sin(1.581) \approx 1.0000, \tag{2.72}$$

$$\frac{\sin(\omega_0 t)}{\omega_0} \approx \frac{1.0000}{3.162} \approx 0.3162. \tag{2.73}$$

Thus

$$\Phi(0.5, 0) \approx \begin{pmatrix} -0.0108 & 0.3162 \\ -3.162 & -0.0108 \end{pmatrix}. \tag{2.74}$$

An initial perturbation $\delta\boldsymbol{x}_0 = (0.1 \text{ rad}, 0)^\mathsf{T}$ (small angular displacement) would evolve as

$$\delta\boldsymbol{x}(0.5) = \Phi(0.5, 0)\delta\boldsymbol{x}_0 \approx \begin{pmatrix} -0.0108 \\ -0.3162 \end{pmatrix}, \tag{2.75}$$

meaning the angle perturbation has nearly oscillated back through zero, and a velocity perturbation of about -0.32 rad/s has developed.

2.7.6 Duhamel Integral With Impulse Control

Suppose we apply an impulse of torque $\delta u(t) = u_0\, \delta(t - t^*)$ at time $t^* = 0.25$ s. The response at time $t = 0.5$ s is

$$\delta\boldsymbol{x}(0.5) = \Phi(0.5, 0.25)\mathbf{B}\, u_0, \tag{2.76}$$

where

$$\Phi(0.5, 0.25) = \Phi(0.25, 0) = \begin{pmatrix} \cos(\sqrt{10} \times 0.25) & \frac{\sin(\sqrt{10}\times 0.25)}{\sqrt{10}} \\ -\sqrt{10}\sin(\sqrt{10} \times 0.25) & \cos(\sqrt{10} \times 0.25) \end{pmatrix}. \tag{2.77}$$

With $\sqrt{10} \times 0.25 \approx 0.7906$ rad:

$$\Phi(0.5, 0.25) \approx \begin{pmatrix} 0.7038 & 0.2499 \\ -0.7905 & 0.7038 \end{pmatrix}. \tag{2.78}$$

If $u_0 = 1$ N·m and $m = 1$ kg, then $\mathbf{B}u_0 = (0, 1)^\mathsf{T}$, and

$$\delta\boldsymbol{x}(0.5) \approx \begin{pmatrix} 0.2499 \\ 0.7038 \end{pmatrix}, \tag{2.79}$$

meaning a torque impulse at quarter-period induces a displacement of about 0.25 rad and velocity of 0.70 rad/s.

Remark 2.8. The numerical values are illustrative. In practice, one would solve this on a computer using ODE solvers (RK4, Runge-Kutta-Fehlberg) or leverage the analytic matrix exponential for this simple 2D system.

2.8 Sensitivity Analysis and Parameter Dependence

2.8.1 Parameter Sensitivity via the Variational Equation

Often we wish to understand how the state depends on a parameter $p \in \mathbb{R}$, where

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, p), \quad x \in \mathbb{R}^n, p \in \mathbb{R}^{n_p}. \tag{2.80}$$

Taking the partial derivative with respect to p:

$$\frac{\partial}{\partial p}\dot{\boldsymbol{x}} = \frac{\partial}{\partial p} f(\boldsymbol{x}, p) = \frac{\partial f}{\partial \boldsymbol{x}} \frac{\partial \boldsymbol{x}}{\partial p} + \frac{\partial f}{\partial p}. \tag{2.81}$$

Let $S(t) := \frac{\partial \boldsymbol{x}(t)}{\partial p}$ be the sensitivity (a matrix of shape $n \times n_p$). Then

$$\dot{S} = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}(t),p} S + \left.\frac{\partial f}{\partial p}\right|_{\boldsymbol{x}(t),p}, \quad S(t_0) = \frac{\partial \boldsymbol{x}(t_0)}{\partial p} = 0 \quad \text{(if } x(t_0) \text{ is independent of } p). \tag{2.82}$$

This is a linear, non-homogeneous ODE in S, with the solution

$$S(t) = \int_{t_0}^{t} \Phi(t,s) \left.\frac{\partial f}{\partial p}\right|_{\boldsymbol{x}(s),p} \mathrm{d}s, \tag{2.83}$$

which is directly an application of the Duhamel formula with $\mathbf{A}(s) = \frac{\partial f}{\partial \boldsymbol{x}}|_{\boldsymbol{x}(s),p}$, $\mathbf{B}(s) = \frac{\partial f}{\partial p}|_{\boldsymbol{x}(s),p}$, and input $\delta \boldsymbol{u} = \mathrm{d}p$.

Computing $S(t)$ allows us to quickly evaluate the sensitivity of any observable (e.g., final position, maximum altitude, minimum distance) to parameter changes without re-simulating the full nonlinear system.

2.8.2 Adjoint Method for Gradient Computation

Now suppose we have a cost functional

$$J(p) = \ell(x(T), p) + \int_{t_0}^{T} \ell_t(\boldsymbol{x}(t), \boldsymbol{u}(t), p)\,\mathrm{d}t, \tag{2.84}$$

and we wish to compute $\frac{\partial J}{\partial p}$.

Rather than propagating sensitivities forward (as in (2.82)), the adjoint method propagates a costate vector λ backward in time. Define

$$\lambda(t) := \left.\frac{\partial J}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}(t)}. \tag{2.85}$$

The adjoint equation is

$$\boxed{\dot{\lambda} = -\frac{\partial f}{\partial \boldsymbol{x}}^T \lambda - \frac{\partial \ell_t}{\partial \boldsymbol{x}}, \quad \lambda(T) = +\left.\frac{\partial \ell}{\partial \boldsymbol{x}}\right|_{x(T)}.} \tag{2.86}$$

Remark 2.9. The terminal sign is forced by the definition (2.85). At $t = T$ the only part of the cost (2.84) that still depends on the state is the terminal term $\ell(\boldsymbol{x}(T), p)$, so $\lambda(T) = \partial J/\partial \boldsymbol{x}|_{\boldsymbol{x}(T)} = +\partial \ell/\partial \boldsymbol{x}$. The plus sign is also what makes the gradient formula (2.87) correct: flipping it negates the first term of $\partial J/\partial p$ while leaving the second alone, so the resulting gradient is neither the true gradient nor its negative, and a descent step computed from it does not reliably decrease J.

Conventions do differ across references — some formulations of Pontryagin's Maximum Principle carry the opposite sign, absorbed into the definition of the Hamiltonian — so verify signs when consulting other sources. The check that settles it in any convention is a finite-difference gradient: perturb one parameter, recompute J, and confirm the adjoint gradient matches to the accuracy of the integrator.

This equation is integrated backward from T to t_0. Once $\lambda(t)$ is known, the gradient is

$$\frac{\partial J}{\partial p} = \int_{t_0}^{T} \left[\lambda^T \frac{\partial f}{\partial p} + \frac{\partial \ell_t}{\partial p}\right] \mathrm{d}t + \left.\frac{\partial \ell}{\partial p}\right|_{x(T)}. \tag{2.87}$$

Design Implication

The adjoint method is far more efficient than forward sensitivity propagation when the cost depends on many parameters but only a few states. Forward sensitivity would require n_p parallel integrations (one per parameter), each of dimension n. The adjoint requires only one backward integration of dimension n, plus quadrature along the trajectory. This is the foundation of gradient-based optimization in optimal control, and it connects directly to the state transition matrix: the propagation of λ is governed by $-\mathbf{A}(t)^T$, the transpose of the linearization.

2.8.3 Connection to Backpropagation

In machine learning, backpropagation through a neural network is exactly the adjoint method applied to the loss function. If the network is viewed as a dynamical system (e.g., via Neural ODEs), then the backward pass computes the adjoint equation $\dot{\lambda} = -\mathbf{A}(t)^T\lambda$, and the gradient is accumulated via quadrature. This is not a coincidence: both variational dynamics and neural network training are governed by the same differential geometry and calculus of variations.

Remark 2.10. Modern automatic differentiation frameworks (PyTorch, JAX, TensorFlow) implement the adjoint method (or its discrete analogue) under the hood when computing gradients of dynamical systems with respect to parameters. This is a striking example of how geometric insights from control theory have been incorporated into machine learning infrastructure.

2.9 Liouville's Formula and Phase Space Volume

Liouville's formula (Theorem 2.4) has a natural geometric interpretation related to the evolution of phase-space volumes.

2.9.1 Interpretation via Divergence

Consider a small volume element δV in state space centered at $\bar{\boldsymbol{x}}(t)$. Under the flow of the nonlinear system (2.1), this volume element evolves as

$$\frac{\delta V(t)}{\delta V(t_0)} = \det(\Phi(t,t_0)) = \exp\left(\int_{t_0}^{t} \operatorname{tr}\mathbf{A}(s)\,\mathrm{d}s\right) = \exp\left(\int_{t_0}^{t} \nabla\cdot f\Big|_{\bar{\boldsymbol{x}}(s)}\,\mathrm{d}s\right). \tag{2.88}$$

Here $\nabla\cdot f = \sum_i \frac{\partial f_i}{\partial x_i} = \operatorname{tr}(\frac{\partial f}{\partial \boldsymbol{x}})$ is the divergence of the vector field f.

2.9.2 Dissipative Systems

If $\operatorname{tr}(\mathbf{A}(t)) < 0$ everywhere along the trajectory, then

$$\det(\Phi(t,t_0)) = \exp\left(\int_{t_0}^{t} \operatorname{tr}\mathbf{A}(s)\,\mathrm{d}s\right) < 1, \tag{2.89}$$

meaning phase-space volumes shrink exponentially. Such systems are called *dissipative* or *contracting*. Most nonlinear systems with friction or damping fall into this category.

2.9.3 Example: Pendulum With Friction

For the pendulum with damping coefficient c:

$$\ddot{\theta} + c\dot{\theta} + \frac{g}{\ell}\sin\theta = \tau/(m\ell^2). \tag{2.90}$$

In first-order form, $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -(g/\ell) & -c \end{pmatrix}$. Thus $\operatorname{tr}(\mathbf{A}) = -c < 0$, so volumes shrink at rate e^{-ct}.

Common Misconception

For numerical integration, tracking $\det(\Phi)$ via Liouville's formula provides a diagnostic: if $\det(\Phi)$ grows unexpectedly, it may indicate numerical instability or an error in the Jacobian computation.

2.10 Numerical Stability and Ill-Conditioning

Computing the STM accurately is challenging, especially for stiff systems (those with widely separated time scales). We discuss the challenges and remedies.

2.10.1 Ill-Conditioning in Stiff Systems

A system is *stiff* if the Jacobian $\mathbf{A}(t)$ has eigenvalues with vastly different magnitudes or real parts. For example, if $\lambda_1 = -1$ and $\lambda_2 = -1000$, the stiffness ratio is $1000 : 1$.

For such systems, naive numerical integration with explicit methods (RK4) becomes extremely slow: the step size must be tiny to resolve the fastest mode (with λ_2), even though the solution is dominated by the slower mode (with λ_1). Implicit methods (BDF, Rosenbrock) handle stiffness better but can still suffer from conditioning issues when computing the STM.

Consider the STM: if a column $\Phi_{\cdot,j}(t, t_0)$ corresponds to an initial condition along an eigenvector of a stable eigenvalue, that column will decay as $e^{\mathrm{Re}(\lambda)t}$. For large times and stable systems, Φ becomes numerically singular (entries underflow to zero), making it difficult to compute Φ^{-1} accurately.

2.10.2 QR-Based Propagation

A robust remedy is to propagate the STM using a QR decomposition:

$$\Phi(t, t_0) = Q(t, t_0)\, R(t, t_0), \tag{2.91}$$

where Q is orthonormal and R is upper triangular. The QR factorization is updated at each step without explicitly computing Φ.

1. **Propagate:** Integrate the ODE $\dot{Q} = \mathbf{A}(t)Q$ with $Q(t_0, t_0) = \mathbf{I}$.

2. **Orthonormalize:** At each step, compute $[Q_{\text{new}}, R_{\text{step}}] = \mathrm{qr}(\tilde{Q})$, where $\tilde{Q}$ is the current approximate matrix from the integrator. Store $R = R \cdot R_{\text{step}}$.

3. **Recover:** At any time, $\Phi = Q \cdot R$ can be reconstructed, but more importantly, R remains well-conditioned (bounded) throughout the integration.

This approach prevents underflow and overflow and maintains numerical stability for arbitrarily long integration times.

2.10.3 Singular Value Monitoring

Another diagnostic is to monitor the singular values of $\Phi(t, t_0)$ as the integration progresses:

$$\Phi = U\,\Sigma\,V^T, \quad \Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_n), \quad \sigma_1 \geq \cdots \geq \sigma_n \geq 0. \tag{2.92}$$

For a stable system, the smallest singular value σ_n decays exponentially, reflecting the contraction in stable directions. If σ_n becomes smaller than machine epsilon ($\sim 10^{-16}$ in double precision), further numerical integration loses information, and the computed STM becomes unreliable. At that point, it is best to restart the integration from a recent checkpoint or use higher precision arithmetic.

Design Implication

Best practices for stable STM computation:

1. Use automatic differentiation to compute $\mathbf{A}(t)$ accurately.
2. For stiff systems, use implicit integrators (BDF or Rosenbrock) rather than explicit ones.
3. For very long integration times, use QR-based propagation to maintain stability.
4. Monitor singular values of Φ and switch to higher precision if needed.
5. When $\det(\Phi)$ becomes very small or very large, verify against Liouville's formula.

2.11 Qualitative Interpretation of the Jacobians

2.11.1 $\mathbf{A}(t)$: Local State Sensitivity

The eigenvalues and singular values of $\mathbf{A}(t)$ reveal the local perturbation landscape:

- **Eigenvalues with positive real part:** indicate locally unstable directions—perturbations that amplify exponentially. If $\lambda = \alpha + i\beta$ with $\alpha > 0$, perturbations grow as $e^{\alpha t}$ while oscillating with frequency β.
- **Eigenvalues with negative real part:** indicate locally contracting directions. Perturbations decay as $e^{\alpha t}$ with $\alpha < 0$.
- **The symmetric part** $\mathrm{sym}(\mathbf{A}) = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ governs instantaneous growth/decay of perturbation energy. The quadratic form $\|\delta\boldsymbol{x}\|^2$ evolves as

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\delta\boldsymbol{x}\|^2 = 2\delta\boldsymbol{x}^T\delta\dot{\boldsymbol{x}} = 2\delta\boldsymbol{x}^T\mathbf{A}\delta\boldsymbol{x} = 2\delta\boldsymbol{x}^T[\mathrm{sym}(\mathbf{A})]\delta\boldsymbol{x} + 2\delta\boldsymbol{x}^T[\mathrm{skew}(\mathbf{A})]\delta\boldsymbol{x}. \tag{2.93}$$

  The skew part $\mathrm{skew}(\mathbf{A}) = \frac{1}{2}(\mathbf{A} - \mathbf{A}^T)$ does not affect energy directly; it causes rotation.

For control design, the eigenvalues of $\mathbf{A}(t)$ tell us which directions are naturally unstable (requiring control action to stabilize) and which are naturally stable (requiring control only for disturbance rejection).

2.11.2 $\mathbf{B}(t)$: Local Control Authority

The matrix $\mathbf{B}(t)$ determines which perturbation directions can be corrected by control action:

- **Column space:** The column space $\mathrm{col}(\mathbf{B}(t))$ is the set of instantaneously reachable perturbation directions. An input $\delta\boldsymbol{u}$ can move the state in any direction in $\mathrm{col}(\mathbf{B}(t))$.
- **Rank deficiency:** If $\mathrm{rank}(\mathbf{B}(t)) < n$, some directions are not directly controllable at that instant (the system is *underactuated*). For example, a 2D quadrotor can only apply forces; it cannot directly exert a torque (without wing asymmetry), so the yaw dynamics are decoupled from translational control.
- **Condition number:** The condition number $\kappa(\mathbf{B}^T\mathbf{B}) = \sigma_{\max}/\sigma_{\min}$ indicates how efficiently different directions can be influenced. A large condition number means some directions require much larger inputs than others, making them "hard to control."
- **Controllability:** Over a finite time horizon, a system is controllable if the Gramian

$$W_c(t_0, t_f) = \int_{t_0}^{t_f} \Phi(t_f, s)\mathbf{B}(s)\mathbf{B}(s)^T\Phi(t_f, s)^T\,\mathrm{d}s \tag{2.94}$$

  is invertible. This integrates the influence of $\mathbf{B}(s)$ over all times, accounting for how control inputs at early times are propagated forward.

Common Misconception

Many practical failures in optimization-based control stem from poor conditioning of $\mathbf{A}(t)$ or $\mathbf{B}(t)$—not from failure of the nominal planner. Monitoring these matrices along a trajectory is essential for robust implementation. Common issues include:

1. High stiffness (large spread in eigenvalues of $\mathbf{A}$) $\Rightarrow$ choose implicit integrators.
2. Low rank in $\mathbf{B}$ $\Rightarrow$ the system is underactuated; ensure the control cost is set appropriately to handle underactuation.
3. Large condition number of $\mathbf{B}^T\mathbf{B}$ $\Rightarrow$ input saturation is likely; include saturation constraints in the optimizer.

2.12 Residuals Under Finite Perturbations

When we use the variational equation to predict the evolution of a finite (not infinitesimal) perturbation $\delta\boldsymbol{x}_0$, the true trajectory $\boldsymbol{x}(t) = \bar{\boldsymbol{x}}(t) + \delta\boldsymbol{x}(t)$ differs from the linear prediction by a residual.

2.12.1 Residual Integral Form

The true state satisfies

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u} = f(\bar{\boldsymbol{x}} + \delta\boldsymbol{x}) + G(\bar{\boldsymbol{x}} + \delta\boldsymbol{x})(\bar{\boldsymbol{u}} + \delta\boldsymbol{u}). \tag{2.95}$$

Expanding as before but keeping second-order terms:

$$\dot{\boldsymbol{x}} = f(\bar{\boldsymbol{x}}) + J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x} + \frac{1}{2}\frac{\partial^2 f}{\partial \boldsymbol{x}^2}(\delta\boldsymbol{x}, \delta\boldsymbol{x}) + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u} + O(\|\delta\boldsymbol{x}\|^2) + \cdots \tag{2.96}$$

The true state perturbation $\delta\boldsymbol{x}_{\text{true}}$ satisfies

$$\dot{\delta\boldsymbol{x}}_{\text{true}} = J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x}_{\text{true}} + \frac{1}{2}\frac{\partial^2 f}{\partial \boldsymbol{x}^2}(\delta\boldsymbol{x}_{\text{true}}, \delta\boldsymbol{x}_{\text{true}}) + O(\|\delta\boldsymbol{x}_{\text{true}}\|^2) + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u}. \tag{2.97}$$

Let $\delta\boldsymbol{x}_{\text{lin}}$ be the solution from the linearized equation:

$$\dot{\delta\boldsymbol{x}}_{\text{lin}} = J_f(\bar{\boldsymbol{x}})\delta\boldsymbol{x}_{\text{lin}} + G(\bar{\boldsymbol{x}})\delta\boldsymbol{u}. \tag{2.98}$$

The residual is $r(\delta\boldsymbol{x}) := \dot{\delta\boldsymbol{x}}_{\text{true}} - \dot{\delta\boldsymbol{x}}_{\text{lin}} = \frac{1}{2}\frac{\partial^2 f}{\partial \boldsymbol{x}^2}(\delta\boldsymbol{x}_{\text{true}}, \delta\boldsymbol{x}_{\text{true}}) + O(\|\delta\boldsymbol{x}_{\text{true}}\|^3)$.
Integrating the error equation:

$$\delta\boldsymbol{x}_{\text{true}}(t) - \delta\boldsymbol{x}_{\text{lin}}(t) = \int_{t_0}^{t} \Phi(t,s)\, r(\delta\boldsymbol{x}_{\text{true}}(s))\,\mathrm{d}s = \frac{1}{2}\int_{t_0}^{t} \Phi(t,s)\frac{\partial^2 f}{\partial \boldsymbol{x}^2}|_{\boldsymbol{x}(s)}(\delta\boldsymbol{x}_{\text{true}}(s), \delta\boldsymbol{x}_{\text{true}}(s))\,\mathrm{d}s + O(\|\delta\boldsymbol{x}_{\text{true}}\|^3). \tag{2.99}$$

2.12.2 Residual Bound

Under C^2 smoothness, the Hessian is bounded:

$$\left\|\frac{\partial^2 f}{\partial \boldsymbol{x}^2}\right\| \le C_H \tag{2.100}$$

in a neighborhood of the nominal trajectory. Thus

$$\|r(\delta\boldsymbol{x})\| \le \tfrac{1}{2}\, C_H\, \|\delta\boldsymbol{x}\|^2. \tag{2.101}$$

For small $\|\delta\boldsymbol{x}\|$, the residual is much smaller than the main term, justifying linearization. As $\|\delta\boldsymbol{x}\|$ grows, the residual becomes significant. Assuming $\|\Phi(t,s)\| \lesssim 1$ (e.g., stable system), we have

$$\|\Delta x(t)\| \lesssim \int_{t_0}^{t} C_H \, \|\delta\boldsymbol{x}_{\text{true}}(s)\|^2 \, \mathrm{d}s. \tag{2.102}$$

If we denote $M := \max_{s\in[t_0,t]} \|\delta\boldsymbol{x}_{\text{true}}(s)\|$, then

$$\|\Delta x(t)\| \lesssim C_H \, M^2 \, (t - t_0). \tag{2.103}$$

This is not a defect of linearization but a *geometric feature*: it measures the curvature of the vector field along the trajectory.

2.12.3 Residual Scaling and Iterative Refinement

The $O(\|\delta\boldsymbol{x}\|^2)$ scaling is why iterative linearization-based methods (DDP, SQP) converge quadratically near a solution: each re-linearization eliminates the dominant residual term, leaving only higher-order curvature. After one iteration, M shrinks quadratically, so after k iterations, the residual is smaller by a factor of 2^{-2^k}, leading to superlinear (quadratic) convergence.

Remark 2.11. This quadratic convergence is one of the key advantages of locally optimal methods like DDP and SQP over open-loop trajectory optimization methods that do not re-linearize: they can refine a solution very quickly once they are in a neighborhood of the optimum.

2.13 Structure-Preserving Numerical Integration

The numerical methods discussed in Section 2.4 (Runge–Kutta methods, Padé approximants) are general-purpose. For specific classes of systems—particularly Hamiltonian mechanical systems and systems evolving on Lie groups—specialized integrators that preserve geometric structure can dramatically improve accuracy and long-term behavior.

2.13.1 Why Structure Preservation Matters

Consider a Hamiltonian system with $H(\boldsymbol{q},\boldsymbol{p}) = T(\boldsymbol{p}) + V(\boldsymbol{q})$. The continuous dynamics preserve the Hamiltonian ($\dot{H} = 0$), the symplectic form, and phase-space volume (Liouville's theorem, Section 2.9). A generic Runge–Kutta integrator preserves *none* of these. Over long simulations, this manifests as:

- **Energy drift**: the numerically computed energy $H(\boldsymbol{q}_k,\boldsymbol{p}_k)$ drifts monotonically upward or downward, even though the true energy is constant.
- **Phase-space expansion**: trajectories spiral outward in phase space, destroying the qualitative behavior of the system.
- **Broken symmetries**: conservation laws (angular momentum, for instance) are violated, leading to unphysical trajectories.

For short simulations (a single golf downswing at 0.5 s), these effects are typically negligible. For repeated simulations, parameter sweeps, or optimization loops that iterate many times, accumulation of drift can corrupt the results.

2.13.2 Symplectic Integrators

A symplectic integrator is a numerical scheme that preserves the symplectic 2-form $\mathrm{d}\boldsymbol{q} \wedge \mathrm{d}\boldsymbol{p}$. The simplest example is the *symplectic Euler* method:

$$\boldsymbol{p}_{k+1} = \boldsymbol{p}_k - h\frac{\partial V}{\partial \boldsymbol{q}}(\boldsymbol{q}_k), \tag{2.104}$$

$$\boldsymbol{q}_{k+1} = \boldsymbol{q}_k + h\frac{\partial T}{\partial \boldsymbol{p}}(\boldsymbol{p}_{k+1}). \tag{2.105}$$

The key property: the map $(\boldsymbol{q}_k, \boldsymbol{p}_k) \mapsto (\boldsymbol{q}_{k+1}, \boldsymbol{p}_{k+1})$ has Jacobian with determinant exactly 1 (for all step sizes h), preserving phase-space volume exactly. Symplectic integrators do not conserve energy exactly, but the energy error remains bounded for all time (it oscillates rather than drifting).

Higher-order symplectic methods (Störmer–Verlet, Yoshida compositions) achieve $O(h^p)$ accuracy while maintaining the symplectic property.

2.13.3 Lie Group Integrators

When the state space includes rotation groups (SO(3) for rigid-body attitude, SE(3) for pose), standard integrators that parameterize rotations via Euler angles or quaternions face singularities or constraint drift. Lie group integrators work directly on the group:

1. Compute the velocity in the Lie algebra $\mathfrak{g}$ (tangent space at the identity).
2. Integrate in $\mathfrak{g}$ using standard methods.
3. Map back to the group via the exponential map: $R_{k+1} = R_k \cdot \exp(\omega_k\, h)$.

This guarantees that $R_{k+1} \in \mathrm{SO}(3)$ exactly (no normalization or re-projection needed). The Crouch-Grossman and Munthe-Kaas methods provide systematic frameworks for constructing Lie group integrators of arbitrary order.

2.13.4 Variational Integrators

Variational integrators derive the discrete equations of motion from a discrete variational principle (discrete Hamilton's principle) rather than discretizing the continuous equations. The resulting integrator automatically preserves the symplectic structure and momentum maps.

For the discrete Lagrangian $L_d(\boldsymbol{q}_k, \boldsymbol{q}_{k+1}) \approx \int_{t_k}^{t_{k+1}} L(\boldsymbol{q}, \dot{\boldsymbol{q}})\,\mathrm{d}t$, the discrete Euler-Lagrange equations yield:

$$D_2 L_d(\boldsymbol{q}_{k-1}, \boldsymbol{q}_k) + D_1 L_d(\boldsymbol{q}_k, \boldsymbol{q}_{k+1}) = 0, \tag{2.106}$$

which implicitly defines $\boldsymbol{q}_{k+1}$ given $\boldsymbol{q}_{k-1}$ and $\boldsymbol{q}_k$.

Design Implication

For trajectory optimization and DDP (Chapter 6), variational integrators have a specific advantage: the STM Φ computed from the variational integrator inherits the symplectic structure. This means the sensitivities used in the Riccati recursion are geometrically consistent, improving convergence of the optimization. For systems on Lie groups (robotics, spacecraft), the combination of Lie group integration with variational principles provides a fully structure-preserving computational framework.

2.14 Chapter Summary

1. **Variational Equation from First Principles:** The variational equation $\dot{\delta \boldsymbol{x}} = \mathbf{A}(t)\delta\boldsymbol{x} + \mathbf{B}(t)\delta\boldsymbol{u}$ emerges from a careful Taylor expansion of the nonlinear dynamics, keeping first-order terms in $\delta\boldsymbol{x}$ and $\delta\boldsymbol{u}$ and dropping $O(\|\delta\boldsymbol{x}\|^2)$ residuals. This is exact for infinitesimal perturbations and an excellent first-order approximation for small but finite perturbations.

2. **State Transition Matrix:** The STM $\Phi(t, t_0)$ is the unique solution to $\dot{\Phi} = \mathbf{A}(t)\Phi$ with $\Phi(t_0, t_0) = \mathbf{I}$. It characterizes the evolution of any perturbation via $\delta\boldsymbol{x}(t) = \Phi(t, t_0)\delta\boldsymbol{x}(t_0)$.

3. **Fundamental Properties:** The STM satisfies three key properties: semigroup ($\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$), invertibility ($\Phi^{-1}(t, t_0) = \Phi(t_0, t)$), and Liouville's formula for the determinant.

4. **Peano-Baker Series:** The STM admits an explicit infinite series solution (Peano-Baker series), which converges uniformly for bounded $\mathbf{A}(t)$. This series naturally handles non-commutativity in the matrix product.

5. **Numerical Integration:** Computing Φ typically requires integrating the $n \times n$ matrix ODE. For time-invariant systems, the matrix exponential can be computed via Padé approximation with scaling and squaring. For stiff systems, QR-based propagation ensures numerical stability.

6. **Variation of Constants (Duhamel) Formula:** When both initial and input perturbations are present, the solution is $\delta\boldsymbol{x}(t) = \Phi(t, t_0)\delta\boldsymbol{x}(t_0) + \int_{t_0}^{t} \Phi(t, s)\mathbf{B}(s)\delta\boldsymbol{u}(s)\mathrm{d}s$. This shows how effects superpose linearly in the tangent space and is fundamental to sensitivity analysis and optimal control.

7. **Discrete-Time Variational Dynamics:** For computational implementation, the continuous-time variational equation is discretized. The discrete STM is a product of Jacobians: $\Phi_{d,\ell,k} = \prod_{j=k}^{\ell-1} \mathbf{A}_j$.

8. **Sensitivity and Adjoint Methods:** Parameter sensitivities $\frac{\partial \boldsymbol{x}}{\partial p}$ satisfy a linearized ODE whose solution is given by the Duhamel formula. The adjoint method computes cost gradients efficiently by propagating a costate backward in time, governed by $\dot{\lambda} = -\mathbf{A}(t)^T \lambda$.

9. **Residuals and Convergence:** Finite perturbations incur residuals of order $O(\|\delta\boldsymbol{x}\|^2)$, reflecting manifold curvature. Iterative re-linearization (as in DDP) eliminates the dominant residual, leading to quadratic convergence.

10. **Numerical Challenges:** Stiff systems and long integration times can render Φ numerically singular. QR-based propagation and singular value monitoring provide robust diagnostics and solutions.

11. **Matrix Interpretation:** $\mathbf{A}(t)$ encodes local sensitivity (eigenvalues reveal stable/unstable directions); $\mathbf{B}(t)$ encodes local control authority (column space reveals reachable directions). Together they are the raw material for all subsequent stability and optimality analyses.

Design Implication

The variational equation is not merely a linear approximation: it is an exact first-order differential-geometric object describing the behavior of tangent vectors along a trajectory in a curved (nonlinear) vector field. The state transition matrix is the parallel transport operator in this geometry, evolving vectors along the trajectory while respecting the local geometry. This perspective connects classical control theory to modern differential geometry and is essential for understanding advanced topics like Riemannian optimization, geometric tracking control, and neural network training dynamics.

Exercises

1. **STM of a linear system.** For the LTI system $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\zeta\omega \end{bmatrix}$ (damped harmonic oscillator), compute $\Phi(t, 0) = e^{\mathbf{A}t}$ analytically using eigenvalue decomposition. Verify the semigroup property $\Phi(t_2, 0) = \Phi(t_2, t_1)\Phi(t_1, 0)$.

2. **Peano-Baker truncation error.** For the system in Exercise 1, compute the first three terms of the Peano-Baker series (2.30). Estimate the truncation error at $t = 1$ and compare with the exact exponential.

3. **Liouville's formula.** Verify Liouville's formula (Theorem 2.4) for the 2D system $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 - 0.1x_2$ by computing both $\det \Phi(t, 0)$ directly and $\exp(\int_0^t \operatorname{tr} \mathbf{A}(s)\, \mathrm{d}s)$.

4. **Sensitivity of a nonlinear system.** For the Van der Pol oscillator $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ with $\mu = 1$, compute the variational equation along the trajectory starting from $(x_0, \dot{x}_0) = (2, 0)$. Integrate the STM numerically for one period. What do the singular values of Φ reveal?

5. **Input sensitivity.** For the control-affine system $\dot{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x} + \mathbf{B}\boldsymbol{u}$ with $\mathbf{A}$, $\mathbf{B}$ from the pendulum example earlier in this chapter, use the variation of constants formula to compute the response to a unit pulse input $\boldsymbol{u}(t) = \delta(t - t_0)$.

6. **Symplectic integrator comparison.** Implement both RK4 and the symplectic Euler method for the undamped pendulum $\ddot{\theta} + \sin\theta = 0$. Integrate for 1000 periods. Plot the energy $H(\theta, \dot{\theta})$ as a function of time for both methods. Explain the qualitative difference.

7. **Discrete-time STM.** For the discrete system $\boldsymbol{x}_{k+1} = \mathbf{A}_k \boldsymbol{x}_k$ with $\mathbf{A}_k$ periodic ($\mathbf{A}_{k+N} = \mathbf{A}_k$), express the one-period STM as a product. What is the relationship between the eigenvalues of this product and the stability of the system?

8. **Magnus expansion.** For a time-varying $\mathbf{A}(t) = \mathbf{A}_0 + \varepsilon \mathbf{A}_1 \sin(\omega t)$, compute the first-order Magnus expansion and compare with the Peano-Baker series at $t = 2\pi/\omega$. Under what conditions does the Magnus expansion converge faster?

Chapter 3

Superposition: From Linear to Affine

A golfer's effort, posture, motion, contact with the ground, and flexible equipment act together. The useful question is not whether one of these explains the swing by itself. It is which mathematical operation allows their effects to be separated, what is held fixed during that operation, and what must be recomputed when the motion changes. This chapter develops that distinction from mechanics, then connects it to energy transfer, constraints, actuator limits, accessibility, and feedback control.

The central result is exact but deliberately local in state: in a declared mechanical model with a fixed contact mode and forces affine in the chosen inputs, input-induced acceleration increments add at the same position and velocity. Entire trajectories generally do not. An instantaneous decomposition can identify how a net joint moment contributes to acceleration now; it cannot, by itself, identify the muscle that supplied that moment, the work done over a downswing, or the result of changing a golfer's movement strategy.

3.1 Three Meanings of Addition

3.1.1 Tangent Vectors and First Variations

At a point on a smooth configuration or state manifold, tangent vectors form a vector space. Their addition is exact. A finite change between two postures, however, is not intrinsically a tangent vector, and vectors attached to different postures cannot be added without a specified identification or transport. Coordinates provide a local representation; they do not make nonlinear geometry disappear.

For a smooth dynamical system, the first variation about a specified reference trajectory satisfies a linear differential equation. Perturbations of that variational equation superpose. Their relationship to finite perturbed trajectories is generally an approximation whose error must be controlled. Thus, exact vector-space algebra and a first-order approximation to a finite response are compatible statements, not competing descriptions.

3.1.2 Exact Input Increments at One State

Theorem 3.1 (Fixed-State Input Superposition)**.** *Let $F(x,u) = f_0(x) + G(x)u$ on a smooth mode, with $u \in \mathbb{R}^m$. At a fixed x, the total derivative is affine in u, while its increment is linear:*

$$\Delta F_x(u) := F(x,u) - F(x,0) = G(x)u. \tag{3.1}$$

Consequently,

$$F(x, u_1 + u_2) = F(x, u_1) + F(x, u_2) - F(x, 0). \tag{3.2}$$

The zero-input anchor is counted once. For any real coefficients, $\Delta F_x(\alpha u_1 + \beta u_2) = \alpha \Delta F_x(u_1) + \beta \Delta F_x(u_2)$. These are algebraic identities wherever the model is defined; a combined input outside the feasible set remains a mathematical evaluation rather than an executable command. Affine interpolation of total derivatives uses coefficients summing to one.

The anchor also depends on the input definition. Zero net joint torque, zero motor voltage, and zero muscle excitation are different interventions. Muscle activation, tendon tension, shaft deflection, and other stored states can retain forces after a command is set to zero. We will use zero applied generalized effort unless another port is explicitly introduced. Known gravity, passive forces, prescribed external loads, and the active constraint model remain present.

3.1.3 Finite Trajectories and the Evolving Baseline

Write $x[u](t)$ for the trajectory from one common initial state under an input history u. In a shared coordinate chart, the appropriate baseline comparison is

$$\epsilon(t) = x[u_1 + u_2](t) - x[u_1](t) - x[u_2](t) + x[0](t). \tag{3.3}$$

Nonlinear dynamics generally give $\epsilon \neq 0$. The zero-input trajectory evolves; it is not generally the initial state or the zero vector. The trajectories sample different f_0 and G, possibly different contacts, and eventually different domains. There can be exceptional input pairs for which this defect vanishes, including symmetry-related responses of a pendulum. One such pair neither proves a general superposition theorem nor disproves nonlinearity.

For a realizable reference $(\bar{x}(t), \bar{u}(t))$, the first variation is

$$\delta\dot{x} = A(t)\delta x + G(\bar{x}(t))\delta u, \qquad A = D_x f_0(\bar{x}) + \sum_j \bar{u}_j D_x G_j(\bar{x}). \tag{3.4}$$

Omitting the derivatives of the input directions gives the wrong linearization when the reference effort is nonzero and G varies with posture. A reference that does not satisfy the dynamics also introduces a residual forcing. In a golf analysis, each of these choices affects the meaning of an estimated sensitivity.

3.2 Mechanical Origins of the Affine Map

3.2.1 Kinetic Energy and a Regular Mass Matrix

Use n independent local coordinates q, with $v = \dot{q}$, and N_b rigid bodies. For a stationary base, each center-of-mass velocity and angular velocity depends linearly on v. Express an angular velocity and its inertia tensor in the same frame, with the inertia taken about that body's center of mass. Then

$$T = \frac{1}{2}\sum_{i=1}^{N_b} \left(m_i \|J_{v,i} v\|^2 + (J_{\omega,i} v)^T I_{C,i} (J_{\omega,i} v) \right) = \frac{1}{2} v^T M(q) v, \tag{3.5}$$

where

$$M = \sum_i \left(m_i J_{v,i}^T J_{v,i} + J_{\omega,i}^T I_{C,i} J_{\omega,i} \right). \tag{3.6}$$

This matrix is symmetric and positive semidefinite. It is positive definite when every nonzero admissible generalized velocity produces positive modeled kinetic energy. Independent regular coordinates and a nondegenerate inertial model supply that property. Massless unconstrained freedoms, redundant coordinates, or singular charts require additional treatment. The number of bodies need not equal the number of coordinates. Quaternion coordinates also require a kinematic map $\dot{q} = N(q)v$ instead of identifying all coordinate rates with independent velocities.

3.2.2 Force, Acceleration, and Connection Coefficients

For the independent-coordinate model, let $U(q)$ be potential energy, $D(q,v)v$ a dissipative force with $v^T D v \geq 0$, and $Q_e(q,v,t)$ a prescribed external generalized force. The equation is

$$M\dot{v} + c(q,v) + \nabla U + Dv = B(q)u + Q_e. \tag{3.7}$$

Here B maps the chosen efforts to generalized forces. It need not be square. Velocity-dependent or time-dependent B also preserves frozen-state input affinity, provided it does not depend on u. Unknown contact reactions are not arbitrary prescribed loads; they will be solved with constraints below.

The Euler–Lagrange equations give the velocity-product force

$$c_i = \sum_{j,k} \Gamma_{ijk} v_j v_k, \qquad \Gamma_{ijk} = \frac{1}{2}\left(\partial_j M_{ik} + \partial_k M_{ij} - \partial_i M_{jk} \right). \tag{3.8}$$

These are Christoffel symbols of the first kind. The second-kind coefficients $\Gamma^i{}_{jk} = \sum_\ell (M^{-1})_{i\ell}\Gamma_{\ell jk}$ act at acceleration level. Adding $\Gamma^i{}_{jk}v_j v_k$ directly to $M\dot{v}$ mixes those levels. One consistent choice is $C_{ij} = \sum_k \Gamma_{ijk}v_k$, so that $Cv = c$ and $\dot{M} - 2C$ is skew symmetric. The matrix C is not unique, whereas the mechanically specified vector c is fixed.

With no potential gradient, damping, or applied nonconstraint force, the free equation is $\ddot{q}^i + \Gamma^i{}_{jk}\dot{q}^j\dot{q}^k = 0$: a geodesic of the kinetic metric. A gravity-driven swing is not such a free geodesic merely because commanded torque is zero. For background on the force-level derivation, see the Modern Robotics Lagrangian dynamics transcript.

Solving the regular equation gives

$$a_0 = M^{-1}(Q_e - c - \nabla U - Dv), \qquad H = M^{-1}B, \qquad \dot{v} = a_0 + Hu. \tag{3.9}$$

Thus $f_0 = \begin{bmatrix} v \\ a_0 \end{bmatrix}$ and $G = \begin{bmatrix} 0 \\ H \end{bmatrix}$. The retained external load stays in a_0. This derivation establishes affinity for the declared effort variables; it does not imply that an arbitrary electrical, neural, or software command enters affinely.

3.2.3 Virtual Work Explains the Force Map

If $\delta p_i = J_{v,i}\delta q$ and the infinitesimal body rotation is $\delta\theta_i = J_{\omega,i}\delta q$, virtual work gives

$$\delta W = \sum_i (F_i^T \delta p_i + N_i^T \delta\theta_i) = Q^T \delta q, \qquad Q = \sum_i (J_{v,i}^T F_i + J_{\omega,i}^T N_i). \tag{3.10}$$

All forces, moments, velocities, and lever arms must use compatible points and frames. Generalized forces are dual to generalized velocities: their pairing is power. At fixed geometry this mapping is linear in physical wrenches. If a wrench is nonlinear in a chosen command, the Jacobian transpose does not remove that nonlinearity. Ideal stationary constraint reactions do no virtual work on allowed displacements; this is why they disappear under a valid reduction, not why their numerical values are independent of actuation.

3.3 One Arm, Three Consistent Formulations

3.3.1 A Fully Specified Uniform-Rod Model

Consider two uniform planar rods with lengths $l_1, l_2 > 0$, masses $m_1, m_2 > 0$, fixed proximal endpoint, and center-of-mass inertias $I_{C,i} = m_i l_i^2/12$. Let q_1 be counterclockwise from the horizontal and q_2 the relative elbow angle. Therefore the absolute angles are q_1 and $q_1 + q_2$, and angular velocities are v_1 and $v_1 + v_2$. These conventions determine every trigonometric sign below. Gravity is the Cartesian vector $(0, -g)$.

Set $e(\theta) = (\cos\theta, \sin\theta)^T$. The centers are

$$p_{C,1} = \frac{l_1}{2}e(q_1), \qquad p_{C,2} = l_1 e(q_1) + \frac{l_2}{2}e(q_1 + q_2). \tag{3.11}$$

Differentiating these positions and adding the rotational energies gives, with $\beta = m_2 l_1 l_2/2$, $d = m_2 l_2^2/3$, and $a = m_1 l_1^2/3 + m_2 l_1^2 + d$,

$$M = \begin{bmatrix} a + 2\beta\cos q_2 & d + \beta\cos q_2 \\ d + \beta\cos q_2 & d \end{bmatrix}, \qquad c = \beta\sin q_2 \begin{bmatrix} -2v_1 v_2 - v_2^2 \\ v_1^2 \end{bmatrix}. \tag{3.12}$$

The potential and its gradient are

$$U = g\left[\left(\frac{m_1 l_1}{2} + m_2 l_1\right)\sin q_1 + \frac{m_2 l_2}{2}\sin(q_1 + q_2)\right], \tag{3.13}$$

$$\nabla U = g\begin{bmatrix} (m_1 l_1/2 + m_2 l_1)\cos q_1 + (m_2 l_2/2)\cos(q_1 + q_2) \\ (m_2 l_2/2)\cos(q_1 + q_2) \end{bmatrix}. \tag{3.14}$$

At a horizontal posture gravity therefore gives a clockwise tendency. A sine formula associated with angles measured from the downward vertical cannot be inserted without changing the coordinate convention. Both joint accelerations appear in each coupled balance; a single-link inertia about the base cannot silently replace the two-link mass matrix.

The determinant is

$$\det M = \frac{m_1 m_2 l_1^2 l_2^2}{9} + m_2^2 l_1^2 l_2^2 \left(\frac{1}{3} - \frac{1}{4}\cos^2 q_2\right) > 0. \tag{3.15}$$

Together with $d > 0$, this verifies positive definiteness for this model, including aligned rods. A task Jacobian can lose rank at alignment while M remains invertible. A condition number, unlike positive definiteness, also depends on parameter values and the scaling chosen for coordinates.

Writing $h = \beta \sin q_2$, a compatible Coriolis matrix is

$$C = \begin{bmatrix} -hv_2 & -h(v_1 + v_2) \\ hv_1 & 0 \end{bmatrix}. \tag{3.16}$$

Direct multiplication recovers c, and $v^T c = \frac{1}{2} v^T \dot{M} v$. The factor of one half in β comes from the second rod's center-of-mass distance. Treating its entire mass as a point at the tip would define a different model and different coefficients.

3.3.2 Newton–Euler Balances Recover the Same Efforts

Let F_{12} be the elbow force on rod 2, F_{01} the base force on rod 1, and τ_2 the counterclockwise actuator moment on rod 2, with reaction $-\tau_2$ on rod 1. The base moment is τ_1. Define $\rho_1 = p_{C,1}$ and $\rho_2 = p_{C,2} - l_1 e(q_1)$. For any proposed joint acceleration, differentiating the center positions gives their Cartesian accelerations, including velocity-product terms. Force balance requires

$$F_{12} = m_2(a_{C,2} - g_{\text{vec}}), \qquad F_{01} = m_1(a_{C,1} - g_{\text{vec}}) + F_{12}. \tag{3.17}$$

With planar cross products interpreted as signed moments, the center-of-mass moment balances give

$$\tau_2 = I_{C,2}(\ddot{q}_1 + \ddot{q}_2) + \rho_2 \times F_{12}, \tag{3.18}$$

$$\tau_1 = I_{C,1}\ddot{q}_1 + \tau_2 + \rho_1 \times F_{01} + (l_1 e(q_1) - \rho_1) \times F_{12}. \tag{3.19}$$

Eliminating the internal forces yields $\tau = M\ddot{q} + c + \nabla U$. Conversely, solving that equation and substituting into these body balances must recover the applied torques. The runnable example is tested in precisely this way, using independently differentiated Cartesian positions. This checks geometry, inertia, gravity signs, and coupling rather than merely evaluating the same mass-matrix formula twice.

These equations used inertial-frame derivatives. In axes rotating with a body, translational balance at its center is $F = m(\dot{v}_C^b + \omega^b \times v_C^b)$, and rotational balance is $N_C = I_C \dot{\omega}^b + \omega^b \times (I_C \omega^b)$. A dot on body components alone omits the transport term. Origins away from the center require the corresponding inertia and acceleration transformations.

3.3.3 Spatial Vectors Preserve the Same Constraints

In body axes at the center of mass, use an angular-first twist $V = (\omega, v_C^b)$ and wrench $W = (N_C, F)$. The pairing $W^T V$ is power. With $[z]_\times w = z \times w$,

$$\mathcal{I} = \operatorname{diag}(I_C, mI_3), \qquad \operatorname{ad}_V = \begin{bmatrix} [\omega]_\times & 0 \\ [v_C^b]_\times & [\omega]_\times \end{bmatrix}, \qquad W = \mathcal{I}\dot{V} - \operatorname{ad}_V^T \mathcal{I} V. \tag{3.20}$$

The lower block contains $m(\dot{v}_C^b + \omega \times v_C^b)$, exactly the force balance above. Wrench transformations are dual to twist transformations; using one transformation for both would generally corrupt power.

For a connected stationary-base chain, stack the body twists as $V = \mathcal{J}(q)v$. Then $\dot{V} = \mathcal{J}\dot{v} + \dot{\mathcal{J}}v$, and the assembled kinetic metric is $M = \mathcal{J}^T \mathcal{I} \mathcal{J}$. Project the body balances with the dual map $\mathcal{J}^T$, retaining their bias terms and joint compatibility. Ideal internal reactions cancel in this projection. Inverting each free body's spatial inertia without enforcing the joints instead simulates disconnected bodies. Generalized actuator efforts also do not uniquely specify an arbitrary collection of free-body wrenches.

Newton–Euler, Lagrange, and spatial-vector methods are consequently different organizations of the same model. Agreement requires the same geometry, inertias, loads, coordinate conventions, and constraints. Their agreement is a verification opportunity, not three independent physical explanations of the swing.

3.4 Power, Work, and Intersegment Transfer

The actuation power is

$$P_u = (Bu)^T v = u^T y, \qquad y = B^T v, \tag{3.21}$$

not generally $u^T v$. At fixed state, $P_u = \sum_j u_j y_j$. If $B = I$, each input is a joint torque and each output its corresponding relative angular velocity. Contributions can be negative: an actuator may absorb mechanical energy while the total motion continues to speed up elsewhere.

For $E = T + U$, differentiating before substituting the dynamics is essential:

$$\dot{E} = v^T M \dot{v} + \frac{1}{2} v^T \dot{M} v + (\nabla U)^T v = u^T B^T v + v^T Q_e - v^T D v. \tag{3.22}$$

The cancellation uses $v^T c = \frac{1}{2} v^T \dot{M} v$, with a positive sign. Coriolis and centrifugal terms are not a dissipative sink. Calling $-(c + \nabla U)^T v$ the total energy rate misses the changing kinetic metric. Gravity exchanges kinetic and potential energy, whereas damping removes mechanical energy in this model.

For the arm, a shared elbow point has the same velocity on both rods. The equal and opposite elbow forces therefore have equal and opposite powers in the combined balance, while each rod can gain or lose energy through that boundary. The actuator moment pair contributes

$$\tau_2(v_1 + v_2) - \tau_2 v_1 = \tau_2 v_2. \tag{3.23}$$

The base force acts at a stationary point and has zero power; the base actuator contributes $\tau_1 v_1$. Summing the two rod energy balances gives $\dot{E} = \tau_1 v_1 + \tau_2 v_2$. On individual rods, joint force power and joint moment power both matter. Their division depends on the chosen subsystem boundary and wrench reference point; a consistent total balance does not.

This is the connection to proximal-to-distal sequencing. Proximal slowing and distal speeding are kinematic observations. Demonstrating transfer requires a segment energy balance with boundary force and moment powers, external work, and any elastic storage. A shaft can temporarily store energy while hands and clubhead change speed. A net wrist moment cannot uniquely identify individual muscle forces or co-contraction. The mechanical accounting can support a proposed mechanism, but the timing pattern alone is not its proof.

Finite channel work is $W_j = \int u_j(t) y_j(t)\, dt$. Along one recorded trajectory these integrals sum exactly. After an intervention changes one channel, all the velocities and possibly the feasible contacts change. Differences between two such experiments are therefore not obtained by adding the original channel works or integrating independent acceleration contributions along their own trajectories.

3.5 Constraints Preserve an Affine Map Within a Regular Mode

3.5.1 Reaction Forces Depend on the Input

For a stationary holonomic constraint $\phi(q) = 0$ with full-row-rank $J = \partial\phi/\partial q$, admissible velocities satisfy $Jv = 0$ and accelerations satisfy $J\dot{v} = -\kappa$, where $\kappa = \dot{J}v$. The configuration manifold is a set of positions; its tangent space is the allowed velocity space at one position. Accelerations form an affine set because of the curvature term. General rolling constraints need not integrate to a configuration equation and must not be classified as holonomic without checking integrability.

Use the reaction convention

$$M\dot{v} = r + Bu + J^T\lambda, \qquad r = Q_e - c - \nabla U - Dv. \tag{3.24}$$

Together with the acceleration constraint this is the regular saddle-point system

$$\begin{bmatrix} M & -J^T \\ J & 0 \end{bmatrix} \begin{bmatrix} \dot{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} r + Bu \\ -\kappa \end{bmatrix}. \tag{3.25}$$

Let $W = JM^{-1}J^T$ and $P = I - M^{-1}J^TW^{-1}J$. The solution is

$$\dot{v} = PM^{-1}r - M^{-1}J^TW^{-1}\kappa + PM^{-1}Bu, \tag{3.26}$$

$$\lambda = -W^{-1}\left(\kappa + JM^{-1}r + JM^{-1}Bu\right). \tag{3.27}$$

Thus the constrained acceleration remains affine in effort, but its slope is $PM^{-1}B$, and its anchor includes curvature. The reactions vary with u. Holding their values fixed while changing effort generally violates the constraint. P is a projection orthogonal in the mass metric, not necessarily in the Euclidean metric.

For a particle of mass m constrained to a circle of radius R, take $\phi = (q^Tq - R^2)/2$. Then $J = q^T$ and $\kappa = v^Tv$. At $q = (R, 0)$ with tangential speed s, the radial acceleration must be $-s^2/R$, even with zero applied force. An additional radial force changes the reaction; an additional tangential force changes the tangential acceleration. This elementary example makes both the offset and the input-dependent reaction visible.

3.5.2 Reduced Coordinates and Contact Feasibility

In a local configuration chart $q = \psi(z)$, let $N = D\psi$ and $w = \dot{z}$. Then $v = Nw$ and $\dot{v} = N\dot{w} + \dot{N}w$. Projection gives

$$(N^TMN)\dot{w} = N^T(r + Bu - M\dot{N}w). \tag{3.28}$$

Dropping $\dot{N}w$ removes the geometry-induced acceleration. An arbitrary null-space basis for a velocity constraint may instead define quasi-velocities; it does not automatically define integrable coordinates z.

Stationary ideal reactions have zero total power because $\lambda^TJv = 0$. For $\phi(q, t) = 0$, the velocity condition is $Jv = -\phi_t$, so reaction power is $-\lambda^T\phi_t$: a moving boundary can supply or absorb energy. Its acceleration equation also needs the corresponding time derivatives.

Ground contact adds inequalities: a normal reaction cannot pull a unilateral surface together, and tangential reactions must satisfy the selected friction law. The affine expression is usable only while its solved reactions remain admissible in the assumed mode. Lift-off, slip, impact, or a change of constraint rank requires the appropriate new equations. A golf swing model with permanently fixed feet must justify that assumption over the analyzed interval; the projection is not a substitute for that check.

3.6 A Pendulum Example That Can Be Reproduced

3.6.1 Frozen-State Arithmetic

Use a point mass $m = 1$ kg at the end of a massless rod of length $l = 1$ m, a frictionless fixed pivot, and $g = 9.81$ m/s^2. Here, unlike the uniform-rod arm, q is measured counterclockwise from the downward vertical. The pivot inertia is $I = ml^2 = 1$ kg m^2, and

$$I\ddot{q} = -mgl\sin q + \tau, \qquad E = \frac{1}{2}I\dot{q}^2 + mgl(1 - \cos q), \qquad \dot{E} = \tau\dot{q}. \tag{3.29}$$

This is a teaching model, not a fitted golf swing. Its standard phase-space and energy structure are described in the MIT Underactuated Robotics pendulum chapter.

At the same state with $q = \pi/6$, the baseline acceleration is -4.905 rad/s^2. Torques of 5, 2.5, and 7.5 N m give respectively 0.095, -2.405, and 2.595 rad/s^2. The identity is

$$2.595 = 0.095 + (-2.405) - (-4.905). \tag{3.30}$$

The incremental gain is $\partial\ddot{q}/\partial\tau = 1/I$, with units (rad/s^2)/(N m). The ratio of total acceleration to total torque is not that gain when gravity contributes an offset. A torque $\tau_g = mgl\sin q$ cancels gravity at the current angle. Maintaining this cancellation as the pendulum moves requires updating the torque, and the actuator's work still appears in the energy balance.

The gravitational contribution is bounded by $g/l = 9.81$ rad/s^2 in magnitude. This simple bound and the sign convention are useful checks on every numerical row before discussing its physical meaning.

3.6.2 Trajectories Under Constant Torques

Start every experiment at $q(0) = \dot{q}(0) = 0$ and hold its torque constant. The program below integrates with fourth-order Runge–Kutta steps no larger than 0.001 s. Table values are rounded to six decimals. Halving the step and comparing with an independently implemented adaptive DOP853 integration both change the sampled state by less than 2×10^{-10} in the stated SI coordinates for the listed inputs and times. These are checks of this example, not universal error bounds for the algorithm.

For $\tau = 5$ N m:

t (s)	q (rad)	$\dot{q}$ (rad/s)	$\ddot{q}$ (rad/s^2)
0.00	0.000000	0.000000	5.000000
0.25	0.148434	1.126321	3.549201
0.50	0.508441	1.613248	0.224337
1.00	1.109439	0.456624	-3.784356
1.50	0.867969	-1.319704	-2.485204
2.00	0.133555	-1.077419	3.693713

At $t = 1$ s the pendulum is still moving in the positive direction but has negative angular acceleration. At $t = 1.5$ s both its velocity and acceleration are negative; the constant positive torque is then absorbing energy. Torque sign, acceleration sign, velocity sign, and power sign answer different questions. The same distinction prevents incorrect interpretations of joint moment or segment deceleration in a swing.

The zero-torque trajectory stays at the downward equilibrium. The odd equation and symmetric initial state imply exactly $q[-5] = -q[5]$ and $\dot{q}[-5] = -\dot{q}[5]$. Therefore the pair $5, -5$ happens to satisfy trajectory addition. To exhibit a nonzero defect, compare the nonsymmetric pair $5, 2.5$ with their sum 7.5:

t (s)	$q[5]$	$q[2.5]$	$q[7.5]$	ϵ_q
0.00	0.000000	0.000000	0.000000	0.000000
0.25	0.148434	0.074215	0.222663	0.000014
0.50	0.508441	0.253779	0.764851	0.002631
1.00	1.109439	0.520651	1.844183	0.214093
1.50	0.867969	0.293486	2.561097	1.399642
2.00	0.133555	0.003088	3.981019	3.844375

The angle and defect entries in this second table are radians on the same unwrapped coordinate. At two seconds the combined input has carried the mass beyond π, while the two smaller-input trajectories have returned near the lower posture. Adding those two positions misses nearly 3.8444 radians. The underlying cause is the different gravitational acceleration sampled along each history, not a failure of forces to add at a fixed state. An angle can be wrapped for display, but wrapping each trajectory independently before subtraction changes this comparison and can create discontinuities.

For any one constant torque, integrating $\dot{E} = \tau\dot{q}$ gives

$$\frac{1}{2}\dot{q}(t)^2 + 9.81(1 - \cos q(t)) - \tau q(t) = 0 \tag{3.31}$$

for these initial conditions. The numerical residual is below 2×10^{-10} J at the sampled times. The work term uses the unwrapped angle; a constant torque performs work on every revolution. Relative error against a zero baseline is undefined, so the tables report absolute differences rather than assigning a misleading zero percent.

3.7 Passivity, Dissipation, and Contraction Are Different Claims

3.7.1 An Energy Inequality at a Declared Port

A system with input u and output y is passive on a stated domain if it has a nonnegative storage function S satisfying

$$S(x(t)) - S(x(0)) \leq \int_0^t u(s)^T y(s)\, ds \tag{3.32}$$

for admissible trajectories. A storage bounded below can be shifted by a constant to be nonnegative. Equality describes a lossless system and can hold away from equilibrium. Positive dissipation is a stronger property than passivity; neither should be substituted for the other. See the definitions and pendulum example in Khalil's passivity lecture.

For the mechanical model, $S = E - E_{\min}$ is available when energy has a lower bound on the domain. The effort port is $(u, B^T v)$. A nonzero external load Q_e must be included as another power port, or its power must be accounted for separately. Omitting it can falsely classify an externally driven subsystem as passive or active. The dissipative assumption is $v^T D v \geq 0$; a general appeal to thermodynamics does not establish that an arbitrary fitted matrix satisfies it.

Bounded input does not imply bounded energy. A unit free mass driven by the constant unit force has $\dot{q} = t$, $q = t^2/2$, and $E = t^2/2$ from rest. It is lossless with $\dot{E} = u\dot{q}$, yet its energy grows without bound. Adding viscous drag bounds the limiting speed under constant force, but position can still drift without bound. A storage that controls velocity alone cannot bound every state variable. Bounded-state and asymptotic-convergence conclusions require suitable storage growth, dissipation or feedback, existence of solutions, and often an invariance or detectability argument.

For a well-posed negative feedback interconnection of two passive systems, choose $u_1 = r - y_2$ and $u_2 = y_1$ with compatible port dimensions. Their inequalities sum to $\dot{S}_1 + \dot{S}_2 \leq r^T y_1$ because internal power cancels. This is the interconnection guarantee. To conclude convergence to a particular equilibrium, one must additionally show that the zero-dissipation invariant set contains only the desired motion in the relevant domain. Algebraic loops, unmodeled delays, saturation, and incompatible ports need their own analysis.

3.7.2 Port-Hamiltonian Structure and Its Limits

In canonical mechanical coordinates $z = (q, p)$, $p = M(q)v$, the Hamiltonian is

$$\mathcal{H}(q, p) = \frac{1}{2} p^T M(q)^{-1} p + U(q). \tag{3.33}$$

With symmetric positive-semidefinite viscous damping and no omitted external port, one representation is

$$\dot{z} = (\mathcal{J} - \mathcal{R})\nabla\mathcal{H} + \mathcal{G}u, \quad \mathcal{J} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathcal{R} = \begin{bmatrix} 0 & 0 \\ 0 & D \end{bmatrix}, \quad \mathcal{G} = \begin{bmatrix} 0 \\ B \end{bmatrix}. \tag{3.34}$$

Since $\mathcal{J}$ is skew symmetric and $\mathcal{R}$ is positive semidefinite,

$$\dot{\mathcal{H}} = -(\nabla\mathcal{H})^T \mathcal{R} \nabla\mathcal{H} + u^T y, \qquad y = \mathcal{G}^T \nabla\mathcal{H} = B^T v. \tag{3.35}$$

This representation exposes power exchange and dissipation. Generalized coordinates and momenta cannot be swapped with (q, v) while retaining the same constant canonical matrix unless the transformation is handled correctly. Nor does arbitrary control-affine dynamics automatically possess this physical Hamiltonian and passive port.

Even a positive-definite dissipation matrix in a more general port-Hamiltonian representation does not alone imply $\dot{\mathcal{H}} \leq -\alpha\mathcal{H}$ with a uniform $\alpha > 0$. For a scalar example, take $\mathcal{H} = x^4/4$, $\mathcal{R} = 1$, and $\dot{x} = -x^3$. Then $\dot{\mathcal{H}} = -x^6$ and $-\dot{\mathcal{H}}/\mathcal{H} = 4x^2$, which approaches zero near the equilibrium. A uniform exponential bound needs an additional relationship between energy and its dissipative gradient.

3.7.3 A Differential Certificate Must Be Checked Separately

Contraction concerns separation of nearby trajectories under the same specified input or closed-loop law. For $\dot{x} = F(x,t)$ and $A = D_x F$, a sufficient differential certificate on a suitable forward-invariant domain is a smooth metric $P(x,t)$ with uniform positive bounds and

$$\dot{P} + A^T P + PA \preceq -2\lambda P, \qquad \lambda > 0. \tag{3.36}$$

Here $\dot{P} = \partial_t P + \sum_k F_k \partial_{x_k} P$ is the derivative along the flow. The inequality makes $\delta x^T P \delta x$ decay; domain and path conditions are then needed to translate this into a statement about distances between trajectories. An energy derivative along one trajectory is a different calculation. For a feedback law, A must include the derivative of the feedback, and robustness needs a quantified perturbation margin in the same metric.

A damped pendulum is passive at its torque–angular-velocity port, yet its upright equilibrium has linearization

$$A_{\mathrm{up}} = \begin{bmatrix} 0 & 1 \\ g/l & -d/I \end{bmatrix}. \tag{3.37}$$

Its determinant is $-g/l < 0$, so one eigenvalue is positive. Damping does not make a domain containing that equilibrium uniformly contracting. The energy Hessian also has a negative angular entry at the upright posture and cannot serve there as a positive-definite contraction metric. Multiple stationary solutions provide another obstruction to uniform incremental convergence on a domain containing them all.

Passivity is therefore useful for power accounting and interconnection, while contraction can certify incremental convergence when its full conditions hold. Neither gives a blanket preference for preserving or cancelling a particular term in a golf controller. A model-based strategy must be evaluated against its actual tracking objective, feasible effort, sensing, uncertainty, and contact assumptions.

3.8 Actuator Limits and the Choice of Input

3.8.1 Feasible Sets Do Not Erase Affinity

At a fixed state, allowed inputs $u \in \mathcal{U}(x)$ produce the acceleration set

$$\mathcal{A}(x) = a_0(x) + H(x)\mathcal{U}(x). \tag{3.38}$$

State-dependent limits change this set, not the affine formula itself. If each scalar channel is independently bounded, $\mathcal{U} = \mathcal{U}_1 \times \cdots \times \mathcal{U}_m$, then $H\mathcal{U}$ is the Minkowski sum of the individual sets $H_j \mathcal{U}_j$. Coupled electrical power limits, muscle-sharing constraints, or contact feasibility may restrict the allowed combinations; summing independently maximized contributions can then predict an unattainable acceleration.

3.8.2 Voltage, Current, and Torque

For an idealized fixed-field DC motor, with armature inductance L, resistance R, positive torque constant k_t, and back-EMF constant k_e,

$$L\dot{i} = V - Ri - k_e\omega, \qquad \tau_m = k_t i. \tag{3.39}$$

These are the electrical and torque relations in the University of Michigan DC motor model. Torque and back-EMF constants have different units even when their numerical SI values coincide for consistent ideal conventions.

If electrical transients can be neglected, the algebraic approximation is

$$\tau_m = \frac{k_t}{R} V - \frac{k_t k_e}{R}\omega. \tag{3.40}$$

With symmetric voltage limits $|V| \le V_{\max}$, its torque interval at a fixed speed is

$$\frac{k_t}{R}(-V_{\max} - k_e\omega) \le \tau_m \le \frac{k_t}{R}(V_{\max} - k_e\omega). \tag{3.41}$$

It is generally offset rather than symmetric about zero. Intersect it with $[-k_t i_{\max}, k_t i_{\max}]$ when a current limit is imposed. These are ideal steady electrical feasibility bounds; thermal limits, available regenerative operation, drive switching, and finite current slew can impose further restrictions. Torque at the load also depends on rotor acceleration, friction, and gearing. For a rigid ideal gear ratio $N = \omega_m/\omega_{\text{load}}$, motor inertia contributes $N^2 J_m$ at the load coordinate. The Modern Robotics actuation transcript explains this reflected-inertia effect.

Keeping current as a state instead of eliminating it gives a voltage-affine extended model. At a frozen (q, v, i), voltage initially changes $\dot{i}$; it does not instantly replace the existing electromagnetic torque. Similarly, zero excitation in a muscle model can leave activation and elastic tension nonzero. An additional actuator state may preserve command affinity while changing the immediate input directions and the meaning of zero-input motion. Whether it does so must be checked from the chosen actuator equations.

3.8.3 Nonsmoothness Is a Separate Modeling Issue

During sliding, a simple dry-friction law has resisting torque $-\tau_c \operatorname{sgn}(\dot{q})$. This is nonsmooth in velocity but does not itself make applied torque enter nonlinearly. At zero velocity, static friction is a reaction in an interval such as $[-\tau_s, \tau_s]$, selected by force balance and the stick/slip conditions; defining $\operatorname{sgn}(0) = 0$ does not reproduce stiction. Smooth approximations may help a numerical method but alter the near-zero behavior and must be validated for the intended time scale.

Saturation composed with a raw command generally makes the raw-command map piecewise affine rather than globally affine. Nonlinear transmission, muscle recruitment, contact switching, or hysteresis can introduce other departures. State the port, state variables, and mode first, then determine which identity survives. Calling every practical limitation a failure of superposition hides the specific assumption that needs repair.

3.9 Lie Brackets, Coupling, and Reachability

3.9.1 The Order of Motions Can Matter

For smooth vector fields, use the convention

$$[f, g](x) = Dg(x)f(x) - Df(x)g(x). \tag{3.42}$$

Comparing short flow compositions in opposite orders produces a second-order difference involving this bracket, with a sign determined by the chosen composition order. This describes how the directions change as the state changes. It does not mean that any signed field, especially the uncontrolled drift, is physically reversible on demand.

For second-order mechanics $f = (v, a_0(q, v))$ and a configuration-dependent input field $g_j = (0, h_j(q))$,

$$[f, g_j] = \begin{bmatrix} -h_j \\ D_q h_j \, v - D_v a_0 \, h_j \end{bmatrix}. \tag{3.43}$$

The bracket has a configuration component, not merely an acceleration in an unactuated coordinate. Even a fully actuated n-coordinate system has at most n immediate input directions in a $2n$-dimensional state space; time integration connects acceleration to velocity and position. For the normalized undamped pendulum, $f = (v, -\sin q)$ and $g = (0, 1)$ give $[f, g] = (-1, 0)$, so g and $[f, g]$ span the two state directions without any intersegment inertia coupling.

3.9.2 Accessibility Is Not Controllability

For smooth driftless systems with reversible controls containing a neighborhood of zero, a bracket-generating distribution yields local connectivity through admissible motions under the hypotheses of the Chow–Rashevskii theorem. The LaValle driftless-system discussion states the setting explicitly. Input bounds and a nonreversible drift require a different conclusion. Accessibility concerns the interior of reachable sets; it does not assert that every nearby state can be reached in arbitrarily short time. Fixed-time accessibility is another distinct property.

A simple illustration is $\dot{x}_1 = u$, $\dot{x}_2 = x_1^2$. Its drift and control brackets span both directions, but x_2 never decreases along a physical trajectory, even with both signs of u available. Starting at the origin, nearby points with negative x_2 are unreachable. This is the counterexample in LaValle's treatment of systems with drift. Algebraic rank alone does not remove that one-way restriction.

For a golf mechanism, a proposed transfer into an unactuated shaft mode must be established from the actual inertia, elastic coupling, forcing, and boundary conditions. Coupling can arise from a spring even with a diagonal mass matrix. Conversely, an off-diagonal mass entry or a named bracket is not evidence that a particular sequencing maneuver is feasible, beneficial, or robust over a finite downswing. Reachability, energy availability, contact feasibility, and the terminal clubhead objective must be checked together.

3.10 Feedback Linearization Has a Local Domain of Validity

For a smooth single-input system $\dot{x} = f(x) + g(x)u$ with scalar output $y = h(x)$, a constant relative degree r on a neighborhood requires $L_g L_f^k h = 0$ there for $k = 0, \dots, r-2$, and $b(x) = L_g L_f^{r-1} h \neq 0$. Then

$$y^{(r)} = L_f^r h + b(x)u, \qquad u = \frac{\nu - L_f^r h}{b(x)} \tag{3.44}$$

gives $y^{(r)} = \nu$ while the feedback and local coordinate transformation remain valid. A decoupling coefficient can vanish elsewhere, the resulting effort can exceed a limit, and the trajectory can leave the coordinate domain. The cancellation is exact for the specified model; its performance under model error requires separate analysis.

If $r < n_x$, the output chain leaves internal states. Their dynamics when the output and its derivatives are held at zero are the zero dynamics. Local asymptotic stability of those dynamics is the usual minimum-phase condition for this construction. Unstable zero dynamics obstruct stable exact tracking in the corresponding regime; they do not say that every trajectory diverges or that every finite-horizon strategy is impossible. If $r = n_x$, there is no internal subsystem to classify. A multi-input version additionally requires a nonsingular decoupling matrix and a compatible vector relative degree.

For the pendulum output q, $r = 2 = n_x$ and $\tau = mgl \sin q + I\nu$ gives $\ddot{q} = \nu$. This does not erase gravitational work from the physical system; the actuator supplies the compensating moment. If torque is bounded, only the corresponding range of ν is feasible at each angle. Tracking a clubhead output in a flexible multibody model is more demanding because internal modes and task singularities can remain after output linearization.

Feedback linearization is a design construction; contraction is a property that a resulting system may or may not possess. They can be used together. A linear time-invariant closed-loop system with a Hurwitz matrix, obtained by exact full-state linearization, has a quadratic contraction metric in its linear coordinates, which can be pulled back locally through a valid transformation. Uniform bounds and the physical domain still need checking. There is no general theorem that one approach is preferable because it preserves a term called drift.

3.11 What This Establishes for the Golf Swing

The instantaneous analysis begins with a common measured or simulated state, a declared effort port, a physically consistent inertial model, and an admissible contact mode. Within that model, each effort increment has an exact acceleration contribution. Contributions may act in coordinates far from the actuated joint because the mechanism couples them. The anchor retains gravity, motion-dependent effects, elastic forces, and any declared external loads; it is not synonymous with effortless or purely passive human motion.

To explain a finite swing, propagate the coupled equations after an intervention and report how the state, reaction forces, power flows, and flexible storage change. A useful argument then has several linked pieces: an induced acceleration describes the immediate mechanical effect; a segment power balance tests the proposed energy pathway; a trajectory calculation tests its accumulated effect; and a feasible-input and contact analysis tests whether the maneuver can occur. Impact and ball flight require their own downstream models before a clubhead change can be interpreted as a distance or accuracy benefit.

None of these mathematical tools independently validates a coaching cue. Net moments estimated from inverse dynamics depend on kinematic differentiation, inertial parameters, external loads, and measurement uncertainty. Muscle-level interpretations require additional physiological information. A rigorous presentation makes the model-based conclusion precise, compares it with observations, and states what alternative mechanisms remain consistent with the evidence.

3.12 Reproducible Verification

The following NumPy-only program generates the two pendulum tables and the uniform-rod matrices used in the independent verification tests. It needs no site-specific package. Arrays use SI units and the angle conventions stated in the function documentation. The integration is for the smooth constant-torque pendulum only; it is not an impact or stick/slip solver.

```
# Superposition chapter example
import numpy as np


def rod_arm(
    q: np.ndarray,
    velocity: np.ndarray,
    lengths: np.ndarray,
    masses: np.ndarray,
    gravity: float = 9.81,
) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
    """Return M, c, and the left-side gravity vector for two uniform rods.

    q[0] is counterclockwise from horizontal; q[1] is the relative elbow angle.
    The proximal endpoint is fixed. SI units are used throughout.
    """
    l1, l2 = lengths
    m1, m2 = masses
    q1, q2 = q
    v1, v2 = velocity
    beta = m2 * l1 * l2 / 2
    distal = m2 * l2**2 / 3
    base = m1 * l1**2 / 3 + m2 * l1**2 + distal
    coupling = beta * np.cos(q2)
    mass = np.array([[base + 2 * coupling, distal + coupling],
                     [distal + coupling, distal]])
    c = beta * np.sin(q2) * np.array([-2 * v1 * v2 - v2**2, v1**2])
    gravity_vector = gravity * np.array([
        (m1 * l1 / 2 + m2 * l1) * np.cos(q1) + m2 * l2 / 2 * np.cos(q1 + q2),
        m2 * l2 / 2 * np.cos(q1 + q2),
    ])
    return mass, c, gravity_vector


def pendulum_trace(
    torque: float, times: np.ndarray, max_step: float = 0.001
) -> np.ndarray:
    """Integrate the unit-mass, unit-length pendulum from rest at q=0.

    Columns are q (rad), velocity (rad/s), acceleration (rad/s^2).
    Times must increase strictly from zero; the angle remains unwrapped.
    """
    times = np.asarray(times, dtype=float)
    if (times.ndim != 1 or times.size == 0 or times[0] != 0
            or not np.all(np.isfinite(times)) or np.any(np.diff(times) <= 0)):
        raise ValueError("Times must increase strictly from zero")
    if not np.isfinite(max_step) or max_step <= 0 or not np.isfinite(torque):
        raise ValueError("Use finite torque and a positive finite step")

```

```
    def rhs(state: np.ndarray) -> np.ndarray:
        return np.array([state[1], torque - 9.81 * np.sin(state[0])])

    state = np.zeros(2)
    result = np.empty((times.size, 3))
    result[0] = [0, 0, torque]
    for index in range(1, times.size):
        duration = times[index] - times[index - 1]
        steps = int(np.ceil(duration / max_step))
        step = duration / steps
        for _ in range(steps):
            k1 = rhs(state)
            k2 = rhs(state + step * k1 / 2)
            k3 = rhs(state + step * k2 / 2)
            k4 = rhs(state + step * k3)
            state += step * (k1 + 2 * k2 + 2 * k3 + k4) / 6
        result[index] = [*state, rhs(state)[1]]
    return result


times = np.array([0.0, 0.25, 0.5, 1.0, 1.5, 2.0])
trajectories = {u: pendulum_trace(u, times) for u in (0.0, 2.5, 5.0, -5.0, 7.5)}
trajectory_defect = trajectories[7.5] - trajectories[5.0] - trajectories[2.5] + trajectories[0.0]
# Inspect trajectories[5.0] and trajectory_defect to reproduce the tables.
```

3.13 Exercises

1. **A Baseline Counted Once.** At $q = \pi/6$ in the unit pendulum, verify the frozen-state identity for torques 5, 2.5, and 7.5 N m. Repeat it about a nonzero reference torque τ_* using increments $\delta\tau$. Explain why total acceleration divided by torque is not the incremental gain.

2. **One Model in Two Formulations.** Derive the uniform-rod mass matrix from the center velocities and center-of-mass rotational energies. Recover the two torque equations from the separate body balances. Show that the determinant is positive for all angles when both lengths and masses are positive.

3. **Conditioning Requires a Scale.** Use $l = (0.8, 0.6)$ m and $m = (1.4, 0.5)$ kg, with both angles in radians. Compute the spectral condition number of M over $q_2 \in [-\pi, \pi]$ and refine any candidate maximum. Recompute after scaling one coordinate by ten. Distinguish physical regularity from numerical conditioning.

4. **Velocity-Product Power.** At $q = (0.4, -0.7)$ rad and $v = (2.1, -0.8)$ rad/s, verify $v^T c = \frac{1}{2} v^T \dot{M} v$ using a centered directional difference of M. Explain why a nonzero value is consistent with zero dissipation.

5. **An Affine Constraint Set.** For $m = 2$ kg, $R = 2$ m, $q = (2, 0)$ m, and $v = (0, 3)$ m/s, compute acceleration and reaction under zero force and under $(4, 6)$ N, without gravity. Verify the radial acceleration and show that the reaction changes with input.

6. **A Trajectory Defect.** Reproduce both pendulum tables, halve the integration step, and check constant-torque work–energy balance. Explain the $5, -5$ symmetry exception and why the $5, 2.5$ pair is a more informative comparison. State the angle convention and avoid relative error against a zero reference.

7. **A Local Nonlinear Counterexample.** Solve $\dot{x} = x^2 + u$ from $x(0) = 0$ for positive constant u, restricting attention to times before the earliest finite-time blow-up. Compare the inputs 1, 2, and 3. Verify fixed-state input affinity despite the nonzero trajectory defect.

8. **A Passive but Unstable Posture.** Derive the damped pendulum's torque-port storage inequality after choosing an energy offset. Compute the upright linearization and energy Hessian. Identify which conclusion prevents global contraction on a domain containing that posture.

9. **Motor Feasibility.** Let $R = 2$ ohm, $k_t = 0.1$ N m/A, $k_e = 0.1$ V s/rad, $V_{\max} = 24$ V, and $i_{\max} = 10$ A. Find the quasistatic torque interval at $\omega = 100$ rad/s. Explain why voltage has no instantaneous effect on torque when current is retained as a state.

10. **Bracket Rank and One-Way Motion.** Compute $[f, g]$ for the normalized pendulum. For $\dot{x}_1 = u$, $\dot{x}_2 = x_1^2$, compute enough brackets to span the plane and explain why negative x_2 remains unreachable from the origin. State why an actuation matrix alone cannot settle the reachability of an unspecified underactuated model.

11. **Cancellation With a Bound.** For a pendulum with $|\tau| \leq \tau_{\max}$, derive the feasible interval of virtual acceleration ν under gravity-compensating feedback. State the relative degree and whether internal zero dynamics exist for the angle output.

12. **A Testable Swing Argument.** Propose a model-based test of the claim that proximal slowing transfers energy to the club. Specify subsystem boundaries, measured quantities, joint force and moment powers, shaft storage, external work, uncertainty, and the intervention used to distinguish transfer from simultaneous active work. Identify what the instantaneous acceleration decomposition can and cannot establish.

The next chapter connects these frozen-state decompositions to induced acceleration analysis in biomechanics. The correspondence is strongest when the inertial model, force channels, constraints, and definition of the zero-input anchor are made explicit in both formulations.

Chapter 4

Induced Acceleration Analysis: Superposition in the Biomechanics Literature

In Plain Language 4.1. A Parallel Discovery

The superposition principle developed in the preceding chapter—that forces map linearly to accelerations at each frozen instant via the inverse mass matrix—was independently developed in the biomechanics community under the name **induced acceleration analysis**. This chapter traces that development and the conditions required before the frameworks can be compared. A shared frozen-state linear map does not make their force partitions, controls, constraints, outputs, or causal interpretations equivalent.

Normative Attribution Record

A publishable attribution must identify the model and revision; engine, solver, and revision; generalized coordinates and reference frame; output point; mass matrix; constraint handling; contact model and mode; complete force partition; residual treatment; and numerical tolerance with closure error. Its identifiability contract must state the measurements or assumptions supporting each term. An algebraic term does not by itself identify anatomical source, neural intent, necessity, sufficiency, or intervention effect. Missing fields make a result **unsupported or unqualified**; an unavailable cross-engine state is reported that way rather than treated as solver parity.
Under $q = Tz$, the same dynamics use $M_z = T^\top M_q T$ and $Q_z = T^\top Q_q$: component values change even though $T\ddot{z} = \ddot{q}$. If $Q = Q_A + Q_B$, then $Q = (Q_A + r) + (Q_B - r)$: residual reassignment changes both terms while preserving their sum. These counterexamples block unique attribution. An intervention must separately re-solve constraints and contact and declare what is held fixed.

4.1 Historical Context and Motivation

Human movement presents an attribution problem: how should a declared model partition its acceleration ledger? Inverse dynamics can estimate net generalized forces, but it does not identify individual muscle forces or a unique causal relationship. IAA adds a forward-model term calculation, not the missing identifiability contract.

[ZG89] addressed this gap in a seminal review that established the theoretical basis for induced acceleration analysis in multi-articular movement. Their central observation was that in a multi-joint system, a muscle force produces torques at the joints it crosses, but through the dynamic coupling inherent in the equations of motion, these torques induce accelerations at *every* joint in the chain. Moreover, the induced accelerations depend not only on the muscle's torque, but on the entire system's mass matrix $\boldsymbol{M}(\boldsymbol{q})$ and its inverse—precisely the quantities that encode the inertial coupling between segments.

This observation motivated a research program spanning more than three decades, applying induced acceleration analysis to walking [ZNK02, ZNK03, NKZ01, KSS97], clinical gait pathology [RK99, Sil18], throwing [HKWO07, HYNO08, HO08], cycling [SRZG93], sit-to-stand transfers [CTC+16], and postural control [Cha11].

4.2 The Formal Framework

4.2.1 Equations of Motion and the Forward Problem

Consider a mechanical system with n generalized coordinates $\boldsymbol{q} \in \mathbb{R}^n$. The equations of motion take the standard form:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{4.1}$$

Here $\boldsymbol{M}(\boldsymbol{q}) \succ 0$ assumes independent minimal coordinates and a regular kinetic-energy metric. Redundant coordinates or active constraints require a declared constrained solve. Generalized-force labels are a model partition; anatomical labels require an actuator map and identifiability contract.

Solving for accelerations:

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[\boldsymbol{\tau} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q})\right] \tag{4.2}$$

In the biomechanics literature, the convention is to move the velocity-dependent and gravity terms to the right side as positive contributions:

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[\boldsymbol{\tau}_{\text{muscle}} + \boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q})\right] \tag{4.3}$$

where $\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = -\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$ collects the velocity-dependent torques (note the sign convention difference) and $\boldsymbol{g}(\boldsymbol{q})$ now represents the gravitational contribution with appropriate sign.

4.2.2 Induced Acceleration Decomposition

The core operation of induced acceleration analysis is the decomposition of the total acceleration into contributions from individual force sources:

$$\ddot{\boldsymbol{q}} = \underbrace{\sum_{k=1}^{m} \boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{\tau}_k}_{\text{muscle-induced}} + \underbrace{\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}})}_{\text{velocity-induced}} + \underbrace{\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{g}(\boldsymbol{q})}_{\text{gravity-induced}} \tag{4.4}$$

Here $\boldsymbol{\tau}_k$ is a declared generalized-force channel. Its mapped term is part of a frozen-state ledger, not an observation of a muscle acting alone.

Theorem 4.1 (Compatibility with Control-Affine Superposition)**.** *The two ledgers can share a linear operator, but their terms are comparable only when plant, state, coordinates, actuator map, constraints, contact, residual treatment, solver qualification, and output match.*

In the control-affine form $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$:

- *The drift $\boldsymbol{f}(\boldsymbol{x})$ corresponds to $\boldsymbol{M}^{-1}(\boldsymbol{q})[\boldsymbol{V}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q})]$ (velocity-induced plus gravity-induced accelerations).*
- *The control columns $\boldsymbol{G}(\boldsymbol{x}) = [\boldsymbol{g}_1(\boldsymbol{x}), \ldots, \boldsymbol{g}_m(\boldsymbol{x})]$ correspond to $\boldsymbol{M}^{-1}(\boldsymbol{q})\boldsymbol{B}$, where $\boldsymbol{B}$ maps control inputs to generalized forces.*
- *A generalized-force channel can match $\boldsymbol{g}_k(\boldsymbol{x})u_k$ only under the same actuator map and partition; neither label identifies a muscle by itself.*

Both frameworks exploit the fact that $\boldsymbol{M}^{-1}(\boldsymbol{q})$ is a linear *operator: the mapping from forces to accelerations is linear at each frozen state, even though the system dynamics are globally nonlinear.*

Proof. Distributing a fixed linear forward operator over $\boldsymbol{\tau} = \sum_k \boldsymbol{\tau}_k$ proves closure for that partition. It does not prove partition uniqueness, coordinate-invariant components, anatomical source, or an intervention effect. $\square$

4.3 Instantaneous Effects, Cumulative Effects, and the Drift Field

One of the most significant conceptual contributions of the induced acceleration literature is the distinction between **instantaneous effects** and **cumulative effects**, developed most clearly by [HYNO08] and [Hir11].

4.3.1 Instantaneous Effects

The mapped term $\boldsymbol{M}^{-1}(\boldsymbol{q}(t))\boldsymbol{\tau}_k(t)$ is a frozen-state quantity under the declared partition. Removing a physical force is a different intervention that can change activation, passive force, constraints, and contact.

- **Direct effects**: A joint torque accelerating its own joint (diagonal elements of $\boldsymbol{M}^{-1}$).
- **Remote effects**: A joint torque accelerating distant joints through dynamic coupling (off-diagonal elements of $\boldsymbol{M}^{-1}$).

4.3.2 Cumulative Effects

The velocity-dependent acceleration $\boldsymbol{M}^{-1}(\boldsymbol{q}(t))\boldsymbol{V}(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))$ depends on the current angular velocities $\dot{\boldsymbol{q}}(t)$, which are the time-integrated result of all previous forces. This term therefore reflects the *cumulative* consequence of the entire force history up to time t.

4.3.3 Mapping to the Drift-Control Decomposition

The correspondence is exact:

IAA Terminology	**Control-Affine Terminology**
Instantaneous muscle-induced acceleration	Control input $\boldsymbol{G}(\boldsymbol{x})\boldsymbol{u}$
Cumulative velocity-dependent acceleration	Drift field $\boldsymbol{f}(\boldsymbol{x})$ (velocity component)
Gravity-induced acceleration	Drift field $\boldsymbol{f}(\boldsymbol{x})$ (gravity component)

The cumulative effect is the drift field: it is what the system would do if all control inputs were set to zero at the current instant. The instantaneous effects are the control inputs: they are what the muscles add on top of the drift at this instant.

A model-reported term-magnitude comparison requires a declared norm, partition, and output. It does not show that a person is on autopilot, identify control authority, or establish cross-engine parity.

4.4 Applications of Induced Acceleration Analysis

4.4.1 Walking: The Zajac–Neptune Framework

The most comprehensive application of induced acceleration analysis has been to human walking. [ZNK02] (Part I) and [ZNK03] (Part II) provided a two-part review that established the methodological framework and derived coordination principles from dynamical simulations.

Key findings include:

1. **Inverse dynamics and IAA have different limits.** Inverse dynamics estimates net generalized forces. IAA can repartition a forward-model ledger, but does not identify causal contributions of individual muscles without an identifiability contract.

2. **Acceleration, power, work, and energy are distinct.** Joint power alone does not identify segment-energy transfer. IAA and induced-power ledgers are also model- and partition-dependent and require constraint, contact, and residual closure.

3. **Biarticular muscles have complex, configuration-dependent roles.** Muscles crossing two joints (such as the gastrocnemius or rectus femoris) can have different—sometimes opposite—effects on the two joints they cross, and these effects change with posture.

[NKZ01] applied this framework to the ankle plantar flexors during walking. Using a forward dynamics simulation driven by 15 individual muscle actuators, they decomposed the contributions of the soleus and gastrocnemius to trunk support (vertical center-of-mass acceleration), forward progression (horizontal center-of-mass acceleration), and swing initiation (mechanical energy delivered to the leg in pre-swing). Their results demonstrated that the soleus delivers its energy primarily to the trunk (providing support and forward progression), while the gastrocnemius delivers its energy primarily to the leg (initiating swing). This functional dissociation between two muscles at the same joint—invisible to inverse dynamics—exemplifies the power of the induced acceleration framework and the principle of superposition applied to individual force sources.

4.4.2 Overarm Throwing: The Hirashima Framework

[HKWO07] and [HYNO08] extended induced acceleration analysis to three-dimensional, 13-degree-of-freedom models of baseball pitching. Their work introduced several methodological advances:

1. **Full 3D induced acceleration analysis.** By writing the equations of motion in generalized coordinates (joint angles) rather than segment coordinates, and premultiplying by the full 13×13 inverse mass matrix, [HYNO08] computed the acceleration induced by each of the 13 generalized forces, plus the velocity-dependent torque, plus gravity, on each of the 13 degrees of freedom.

2. **Decomposition of velocity-dependent torques.** [HYNO08] further decomposed the velocity-dependent torque into contributions from the angular velocity of each individual segment, tracing the kinematic source of each cumulative effect. This allowed them to determine which segment's motion was the "origin" of the velocity-dependent torque acting at distant joints.

3. **Correction of erroneous methods in the literature.** [HO08] demonstrated that previous studies had incorrectly analyzed multi-joint dynamics by treating each segment's equation of motion independently (dividing by the diagonal inertia term) rather than solving the coupled system through the full inverse mass matrix. This error led to incorrect attributions of muscle function and is equivalent, in our framework, to neglecting the off-diagonal elements of $\boldsymbol{M}^{-1}$—that is, ignoring the dynamic coupling that is the very phenomenon induced acceleration analysis is designed to capture.

The key finding from the throwing studies was that the kinetic chain operates through a conversion of instantaneous muscle effects (at proximal joints, in the early phase) into cumulative velocity-dependent effects (at distal joints, in the late phase). The proximal muscles create angular velocities that persist in the system as velocity-dependent torques, which then accelerate the distal joints without requiring large distal muscle forces. This is the kinetic chain reinterpreted through the lens of superposition: proximal forces create the drift field that drives the distal motion.

4.4.3 Postural Control and the Induced Acceleration Index

[Cha11] introduced the **induced acceleration index** (IAI), a dimensionless quantity that captures the configuration-dependent coupling potential between joints:

$$\mathrm{IAI}_{j \to k} = \left| \boldsymbol{M}_{kk}^{-1} \boldsymbol{M}_{kj} \right| \tag{4.5}$$

For a two-joint model of quiet standing (ankle and hip), the IAI revealed that the ankle has over 12 times the ability to accelerate the hip compared to the reverse. This asymmetry depends only on the system's inertial properties and configuration, not on the forces applied, and provides a structural explanation for the dominance of the ankle strategy in postural control.

The IAI is a scalar summary of the off-diagonal structure of $\boldsymbol{M}^{-1}(\boldsymbol{q})$. In terms of the control-affine framework, it quantifies how the control authority at one joint propagates to other joints through the dynamic coupling. The IAI is useful for any application where the question is: "given the current configuration, how effective is a torque at joint A at producing acceleration at joint B?"

4.4.4 Clinical Applications

Induced acceleration analysis has been applied to clinical gait analysis, where it provides individualized diagnostic information about the sources of movement dysfunction:

- **Stiff-legged gait:** [RK99] showed that reduced knee flexion in swing-phase gait after stroke can result from impairments at the hip, knee, or ankle, with the specific cause varying between patients. Induced acceleration analysis, by decomposing knee acceleration into contributions from each joint's torque, identified the patient-specific impairment pattern and informed individualized rehabilitation protocols.
- **Sit-to-stand transfers:** [CTC+16] reported positive gluteus-maximus and soleus terms and an opposing quadriceps term for selected center-of-mass outputs in a 46-degree-of-freedom, 194-actuator model. These are model-reported terms, not uniquely identified anatomical sources.
- **Electrical stimulation cycling:** [SRZG93] used a musculoskeletal model to analyze electrically stimulated leg cycling for spinal cord injury rehabilitation, identifying how seat configuration and stimulation timing affect the ability of stimulated muscles to produce the desired pedaling motion.

4.5 Methodological Considerations

4.5.1 Ground Contact and Constraint Forces

In movements involving ground contact (walking, standing, the golf stance), the ground reaction force must be included in the induced acceleration analysis. Because the ground reaction force is itself a constraint force that depends on all other forces in the system, decomposing it into individual muscle contributions requires special treatment. [NKZ01] addressed this by computing each muscle's contribution to the ground reaction force through a perturbation approach: remove one muscle force, re-integrate the equations of motion over one time step, and compare the resulting ground reaction force to the original.

[Sil18] reviewed the various computational approaches and their assumptions. The choice of ground contact model (rigid constraint, viscoelastic elements, rolling surface) affects the decomposition and must be reported and validated.

4.5.2 Model Dependence

[HYNO08] noted that the results of induced acceleration analysis can depend on the number of segments and degrees of freedom in the model. Adding trunk motion or lower extremity dynamics can change the interpretation of upper extremity muscle function. The induced acceleration framework is exact given the model, but the model is always an approximation of the biological system. Validation against experimental kinematics (by integrating the induced accelerations forward and comparing to observed motion) is an essential step, as demonstrated by [HYNO08] and [RK99].

4.6 Synthesis: Two Communities, One Mathematical Structure

IAA and control-affine analysis can share a frozen-state linear map. Their answers are not mathematically identical unless plant, coordinates, force and input partitions, constraints, contact mode, residual treatment, solver qualification, and output are identical.

Recognizing this equivalence has several benefits:

1. **Terminology bridge.** Researchers in biomechanics can draw on the extensive control-theoretic literature on controllability, observability, and optimal control for affine systems. Researchers in control theory can draw on the rich experimental database of induced acceleration results for biological systems.
2. **Conceptual unification.** The drift-control distinction in control theory provides a natural framework for organizing the biomechanical findings: instantaneous muscle effects are control inputs; cumulative velocity-dependent effects are the drift field; drift dominance in fast movements is a statement about the relative magnitudes of these two contributions.

3. **Methodological rigor.** The control-affine framework's emphasis on the *instantaneous* nature of superposition (Chapter 3) provides a clear warning against the trajectory-superposition errors identified by [HO08] in the biomechanics literature.

4. **Broader applicability.** The same mathematical structure applies to any multi-body mechanical system: robotic manipulators, prosthetic limbs, sports equipment, and any system governed by Lagrangian or Newtonian mechanics.

Common Misconception

Superposition is instantaneous, not global. Induced acceleration analysis decomposes the *acceleration at one instant* into individual contributions. It does not decompose the *trajectory over time* into individual contributions, because the system dynamics are nonlinear. The cumulative effect (drift field) at time t depends on the entire history of forces up to t, and there is no general way to attribute portions of the trajectory to individual muscles over extended time intervals.

Chapter 5

Contraction Theory and Metric Certificates

In Plain Language 5.1. Shrinking Mistakes

Imagine running the same robot motion twice with slightly different starting conditions. If the system is "contracting," those two runs quickly converge to the same trajectory—small mistakes do not snowball. This chapter develops the mathematical machinery for certifying that property: a metric (local ruler) that provably shrinks along the flow.

5.1 Historical Context and the Shift to Trajectory Stability

5.1.1 From Lyapunov Point Stability to Trajectory Convergence

Classical Lyapunov stability, as developed in the late 1800s and formalized throughout the twentieth century, addresses a foundational question: *does the system return to an equilibrium point after small perturbations?* This point-stability perspective is powerful for understanding equilibria, but it misses a central feature of many practical systems: the behavior of interest is often a *moving trajectory*, not a rest point.

A robot executing a reaching task, a vehicle tracking a path, a biological motor system performing a skilled movement—these systems are intrinsically *trajectory-focused*. A Lyapunov function certifying stability at one point does not explain why nearby trajectories diverge under perturbations or what determines their convergence rate.

The concept of contraction—trajectories converging robustly—is fundamentally dual to the concept of controllability in Agrachev and Sachkov's geometric framework [AS04, Ch. 4]. While controllability asks *what is reachable* from a given initial condition via forward-time controls, contraction analysis asks *do trajectories converge* and thus provides a certificate of robustness and structural stability. Both are geometric properties: controllability relies on Lie bracket accessibility in the forward direction, while contraction provides a metric-based certificate of asymptotic behavior.

Contraction theory reframes the question: **do all nearby trajectories converge to each other, regardless of initial condition?** This is a genuinely different property, and it provides the certificate we need.

5.1.2 The Lohmiller–Slotine Framework (1998)

The modern contraction theory for nonlinear systems was formalized by **Lohmiller and Slotine** in 1998 [LS98]. Their key insight was to study how *tangent vectors* between trajectories evolve under the flow [MS17]. If all such vectors shrink in a suitably chosen metric, then the geodesic distance between any two trajectories shrinks exponentially.

The framework bridges classical results in differential geometry (Riemannian metrics, geodesic flows) with modern control and stability analysis. By allowing the metric to vary in space and time, Lohmiller–Slotine generalized earlier work and made contraction a practical tool for nonlinear system analysis and design.

5.1.3 Demidovich's Condition (1961): An Historical Precursor

The mathematical seeds of contraction theory are rooted in work by **Demidovich** in 1961 [Dem61], who studied conditions for *global asymptotic stability* based on the symmetric part of the Jacobian. Demidovich showed that if the symmetric part of the system's Jacobian is uniformly negative definite, all nearby solutions converge to a single trajectory. This foundational result [Kha02] predated modern contraction theory by decades and provides important context for understanding contemporary approaches.

Although Demidovich did not use the language of Riemannian metrics or contraction, his condition is in fact a special case of modern contraction theory: it corresponds to contraction in the *Euclidean metric* (identity matrix). The modern framework generalizes Demidovich's insight by relaxing the constant-metric requirement, allowing the "ruler" to change with position and time.

5.1.4 Connection to Modern Research

Contraction theory has since become a cornerstone of:

- **Incremental stability and synchronization**: studying how coupled oscillators, networked systems, and consensus algorithms achieve agreement.
- **Control design**: synthesis of feedback laws that guarantee contraction, leading to robust and verifiable controllers.
- **Learning-based control**: verification of neural network controllers via metric composition and convex optimization.
- **Robotics and autonomous systems**: design of feedback laws that are tolerant to perturbations and external disturbances.

5.2 From Equilibrium Stability to Trajectory Stability

Classical Lyapunov stability asks: does a system return to an equilibrium point after a perturbation? Contraction theory asks a more general question: do *all* nearby trajectories converge to each other, regardless of where they start?

This generalization is essential for systems where the behavior of interest is a *moving trajectory*—a robot executing a task, a vehicle following a path, a biological organism performing a skilled movement—rather than a static rest point.

Geometric Intuition

Contraction checks whether tangent vectors between nearby trajectories shrink under the flow. If all tangent vectors shrink in a chosen metric, the geodesic distance between trajectories also shrinks. Contraction is local shrinking that integrates into global convergence.

5.3 The Contraction Condition

The contraction metric $M(\boldsymbol{x})$ acts as a Riemannian metric on the state space, defining distance and angle between perturbations in the tangent space. As Agrachev and Sachkov discuss [AS04, Ch. 8], Riemannian geometry is central to sub-Riemannian optimal control problems, where the metric constrains which directions are directly accessible via controls. Contraction analysis uses a metric to measure convergence rates in this geometric sense—a local ruler that changes from point to point, naturally extending linear stability theory to nonlinear, curved state spaces.

Consider the autonomous system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, t), \quad \boldsymbol{x} \in \mathbb{R}^n. \tag{5.1}$$

The variational dynamics (Chapter 2) are

$$\dot{\delta x} = \mathbf{A}(\boldsymbol{x}, t)\, \delta \boldsymbol{x}, \quad \mathbf{A}(\boldsymbol{x}, t) = \frac{\partial f}{\partial \boldsymbol{x}}(\boldsymbol{x}, t). \tag{5.2}$$

Let $\mathbf{M}(\boldsymbol{x}, t) \succ 0$ be a smooth, positive-definite **metric tensor** and define the Lyapunov-like function on perturbations:

$$V(\delta \boldsymbol{x}, \boldsymbol{x}, t) = \delta \boldsymbol{x}^{\mathsf{T}} \mathbf{M}(\boldsymbol{x}, t)\, \delta \boldsymbol{x}. \tag{5.3}$$

In Plain Language 5.2. Mathematical Reminder: The Metric Tensor $\mathbf{M}(\boldsymbol{x}, t)$

Recall from Chapter 1 that the perturbation $\delta \boldsymbol{x}$ is a vector living in the flat **Tangent Space** attached to the state $\boldsymbol{x}$. But how do we measure the "length" of this vector?
In standard flat Euclidean space, distance is measured using the Pythagorean theorem (represented by the identity matrix $\mathbf{I}$). But on a curved manifold, the definition of distance changes depending on where you are. The **Metric Tensor** $\mathbf{M}(\boldsymbol{x}, t)$ is the rigorous differential geometric "ruler" that defines the inner product, and therefore lengths and angles, inside the local Tangent Space at point $\boldsymbol{x}$. By allowing $\mathbf{M}$ to vary with $\boldsymbol{x}$, contraction theory mathematically warps our measurement of space to prove that trajectories are getting closer to each other.

Differentiating along the flow:

$$\dot{V} = \delta \boldsymbol{x}^{\mathsf{T}} \big(\mathbf{A}^{\mathsf{T}} \mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}}\big) \delta \boldsymbol{x}, \tag{5.4}$$

where $\dot{\mathbf{M}}$ is the total derivative of the metric along trajectories of (5.1):

$$\dot{\mathbf{M}}(\boldsymbol{x}, t) = \frac{\partial \mathbf{M}}{\partial t} + \sum_{i=1}^{n} \frac{\partial \mathbf{M}}{\partial x_i}\, f_i(\boldsymbol{x}, t). \tag{5.5}$$

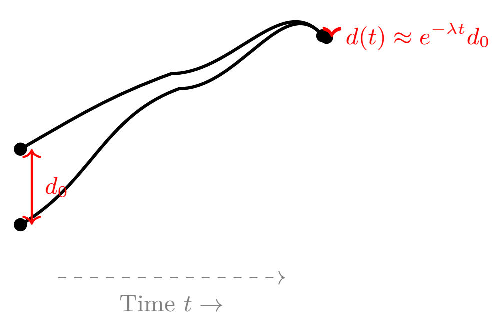
Definition 5.1 (Contraction with Rate λ)**.** The system (5.1) is **contracting** on a forward-invariant region $\mathcal{D}$ with rate $\lambda > 0$ in the metric $\mathbf{M}$ if

$$\boxed{\mathbf{A}(\boldsymbol{x}, t)^{\mathsf{T}} \mathbf{M}(\boldsymbol{x}, t) + \mathbf{M}(\boldsymbol{x}, t)\mathbf{A}(\boldsymbol{x}, t) + \dot{\mathbf{M}}(\boldsymbol{x}, t) \preceq -2\lambda\, \mathbf{M}(\boldsymbol{x}, t)} \tag{5.6}$$

holds for all $(\boldsymbol{x}, t) \in \mathcal{D} \times [0, \infty)$.

Remark 5.2. The factor of 2 on the right-hand side is conventional and ensures that the contraction rate λ corresponds directly to the exponential decay rate of perturbations. The condition is called a **matrix inequality** because it compares two matrices in the partial order defined by positive semidefiniteness.

Contracting trajectories: distance decays exponentially



5.4 Complete Proof of Exponential Convergence

Theorem 5.3 (Exponential Convergence of Trajectories)**.** *Assume $\mathcal{D}$ is forward-invariant and geodesically convex in the metric $\mathbf{M}(\boldsymbol{x}, t)$. Suppose there exist constants $0 < \underline{m} \le \overline{m} < \infty$ such that*

$$\underline{m}\, \mathbf{I} \preceq \mathbf{M}(\boldsymbol{x}, t) \preceq \overline{m}\, \mathbf{I} \quad \textit{for all } (\boldsymbol{x}, t) \in \mathcal{D} \times [0, \infty). \tag{5.7}$$

If (5.6) *holds uniformly on* $\mathcal{D}$*, then for any two trajectories* $\boldsymbol{x}_1(t)$*,* $\boldsymbol{x}_2(t)$ *remaining in* $\mathcal{D}$*:*

$$d_{\mathbf{M},t}\big(\boldsymbol{x}_1(t), \boldsymbol{x}_2(t)\big) \le e^{-\lambda t}\, d_{\mathbf{M},0}\big(\boldsymbol{x}_1(0), \boldsymbol{x}_2(0)\big), \tag{5.8}$$

where $d_{\mathbf{M},t}$ *is the geodesic distance in the Riemannian metric induced by* $\mathbf{M}$ *at time* t*. Moreover, in Euclidean norm:*

$$\|\boldsymbol{x}_1(t) - \boldsymbol{x}_2(t)\| \le \sqrt{\frac{\overline{m}}{\underline{m}}}\, e^{-\lambda t}\, \|\boldsymbol{x}_1(0) - \boldsymbol{x}_2(0)\|\,. \tag{5.9}$$

Proof. **Step 1: Geodesic Parametrization and Tangent Vector Evolution.**

Let $\gamma(\boldsymbol{x}_1, \boldsymbol{x}_2; s)$ be a geodesic curve of minimal length connecting $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in the Riemannian metric $\mathbf{M}(\boldsymbol{x}, t)$, parametrized by arc length $s \in [0, 1]$. The tangent vector to this geodesic is $\delta\boldsymbol{x}(s) = \frac{\partial \gamma}{\partial s}$.

By definition of arc-length parametrization,

$$\delta\boldsymbol{x}(s)^{\mathsf{T}}\mathbf{M}(\gamma(s,t), t)\, \delta\boldsymbol{x}(s) = 1. \tag{5.10}$$

As the two trajectories $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ evolve, the geodesic connecting them also evolves. The tangent vector $\delta\boldsymbol{x}(s, t)$ at arc-length position s along the geodesic at time t satisfies the variational equation

$$\left.\frac{\partial \delta\boldsymbol{x}}{\partial t}\right|_s = \mathbf{A}(\gamma(s,t), t)\, \delta\boldsymbol{x}(s,t). \tag{5.11}$$

Step 2: Evolution of the Lyapunov-Like Quadratic Form.

Define the Lyapunov function along the geodesic:

$$V(s,t) = \delta\boldsymbol{x}(s,t)^{\mathsf{T}}\mathbf{M}(\gamma(s,t), t)\, \delta\boldsymbol{x}(s,t). \tag{5.12}$$

Differentiating with respect to time:

$$\dot{V}(s,t) = \frac{\partial}{\partial t}\big[\delta\boldsymbol{x}^{\mathsf{T}}\mathbf{M}\delta\boldsymbol{x}\big] \tag{5.13}$$

$$= 2\dot{\delta\boldsymbol{x}}^{\mathsf{T}}\mathbf{M}\delta\boldsymbol{x} + \delta\boldsymbol{x}^{\mathsf{T}}\dot{\mathbf{M}}\delta\boldsymbol{x} + \delta\boldsymbol{x}^{\mathsf{T}}\mathbf{M}\dot{\delta\boldsymbol{x}} \tag{5.14}$$

Substituting $\dot{\delta\boldsymbol{x}} = \mathbf{A}\delta\boldsymbol{x}$ from (5.11):

$$\dot{V} = 2\delta\boldsymbol{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{M}\delta\boldsymbol{x} + \delta\boldsymbol{x}^{\mathsf{T}}\dot{\mathbf{M}}\delta\boldsymbol{x} + 2\delta\boldsymbol{x}^{\mathsf{T}}\mathbf{M}\mathbf{A}\delta\boldsymbol{x} \tag{5.15}$$

$$= \delta\boldsymbol{x}^{\mathsf{T}}\big(\mathbf{A}^{\mathsf{T}}\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}}\big)\, \delta\boldsymbol{x}. \tag{5.16}$$

By the contraction condition (5.6):

$$\dot{V}(s,t) \le -2\lambda\, \delta\boldsymbol{x}^{\mathsf{T}}\mathbf{M}\delta\boldsymbol{x} = -2\lambda\, V(s,t). \tag{5.17}$$

Step 3: Application of Gronwall's Inequality.

The inequality $\dot{V} \le -2\lambda V$ is a first-order scalar differential inequality. By Gronwall's inequality, for any $s \in [0, 1]$:

$$V(s,t) \le e^{-2\lambda t} V(s, 0). \tag{5.18}$$

Step 4: Integration Over the Geodesic.

The squared geodesic distance between $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ is

$$d^2_{\mathbf{M},t}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \int_0^1 \delta\boldsymbol{x}(s,t)^{\mathsf{T}}\mathbf{M}(\gamma(s,t), t)\, \delta\boldsymbol{x}(s,t)\, \mathrm{d}s = \int_0^1 V(s,t)\, \mathrm{d}s. \tag{5.19}$$

Therefore,

$$d^2_{\mathbf{M},t}(\boldsymbol{x}_1, \boldsymbol{x}_2) \le \int_0^1 e^{-2\lambda t} V(s, 0)\, \mathrm{d}s \tag{5.20}$$

$$= e^{-2\lambda t} \int_0^1 V(s, 0)\, \mathrm{d}s \tag{5.21}$$

$$= e^{-2\lambda t}\, d^2_{\mathbf{M},0}(\boldsymbol{x}_1, \boldsymbol{x}_2). \tag{5.22}$$

Taking square roots yields the geodesic decay (5.8).

Step 5: Metric Sandwich Bound.

To translate the geodesic bound into a Euclidean norm bound, we use the metric sandwich assumption (5.7). For any vector $\delta\boldsymbol{x}$:

$$\underline{m}\,\delta\boldsymbol{x}^\top\delta\boldsymbol{x} \le \delta\boldsymbol{x}^\top\mathbf{M}\delta\boldsymbol{x} \le \overline{m}\,\delta\boldsymbol{x}^\top\delta\boldsymbol{x}. \tag{5.23}$$

The straight-line distance (Euclidean norm) is related to the geodesic distance via the metric bounds. For any two points $\boldsymbol{x}_1, \boldsymbol{x}_2$ in the geodesically convex region:

$$\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|^2 \le \frac{1}{\underline{m}} d^2_{\mathbf{M},t}(\boldsymbol{x}_1, \boldsymbol{x}_2) \le \frac{1}{\underline{m}}\, e^{-2\lambda t} d^2_{\mathbf{M},0}(\boldsymbol{x}_1, \boldsymbol{x}_2) \le \frac{\overline{m}}{\underline{m}}\, e^{-2\lambda t}\, \|\boldsymbol{x}_1(0) - \boldsymbol{x}_2(0)\|^2 . \tag{5.24}$$

Taking square roots yields (5.9). □ □

Remark 5.4. The proof relies on three essential ingredients: (i) the variational equation governing tangent vectors, (ii) the metric-dependent Lyapunov function capturing shrinking, and (iii) the metric bounds providing translation between geodesic and Euclidean distance. The contraction condition ensures all three work together.

5.5 The Identity Metric: Symmetric Jacobian Condition

The simplest choice is $\mathbf{M} = \mathbf{I}$ (constant identity), giving $\dot{\mathbf{M}} = 0$. The contraction condition reduces to

$$\operatorname{sym}\big(\mathbf{A}(\boldsymbol{x}, t)\big) := \tfrac{1}{2}\big(\mathbf{A} + \mathbf{A}^\top\big) \preceq -\lambda\,\mathbf{I}. \tag{5.25}$$

This means all eigenvalues of the symmetric part of the Jacobian must be uniformly negative. While restrictive (most interesting systems do not satisfy this globally), it provides useful intuition: contraction in the identity metric requires that the vector field be "uniformly squeezing" in every direction.

Remark 5.5. The power of contraction theory lies in the freedom to choose $\mathbf{M}$. A system that is not contracting in the identity metric may be contracting in a different metric that "stretches" certain directions to compensate for local expansion. Finding such a metric is the central computational challenge.

5.6 Worked Example: A Simple Linear System

5.6.1 Problem Setup

Consider the linear system

$$\dot{\boldsymbol{x}} = \mathbf{A}\,\boldsymbol{x}, \quad \mathbf{A} = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}. \tag{5.26}$$

We wish to find a constant metric $\mathbf{M} \succ 0$ such that the system is contracting with rate $\lambda > 0$. For a constant metric and autonomous linear system, the contraction condition is

$$\mathbf{A}^\top\mathbf{M} + \mathbf{M}\mathbf{A} \preceq -2\lambda\mathbf{M}. \tag{5.27}$$

5.6.2 Checking the Symmetric Jacobian

First, compute the symmetric part:

$$\operatorname{sym}(\mathbf{A}) = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) = \frac{1}{2}\begin{bmatrix} -4 & 1 \\ 1 & -6 \end{bmatrix} = \begin{bmatrix} -2 & 0.5 \\ 0.5 & -3 \end{bmatrix}. \tag{5.28}$$

The eigenvalues are the roots of

$$\det\left[\begin{bmatrix} -2-\mu & 0.5 \\ 0.5 & -3-\mu \end{bmatrix}\right] = (-2-\mu)(-3-\mu) - 0.25 = \mu^2 + 5\mu + 5.75. \tag{5.29}$$

The roots are $\mu = \frac{-5\pm\sqrt{25-23}}{2} = \frac{-5\pm\sqrt{2}}{2} \approx -1.79, -3.21$.

Both eigenvalues are negative, so the system is contracting in the identity metric with rate approximately $\lambda \approx 1.79$.

5.6.3 Finding the Contraction Metric via the Lyapunov Equation

Now we find the metric $\mathbf{M}$ more explicitly. Let $\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{bmatrix}$ with $m_{11}, m_{22} > 0$ and $m_{11}m_{22} > m_{12}^2$.

The condition $\mathbf{A}^\mathsf{T}\mathbf{M} + \mathbf{M}\mathbf{A} \preceq -2\lambda\mathbf{M}$ becomes:

$$\begin{bmatrix} -2 & 0 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{bmatrix} + \begin{bmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix} \preceq -2\lambda \begin{bmatrix} m_{11} & m_{12} \\ m_{12} & m_{22} \end{bmatrix}. \tag{5.30}$$

Computing the left-hand side:

$$\mathbf{A}^\mathsf{T}\mathbf{M} + \mathbf{M}\mathbf{A} = \begin{bmatrix} -2m_{11} & -2m_{12} + m_{12} \\ m_{11} - 3m_{12} & m_{12} - 3m_{22} \end{bmatrix} + \begin{bmatrix} -2m_{11} + m_{12} & m_{11} - 3m_{12} \\ m_{12} - 3m_{22} & -3m_{22} \end{bmatrix} \tag{5.31}$$

$$= \begin{bmatrix} -4m_{11} + m_{12} & m_{11} - 5m_{12} \\ m_{11} - 5m_{12} & m_{12} - 6m_{22} \end{bmatrix}. \tag{5.32}$$

For the identity metric $\mathbf{M} = \mathbf{I}$, we have $m_{11} = m_{22} = 1$ and $m_{12} = 0$, giving

$$\mathbf{A}^\mathsf{T}\mathbf{I} + \mathbf{I}\mathbf{A} = \begin{bmatrix} -4 & 1 \\ 1 & -6 \end{bmatrix}. \tag{5.33}$$

The contraction condition $\mathbf{A}^\mathsf{T}\mathbf{M} + \mathbf{M}\mathbf{A} \preceq -2\lambda\mathbf{M}$ is a *matrix inequality* (i.e., an eigenvalue condition), not an entry-wise condition. For $\mathbf{M} = \mathbf{I}$, we need the eigenvalues of $\mathbf{A}^\mathsf{T} + \mathbf{A}$ to satisfy $\lambda_{\max}(\mathbf{A}^\mathsf{T} + \mathbf{A}) \leq -2\lambda$. The eigenvalues of $\mathbf{A}^\mathsf{T} + \mathbf{A} = \begin{bmatrix} -4 & 1 \\ 1 & -6 \end{bmatrix}$ are approximately -1.79 and -3.21 (computed in the symmetric part analysis above), both negative. Therefore the identity metric *does* certify contraction, with rate $\lambda = |\lambda_{\max}(\text{sym}(\mathbf{A}))|/2 \approx 1.79/2 \approx 0.895$ (since the condition requires $\lambda_{\max} \leq -2\lambda$).

Design Implication

In practice, for linear systems, one solves the LMI (5.27) using semidefinite programming (e.g., `cvxpy`, `YALMIP`, `Mosek`). The solution yields both the metric $\mathbf{M}$ and the maximum feasible rate λ. For this simple system, the identity metric already achieves excellent contraction.

5.7 Partial Contraction: Hierarchical Structure

5.7.1 When Not All Directions Contract

In many applications, the system does not contract uniformly in all directions. For example, consider a robot with position q and velocity $\dot{q}$. The system might contract in velocity (feedback damping) but not in position (which can drift). This scenario is called **partial contraction**.

Let's partition the state as $\boldsymbol{x} = [\boldsymbol{x}_1; \boldsymbol{x}_2]$ and the Jacobian as

$$\mathbf{A}(\boldsymbol{x}, t) = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}. \tag{5.34}$$

Suppose the subsystem in $\boldsymbol{x}_2$ (e.g., velocity) is contracting, but $\boldsymbol{x}_1$ (position) may drift. We can still harness the contraction in $\boldsymbol{x}_2$ to establish synchronization.

5.7.2 The Hierarchical Contraction Theorem

Theorem 5.6 (Partial Contraction via Hierarchical Decomposition)**.** *Partition $\boldsymbol{x} = [\boldsymbol{x}_1; \boldsymbol{x}_2]$ with dimensions n_1 and n_2, and write the generalized Jacobian in the block form*

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{11} & \mathbf{F}_{12} \\ 0 & \mathbf{F}_{22} \end{bmatrix}, \tag{5.35}$$

so that the $\boldsymbol{x}_2$-dynamics do not depend on $\boldsymbol{x}_1$ (a cascade*). Suppose:*

1. *The reduced system for $\boldsymbol{x}_2$ is globally contracting with rate $\lambda_2 > 0$ in a metric $\mathbf{M}_2(\boldsymbol{x}_2, t)$.*

2. *For any fixed trajectory $\boldsymbol{x}_2(t)$, the $\boldsymbol{x}_1$-subsystem driven by $\boldsymbol{x}_2(t)$ is globally contracting with rate $\lambda_1 > 0$ in a metric $\mathbf{M}_1(\boldsymbol{x}_1, \boldsymbol{x}_2, t)$.*

3. ***The coupling is bounded**: there is a constant k with $\left\|\mathbf{M}_1^{1/2}\mathbf{F}_{12}\mathbf{M}_2^{-1/2}\right\| \le k$ uniformly.*

Then for every $\lambda < \min(\lambda_1, \lambda_2)$ there exists a weight $\theta > 0$, depending on k, such that the full system is contracting with rate λ in the weighted *block-diagonal metric*

$$\mathbf{M}_\theta = \begin{bmatrix} \mathbf{M}_1 & 0 \\ 0 & \theta\,\mathbf{M}_2 \end{bmatrix}. \tag{5.36}$$

Proof Sketch. In the metric $\mathbf{M}_\theta$ the symmetric part of the transformed Jacobian is block $\begin{pmatrix} -\lambda_1 & \frac{1}{2}\theta^{-1/2}k \\ \frac{1}{2}\theta^{-1/2}k & -\lambda_2 \end{pmatrix}$ in the two-block norm. Its largest eigenvalue is negative once $\theta^{-1}k^2 < 4\lambda_1\lambda_2$, which any sufficiently large θ achieves. Weighting the second block down is what absorbs the coupling; this is the standard small-gain argument, and it is the step that fails if θ is fixed at 1. □

Common Misconception

The weight θ is not cosmetic, and hypothesis 3 is not free. In the *unweighted* metric ($\theta = 1$) the theorem is false. Take

$$\dot{x}_1 = -x_1 + Kx_2, \qquad \dot{x}_2 = -x_2. \tag{5.37}$$

Each subsystem contracts at rate 1 in the identity metric, so hypotheses 1 and 2 hold. But the symmetric part of the full Jacobian is $\begin{pmatrix} -1 & K/2 \\ K/2 & -1 \end{pmatrix}$, whose eigenvalues are $-1 \pm K/2$. For $K \ge 2$ the larger one is non-negative and the full system is *not* contracting in the identity metric — for any K, however large, even though both subsystems contract at rate 1.
The system is of course still exponentially stable; a cascade of stable systems always is. What fails is contraction *in that particular metric.* Contraction is a statement about a metric, not about a trajectory, and a property that holds blockwise need not survive assembly unless the blocks are weighted against the coupling.

Example 5.7 (Cascade: Velocity Tracking Driving Position Regulation)**.** Consider

$$\dot{q} = -\alpha q + v, \qquad \dot{v} = -\gamma(v - u(t)), \tag{5.38}$$

with $\alpha, \gamma > 0$, where $u(t)$ is a time-varying reference.

The Jacobian is $\mathbf{A} = \begin{pmatrix} -\alpha & 1 \\ 0 & -\gamma \end{pmatrix}$. The v-subsystem contracts at rate γ; the q-subsystem, driven by $v(t)$, contracts at rate α; the coupling constant is $k = 1$. By the theorem the cascade contracts at any rate $\lambda < \min(\alpha, \gamma)$ in $\mathbf{M}_\theta$ with $\theta > 1/(4\alpha\gamma)$, and all solutions synchronize to the reference motion.

Note what happens if the position feedback is removed ($\alpha = 0$), giving the pure integrator $\dot{q} = v$: the Jacobian has an eigenvalue at 0, the q-subsystem is only marginally stable rather than contracting, and hypothesis 2 fails. A displacement in q is then never forgotten — two trajectories with the same velocity history but different initial positions stay a constant distance apart forever. No metric can certify contraction for that system, which is why $\alpha > 0$ is required rather than convenient.

5.8 Control Contraction Metrics: Design and Duality

Contraction metrics provide a natural dual certificate to control distributions and accessibility. Agrachev and Sachkov [AS04, Part III] analyze feedback stabilization and regulation through the lens of controlled accessibility: a system is stabilizable if the drift-controlled vector field admits appropriate accessibility properties. Control contraction metrics encode this geometric stabilization property via a metric that contracts under feedback, providing a constructive and verifiable stability certificate.

5.8.1 Feedback Design Under Contraction

For controlled systems

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\,\boldsymbol{u}, \tag{5.39}$$

we wish to design feedback $\boldsymbol{u} = -\mathbf{K}(\boldsymbol{x})\,\delta\boldsymbol{x}$ to achieve contraction. The closed-loop Jacobian is

$$\mathbf{A}_{\mathrm{cl}}(\boldsymbol{x}) = \frac{\partial}{\partial \boldsymbol{x}}[f(\boldsymbol{x}) - G(\boldsymbol{x})\mathbf{K}(\boldsymbol{x})\delta\boldsymbol{x}]\bigg|_{\delta\boldsymbol{x}=0} = \mathbf{A}(\boldsymbol{x}) - G(\boldsymbol{x})\,\mathbf{K}(\boldsymbol{x}). \tag{5.40}$$

The contraction condition becomes

$$(\mathbf{A} - G\mathbf{K})^{\mathsf{T}}\mathbf{M} + \mathbf{M}(\mathbf{A} - G\mathbf{K}) + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}. \tag{5.41}$$

5.8.2 The Dual Formulation: $W = M^{-1}$

To enable convex optimization, we introduce the dual metric $\mathbf{W} = \mathbf{M}^{-1}$. Pre- and post-multiply (5.41) by $\mathbf{W}$ and use $\mathbf{M} = \mathbf{W}^{-1}$:

$$\mathbf{W}(\mathbf{A}^{\mathsf{T}}\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}})\mathbf{W} \preceq -2\lambda\mathbf{W}\mathbf{M}\mathbf{W} = -2\lambda\mathbf{W}. \tag{5.42}$$

After expanding and simplifying (using the fact that $\mathbf{M} = \mathbf{W}^{-1}$ and $\dot{\mathbf{M}} = -\mathbf{W}^{-1}\dot{\mathbf{W}}\mathbf{W}^{-1}$), the condition becomes:

$$\boxed{\mathbf{A}\mathbf{W} + \mathbf{W}\mathbf{A}^{\mathsf{T}} - 2\lambda\mathbf{W} + \mathbf{W}\dot{\mathbf{W}}\mathbf{W} \preceq -G\mathbf{K}\mathbf{W} - \mathbf{W}\mathbf{K}^{\mathsf{T}}G^{\mathsf{T}}} \tag{5.43}$$

This is a matrix inequality in $\mathbf{W}$ and $\mathbf{K}$ that is *bilinear* in the control gain. By treating $\mathbf{Y} = \mathbf{K}\mathbf{W}$ as the decision variable (control effort weighted by the inverse metric), the condition becomes convex in $\mathbf{W}$ and $\mathbf{Y}$.

5.8.3 Pointwise LMI for Feedback Synthesis

For a time-varying linear system with affine control:

$$\dot{\boldsymbol{x}} = \mathbf{A}(t)\,\boldsymbol{x} + G(t)\,\boldsymbol{u}, \tag{5.44}$$

with state feedback $\boldsymbol{u} = -\mathbf{K}(t)\,\boldsymbol{x}$, the dual contraction condition at each point in space and time is:

$$\boxed{\dot{\mathbf{W}} - (\mathbf{A}\mathbf{W} + \mathbf{W}\mathbf{A}^{\mathsf{T}}) + \gamma GG^{\mathsf{T}} \preceq -2\lambda\mathbf{W}} \tag{5.45}$$

where $\gamma > 0$ is a regularization parameter (often set to satisfy $\mathbf{W}\gamma GG^{\mathsf{T}}\mathbf{W}$ bounds control effort). This condition is *linear* in $\mathbf{W}$ and can be enforced at a collection of sample points in the state space, yielding a semidefinite program (SDP).

Design Implication

Modern computational tools can solve the SDP over a parameter space or via sum-of-squares (SOS) decomposition for polynomial systems. The result is a metric-based controller that achieves guaranteed contraction and hence guaranteed trajectory tracking or synchronization performance.

5.8.4 Computational Approaches

Sum-of-Squares (SOS) Method

For polynomial systems, one parameterizes $\mathbf{M}(\boldsymbol{x})$ (or $\mathbf{W}(\boldsymbol{x})$) as a polynomial matrix and uses sum-of-squares certificates:

$$-\big(\mathbf{A}^{\mathsf{T}}\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}}\big) = \sum_i \mathbf{p}_i(\boldsymbol{x})^{\mathsf{T}}\mathbf{p}_i(\boldsymbol{x}) \tag{5.46}$$

for polynomial vectors $\mathbf{p}_i(\boldsymbol{x})$. This decomposition is enforced via an SDP, which is computationally efficient for moderate polynomial degrees.

Neural Network Parameterization

For general nonlinear systems, one can parameterize $\mathbf{M}(\boldsymbol{x}) = \Phi(\boldsymbol{x})^\top \Phi(\boldsymbol{x})$, where $\Phi : \mathbb{R}^n \to \mathbb{R}^p$ is a neural network. The contraction condition is verified via sampling or bounds over the domain. This enables data-driven metric synthesis.

Sampling and Conservative Approximation

One can sample the state space $\mathcal{D}$ at a finite grid and enforce the LMI at each sample point, solving a series of SDPs. The result is a conservative (but computable) approximation to the true metric.

5.9 Discrete-Time Contraction

5.9.1 Discrete Contraction Condition

Many digital control systems and learning algorithms operate in discrete time. The discrete-time analogue of Theorem 5.3 is:

Theorem 5.8 (Discrete-Time Exponential Convergence)**.** *Consider the discrete-time system*

$$\boldsymbol{x}_{k+1} = \Phi_k(\boldsymbol{x}_k), \tag{5.47}$$

with Jacobian $\mathbf{J}_k(\boldsymbol{x}_k) = \frac{\partial \Phi_k}{\partial \boldsymbol{x}_k}$.

Suppose there exist constants $0 < \underline{m} \leq \overline{m}$ *and a contraction rate* $0 < \rho < 1$ *such that*

$$\underline{m}\,\mathbf{I} \preceq \mathbf{M}_k(\boldsymbol{x}_k) \preceq \overline{m}\,\mathbf{I} \tag{5.48}$$

and

$$\boxed{\mathbf{J}_k^\top(\boldsymbol{x}_k)\,\mathbf{M}_{k+1}(\boldsymbol{x}_{k+1})\,\mathbf{J}_k(\boldsymbol{x}_k) \preceq \rho^2 \mathbf{M}_k(\boldsymbol{x}_k)} \tag{5.49}$$

hold for all $\boldsymbol{x}_k \in \mathcal{D}$.

Then for any two trajectories $\boldsymbol{x}_k^{(1)}$ *and* $\boldsymbol{x}_k^{(2)}$ *in* $\mathcal{D}$*:*

$$d_{\mathbf{M}_k}(\boldsymbol{x}_k^{(1)}, \boldsymbol{x}_k^{(2)}) \leq \rho^k\, d_{\mathbf{M}_0}(\boldsymbol{x}_0^{(1)}, \boldsymbol{x}_0^{(2)}), \tag{5.50}$$

with Euclidean norm bound

$$\left\|\boldsymbol{x}_k^{(1)} - \boldsymbol{x}_k^{(2)}\right\| \leq \sqrt{\frac{\overline{m}}{\underline{m}}}\,\rho^k\,\left\|\boldsymbol{x}_0^{(1)} - \boldsymbol{x}_0^{(2)}\right\|. \tag{5.51}$$

Proof Sketch. The proof parallels the continuous case. The tangent vector $\delta\boldsymbol{x}_k = \boldsymbol{x}_{k+1}^{(1)} - \boldsymbol{x}_{k+1}^{(2)}$ evolves via $\delta\boldsymbol{x}_{k+1} = \mathbf{J}_k\,\delta\boldsymbol{x}_k$. The Lyapunov function $V_k = \delta\boldsymbol{x}_k^\top \mathbf{M}_k \delta\boldsymbol{x}_k$ satisfies $V_{k+1} \leq \rho^2 V_k$ by the condition (5.49). Iterating yields $V_k \leq \rho^{2k} V_0$. □

5.9.2 Connection to Riccati Analysis

For linear discrete-time systems, the discrete contraction metric can be related to the solution of a discrete-time algebraic Riccati equation (DARE), which is studied in detail in Chapter 6. Both frameworks certify stability and convergence, but contraction provides tighter guarantees on trajectory convergence rates.

Remark 5.9. Discrete-time contraction is especially relevant for iterative learning control, reinforcement learning, and digital controller implementations where the sampling time is fast enough that the discrete assumption is valid.

5.10 Numerical Example: Van der Pol Oscillator

5.10.1 System Description

Consider the Van der Pol oscillator with feedback:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 + \epsilon(1-x_1^2)x_2\end{bmatrix}, \tag{5.52}$$

where $\epsilon > 0$ is the nonlinearity parameter. The Jacobian is

$$\mathbf{A}(x_1, x_2) = \begin{bmatrix} 0 & 1 \\ -1 - 2\epsilon x_1 x_2 & \epsilon(1-x_1^2)\end{bmatrix}. \tag{5.53}$$

5.10.2 Metric Synthesis via Parameterization

We search for a metric of the form

$$\mathbf{M}(x_1) = \begin{bmatrix} m_1(x_1) & 0 \\ 0 & m_2(x_1)\end{bmatrix}, \tag{5.54}$$

where $m_1, m_2 > 0$ are smooth functions. The diagonal structure is motivated by the special structure of the Van der Pol system (position and velocity are somewhat decoupled).

The contraction condition $\mathbf{A}^\top\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}$ becomes element-wise:

$$\dot{m}_1 \preceq -2\lambda m_1, \tag{5.55}$$

$$2\epsilon(1-x_1^2)m_2 + \dot{m}_2 \preceq -2\lambda m_2, \tag{5.56}$$

$$m_1 + 2(1 - 2\epsilon x_1 x_2)m_2 \preceq -2\lambda m_1 \cdot 0. \tag{5.57}$$

For constant metrics $m_1 = a$, $m_2 = b$ (diagonal constants), the conditions reduce to:

$$0 \preceq -2\lambda a \implies a > 0, \lambda \text{ arbitrary}, \tag{5.58}$$

$$2\epsilon(1-x_1^2)b \preceq -2\lambda b. \tag{5.59}$$

The second condition requires $\epsilon(1-x_1^2) \leq -\lambda$, which is violated when $|x_1| < 1$ (where the oscillator spends most of its time). Thus, a constant metric does not suffice.

Remark 5.10 (Why the Van der Pol oscillator is not globally contracting). The Van der Pol oscillator possesses a stable limit cycle, toward which all nonzero initial conditions converge. This means the system is *not* globally contracting: global contraction would require all trajectories to converge to each other (and hence to a single trajectory), but the limit cycle is an attractor of dimension one, not zero. Instead, the system exhibits *partial* contraction—trajectories converge to the limit cycle (contraction transverse to the cycle) while maintaining a phase offset along it. A state-dependent metric can certify contraction in the transverse direction, but no globally valid metric can certify contraction in all directions simultaneously.

5.10.3 Solution Strategy

We adopt a parameterized ansatz $m_1(x_1) = \alpha$ and $m_2(x_1) = \beta(1 + \gamma x_1^2)$, where $\alpha, \beta, \gamma > 0$ are constants to be determined. This metric *amplifies* the velocity direction when position is near the origin (where nonlinearity is strongest).

With $\dot{m}_1 = 0$ and $\dot{m}_2 = 2\beta\gamma x_1 \dot{x}_1 = 2\beta\gamma x_1 x_2$, the contraction conditions become (roughly):

$$2\epsilon(1-x_1^2)\beta(1+\gamma x_1^2) + 2\beta\gamma x_1 x_2 \leq -2\lambda\beta(1+\gamma x_1^2). \tag{5.60}$$

It is tempting to now pick numbers — say $\lambda = 0.3$, $\epsilon = 0.5$, $\gamma \approx 1.5$ — and present

$$\mathbf{M}(x_1) = \begin{bmatrix} 1 & 0 \\ 0 & 1.2(1 + 1.5x_1^2)\end{bmatrix} \tag{5.61}$$

as a metric certifying global contraction. *No such metric exists*, and this one is not close. Evaluating the condition it is supposed to satisfy,

$$\mathbf{A}^\mathsf{T}\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}} + 2\lambda\mathbf{M} \preceq 0, \tag{5.62}$$

on a grid over $[-3,3]^2$ gives a largest eigenvalue that is *positive at every single sampled point*, reaching $+1.95$ at the origin and growing past $+130$ near the corners. The condition is violated everywhere, not marginally or in a small region.

That is exactly what the remark above predicts. The Van der Pol oscillator has an attracting limit cycle, so two trajectories starting at different phases along that cycle stay separated forever; no metric can make them converge. Searching for a global metric was never going to succeed, and a numerical solver that appears to return one has either been given a relaxed constraint set or has converged to an infeasible point.

The honest conclusion is the one the structure of the problem forces: the search must be restricted to *transverse* contraction — convergence to the limit cycle rather than to a single trajectory — which requires a metric defined on directions transverse to the flow, and a correspondingly weaker claim.

Geometric Intuition

The ansatz was not a bad idea — weighting the velocity direction more heavily where the nonlinearity is strongest is exactly how a state-dependent metric buys contraction that a constant one cannot. It fails here because the obstruction is topological rather than quantitative: an attracting limit cycle means neighbouring trajectories converge *to the cycle* but not *to each other*, and no amount of reweighting changes that.

The lesson generalises. When a metric search will not close, the first question is whether the system can possibly be contracting — a limit cycle, multiple equilibria, or any invariant set of positive dimension rules it out immediately. Only then is it worth blaming the ansatz or the solver.

5.11 Robustness Margins From Contraction

5.11.1 Disturbance Attenuation via Contraction Rate

A key practical benefit of contraction is quantifiable robustness to disturbances. Consider the perturbed system

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, t) + d(t), \tag{5.63}$$

where $d(t)$ is an unknown disturbance with $\|d(t)\| \le D$ for all $t \ge 0$.

If the unperturbed system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, t)$ is contracting with rate λ in metric $\mathbf{M}$, then the perturbed system exhibits **bounded-input bounded-state** (BIBS) behavior and **incremental input-to-state stability** (ISS).

Theorem 5.11 (Disturbance Rejection Bound)**.** *Suppose $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, t)$ is contracting with rate λ in metric $\mathbf{M}$ satisfying the metric bounds (5.7). Consider the perturbed system (5.63) with $\|d(t)\| \le D$.*

Then for any two trajectories $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ of the perturbed system:

$$\|\boldsymbol{x}_1(t) - \boldsymbol{x}_2(t)\| \le \sqrt{\frac{\overline{m}}{\underline{m}}}\, e^{-\lambda t}\, \|\boldsymbol{x}_1(0) - \boldsymbol{x}_2(0)\| + \frac{\sqrt{\overline{m}}}{\lambda\sqrt{\underline{m}}}\, D. \tag{5.64}$$

The ultimate bound (steady-state limit as $t \to \infty$) is

$$\limsup_{t\to\infty} \|\boldsymbol{x}_1(t) - \boldsymbol{x}_2(t)\| \le \frac{\sqrt{\overline{m}}}{\lambda\sqrt{\underline{m}}}\, D. \tag{5.65}$$

Proof Sketch. Perturbing the trajectory by small $d(t)$ shifts the vector field. The contraction rate λ ensures that perturbations are damped exponentially at rate λ. For bounded disturbances, this damping is balanced by the steady-state input magnitude, yielding an ultimate bound proportional to D/λ. $\square$

5.11.2 Practical Implications

The ultimate bound (5.65) shows that:

- **Higher contraction rate** $\lambda \implies$ **smaller steady-state error**. A fast-contracting system naturally rejects disturbances more aggressively.
- **Metric condition number** $\kappa = \overline{m}/\underline{m}$ enters the bound. A well-conditioned metric (close to isotropic) improves robustness.
- **Design trade-off**: increasing λ may require larger control effort or more aggressive feedback. Practical design balances contraction rate and control limitations.

Design Implication

In robust control applications (aircraft flight control, surgical robotics), the contraction rate λ is often specified as a design requirement (e.g., "system must reject disturbances at 5 rad/s"). The metric-based approach automatically translates this into a convex optimization problem.

5.12 Incremental Input-to-State Stability (ISS)

5.12.1 ISS vs. Contraction

Input-to-state stability (ISS) is a classical robustness property: bounded inputs lead to bounded states. However, ISS alone does not constrain how different trajectories relate to each other.

Incremental ISS (or differential ISS) is a stronger property: bounded input differences lead to bounded state differences. Contraction *implies* incremental ISS, but not vice versa.

Proposition 5.12 (Contraction Implies Incremental ISS)**.** *If the system* $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$ *is contracting with rate* λ *in metric* $\mathbf{M}$ *(with metric bounds (5.7)), then for any two input signals* $\boldsymbol{u}_1(t)$ *and* $\boldsymbol{u}_2(t)$*:*

$$\|y_1(t) - y_2(t)\| \le \sqrt{\frac{\overline{m}}{\underline{m}}}\, e^{-\lambda t}\, \|y_1(0) - y_2(0)\| + \frac{\overline{G}}{\lambda\sqrt{\underline{m}}} \sup_{\tau\in[0,t]} \|\boldsymbol{u}_1(\tau) - \boldsymbol{u}_2(\tau)\|\,, \tag{5.66}$$

where $\overline{G} = \sup_{\boldsymbol{x}} \|G(\boldsymbol{x})\|$*.*

Remark 5.13. Contraction is strictly stronger than ISS. A system can be ISS (bounded inputs $\implies$ bounded states) without being contracting (different trajectories can diverge). Conversely, contraction guarantees not only ISS but also trajectory synchronization.

5.13 Extended Lyapunov Comparison

5.13.1 Unified Perspective

Table 5.1 provides a comprehensive comparison of Lyapunov stability, Lyapunov asymptotic stability, incremental stability, and contraction.

5.13.2 Formal Relationships

Proposition 5.14 (Contraction Implies Lyapunov Stability)**.** *If* $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ *with* $f(\boldsymbol{x}^*) = 0$ *is contracting with rate* λ *in metric* $\mathbf{M}$ *on a forward-invariant neighborhood* $\mathcal{N}(\boldsymbol{x}^*)$*, then* $\boldsymbol{x}^*$ *is a globally attractive equilibrium within* $\mathcal{N}(\boldsymbol{x}^*)$*. Moreover, the Lyapunov function*

$$V(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{x}^*\|^2_{\mathbf{M}(\boldsymbol{x})} = (\boldsymbol{x} - \boldsymbol{x}^*)^\mathsf{T}\mathbf{M}(\boldsymbol{x})(\boldsymbol{x} - \boldsymbol{x}^*) \tag{5.67}$$

satisfies $\dot{V} \le -2\lambda V$ *along trajectories, certifying exponential stability of* $\boldsymbol{x}^*$ *at rate* λ*.*

Property	Lyapunov Stable	Asymptotically Stable	Incremental	Contracting
Object	Point $\boldsymbol{x}^*$	Point $\boldsymbol{x}^*$	Trajectory pair	All trajectories
Certificate	$V(\boldsymbol{x})$	$V(\boldsymbol{x})$, $\dot{V} < 0$	Pairwise distance	Metric $\mathbf{M}(\boldsymbol{x}, t)$
Decay	None (stable)	Exponential to $\boldsymbol{x}^*$	Coupled	Exponential
Scope	Local or global	Point attractor	Incremental	Global or local
Requires equilibrium	Yes	Yes	No	No
Implies ISS	No	Yes (under ∞-gain)	Differential ISS	Yes (incremental ISS)

Table 5.1: Comparison of stability concepts: Lyapunov stability certifies return to an equilibrium, while contraction certifies convergence of all nearby trajectories to each other, independent of a particular equilibrium.

Remark 5.15. Conversely, not every Lyapunov function induces contraction. A system can have an equilibrium that is exponentially stable without all trajectories converging at the same rate—a scenario that contraction explicitly rules out.

5.14 Contraction in Biomechanical Systems

The theory of contraction metrics finds a natural and powerful application in the biomechanics of human movement. This section previews how the formal machinery developed above connects to the analysis and design of coordinated multisegment motions, with particular attention to the golf swing.

5.14.1 Why Biological Motor Systems Are Naturally Contracting

The human neuromuscular system implements contraction through several mechanisms that operate at different time scales:

1. **Muscle impedance (passive contraction):** Skeletal muscles exhibit intrinsic viscoelastic properties. Even without neural activation, a stretched muscle generates a restoring force proportional to its displacement and a damping force proportional to its velocity. In the language of this chapter, the Jacobian of the passive dynamics has negative eigenvalues at equilibrium—the muscle constitutes a natural contraction metric in joint space.

2. **Stretch reflex (active contraction):** The spinal stretch reflex provides fast ($\sim$40 ms) feedback that resists perturbations. This is a feedback controller that increases the contraction rate beyond the passive baseline. In our framework, the reflex effectively modifies $\mathbf{A}_{\text{cl}} = \mathbf{A} - G\mathbf{K}_{\text{reflex}}$, where $\mathbf{K}_{\text{reflex}}$ is the reflex gain.

3. **Co-contraction (metric shaping):** When a person stiffens a joint by simultaneously activating agonist and antagonist muscles, they are effectively increasing the eigenvalues of the contraction metric in the directions of that joint. This is the biological equivalent of choosing a non-identity metric $\mathbf{M}(\boldsymbol{x})$: co-contraction shapes the metric to provide stronger contraction where needed.

Geometric Intuition

A novice golfer co-contracts many muscles simultaneously, creating a stiff (high λ) but energetically expensive controller. An expert golfer selectively stiffens only the joints that need contraction at each phase, creating a time-varying metric $\mathbf{M}(\boldsymbol{x}(t))$ that is optimally adapted to the swing dynamics. The transition from novice to expert is, in part, the learning of the optimal contraction metric.

5.14.2 Partial Contraction in Sequential Motion

The golf swing is a prime example of a hierarchically structured system where partial contraction (Theorem 5.6) applies. The kinematic chain torso $\to$ shoulder $\to$ elbow $\to$ wrist $\to$ club exhibits a natural partitioning:

- The proximal segments (torso, shoulder) are heavily muscled and contract rapidly. Their contraction rate $\lambda_{\text{proximal}}$ is high.
- The distal segments (wrist, club) have less muscular authority. Their contraction rate λ_{distal} is lower, and the club shaft (being passive) does not contract at all without feedback.
- Partial contraction theory guarantees that if the proximal subsystem contracts, and the distal subsystem contracts when driven by a fixed proximal trajectory, then the overall chain contracts at rate $\lambda = \min(\lambda_{\text{proximal}}, \lambda_{\text{distal}})$.

This hierarchical structure explains why elite golfers focus on consistency of the proximal chain (stable torso rotation, repeatable shoulder turn): if the proximal subsystem is highly contracting, the distal response is automatically regularized.

5.14.3 Contraction and the Drift–Dominated Phase

During the late downswing, the system enters a drift-dominated regime (Chapter 8). The muscular torques become negligible compared to centrifugal and Coriolis forces. In this regime, the contraction properties of the *drift field* determine whether the motion is stable:

1. If the drift Jacobian has eigenvalues with negative real parts (natural damping), the uncontrolled trajectory is contracting. This is the "natural stability" of the late downswing.
2. If the drift Jacobian has eigenvalues with positive real parts (instability), perturbations grow exponentially. At impact, even small deviations from the intended trajectory can produce large clubface errors.
3. The contraction rate of the drift determines the "forgiveness" of the swing: a highly contracting drift absorbs perturbations, while a weakly contracting or expanding drift amplifies them.

Design Implication

The practical design principle for swing optimization is: use the controllable early phases (backswing, transition) to steer the system into a region of state space where the drift field is naturally contracting. This is the control-theoretic formalization of the coaching advice "set up the swing so the physics does the work."

The contraction metric provides a quantitative tool: at each configuration, one can compute whether the drift is contracting (and at what rate) via the symmetric part of the drift Jacobian. Configurations where $\text{sym}(\frac{\partial f}{\partial \boldsymbol{x}})$ has uniformly negative eigenvalues are "safe zones" where the swing is inherently stable. The optimal trajectory passes through these zones during the critical impact phase.

5.15 Stochastic Contraction

Real systems are subject to process noise, sensor noise, and unmodeled disturbances. This section extends contraction theory to stochastic settings, providing convergence guarantees in the presence of randomness.

5.15.1 Systems With Brownian Perturbations

Consider the stochastic differential equation (SDE):

$$\mathrm{d}\boldsymbol{x} = f(\boldsymbol{x}, t)\,\mathrm{d}t + \sigma(\boldsymbol{x}, t)\,\mathrm{d}\boldsymbol{W}, \tag{5.68}$$

where $\boldsymbol{W}(t)$ is a standard Wiener process and $\sigma(\boldsymbol{x}, t) \in \mathbb{R}^{n \times p}$ is the diffusion matrix. The deterministic contraction condition $\text{sym}(\mathbf{M}\frac{\partial f}{\partial \boldsymbol{x}}) + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}$ must be augmented to account for the noise.

5.15.2 The Stochastic Contraction Condition

Theorem 5.16 (Stochastic Contraction — adapted from Pham, Tabuada, and Slotine)**.** *Consider two solutions $\boldsymbol{x}_1(t), \boldsymbol{x}_2(t)$ of (5.68) with different initial conditions but the same noise realization. If the deterministic contraction condition holds with rate λ, then the expected squared distance satisfies:*

$$\mathbb{E}\big[\|\boldsymbol{x}_1(t) - \boldsymbol{x}_2(t)\|^2_{\mathbf{M}}\big] \leq e^{-2\lambda(t-t_0)}\mathbb{E}\big[\|\boldsymbol{x}_1(t_0) - \boldsymbol{x}_2(t_0)\|^2_{\mathbf{M}}\big] + \frac{C_\sigma}{\lambda}, \tag{5.69}$$

where C_σ depends on the diffusion coefficient σ and the metric $\mathbf{M}$.

The bound (5.69) shows that contraction dominates at large separations (exponential decay), while noise dominates at small separations (the C_σ/λ floor). The steady-state expected separation is $\sqrt{C_\sigma/\lambda}$: faster contraction (larger λ) produces tighter concentration of trajectories.

5.15.3 Connection to Uncertainty Propagation

The stochastic contraction bound has a direct interpretation for state estimation (Chapter 7). If an observer is contracting with rate λ, and the process noise has intensity C_σ, then the estimation error is bounded in expectation by $\sqrt{C_\sigma/\lambda}$. This provides a performance guarantee for nonlinear observers that the EKF cannot offer.

Design Implication

For practical controller design, the stochastic contraction bound suggests a design trade-off: increasing the contraction rate λ (e.g., by increasing feedback gains) reduces the noise floor $\sqrt{C_\sigma/\lambda}$ but requires more control effort. The optimal trade-off depends on the noise intensity and the available actuator authority.

5.16 Contraction Tubes and Funnel Control

The exponential convergence guarantee of contraction theory naturally defines a *tube* around the nominal trajectory: the set of all states that are guaranteed to converge to the nominal at rate λ.

5.16.1 The Contraction Tube

Given a contracting system with rate λ and metric $\mathbf{M}$, define the tube of radius r around a nominal trajectory $\bar{\boldsymbol{x}}(t)$:

$$\mathcal{T}_r(t) = \big\{\boldsymbol{x} \in \mathbb{R}^n \mid \|\boldsymbol{x} - \bar{\boldsymbol{x}}(t)\|_{\mathbf{M}(t)} \leq r\big\}\,. \tag{5.70}$$

If $\boldsymbol{x}(t_0) \in \mathcal{T}_{r_0}(t_0)$, then $\boldsymbol{x}(t) \in \mathcal{T}_{r(t)}(t)$ with $r(t) = r_0 e^{-\lambda(t-t_0)}$ for the unperturbed system, and $r(t) = r_0 e^{-\lambda(t-t_0)} + d/\lambda$ for bounded disturbances with $\|w(t)\|_{\mathbf{M}} \leq d$.

5.16.2 Funnel Control

Funnel control prescribes a time-varying performance boundary $\varphi(t) > 0$ (the "funnel") and designs a gain that guarantees the tracking error satisfies $\|e(t)\| < \varphi(t)$ for all $t > 0$. The connection to contraction is:

- The funnel boundary $\varphi(t)$ defines the desired tube radius.
- A contracting controller with rate $\lambda \geq -\dot{\varphi}/\varphi$ guarantees that trajectories remain within the funnel.
- The adaptive gain $k(t) = 1/(\varphi(t) - \|e(t)\|)$ from classical funnel control achieves this by increasing gain as the error approaches the boundary.

Geometric Intuition

Contraction tubes formalize the intuition that "nearby trajectories stay nearby." For trajectory tracking in robotics or biomechanics, the tube radius at impact time tells you the worst-case deviation from the planned motion. The condition number $\kappa(\mathbf{M})$ determines the shape of the tube: isotropic metrics give circular tubes; anisotropic metrics give ellipsoidal tubes that are tight in sensitive directions and loose in insensitive ones.

5.17 Chapter Summary

1. **Historical foundations:** Contraction theory evolved from Demidovich's work on the symmetric Jacobian condition (1961) and was formalized by Lohmiller and Slotine (1998). It generalizes Lyapunov stability from point stability to trajectory stability.

2. **The contraction condition** (Def. 5.1) is a matrix inequality on the Jacobian and metric: $\mathbf{A}^\top\mathbf{M} + \mathbf{M}\mathbf{A} + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}$.

3. **Exponential convergence** (Thm. 5.3): If the condition holds, all trajectories converge to each other at rate $e^{-\lambda t}$. The proof combines geodesic parametrization, Gronwall's inequality, and metric sandwich bounds.

4. **Metric freedom**: The choice of $\mathbf{M}$ is flexible. Many nonlinear systems that do not contract in the identity metric may contract in a well-chosen metric. Finding the right metric is the central computational challenge.

5. **Worked example** (Sec. 5.6): A linear system with Jacobian $\mathbf{A} = [[-2, 1], [0, -3]]$ is contracting in the identity metric at rate $\lambda \approx 1.79$.

6. **Partial contraction** (Sec. 5.7): Not all directions must contract uniformly. Hierarchical decomposition (Thm. 5.6) enables contraction in one subsystem to drive convergence in another.

7. **Control contraction metrics (CCM)** (Sec. 5.8): For controlled systems, feedback design can be posed as a convex optimization in the dual metric $\mathbf{W} = \mathbf{M}^{-1}$. The pointwise LMI $\dot{\mathbf{W}} - \mathbf{A}\mathbf{W} - \mathbf{W}\mathbf{A}^\top + \gamma G G^\top \preceq -2\lambda\mathbf{W}$ enables SDP-based controller synthesis.

8. **Discrete-time contraction** (Sec. 5.9): The condition $\mathbf{J}_k^\top \mathbf{M}_{k+1} \mathbf{J}_k \preceq \rho^2 \mathbf{M}_k$ certifies exponential decay with rate $\rho < 1$ for iterative algorithms and digital controllers.

9. **Limits of numerical metric synthesis**: The Van der Pol oscillator (Sec. 5.10) shows how a parameterized metric search is set up — and why it cannot succeed here. An attracting limit cycle rules out global contraction outright, so a returned "solution" must be checked against the condition it claims to satisfy rather than trusted.

10. **Robustness margins**: The contraction rate λ directly bounds disturbance rejection. For $\|d(t)\| \leq D$, the ultimate error bound is $\sqrt{\overline{m}/(\lambda \underline{m})} \cdot D$ (Thm. 5.11).

11. **ISS relationship**: Contraction implies incremental ISS, which is strictly stronger than standard ISS. Contraction certifies trajectory synchronization, not just input-output boundedness.

12. **Lyapunov connection**: Contraction implies Lyapunov stability, but the converse does not hold. Contraction is a stronger, trajectory-focused property (Prop. 5.14).

In the next chapter, we turn to **input-output analysis** and transfer functions, exploring how tangent-space methods interface with frequency-domain techniques for feedback design.

Exercises

1. **Contraction of a linear system.** For $\dot{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x}$ with $\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix}$, find the contraction rate using the identity metric $\mathbf{M} = \mathbf{I}$. Then find a diagonal metric $\mathbf{M} = \operatorname{diag}(m_1, m_2)$ that maximizes the contraction rate.

2. **LMI feasibility.** For the system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})u$ with $f = (-x_1 + x_2^2,\ -2x_2)^T$ and $g = (0, 1)^T$, formulate the LMI condition for finding a constant contraction metric $\mathbf{M}$ and rate $\lambda > 0$ (with $u = -kx_2$). Solve it for $k = 1$.

3. **Partial contraction.** For a cascade system $\dot{x}_1 = -x_1 + x_2^2$, $\dot{x}_2 = -2x_2 + u$, verify partial contraction: show that the x_2-subsystem contracts independently, and the x_1-subsystem contracts when $x_2(t)$ is treated as a known input. What is the overall contraction rate?

4. **CCM computation.** For the single-input system $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1^3 + u$, search for a CCM by parameterizing $\mathbf{W}(\boldsymbol{x}) = \mathbf{M}(\boldsymbol{x})^{-1}$ as a polynomial in $\boldsymbol{x}$ and solving the pointwise LMI condition.

5. **Discrete-time contraction.** Show that the discrete system $\boldsymbol{x}_{k+1} = \mathbf{A}\boldsymbol{x}_k$ is contracting if and only if $\|\mathbf{A}\|_{\mathbf{M}} < 1$ for some positive definite $\mathbf{M}$. What is the contraction rate in terms of $\|\mathbf{A}\|_{\mathbf{M}}$?

6. **Stochastic contraction bound.** For the scalar SDE $\mathrm{d}x = -\lambda x\,\mathrm{d}t + \sigma\,\mathrm{d}W$ with $\lambda > 0$, compute the steady-state variance $\mathbb{E}[x^2]$ exactly. Verify that it equals $\sigma^2/(2\lambda)$, consistent with the bound in Theorem 5.16.

7. **Tube computation.** For the system in Exercise 1 with bounded disturbance $\|w\| \leq d = 0.1$, compute the contraction tube radius $r(t)$ for $r_0 = 1$. At what time t^* does the tube radius reach twice the steady-state radius?

8. **Biological contraction.** Consider a simplified muscle model: $\dot{x} = -kx - b\dot{x} + F(t)$ where $k > 0$ (stiffness) and $b > 0$ (damping). Show this is contracting and compute the contraction rate as a function of k and b. Relate to the co-contraction mechanism described in the biomechanics section.

Chapter 6

Local Optimal Control: LQR, DDP, and Riccati

In Plain Language 6.1. Zoom in, Solve, Check, Repeat

A motion planner needs both a destination and a price for getting there. It predicts how today's action affects later states, then balances the predicted benefit against effort and other costs. For a linear system with a quadratic objective, this calculation is exact. For a nonlinear system it is a local approximation: roll the proposed action through the full dynamics and check whether the promised improvement actually occurs.

Three questions organize this chapter: what is optimal for a specified model and cost, whether an algorithm finds a stationary solution, and whether the resulting feedback stabilizes the physical system. An answer to one does not automatically answer the others.

6.1 Bellman and Pontryagin: Two Views of the Same Objective

For a fixed initial state, consider

$$\min_{u_0,\dots,u_{N-1}} J = \ell_N(x_N) + \sum_{k=0}^{N-1} \ell_k(x_k, u_k), \qquad x_{k+1} = f_k(x_k, u_k). \tag{6.1}$$

Here $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, and f_k is a discrete transition, potentially obtained by integrating a continuous model. Its derivatives must match that transition and its time step.

Bellman's principle gives

$$V_k(x) = \min_u \{\ell_k(x, u) + V_{k+1}(f_k(x, u))\}, \qquad V_N = \ell_N. \tag{6.2}$$

The future value makes this different from repeatedly minimizing today's stage cost. With linear dynamics, quadratic costs, and unconstrained controls, this recursion reduces to a Riccati recursion [Bel57].

For Pontryagin's conditions, choose the Lagrangian sign convention

$$\mathcal{L} = \ell_N(x_N) + \sum_{k=0}^{N-1} \{\ell_k(x_k, u_k) + \lambda_{k+1}^T [f_k(x_k, u_k) - x_{k+1}]\}. \tag{6.3}$$

With $H_k = \ell_k + \lambda_{k+1}^T f_k$, stationarity gives

$$0 = \ell_{k,u} + f_{k,u}^T \lambda_{k+1}, \tag{6.4}$$

$$\lambda_k = \ell_{k,x} + f_{k,x}^T \lambda_{k+1}, \quad \lambda_N = \ell_{N,x}. \tag{6.5}$$

In continuous time, the normal minimization convention $H = L + \lambda^T f$ uses $\dot{x} = H_\lambda$, $\dot{\lambda} = -H_x$, and a control minimizing H. The alternative maximum convention changes the sign of the running-cost term. The name "maximum principle" does not justify mixing conventions. For general discrete nonlinear dynamics the displayed stationarity conditions are necessary first-order conditions; they do not establish a global Hamiltonian minimum [PBGM62].

Where the value function is differentiable, the costate trajectory is the gradient of the value function along an optimal trajectory: $\lambda_k = \nabla_x V_k(x_k)$. Thus $V = \frac{1}{2}x^T Sx$ gives $\lambda = Sx$, whereas $V = x^T Sx$ gives $\lambda = 2Sx$. A costate is a covector of marginal cost, not the scalar cost itself. Constraints, nondifferentiable values, and abnormal extremals require the corresponding more general conditions.

6.2 A Dimensionally Consistent Local Model

Let $(\bar{x}_k, \bar{u}_k)$ be dynamically feasible and write $\delta x = x - \bar{x}$, $\delta u = u - \bar{u}$. Derivatives below are evaluated on that trajectory. Use column gradients, $\ell_{ux} \in \mathbb{R}^{m\times n}$, and $\ell_{xu} = \ell_{ux}^T$:

$$\ell \simeq \ell_0 + \ell_x^T \delta x + \ell_u^T \delta u + \frac{1}{2}\begin{bmatrix}\delta x\\ \delta u\end{bmatrix}^T \begin{bmatrix}\ell_{xx} & \ell_{xu}\\ \ell_{ux} & \ell_{uu}\end{bmatrix}\begin{bmatrix}\delta x\\ \delta u\end{bmatrix}. \tag{6.6}$$

Retain the linear terms: they drive improvement when the nominal trajectory is not stationary. Dropping them generally removes the feedforward correction. The first-order dynamics are

$$\delta x_{k+1} \simeq A_k \delta x_k + B_k \delta u_k, \qquad A_k = f_{k,x}, \quad B_k = f_{k,u}. \tag{6.7}$$

Even an exact Jacobian gives a tangent approximation to a finite perturbation. An infeasible nominal trajectory introduces a defect $f_k(\bar{x}_k, \bar{u}_k) - \bar{x}_{k+1}$ that a defect-aware method must carry.

Write the next local value as $V_{k+1} \simeq V_0 + s^T \delta x_{k+1} + \frac{1}{2}\delta x_{k+1}^T S \delta x_{k+1}$. The action value $Q = \ell + V_{k+1} \circ f_k$ has derivatives

$$Q_x = \ell_x + A^T s, \qquad Q_u = \ell_u + B^T s, \tag{6.8}$$

$$Q_{xx} = \ell_{xx} + A^T SA + \sum_i s_i f_{xx}^i, \tag{6.9}$$

$$Q_{uu} = \ell_{uu} + B^T SB + \sum_i s_i f_{uu}^i, \tag{6.10}$$

$$Q_{ux} = \ell_{ux} + B^T SA + \sum_i s_i f_{ux}^i. \tag{6.11}$$

The superscript i selects a component of f. The sums contract the value gradient with dynamics second derivatives. Full DDP retains these terms; iLQR omits them. Both methods retain the complete cost Hessian, including ℓ_{ux}, and the first-derivative coupling $B^T SA$. Velocity dependence does not itself require full DDP: iLQR still linearizes those dynamics.

6.3 The Discrete Riccati Recursion

Suppose $Q_{uu} \succ 0$. Invertibility alone gives only a stationary control, possibly a maximum or saddle. Completing the square gives

$$\delta u^* = k + K\delta x, \tag{6.12}$$

$$k = -Q_{uu}^{-1} Q_u, \qquad K = -Q_{uu}^{-1} Q_{ux}. \tag{6.13}$$

Negative signs are included in k and K throughout this chapter. Implement the inverses with factorization and linear solves. The unregularized update is

$$s_k = Q_x - Q_{xu} Q_{uu}^{-1} Q_u, \tag{6.14}$$

$$S_k = Q_{xx} - Q_{xu} Q_{uu}^{-1} Q_{ux}. \tag{6.15}$$

Initialize $s_N = \ell_{N,x}$ and $S_N = \ell_{N,xx}$. Both gradient and curvature must be propagated. A large eigenvalue of Q_{uu} penalizes a control change; for fixed gradient it reduces the step in that direction.

For linear dynamics and stage cost $\ell_k = \frac{1}{2}(x^T Q_k x + 2u^T N_k x + u^T R_k u)$, with $N_k \in \mathbb{R}^{m\times n}$, the exact recursion is

$$H_k = R_k + B_k^T S_{k+1} B_k, \quad D_k = N_k + B_k^T S_{k+1} A_k, \tag{6.16}$$

$$K_k = -H_k^{-1} D_k, \tag{6.17}$$

$$S_k = Q_k + A_k^T S_{k+1} A_k - D_k^T H_k^{-1} D_k, \quad S_N = Q_N. \tag{6.18}$$

A sufficient convexity assumption is that the joint stage-cost block $\begin{bmatrix} Q_k & N_k^T \\ N_k & R_k \end{bmatrix}$ is positive semidefinite, $R_k \succ 0$, and $Q_N \succeq 0$. For $N_k = 0$ this is the familiar $Q_k \succeq 0$, $R_k \succ 0$ setting. The feedback minimizes this finite-horizon linear-quadratic problem globally. Applying it to a nonlinear system has a local approximation scope.

An independent check, including the cross-cost terms, is

$$S_k = Q_k + K_k^T N_k + N_k^T K_k + K_k^T R_k K_k + (A_k + B_k K_k)^T S_{k+1}(A_k + B_k K_k). \tag{6.19}$$

Let $A = B = Q = R = 1$, $N = 1/2$, and $S_{k+1} = 1$. Then $H = 2$, $D = 3/2$, $K = -3/4$, and $S_k = 7/8$. Both formulas give 7/8; omitting the two cross-cost terms does not.

6.4 Controllability, Finite Cost, and Stabilization

Finite-horizon LQR does not require controllability. Under the positivity assumptions above, $H_k \succ 0$, including when $B_k = 0$. The optimizer then computes unavoidable state cost. Controllability asks whether arbitrary state transfers can be achieved. A hard terminal equality can be infeasible even though unconstrained finite-horizon LQR has a finite solution.

For a continuous time-varying linearization, endpoint reachability uses

$$W_c(t_0, T) = \int_{t_0}^{T} \Phi(T, s) B(s) B(s)^T \Phi(T, s)^T \, ds, \tag{6.20}$$

where Φ is the state-transition matrix of $A(t)$. A frozen-time rank test on $[B, AB, \ldots]$ is not a general substitute. Small Gramian eigenvalues indicate expensive transfers under the specified state units and input metric; raw condition numbers change under arbitrary rescaling.

For infinite-horizon time-invariant LQR with $Q \succeq 0$, $R \succ 0$, and no cross term, stabilizability of (A, B) and detectability of $(Q^{1/2}, A)$ are standard sufficient assumptions for the stabilizing Riccati solution. Controllability is stronger than necessary. A finite-horizon optimum alone does not guarantee asymptotic stability.

6.5 Differential Dynamic Programming and iLQR

DDP repeatedly constructs the local action value and updates the nominal trajectory [JM70]. A practical implementation follows this sequence.

1. Specify initial state, model, integration method, horizon, cost, and constraints. Roll out nominal controls to obtain a feasible state sequence.

2. Compute cost gradients and Hessians and dynamics derivatives. Full DDP adds the three dynamics-Hessian contractions; iLQR omits those contractions.

3. Sweep backward from the terminal gradient and Hessian. Ensure positive control curvature, increasing regularization and restarting when necessary.

4. From the same initial state, roll out $u_k^{\text{trial}} = \bar{u}_k + \alpha k_k + K_k(x_k^{\text{trial}} - \bar{x}_k)$ and $x_{k+1}^{\text{trial}} = f_k(x_k^{\text{trial}}, u_k^{\text{trial}})$.

5. Compare actual and predicted cost changes and accept sufficient decrease. If no trial is acceptable, increase regularization or report failure.

6. Check stationarity, feasibility, cost change, and step size. A failed line search or tiny regularized step is not a certificate of convergence.

When choosing a policy, regularization may replace Q_{uu} by $Q_{uu} + \mu I$. Choose μ large enough to obtain positive definiteness. To evaluate that policy in the original quadratic model, use general substitution:

$$s_k = Q_x + K^T Q_u + Q_{xu} k + K^T Q_{uu} k, \tag{6.21}$$

$$S_k = Q_{xx} + Q_{xu} K + K^T Q_{ux} + K^T Q_{uu} K. \tag{6.22}$$

Mixing regularized gains with an unqualified Schur-complement update silently changes the model. State the regularization convention implemented.

Both methods use local Taylor approximations. Solving a quadratic subproblem exactly does not make that approximation globally exact or establish a global optimum. Repeated relinearization updates the approximation where it is used. For dense matrices, both methods propagate an $n \times n$ value Hessian and factor an $m \times m$ control Hessian. Standard dense algebra costs on the order of $N(n^3 + n^2 m + nm^2 + m^3)$, before derivative evaluation. Full DDP adds dynamics second derivatives or their contractions. iLQR does not reduce the whole backward pass to $O(Nm^3)$. Compare measured runtime at matched tolerances; there is no universal hundredfold speed advantage.

6.6 What Local Convergence Requires

Eliminating states in an unconstrained shooting problem yields a reduced objective $J(U)$ in stacked controls. Exact Newton solves $\nabla^2 J(U)\Delta U = -\nabla J(U)$. With positive-definite Hessian at a stationary point, locally Lipschitz Hessian, sufficiently close initialization, and eventual full unregularized steps, its standard local result is

$$\|U^{(i+1)} - U^*\| \leq C\|U^{(i)} - U^*\|^2. \tag{6.23}$$

This measures error from the solution, not successive step size. A strict minimum need not have a positive-definite Hessian; twice continuous differentiability alone does not imply a Lipschitz Hessian.

Full second-order DDP has a closely related local quadratic-convergence theory under suitable smoothness, nonsingular positive control curvature, and step acceptance assumptions [JM70]. Its nonlinear feedback rollout need not equal an open-loop Newton step away from the solution. Holding each costate fixed and independently minimizing its Hamiltonian misses the dependence of later states and costates on that control.

iLQR uses an approximate reduced Hessian. Superlinear convergence is not a general consequence: linear convergence is possible, and faster rates need additional conditions on omitted terms. An exact Jacobian alone establishes neither quadratic nor global convergence. Globalization helps find acceptable steps without guaranteeing the desired local minimum.

6.7 A Reproducible Pendulum Example

For a point mass m on a massless rod of length l, viscous damping $b \geq 0$, torque τ, and angle measured from downward vertical,

$$ml^2\ddot{\theta} = -mgl\sin\theta - b\dot{\theta} + \tau. \tag{6.24}$$

Hanging is $\theta = 0$ and upright is $\theta = \pi$. With $m = 1\,\mathrm{kg}$, $l = 1\,\mathrm{m}$, $g = 10\,\mathrm{m/s^2}$, $b = 0.1\,\mathrm{N\,m\,s}$,

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -10\sin\theta - 0.1\omega + \tau. \tag{6.25}$$

The energy $E = \frac{1}{2}ml^2\omega^2 + mgl(1-\cos\theta)$ obeys $\dot{E} = \tau\omega - b\omega^2$, independently checking the damping sign.

A swing-up angle penalty $1 + \cos\theta$ is minimized at upright. An illustrative running cost is $L = 1 + \cos\theta + 0.1\omega^2 + 0.01\tau^2$. Near upright, an unwrapped local angle chart allows the balancing cost $100(\theta - \pi)^2 + 10\omega^2 + 0.1\tau^2$. These weighted objectives are not energy or metabolic expenditure. A step h gives a running-cost approximation hL; changing h without scaling it changes the trade-off with terminal cost.

An independently reproducible calculation is infinite-horizon LQR balance for $e = \theta - \pi$ with

$$A = \begin{bmatrix} 0 & 1 \\ 10 & -0.1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad Q = \operatorname{diag}(100, 10), \quad R = 0.1. \tag{6.26}$$

For cost $\int_0^\infty (x^T Q x + R\tau^2)dt$, feedback is $\tau = -R^{-1}B^T P x$. Write the symmetric P using entries p_{11}, p_{12}, p_{22}. Its stabilizing solution satisfies

$$p_{12} = 1 + \sqrt{11}, \tag{6.27}$$

$$p_{22} = \frac{-0.2 + \sqrt{0.04 + 40(2p_{12} + 10)}}{20}, \tag{6.28}$$

$$p_{11} = 0.1p_{12} - 10p_{22} + 10p_{12}p_{22}. \tag{6.29}$$

The position and velocity feedback gains are approximately 43.166 and 13.551. The closed-loop polynomial is $s^2 + (0.1 + 13.551)s + (43.166 - 10)$, with negative real-part roots. This establishes linearized local balance, not swing-up or robustness to arbitrary torque limits.

To report a swing-up experiment, publish initial state, integration scheme and step, horizon, bounds, objective, initial controls, solver version, derivative checks, stopping tolerances, and trajectories. Check energy residuals and step convergence. The previous iteration and gain tables did not specify such a reproducible run; they are not evidence for the corrected model.

6.8 Constraints and the Meaning of the Cost

Clamping a trial torque respects box bounds for that trial but does not solve the bound-constrained local problem. Use a box-constrained backward solve or another constrained optimizer. Augmented Lagrangian methods update multipliers as well as penalties; a pure penalty method should not be given that name.

A soft penalty $\rho \max(0, g(x))^2/2$ does not guarantee $g(x) \le 0$ at finite ρ. A logarithmic barrier $-\mu \log[-g(x)]$, $\mu > 0$, tends to positive infinity when approaching the boundary from inside. It is undefined outside, so the solver must reject infeasible trials. Neither treatment automatically enforces constraints between integration nodes.

Weights encode the question. Normalize by meaningful scales and distinguish impact accuracy, peak load, squared torque, work, and metabolic cost. Squared torque and work have different dimensions and favor different actions. Examine sensitivity to weights. Increasing an effort weight is not a universal method for increasing damping: check poles, constraints, and nonlinear response.

6.9 Continuous-Time Riccati and a Counterexample

The Hamilton–Jacobi–Bellman equation is

$$-V_t = \min_u \{L + \nabla V^T f\}, \qquad V(T, x) = \ell_T(x). \tag{6.30}$$

For $\dot{x} = Ax + Bu$, $L = \frac{1}{2}(x^T Q x + u^T R u)$, and $V = \frac{1}{2}x^T S x$,

$$-\dot{S} = Q + A^T S + SA - SBR^{-1}B^T S, \quad S(T) = Q_T, \quad u^* = -R^{-1}B^T Sx. \tag{6.31}$$

The time-invariant algebraic equation sets $\dot{S} = 0$ and requires the stabilizing solution under the assumptions stated earlier. For $\dot{x} = x$, $B = 0$, $Q = 1$, $Q_T = 0$, the finite-horizon solution is $S(t) = (e^{2(T-t)} - 1)/2$. It is finite despite uncontrollability and instability. The infinite-horizon cost diverges for nonzero initial state. Finite cost, reachability, and stabilization are different properties.

6.10 Model Predictive Control and Terminal Guarantees

MPC repeatedly solves a finite-horizon problem from the measured state, applies its first control, and replans. DDP and iLQR are possible solvers, alongside other nonlinear and quadratic-programming methods.

Warm-start by shifting old controls one relative horizon index, appending terminal control, and rolling out states from the new measurement. Disturbances and changing active constraints can make this guess poor.

A standard nominal stability argument assumes admissible terminal feedback κ_f, an invariant terminal set $\mathcal{X}_f$, and

$$V_f(f(x, \kappa_f(x))) - V_f(x) \leq -\ell(x, \kappa_f(x)), \qquad x \in \mathcal{X}_f. \tag{6.32}$$

With initial feasibility, the terminal constraint, positive-definite stage cost relative to the target, and appropriate regularity, a shifted candidate establishes recursive feasibility and decrease of the optimal value. Exact optimization, or a suboptimal solver preserving that feasibility and decrease, supports the stability conclusion. An arbitrary finite-horizon local solve does not supply those guarantees.

An LQR quadratic is a candidate terminal cost; nonlinear decrease and input admissibility must be checked on the terminal set. Lyapunov decrease toward one equilibrium differs from contraction between trajectories. A differential metric $M(x,t)$ needs uniform positive-definiteness bounds and an inequality involving $\dot{M} + A_{\text{cl}}^T M + M A_{\text{cl}}$ throughout the region. A cost Hessian is not automatically such a certificate.

6.11 Connecting the Calculation to the Golf Swing

Geometry changes the dynamics Jacobians; accumulated motion sets the state from which the next action operates; joint impedance changes perturbation response; and impact accuracy changes terminal error weights. Riccati propagation connects these effects across time. A small action can matter because it changes a later, highly weighted impact variable. Maximizing instantaneous acceleration is a different objective.

Compare stiff and compliant strategies under the same actuator limits, starting state, impact target, perturbation model, and cost definition. Include the mechanics and energetic cost of changing stiffness. Otherwise an apparent advantage may reflect extra actuator authority or a different objective.

An optimal torque history is a candidate explanation to test. It does not identify muscle co-contraction, prove that a golfer uses an LQR controller, or establish the nervous system's objective. Compare predicted kinematics, forces, perturbation responses, and uncertainty with measurements. This preserves the useful connection between geometry and control without turning an optimizer into an unsupported physiological claim.

Exercises

1. Verify the 7/8 Riccati example with both formulas and show the error from dropping N while retaining the original cost.

2. Substitute the pendulum matrix P into the algebraic Riccati equation; check residual, positive definiteness, and closed-loop roots.

3. Check a discrete pendulum Jacobian and Hessian contractions against finite differences at nonzero states.

4. Compare iLQR and DDP with the same cross-coupled cost, reporting stationarity residual, feasibility, objective, and runtime from several guesses.

5. Explain why a hard endpoint constraint changes feasibility in the uncontrollable finite-horizon counterexample.

6. For $\dot{x} = -x + u$ and cost $\int (x^2 + u^2)dt$, verify $P = -1 + \sqrt{2}$ solves $1 - 2P - P^2 = 0$. Substitute $V = Px^2$, $u^* = -Px$ into $0 = x^2 + (u^*)^2 + 2Px(-x + u^*)$.

Chapter 7

The Stability–Optimality Duality

In Plain Language 7.1. Two Sides of the Same Coin

The same matrix that LQR uses to score future cost can also serve as a ruler proving that small errors shrink under the feedback law. This chapter reveals that optimality and stability are not separate concerns but two views of the same geometric object: the Riccati solution acting simultaneously as cost curvature and contraction metric. This connection, recognized in the 1960s by Kalman and formalized by Willems, remains one of the most powerful principles in modern control: the controller that minimizes energy consumption guarantees robustness, and the controller that maximizes robustness often minimizes energy. Duality is why we can design once and verify both objectives simultaneously.

7.1 The Central Bridge

Chapters 5 and 6 developed two apparently independent threads:

- **Contraction theory**: find a metric $\mathbf{M}$ such that $\mathbf{A}_{\text{cl}}^{\mathsf{T}}\mathbf{M} + \mathbf{M}\mathbf{A}_{\text{cl}} + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}$.
- **Optimal control**: find a gain $\mathbf{K}$ via the Riccati equation, producing a cost-to-go matrix $\mathbf{S}$.

The duality theorem shows these are the same computation under the right interpretation.

The Riccati solution $S(\boldsymbol{x})$ is the Hessian of the value function in the symplectic structure of the Hamiltonian formulation. In Agrachev and Sachkov's framework [AS04, Part IV], the costate-adjoint dynamics and the Riccati equation arise naturally from the symplectic geometry of the cotangent bundle. The value function $V^*(\boldsymbol{x})$ satisfies the Hamilton–Jacobi–Bellman equation, and its Hessian $\nabla^2 V^*$ encodes the local cost-to-go curvature. This Hessian is precisely the Riccati matrix, connecting optimization and stability through the geometry of the Hamiltonian flow on the symplectic manifold.

Geometric Intuition

Optimal control builds a local quadratic "cost bowl" around the trajectory; contraction checks whether perturbations shrink under a metric "bowl." Duality appears when those bowls coincide—the Riccati matrix $\mathbf{S}$ serves double duty as both value curvature and contraction metric for the closed-loop tangent dynamics.

7.2 Historical Context: Discovering the Duality

The connection between Lyapunov stability and optimal control emerged gradually in the mid-twentieth century. Lyapunov's foundational work (1892) established that a suitable energy-like function can certify stability; nearly 70 years later, Kalman's pivotal contributions recognized that the optimal control problem and the stability problem share a common certificate.

7.2.1 Kalman's Insight (1960)

In his seminal papers on linear-quadratic regulation [Kal60], Kalman showed that the unique positive-definite solution $\mathbf{S}$ of the algebraic Riccati equation (ARE)

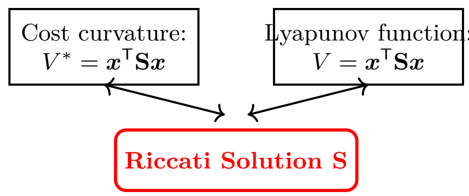
$$\mathbf{0} = \mathbf{Q} + \mathbf{A}^{\mathsf{T}}\mathbf{S} + \mathbf{S}\mathbf{A} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{S} \tag{7.1}$$

is simultaneously:

1. The **cost curvature** matrix: the optimal cost-to-go for a state $\boldsymbol{x}$ is exactly $V^\star(\boldsymbol{x}) = \boldsymbol{x}^\mathsf{T}\mathbf{S}\boldsymbol{x}$.

2. A **Lyapunov function** for the closed-loop system $\dot{\boldsymbol{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\boldsymbol{x}$ with $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$.

This insight unified two previously separate fields: *the stability guarantee comes free from optimizing the cost.* This was conceptually revolutionary. Before Kalman, engineers solved LQR problems for one reason (minimize energy) and separately checked stability (find a Lyapunov function). Afterward, they solved it once and received both certifications.



One solution serves both purposes

7.2.2 Dissipation Inequalities and Willems' Framework (1971)

Jan Willems formalized and generalized Kalman's observation into the language of *dissipation inequalities.* His work (Willems, 1972) showed that a system is stable if and only if there exists a positive-definite matrix $\mathbf{S}$ and a cost function $\ell(x, u)$ such that the system "dissipates" energy at a rate determined by $\mathbf{S}$:

$$V_{k+1} - V_k \leq -\ell(x_k, u_k), \tag{7.2}$$

where $V_k = x_k^\mathsf{T}\mathbf{S}x_k$. The LQR problem is precisely the one where the cost function ℓ is the stage cost $x^\mathsf{T}\mathbf{Q}x + u^\mathsf{T}\mathbf{R}u$, and the dissipation inequality becomes an equality:

$$V_{k+1} - V_k = -\ell(x_k, u_k). \tag{7.3}$$

Willems' perspective recast the Riccati solution as a **dissipation certificate**: proof that the system loses energy (in the **S**-weighted sense) at a rate equal to the optimal cost.

7.2.3 Robustness Margins From LQR

Anderson and Moore's foundational book (1971) and subsequent work by Doyle and Stein emphasized that LQR not only optimizes the cost but also delivers remarkable robustness properties *for free*:

Proposition 7.1 (Guaranteed Robustness Margins of LQR)**.** *The continuous-time LQR controller* $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$ *derived from the ARE guarantees:*

1. ***Gain margin**: multiplicative perturbations to the control input satisfying* $\|\Delta\mathbf{K}\|$ *remain stable for any* $\Delta\mathbf{K}$ *with* $\|\Delta\mathbf{K}\| < \|\mathbf{K}\|$.

2. ***Strict gain margin**: even more precisely, the loop remains stable if* $\mathbf{K} \to \eta\mathbf{K}$ *for any* $\eta \in (1/2, \infty)$.

3. ***Phase margin**: the loop margin under phase perturbations is at least* $60°$ *on both sides.*

4. ***Disk margin**: the closed-loop poles lie outside a disk of radius* $1/2$ *centered at* $(-1, 0)$.

Proof Sketch (Completion of Squares). The ARE states $\mathbf{A}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A} + \mathbf{Q} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} = 0$, with $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$. Consider the perturbed controller $\boldsymbol{u} = -\eta\mathbf{K}\boldsymbol{x}$ for $\eta > 0$. Using $V = \boldsymbol{x}^\mathsf{T}\mathbf{S}\boldsymbol{x}$:

$$\dot{V} = \boldsymbol{x}^\mathsf{T}[(\mathbf{A} - \eta\mathbf{B}\mathbf{K})^\mathsf{T}\mathbf{S} + \mathbf{S}(\mathbf{A} - \eta\mathbf{B}\mathbf{K})]\boldsymbol{x}. \tag{7.4}$$

Substituting the ARE ($\mathbf{A}^\top\mathbf{S} + \mathbf{S}\mathbf{A} = -\mathbf{Q} + \mathbf{K}^\top\mathbf{R}\mathbf{K}$) and noting $\mathbf{S}\mathbf{B}\mathbf{K} = \mathbf{K}^\top\mathbf{R}\mathbf{K}$:

$$\begin{aligned}\dot{V} &= \boldsymbol{x}^\top[-\mathbf{Q} + \mathbf{K}^\top\mathbf{R}\mathbf{K} - 2\eta\mathbf{K}^\top\mathbf{R}\mathbf{K}]\boldsymbol{x} \\ &= -\boldsymbol{x}^\top[\mathbf{Q} + (2\eta - 1)\mathbf{K}^\top\mathbf{R}\mathbf{K}]\boldsymbol{x}. \end{aligned} \tag{7.5}$$

Since $\mathbf{Q} \succeq 0$ and $\mathbf{R} \succ 0$, the expression is negative definite whenever $2\eta - 1 > 0$, i.e., $\eta > 1/2$. For $\eta \to \infty$, the quadratic term grows without bound, so stability is maintained for all $\eta \in (1/2, \infty)$.

Remark 7.2. This result is specific to continuous-time LQR (SISO: gain margin $[1/2, \infty)$, phase margin $\pm 60°$). Discrete-time LQR does not enjoy such generous margins; the relevant concept there is the *disk margin*, which depends on the sampling rate and system dynamics. The two settings should not be conflated.

□

This proposition is not academic: it explains why tuning an LQR controller by scaling the gain (increasing η to be more aggressive) fails only when $\eta \leq 1/2$—and by then, the system is already far too aggressive for practical use.

7.3 Discrete-Time Duality: The Value Decrease Identity

7.3.1 The Core Derivation

Under the optimal feedback $\delta\boldsymbol{u}_k = -\mathbf{K}_k\delta\boldsymbol{x}_k$, the closed-loop perturbation dynamics are

$$\delta\boldsymbol{x}_{k+1} = (\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)\,\delta\boldsymbol{x}_k. \tag{7.6}$$

Define the quadratic Lyapunov candidate

$$V_k = \delta\boldsymbol{x}_k^\top\mathbf{S}_k\delta\boldsymbol{x}_k, \tag{7.7}$$

where $\mathbf{S}_k$ is the Riccati solution from (6.18):

$$\mathbf{S}_k = \mathbf{Q}_k + \mathbf{A}_k^\top\mathbf{S}_{k+1}\mathbf{A}_k - \mathbf{A}_k^\top\mathbf{S}_{k+1}\mathbf{B}_k(\mathbf{R}_k + \mathbf{B}_k^\top\mathbf{S}_{k+1}\mathbf{B}_k)^{-1}\mathbf{B}_k^\top\mathbf{S}_{k+1}\mathbf{A}_k. \tag{7.8}$$

Introduce the gain matrix

$$\mathbf{K}_k = (\mathbf{R}_k + \mathbf{B}_k^\top\mathbf{S}_{k+1}\mathbf{B}_k)^{-1}\mathbf{B}_k^\top\mathbf{S}_{k+1}\mathbf{A}_k, \tag{7.9}$$

which allows us to rewrite the Riccati equation in the **alternative (or canonical) form**:

$$\mathbf{S}_k = \mathbf{Q}_k + \mathbf{K}_k^\top\mathbf{R}_k\mathbf{K}_k + (\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)^\top\mathbf{S}_{k+1}(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k), \tag{7.10}$$

which is a key form that appears throughout duality theory.

Now we compute $V_{k+1} - V_k$ step by step.

Theorem 7.3 (Value Decrease = Contraction Certificate)**.** *Under the LQR-optimal feedback with gain defined by* (7.9)*:*

$$\boxed{V_{k+1} - V_k = -\delta\boldsymbol{x}_k^\top\big(\mathbf{Q}_k + \mathbf{K}_k^\top\mathbf{R}_k\mathbf{K}_k\big)\delta\boldsymbol{x}_k \leq -\alpha\left\|\delta\boldsymbol{x}_k\right\|^2} \tag{7.11}$$

for some $\alpha > 0$ (the smallest eigenvalue of $\mathbf{Q}_k + \mathbf{K}_k^\top\mathbf{R}_k\mathbf{K}_k$).

Complete Step-by-Step Derivation. **Step 1: Compute V_{k+1} using the closed-loop dynamics**. From (7.6):

$$\delta\boldsymbol{x}_{k+1} = (\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)\,\delta\boldsymbol{x}_k, \tag{7.12}$$

$$\begin{aligned} V_{k+1} &= \delta\boldsymbol{x}_{k+1}^\top\mathbf{S}_{k+1}\delta\boldsymbol{x}_{k+1} \\ &= \delta\boldsymbol{x}_k^\top(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)^\top\mathbf{S}_{k+1}(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)\delta\boldsymbol{x}_k. \end{aligned} \tag{7.13}$$

Step 2: Write $V_{k+1} - V_k$. We have $V_k = \delta\boldsymbol{x}_k^\top\mathbf{S}_k\delta\boldsymbol{x}_k$, so:

$$V_{k+1} - V_k = \delta\boldsymbol{x}_k^\top\big[(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k)^\top\mathbf{S}_{k+1}(\mathbf{A}_k - \mathbf{B}_k\mathbf{K}_k) - \mathbf{S}_k\big]\delta\boldsymbol{x}_k. \tag{7.14}$$

Step 3: Invoke the Riccati alternative form. Equation (7.10) states:

$$\mathbf{S}_k = \mathbf{Q}_k + \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k + (\mathbf{A}_k - \mathbf{B}_k \mathbf{K}_k)^\top \mathbf{S}_{k+1} (\mathbf{A}_k - \mathbf{B}_k \mathbf{K}_k). \tag{7.15}$$

Therefore:

$$(\mathbf{A}_k - \mathbf{B}_k \mathbf{K}_k)^\top \mathbf{S}_{k+1} (\mathbf{A}_k - \mathbf{B}_k \mathbf{K}_k) = \mathbf{S}_k - \mathbf{Q}_k. \tag{7.16}$$

Step 4: Substitute back:

$$\begin{aligned} V_{k+1} - V_k &= \delta \boldsymbol{x}_k^\top [(\mathbf{S}_k - \mathbf{Q}_k) - \mathbf{S}_k] \delta \boldsymbol{x}_k \\ &= \delta \boldsymbol{x}_k^\top [-\mathbf{Q}_k] \delta \boldsymbol{x}_k && (7.17) \\ &= -\delta \boldsymbol{x}_k^\top \mathbf{Q}_k \delta \boldsymbol{x}_k. && (7.18) \end{aligned}$$

Step 5: Add the control cost. The optimal value function satisfies a fundamental recursion:

$$V_k = x_k^\top \mathbf{Q}_k x_k + u_k^\top \mathbf{R}_k u_k + V_{k+1}. \tag{7.19}$$

Rearranging for perturbations:

$$V_{k+1} - V_k = -\delta \boldsymbol{x}_k^\top \mathbf{Q}_k \delta \boldsymbol{x}_k - \delta \boldsymbol{u}_k^\top \mathbf{R}_k \delta \boldsymbol{u}_k. \tag{7.20}$$

With the optimal control $\delta \boldsymbol{u}_k = -\mathbf{K}_k \delta \boldsymbol{x}_k$:

$$\begin{aligned} V_{k+1} - V_k &= -\delta \boldsymbol{x}_k^\top \mathbf{Q}_k \delta \boldsymbol{x}_k - (-\mathbf{K}_k \delta \boldsymbol{x}_k)^\top \mathbf{R}_k (-\mathbf{K}_k \delta \boldsymbol{x}_k) \\ &= -\delta \boldsymbol{x}_k^\top \mathbf{Q}_k \delta \boldsymbol{x}_k - \delta \boldsymbol{x}_k^\top \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k \delta \boldsymbol{x}_k && (7.21) \\ &= -\delta \boldsymbol{x}_k^\top (\mathbf{Q}_k + \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k) \delta \boldsymbol{x}_k. && (7.22) \end{aligned}$$

Since $\mathbf{Q}_k \succ 0$ and $\mathbf{R}_k \succ 0$ by assumption (Assumption from Chapter 6), we have $\mathbf{Q}_k + \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k \succ 0$. Let $\alpha_k = \lambda_{\min}(\mathbf{Q}_k + \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k)$. Then:

$$V_{k+1} - V_k \leq -\alpha_k \left\| \delta \boldsymbol{x}_k \right\|^2. \tag{7.23}$$

□

Remark 7.4 (Interpretation). The value decrease identity reveals the fundamental bargain of LQR: the cost is paid in two parts at each step:

1. **State penalty**: $\delta \boldsymbol{x}_k^\top \mathbf{Q}_k \delta \boldsymbol{x}_k$ penalizes the state deviation from the reference.
2. **Control penalty**: $\delta \boldsymbol{u}_k^\top \mathbf{R}_k \delta \boldsymbol{u}_k = \delta \boldsymbol{x}_k^\top \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k \delta \boldsymbol{x}_k$ penalizes the energy cost of the optimal control action.

The sum of these two terms is the rate at which the Lyapunov function (Riccati-weighted squared norm) decreases. This is why LQR is so well-behaved: every unit of perturbation energy is forced to be paid down via the optimal control.

7.3.2 Extracting Exponential Decay

Suppose there exist uniform bounds:

$$\mathbf{Q}_k + \mathbf{K}_k^\top \mathbf{R}_k \mathbf{K}_k \succeq \alpha \mathbf{I}, \qquad \underline{s}\,\mathbf{I} \preceq \mathbf{S}_k \preceq \overline{s}\,\mathbf{I}, \tag{7.24}$$

for constants $\alpha, \underline{s}, \overline{s} > 0$ with $\alpha < \overline{s}$.

Then from (7.11):

$$V_{k+1} \leq V_k - \alpha \left\| \delta \boldsymbol{x}_k \right\|^2. \tag{7.25}$$

Since $\underline{s} \left\| \delta \boldsymbol{x}_k \right\|^2 \leq V_k = \delta \boldsymbol{x}_k^\top \mathbf{S}_k \delta \boldsymbol{x}_k \leq \overline{s} \left\| \delta \boldsymbol{x}_k \right\|^2$, we have $\left\| \delta \boldsymbol{x}_k \right\|^2 \leq V_k / \underline{s}$, giving:

$$V_{k+1} \leq V_k - \frac{\alpha}{\overline{s}} V_k = \left(1 - \frac{\alpha}{\overline{s}} \right) V_k. \tag{7.26}$$

Define the decay rate:

$$\rho := 1 - \frac{\alpha}{\overline{s}} \in (0, 1). \tag{7.27}$$

Iterating the inequality:

$$V_k \le \rho^k V_0, \tag{7.28}$$

and converting to Euclidean norm via the bounds on $\mathbf{S}_k$:

$$\underline{s}\,\|\delta \boldsymbol{x}_k\|^2 \le V_k \le \rho^k V_0 \le \rho^k \overline{s}\,\|\delta \boldsymbol{x}_0\|^2, \tag{7.29}$$

$$\|\delta \boldsymbol{x}_k\|^2 \le \frac{\overline{s}}{\underline{s}} \rho^k \|\delta \boldsymbol{x}_0\|^2. \tag{7.30}$$

This is a **discrete contraction inequality** with the Riccati matrix as the metric. The contraction rate ρ is determined by the smallest eigenvalue of the stage cost and the condition number of the value matrix.

Remark 7.5 (Decay Rate Interpretation). The decay rate $\rho = 1 - \alpha/\overline{s}$ reveals a trade-off:

- **Larger** α (stronger stage cost): faster decay.
- **Larger** $\overline{s}$ (larger Riccati solution): slower decay, but more robust (larger basin of attraction).
- The condition number $\kappa_S := \overline{s}/\underline{s}$ appears in the Euclidean norm bound: worse conditioning means a larger factor in front of the exponential decay.

Good LQR design balances aggressiveness (large α) with robustness (well-conditioned $\mathbf{S}$).

7.4 Worked Numerical Example: The Discrete Pendulum

We now apply the duality theory to a concrete system: the linearized discrete-time inverted pendulum from Chapter 6.

7.4.1 System Setup

$$\mathbf{A} = \begin{bmatrix} 1 & 0.1 \\ 0.981 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{R} = 0.1. \tag{7.31}$$

We apply the discrete-time Riccati recursion backward until it reaches its fixed point.

Remark 7.6 (Every number in this section is computed, not transcribed). The values below are produced by `scripts/generate_worked_examples.py`, which solves the Riccati equation with `src/affine_control/lqr.py` and writes a LaTeX fragment this chapter `\input`s. CI regenerates the fragment and fails if it differs from what is committed.

That machinery exists because an earlier revision of this section did not have it. Its $\mathbf{S}$ was not a solution of the Riccati equation — the residual norm was 16.25 rather than zero — and the gain that followed produced a closed loop with spectral radius 1.043, printed directly beneath the sentence "Both eigenvalues lie inside the unit circle, confirming stability." The system was unstable and the text said otherwise.

7.4.2 Riccati Solution

The recursion converges to

$$\mathbf{S} = \begin{bmatrix} 94.9386 & 20.7112 \\ 20.7112 & 7.3663 \end{bmatrix}, \tag{7.32}$$

which satisfies the discrete algebraic Riccati equation to a residual norm below $1e-10$ — in practice it lands at machine precision, but the published figure is a bound the generator enforces, because the residual itself is a difference of nearly-equal floating-point numbers and its exact value varies with the machine. Whether the residual is small at all is the first property worth checking about any matrix presented as a Riccati solution.

The eigenvalues of $\mathbf{S}$ are $\lambda_{\max}(\mathbf{S}) = 99.5898$ and $\lambda_{\min}(\mathbf{S}) = 2.7150$, both positive as they must be, giving a condition number

$$\kappa(\mathbf{S}) = \frac{\lambda_{\max}(\mathbf{S})}{\lambda_{\min}(\mathbf{S})} = \frac{99.5898}{2.7150} = 36.6808. \tag{7.33}$$

7.4.3 Optimal Gain

From the Riccati recursion, the steady-state optimal gain is:

$$\mathbf{K} = (\mathbf{R} + \mathbf{B}^\mathsf{T}\mathbf{S}\mathbf{B})^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}\mathbf{A}. \tag{7.34}$$

Computing step-by-step:

$$\begin{aligned}
\mathbf{B}^\mathsf{T}\mathbf{S} &= \begin{bmatrix} 2.5458 & 0.8402 \end{bmatrix}, && (7.35)\\
\mathbf{B}^\mathsf{T}\mathbf{S}\mathbf{A} &= \begin{bmatrix} 3.3700 & 1.0948 \end{bmatrix}, && (7.36)\\
\mathbf{B}^\mathsf{T}\mathbf{S}\mathbf{B} &= 0.09675, && (7.37)\\
\mathbf{R} + \mathbf{B}^\mathsf{T}\mathbf{S}\mathbf{B} &= 0.19675, && (7.38)\\
\mathbf{K} &= \frac{1}{0.19675}\begin{bmatrix} 3.3700 & 1.0948 \end{bmatrix} = \begin{bmatrix} 17.1287 & 5.5643 \end{bmatrix}. && (7.39)
\end{aligned}$$

The optimal feedback is therefore $u_k = -17.1287\,\theta_k - 5.5643\,\dot{\theta}_k$: much more aggressive on the angle than on the angular velocity, which is what $\mathbf{Q} = \operatorname{diag}(10, 1)$ asks for.

7.4.4 Closed-Loop Dynamics

The closed-loop system is:

$$\begin{aligned}
\mathbf{A}_{\text{cl}} &= \mathbf{A} - \mathbf{B}\mathbf{K} && (7.40)\\
&= \begin{bmatrix} 1 & 0.1 \\ 0.981 & 1 \end{bmatrix} - \begin{bmatrix} 0.0856 & 0.0278 \\ 1.7129 & 0.5564 \end{bmatrix} = \begin{bmatrix} 0.9144 & 0.0722 \\ -0.7319 & 0.4436 \end{bmatrix}. && (7.41)
\end{aligned}$$

The eigenvalues of $\mathbf{A}_{\text{cl}}$ are:

$$\lambda_1(\mathbf{A}_{\text{cl}}) = 0.6281, \quad \lambda_2(\mathbf{A}_{\text{cl}}) = 0.7298, \tag{7.42}$$

so the spectral radius is $0.7298 < 1$ and the closed loop is stable. The dominant eigenvalue 0.7298 sets the asymptotic decay per step in the Euclidean sense, without the Riccati metric.

Common Misconception

Check this rather than trusting it. A 2×2 closed loop makes the check almost free: the eigenvalues satisfy $\lambda^2 - \operatorname{tr}(\mathbf{A}_{\text{cl}})\lambda + \det(\mathbf{A}_{\text{cl}}) = 0$, so both lie inside the unit circle exactly when $|\operatorname{tr}| < 1 + \det$ and $|\det| < 1$. A gain derived from a matrix that is *almost* a Riccati solution can easily fail that test while looking entirely plausible, and the failure is silent — nothing about the printed matrix announces it.

7.4.5 Stage Cost Matrix

The stage cost matrix is:

$$\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K} = \begin{bmatrix} 39.3393 & 9.5310 \\ 9.5310 & 4.0962 \end{bmatrix}. \tag{7.43}$$

Its eigenvalues are $\lambda_{\text{max}} = 41.7517$ and $\lambda_{\text{min}} = 1.6838$, so $\alpha = 1.6838$.

7.4.6 Contraction Rate From Riccati Bounds

From the Riccati bounds:

$$\rho = 1 - \frac{\alpha}{\bar{s}} = 1 - \frac{1.6838}{99.5898} = 0.9831. \tag{7.44}$$

The Euclidean norm decay is bounded by:

$$\|\delta \boldsymbol{x}_k\| \le \sqrt{\kappa(\mathbf{S})}\,\rho^{k/2}\,\|\delta \boldsymbol{x}_0\| = 6.0565\,(0.9831)^{k/2}\,\|\delta \boldsymbol{x}_0\|. \tag{7.45}$$

The prefactor $\sqrt{\kappa(\mathbf{S})} = 6.0565$ exceeds one, so for small k this bound is weaker than saying nothing at all — an artifact of measuring decay in a norm that is not adapted to the dynamics. The contraction is stated cleanly in the Riccati-weighted norm $V_k = \delta \boldsymbol{x}_k^\mathsf{T} \mathbf{S} \delta \boldsymbol{x}_k$:

$$V_k \leq \rho^k V_0 = (0.9831)^k V_0. \tag{7.46}$$

7.4.7 Bound Versus Trajectory

Starting from $\delta \boldsymbol{x}_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}^\mathsf{T}$ (unit angle error), the table below simulates the closed loop and places the bound beside it.

Step k	V_k	$\Vert\delta \boldsymbol{x}_k\Vert$	V_k/V_{k-1} (actual)	ρ^k (bound)
0	94.9386	1.0000	—	1.0000
1	55.5992	1.1712	0.5856	0.9831
2	33.2717	1.2654	0.5984	0.9665
3	19.9314	1.2015	0.5990	0.9501
4	11.8393	1.0561	0.5940	0.9341
5	6.9497	0.8847	0.5870	0.9183
6	4.0294	0.7177	0.5798	0.9027
7	2.3094	0.5693	0.5731	0.8875
8	1.3099	0.4441	0.5672	0.8725
9	0.7363	0.3421	0.5621	0.8577
10	0.4107	0.2610	0.5578	0.8432

Table 7.1: Simulated evolution of the value function under optimal feedback, with the Riccati bound ρ^k alongside. V_k falls to $0.0043V_0$ after ten steps while the bound only guarantees $0.8432V_0$ — the bound is roughly two orders of magnitude loose here.

Two features of that table are worth dwelling on, because both were misrepresented in an earlier revision of this section.

First, **the per-step ratio is not constant**. It moves between roughly 0.56 and 0.60 as the trajectory rotates through the state space, settling toward the square of the dominant closed-loop eigenvalue. A table whose decay-ratio column is constant to three decimals is not a simulation — it is ρ^k evaluated and presented as one, which is what the previous version of this table did.

Second, **the Euclidean norm initially grows**, from 1.000 to about 1.27 by step two, before decaying. That is not a contradiction: contraction is asserted in the $\mathbf{S}$-weighted norm, and V_k decreases monotonically throughout. A non-normal $\mathbf{A}_{\mathrm{cl}}$ routinely produces exactly this transient growth in the Euclidean norm, and a hand-filled table that shows $\Vert\delta \boldsymbol{x}_k\Vert$ decreasing monotonically from the first step has quietly assumed the conclusion.

7.5 Continuous-Time Duality: Detailed Derivation

The connection between dissipativity (Willems' framework) and accessibility can be understood through Agrachev and Sachkov's analysis [AS04, Ch. 4]: a system is dissipative if it admits storage functions capturing energy-like bounds, while accessibility asks whether the reachable set is large. The Riccati solution acts as a storage function from the control perspective, with the contraction rate (dissipation margin) dual to how richly the control distribution generates motion. Systems with high controllability—where Lie brackets unlock new directions—admit low-dissipation storage functions, reflecting the geometric intuition that accessible systems can freely exchange energy without accumulation.

7.5.1 Setup and Objectives

For the continuous-time system:

$$\dot{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x} + \mathbf{B}\boldsymbol{u}, \tag{7.47}$$

with cost:

$$J = \int_0^\infty (\boldsymbol{x}^\mathsf{T}\mathbf{Q}\boldsymbol{x} + \boldsymbol{u}^\mathsf{T}\mathbf{R}\boldsymbol{u})\, dt, \tag{7.48}$$

the optimal control is:

$$\boldsymbol{u}^\star = -\mathbf{K}\boldsymbol{x} = -\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}\boldsymbol{x}, \tag{7.49}$$

where $\mathbf{S}$ is the solution to the algebraic Riccati equation (7.1).

For time-varying systems or finite-horizon problems, we use the dynamic Riccati equation (DRE):

$$-\dot{\mathbf{S}} = \mathbf{A}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} + \mathbf{Q}, \tag{7.50}$$

with terminal condition $\mathbf{S}(T) = \mathbf{S}_f$ (typically zero or a penalty on final state).

7.5.2 Value Function Time Derivative

Consider perturbations $\delta\boldsymbol{x}$ in the closed-loop system:

$$\dot{\delta\boldsymbol{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\,\delta\boldsymbol{x} = \mathbf{A}_{\text{cl}}\,\delta\boldsymbol{x}, \tag{7.51}$$

where $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$.

The quadratic value function is:

$$V(t) = \delta\boldsymbol{x}(t)^\mathsf{T}\mathbf{S}(t)\delta\boldsymbol{x}(t). \tag{7.52}$$

Taking the time derivative:

$$\begin{aligned}
\dot{V} &= \frac{d}{dt}[\delta\boldsymbol{x}^\mathsf{T}\mathbf{S}\delta\boldsymbol{x}] \\
&= \dot{\delta\boldsymbol{x}}^\mathsf{T}\mathbf{S}\delta\boldsymbol{x} + \delta\boldsymbol{x}^\mathsf{T}\dot{\mathbf{S}}\delta\boldsymbol{x} + \delta\boldsymbol{x}^\mathsf{T}\mathbf{S}\dot{\delta\boldsymbol{x}} \\
&= (\mathbf{A}_{\text{cl}}\delta\boldsymbol{x})^\mathsf{T}\mathbf{S}\delta\boldsymbol{x} + \delta\boldsymbol{x}^\mathsf{T}\dot{\mathbf{S}}\delta\boldsymbol{x} + \delta\boldsymbol{x}^\mathsf{T}\mathbf{S}(\mathbf{A}_{\text{cl}}\delta\boldsymbol{x}) \\
&= \delta\boldsymbol{x}^\mathsf{T}[\mathbf{A}_{\text{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\text{cl}} + \dot{\mathbf{S}}]\delta\boldsymbol{x}.
\end{aligned} \tag{7.53}$$

7.5.3 Simplifying Using the Riccati Equation

Recall that $\mathbf{A}_{\text{cl}} = \mathbf{A} - \mathbf{B}\mathbf{K}$. The key is to show that:

$$\mathbf{A}_{\text{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\text{cl}} + \dot{\mathbf{S}} = -(\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K}). \tag{7.54}$$

To prove this, we expand the left-hand side:

$$\begin{aligned}
&\mathbf{A}_{\text{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\text{cl}} + \dot{\mathbf{S}} \\
&= (\mathbf{A} - \mathbf{B}\mathbf{K})^\mathsf{T}\mathbf{S} + \mathbf{S}(\mathbf{A} - \mathbf{B}\mathbf{K}) + \dot{\mathbf{S}} \\
&= \mathbf{A}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A} - \mathbf{K}^\mathsf{T}\mathbf{B}^\mathsf{T}\mathbf{S} - \mathbf{S}\mathbf{B}\mathbf{K} + \dot{\mathbf{S}}.
\end{aligned} \tag{7.55}$$

Now recall the DRE (7.50):

$$-\dot{\mathbf{S}} = \mathbf{A}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} + \mathbf{Q}. \tag{7.56}$$

Rearranging:

$$\mathbf{A}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A} + \dot{\mathbf{S}} = -\mathbf{Q} + \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}. \tag{7.57}$$

Substituting into (7.55):

$$\begin{aligned}
&\mathbf{A}_{\text{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\text{cl}} + \dot{\mathbf{S}} \\
&= [-\mathbf{Q} + \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}] - \mathbf{K}^\mathsf{T}\mathbf{B}^\mathsf{T}\mathbf{S} - \mathbf{S}\mathbf{B}\mathbf{K}.
\end{aligned} \tag{7.58}$$

Now use $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$ and the symmetry $\mathbf{R} = \mathbf{R}^\mathsf{T}$:

$$\mathbf{K}^\mathsf{T}\mathbf{B}^\mathsf{T}\mathbf{S} = \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}, \tag{7.59}$$

$$\mathbf{S}\mathbf{B}\mathbf{K} = \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}. \tag{7.60}$$

Hence $\mathbf{K}^\mathsf{T}\mathbf{B}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{B}\mathbf{K} = 2\mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$. Note also that $\mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K} = \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$.

Substituting back:

$$\begin{aligned}\mathbf{A}_{\mathrm{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\mathrm{cl}} + \dot{\mathbf{S}} &= -\mathbf{Q} + \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} - 2\mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} \\ &= -\mathbf{Q} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S} \\ &= -\mathbf{Q} - \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K}. \end{aligned} \tag{7.61}$$

Therefore:

$$\dot{V} = \delta\boldsymbol{x}^\mathsf{T}[\mathbf{A}_{\mathrm{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\mathrm{cl}} + \dot{\mathbf{S}}]\delta\boldsymbol{x} = -\delta\boldsymbol{x}^\mathsf{T}(\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K})\delta\boldsymbol{x}. \tag{7.62}$$

7.5.4 Exponential Decay in Continuous Time

If $\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K} \succeq \alpha\mathbf{I}$ and $\underline{s}\mathbf{I} \preceq \mathbf{S} \preceq \overline{s}\mathbf{I}$, then:

$$\dot{V} \leq -\alpha\left\|\delta\boldsymbol{x}\right\|^2 \leq -\frac{\alpha}{\overline{s}}V. \tag{7.63}$$

Solving the differential inequality:

$$V(t) \leq e^{-\lambda t}V(0), \quad \text{where} \quad \lambda = \frac{\alpha}{\overline{s}}. \tag{7.64}$$

Converting to Euclidean norm:

$$\left\|\delta\boldsymbol{x}(t)\right\|^2 \leq \frac{\overline{s}}{\underline{s}}e^{-2\lambda t}\left\|\delta\boldsymbol{x}(0)\right\|^2, \tag{7.65}$$

or:

$$\left\|\delta\boldsymbol{x}(t)\right\| \leq \sqrt{\frac{\overline{s}}{\underline{s}}}\,e^{-\lambda t}\left\|\delta\boldsymbol{x}(0)\right\|. \tag{7.66}$$

7.5.5 Comparison With Contraction LMI

Comparing (7.62) with the contraction condition from Chapter 5:

$$\mathbf{A}_{\mathrm{cl}}^\mathsf{T}\mathbf{M} + \mathbf{M}\mathbf{A}_{\mathrm{cl}} + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}, \tag{7.67}$$

we see that setting $\mathbf{M} = \mathbf{S}$ and $\lambda = \alpha/(2\overline{s})$ (so that $2\lambda = \alpha/\overline{s}$) gives:

$$\mathbf{A}_{\mathrm{cl}}^\mathsf{T}\mathbf{S} + \mathbf{S}\mathbf{A}_{\mathrm{cl}} + \dot{\mathbf{S}} = -(\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K}) \preceq -\alpha\mathbf{I} \preceq -\frac{\alpha}{\overline{s}}\mathbf{S}. \tag{7.68}$$

This is precisely the contraction condition: **the Riccati solution simultaneously certifies optimality and contraction**.

7.6 Robustness Margins From LQR

7.6.1 Gain Margin

Theorem 7.7 (Gain Margin of Continuous-Time LQR)**.** *The continuous-time LQR controller with gain* $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{S}$ *guarantees stability under any multiplicative perturbation* $\mathbf{K}_\eta = \eta\mathbf{K}$ *for* $\eta \in (1/2, \infty)$.

Proof. Consider the perturbed closed-loop system $\dot{\boldsymbol{x}} = (\mathbf{A} - \eta\mathbf{BK})\boldsymbol{x}$. Using the value function $V = \boldsymbol{x}^{\mathsf{T}}\mathbf{S}\boldsymbol{x}$:

$$\begin{aligned}\dot{V} &= \boldsymbol{x}^{\mathsf{T}}[(\mathbf{A} - \eta\mathbf{BK})^{\mathsf{T}}\mathbf{S} + \mathbf{S}(\mathbf{A} - \eta\mathbf{BK})]\boldsymbol{x} \\ &= \boldsymbol{x}^{\mathsf{T}}[\mathbf{A}^{\mathsf{T}}\mathbf{S} + \mathbf{SA} - \eta(\mathbf{K}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}\mathbf{S} + \mathbf{SBK})]\boldsymbol{x}.\end{aligned} \tag{7.69}$$

From the ARE:

$$\mathbf{A}^{\mathsf{T}}\mathbf{S} + \mathbf{SA} = -\mathbf{Q} - \mathbf{SBR}^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{S}. \tag{7.70}$$

And we showed:

$$\mathbf{K}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}\mathbf{S} + \mathbf{SBK} = 2\mathbf{SBR}^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{S} = 2\mathbf{K}^{\mathsf{T}}\mathbf{RK}. \tag{7.71}$$

Thus:

$$\begin{aligned}\dot{V} &= \boldsymbol{x}^{\mathsf{T}}[-\mathbf{Q} - \mathbf{SBR}^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{S} - 2\eta\mathbf{K}^{\mathsf{T}}\mathbf{RK}]\boldsymbol{x} \\ &= -\boldsymbol{x}^{\mathsf{T}}[\mathbf{Q} + (2\eta)\mathbf{K}^{\mathsf{T}}\mathbf{RK} + \mathbf{SBR}^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{S}]\boldsymbol{x}.\end{aligned} \tag{7.72}$$

The condition $\dot{V} < 0$ requires the bracketed term to be positive definite. Assuming $\eta > 1/2$, we have $2\eta > 1$, which means the positive definite matrix $2\eta\mathbf{K}^{\mathsf{T}}\mathbf{RK}$ dominates, and $\dot{V} < 0$ is maintained. □

7.6.2 Phase Margin

Theorem 7.8 (Phase Margin of Continuous-Time LQR)**.** *The continuous-time LQR controller guarantees a phase margin of at least* $60°$ *on both sides of the nominal open-loop transfer function.*

The proof uses the Nyquist criterion and the Riccati-induced positive real condition on the return difference $\mathbf{I} + \mathbf{K}(\mathbf{sI} - \mathbf{A})^{-1}\mathbf{B}$. We omit the technical details but note the key insight: the Riccati solution makes the closed-loop system "strictly positive real," which bounds phase perturbations.

7.6.3 Disk Margin

The disk margin, or μ-margin, quantifies robustness in the complex plane. For LQR, the loop transfer matrix satisfies a disk margin condition: the poles of the closed-loop system remain in a half-plane with guaranteed clearance from the origin. This is a consequence of the Riccati solution being positive definite and the ARE's structure.

7.6.4 Implication for Design

These robustness margins explain why LQR is ubiquitously used in practice: not only does it minimize a well-defined cost, but it automatically provides guardrails against model uncertainty, sensor noise, and actuator saturation. The engineer need not separately verify robustness—the LQR computation provides it.

7.7 Condition Number and Its Effect on Contraction Rate

7.7.1 Definition and Interpretation

Define the condition number of the Riccati solution:

$$\kappa(\mathbf{S}) := \frac{\lambda_{\max}(\mathbf{S})}{\lambda_{\min}(\mathbf{S})} = \frac{\overline{s}}{\underline{s}}. \tag{7.73}$$

A well-conditioned matrix has $\kappa(\mathbf{S}) \approx 1$; a poorly conditioned matrix has $\kappa(\mathbf{S}) \gg 1$.

7.7.2 Ill-Conditioning Signals Loss of Controllability

When $\kappa(\mathbf{S})$ is large, it often indicates that:

1. The system is nearly uncontrollable in some direction (e.g., the angular velocity of the pendulum is hard to control).
2. The Riccati matrix has very different scales along its principal directions.
3. Perturbations in one direction (corresponding to $\lambda_{\max}(\mathbf{S})$) are much more costly than in another (corresponding to $\lambda_{\min}(\mathbf{S})$).

In the limit of complete uncontrollability, $\lambda_{\min}(\mathbf{S}) \to 0$, and $\kappa(\mathbf{S}) \to \infty$.

7.7.3 Condition Number and Contraction Rate

From the discrete-time analysis:

$$\rho = 1 - \frac{\alpha}{\overline{s}} = 1 - \frac{\alpha}{\kappa(\mathbf{S})\underline{s}}. \tag{7.74}$$

As $\kappa(\mathbf{S})$ increases, $\overline{s}$ increases (for fixed $\underline{s}$ and α), and the contraction rate ρ increases (approaches 1), meaning convergence slows. More precisely:

$$1 - \rho = \frac{\alpha}{\overline{s}} \approx \frac{\alpha}{\kappa(\mathbf{S})\underline{s}}. \tag{7.75}$$

The number of time steps to reduce the value by a factor of e is:

$$\tau = \frac{1}{\ln(1/\rho)} \approx \frac{1}{\alpha/\overline{s}} = \frac{\overline{s}}{\alpha} = \kappa(\mathbf{S})\frac{\underline{s}}{\alpha}. \tag{7.76}$$

Conclusion: A large condition number $\kappa(\mathbf{S})$ is a warning sign. It means the system has directions that are hard to control, and convergence is slower. The designer should either:

1. Increase the state penalty $\mathbf{Q}$ in the weakly controllable direction to reduce $\overline{s}/\underline{s}$.
2. Accept slower convergence but maintain open-loop robustness.
3. Redesign the system (e.g., add actuators) to improve controllability.

7.8 Connections to H-Infinity and Robust Control

7.8.1 The H-Infinity Problem

The H^∞ optimal control problem is: given a system with disturbances $\mathbf{w}$,

$$\dot{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x} + \mathbf{B}_u\boldsymbol{u} + \mathbf{B}_w\mathbf{w}, \tag{7.77}$$

with controlled output:

$$\mathbf{z} = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{D}_{uu} \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ \mathbf{D}_{uw} \end{bmatrix} \mathbf{w}, \tag{7.78}$$

find a feedback controller $\mathbf{K}$ that minimizes the L_2-induced norm from disturbances to outputs:

$$\gamma^\star = \min_{\mathbf{K}} \max_{\mathbf{w}\neq 0} \frac{\|\mathbf{z}\|_{L_2}}{\|\mathbf{w}\|_{L_2}}. \tag{7.79}$$

7.8.2 The H-Infinity Riccati Equation

The optimal H^∞ controller satisfies a Riccati equation similar in structure to the LQR ARE:

$$0 = \mathbf{A}^\mathsf{T}\mathbf{S}_\infty + \mathbf{S}_\infty\mathbf{A} + \mathbf{C}_1^\mathsf{T}\mathbf{C}_1 - \mathbf{S}_\infty(\mathbf{B}_u\mathbf{D}_{uu}^{-1}\mathbf{B}_u^\mathsf{T} - \gamma^{-2}\mathbf{B}_w\mathbf{B}_w^\mathsf{T})\mathbf{S}_\infty, \tag{7.80}$$

where γ is the worst-case L_2 gain.

7.8.3 Duality in the Robust Setting

The duality theorem extends to H^∞ control: the Riccati solution $\mathbf{S}_\infty$ serves as both:

1. **Optimal performance certificate**: the value function $V = \boldsymbol{x}^\top \mathbf{S}_\infty \boldsymbol{x}$ guarantees that the L_2 gain from disturbances to outputs is at most γ.

2. **Contraction metric under disturbances**: the time-derivative of V is negative even under the worst-case disturbance, ensuring exponential stability.

The key insight is the disturbance direction in the Riccati equation: $\gamma^{-2}\mathbf{B}_w\mathbf{B}_w^\top$ enters with a *negative* sign (compared to the control input $\mathbf{B}_u$ which is positive). This asymmetry means the controller is simultaneously designed to:

- Minimize control energy (via $\mathbf{B}_u$).

- Maximize disturbance rejection (via $\mathbf{B}_w$ with the worst-case scaling).

7.8.4 Worked Example: Pendulum With Wind Disturbance

Consider the pendulum with a wind force disturbance w entering as:

$$\ddot{\theta} = \theta - u + w. \tag{7.81}$$

The H^∞ design seeks a controller that minimizes the gain from wind w to angle tracking error θ. The Riccati solution provides a controller that is simultaneously as energy-efficient as possible while rejecting the worst-case wind disturbance. The duality theorem guarantees that this single controller solves both the performance and robustness objectives.

7.9 Time-Varying vs. Steady-State Duality

7.9.1 The Discrete Riccati Equation (DRE)

For finite-horizon problems (e.g., trajectory optimization over a T-step horizon), the cost-to-go matrix evolves backward according to the discrete Riccati recursion:

$$\mathbf{S}_k = \mathbf{Q}_k + \mathbf{A}_k^\top \mathbf{S}_{k+1}\mathbf{A}_k - \mathbf{A}_k^\top \mathbf{S}_{k+1}\mathbf{B}_k(\mathbf{R}_k + \mathbf{B}_k^\top \mathbf{S}_{k+1}\mathbf{B}_k)^{-1}\mathbf{B}_k^\top \mathbf{S}_{k+1}\mathbf{A}_k, \tag{7.82}$$

with $\mathbf{S}_T = \mathbf{S}_f$ (a terminal cost, often zero).

At each time k, the contraction property holds in the tangent space:

$$\|\delta \boldsymbol{x}_{k+1}\|_{\mathbf{S}_{k+1}} \leq \rho_k \|\delta \boldsymbol{x}_k\|_{\mathbf{S}_k}, \tag{7.83}$$

where $\rho_k = 1 - \alpha_k/\overline{s}_k$ is *time-dependent*.

7.9.2 The Algebraic Riccati Equation (ARE)

For infinite-horizon, time-invariant LQR, the Riccati recursion converges to a fixed point, the ARE:

$$0 = \mathbf{Q} + \mathbf{A}^\top\mathbf{S} + \mathbf{S}\mathbf{A} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top\mathbf{S}. \tag{7.84}$$

The solution $\mathbf{S}$ is *constant* in time, and the contraction rate ρ is also constant:

$$\|\delta \boldsymbol{x}_k\|_{\mathbf{S}} \leq \rho^k \|\delta \boldsymbol{x}_0\|_{\mathbf{S}}, \tag{7.85}$$

with $\rho = 1 - \alpha/\overline{s}$ independent of k.

7.9.3 Geometric Interpretation

1. **Time-varying case (DRE)**: The Riccati matrix $\mathbf{S}_k$ changes at each step, reflecting the changing "cost-to-go" as we move backward in time from the final state. The tangent space metric is *adapted to the current time*: at times near the terminal condition, the metric emphasizes directions that are expensive to steer; far from the terminal, the metric is influenced by asymptotic stability.

2. **Steady-state case (ARE)**: The Riccati matrix reaches a time-invariant steady state. The metric no longer changes, and the contraction property is uniform in time. This is appropriate for infinite-horizon problems or problems where the horizon is much longer than the natural time scale of the system.

In both cases, the duality theorem holds: the Riccati solution (time-varying or steady-state) is simultaneously the optimal cost-to-go and the contraction metric.

7.10 Tangent Bundle Geometry and Geodesics

7.10.1 The Tangent Bundle

A trajectory on a state-space manifold is a curve $\gamma(t) = (x_1(t), \ldots, x_n(t))$. At each point x on the trajectory, the **tangent space** T_xM is the space of infinitesimal perturbation vectors $\delta\boldsymbol{x}$. The **tangent bundle** TM is the union of all tangent spaces:

$$TM = \bigcup_{x \in M} T_xM = \{(x, \delta\boldsymbol{x}) : x \in M, \delta\boldsymbol{x} \in \mathbb{R}^n\}. \tag{7.86}$$

7.10.2 Riemannian Metric on the Tangent Bundle

A Riemannian metric is a positive-definite inner product on the tangent space at each point. In our context, the Riccati solution $\mathbf{S}(x)$ (which may depend on the state x for nonlinear systems, but is constant for linear systems) defines a Riemannian metric:

$$\langle \delta\boldsymbol{x}_1, \delta\boldsymbol{x}_2 \rangle_{\mathbf{S}} = \delta\boldsymbol{x}_1^{\mathsf{T}} \mathbf{S}\, \delta\boldsymbol{x}_2. \tag{7.87}$$

The length of a tangent vector is:

$$\|\delta\boldsymbol{x}\|_{\mathbf{S}} = \sqrt{\delta\boldsymbol{x}^{\mathsf{T}} \mathbf{S}\, \delta\boldsymbol{x}}. \tag{7.88}$$

7.10.3 Geodesics and Optimal Control

In Riemannian geometry, a **geodesic** is a curve of shortest length (in the metric sense). For our system, the optimal trajectories (those minimizing the LQR cost) are precisely the **geodesics** with respect to the Riccati metric.

Here's why: the LQR cost functional

$$J = \int_0^T \left(\boldsymbol{x}^{\mathsf{T}} \mathbf{Q} \boldsymbol{x} + \boldsymbol{u}^{\mathsf{T}} \mathbf{R} \boldsymbol{u} \right) dt \tag{7.89}$$

can be rewritten, using the Riccati solution, as a metric-weighted length:

$$J \approx \int_0^T \|\dot{\gamma}\|_{\mathbf{S}} \; dt, \tag{7.90}$$

where the precise relationship depends on the details of the derivation. The Euler-Lagrange equations for minimizing this metric-weighted path length are equivalent to the optimal control equations.

7.10.4 Convergence of Neighboring Trajectories

A key property of geodesics is that they are *stable* (in a local sense): neighboring geodesics maintain bounded separation. In our setting, the contraction property guarantees stronger convergence:

$$\|\Delta\gamma(t)\|_{\mathbf{S}} \leq \rho^{t/\tau}\|\Delta\gamma(0)\|_{\mathbf{S}}, \tag{7.91}$$

where $\Delta\gamma$ is the difference between two nearby optimal trajectories, and τ is the natural time scale.

This is a remarkable property: not only are the optimal trajectories geodesics (shortest paths), but *they also attract nearby geodesics exponentially fast.* This is what makes the LQR solution so robust and widely applicable.

7.10.5 Visualization in 2D

For a 2D system (e.g., the pendulum with angle and angular velocity), the Riccati metric defines an elliptical "distance metric." The level sets of the Lyapunov function $V = \delta\boldsymbol{x}^{\mathsf{T}}\mathbf{S}\,\delta\boldsymbol{x} = 1$ form ellipses. The principal axes of the ellipse are the eigenvectors of $\mathbf{S}$, with radii proportional to $1/\sqrt{\lambda_i(\mathbf{S})}$.

A smaller (more negative) eigenvalue means a longer radius in that direction: perturbations along that eigenvector are "cheaper" in the metric sense. Conversely, a larger eigenvalue means a shorter radius: perturbations are expensive.

Under the optimal feedback, all perturbation ellipses shrink at the rate ρ per time step, and the ellipses align with the eigenvectors of $\mathbf{A}_{\mathrm{cl}}$.

7.11 Practical Design With Duality

7.11.1 Integrated Design Workflow

The duality theorem suggests a unified design methodology:

Design Implication

Integrated Stability-Optimality Design:

1. **Model the system**: discretize or linearize to get $(\mathbf{A}, \mathbf{B})$.
2. **Choose weights**: select $\mathbf{Q}$ and $\mathbf{R}$ based on performance objectives (state tracking, control effort).
3. **Solve the Riccati equation**: compute $\mathbf{S}$ (via the ARE or DRE).
4. **Extract the gain**: $\mathbf{K} = (\mathbf{R} + \mathbf{B}^\top\mathbf{S}\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{S}\mathbf{A}$.
5. **Verify contraction margins**:
   - Compute $\alpha = \lambda_{\min}(\mathbf{Q} + \mathbf{K}^\top\mathbf{R}\mathbf{K})$.
   - Compute $\overline{s} = \lambda_{\max}(\mathbf{S})$ and $\underline{s} = \lambda_{\min}(\mathbf{S})$.
   - Contraction rate: $\rho = 1 - \alpha/\overline{s}$.
   - Condition number: $\kappa(\mathbf{S}) = \overline{s}/\underline{s}$.
6. **Decision point**:
   - If ρ is close to 1 (say, $\rho > 0.9$) or $\kappa(\mathbf{S})$ is large (say, > 100), the system has weak controllability. Increase $\mathbf{Q}$ in the hard-to-control direction and re-solve.
   - If ρ is small (say, < 0.3) and $\kappa(\mathbf{S})$ is reasonable, the design is aggressive and robust. Proceed to implementation.
   - Otherwise, the design is balanced. Implement and tune on hardware if needed.
7. **Robustness certification**: the gain margins $[1/2, \infty)$ and phase margin $\geq 60°$ apply automatically.

7.11.2 When Duality Breaks Down

The duality requires several conditions:

Common Misconception

Limitations of the Duality:

1. **Bounded Riccati**: If the system loses controllability near a trajectory, the Riccati matrix can become unbounded or ill-conditioned. This is detected by monitoring $\kappa(\mathbf{S})$. If $\kappa(\mathbf{S}) > 10^6$ or $\lambda_{\min}(\mathbf{S}) < 10^{-6}$, the system is nearly uncontrollable.
2. **Meaningful weights**: The weights $\mathbf{Q}$ and $\mathbf{R}$ must reflect true performance objectives. If weights are chosen arbitrarily (e.g., to make the Riccati well-conditioned), the resulting controller may be suboptimal in practice.
3. **Linearity and perturbations**: The duality holds exactly in the tangent space (linearized dynamics). For large perturbations or highly nonlinear systems, the linear theory breaks down. Nonlinear generalizations (e.g., differential games with Bellman equations) exist but are more complex.
4. **Measurement and state estimation**: The duality assumes perfect state knowledge ($\boldsymbol{u} = -\mathbf{K}\boldsymbol{x}$). With measurement noise or partial observability, a separate state estimator (Kalman filter) is needed, and the duality extends in a more subtle way (separation principle).
5. **Sampling and discretization**: When continuous-time controllers are implemented on digital hardware, discretization errors can break the duality. Careful Tustin or zero-order-hold discretization is required.

7.12 The Estimation–Control Duality

The duality developed so far connects stability to optimality on the *control* side. But the Riccati equation has a mirror image: the *estimation* Riccati, which determines how uncertainty propagates through the system and how observations reduce it. This section develops the estimation side of the duality.

7.12.1 The Dual Riccati Equation

Consider the system with process noise and output measurement:

$$\dot{\boldsymbol{x}} = \mathbf{A}(t)\boldsymbol{x} + \mathbf{B}(t)\boldsymbol{u} + \mathbf{G}_w w, \tag{7.92}$$

$$\boldsymbol{y} = \mathbf{C}(t)\boldsymbol{x} + v, \tag{7.93}$$

where $w \sim \mathcal{N}(0, \mathbf{W})$ and $v \sim \mathcal{N}(0, \mathbf{V})$ are process and measurement noise. The *estimation Riccati equation* governs the covariance $\mathbf{P}(t)$ of the optimal (Kalman) state estimate:

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^T + \mathbf{G}_w\mathbf{W}\mathbf{G}_w^T - \mathbf{P}\mathbf{C}^T\mathbf{V}^{-1}\mathbf{C}\mathbf{P}. \tag{7.94}$$

Compare with the control Riccati:

$$-\dot{\mathbf{S}} = \mathbf{A}^T\mathbf{S} + \mathbf{S}\mathbf{A} + \mathbf{Q} - \mathbf{S}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{S}. \tag{7.95}$$

The duality is exact: the estimation Riccati for $(\mathbf{A}, \mathbf{C}, \mathbf{W}, \mathbf{V})$ is the control Riccati for $(\mathbf{A}^T, \mathbf{C}^T, \mathbf{G}_w\mathbf{W}\mathbf{G}_w^T, \mathbf{V})$. The estimation gain $\mathbf{L} = \mathbf{P}\mathbf{C}^T\mathbf{V}^{-1}$ is the dual of the control gain $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{S}$.

7.12.2 The Extended Kalman Filter as Tangent-Space Estimation

For nonlinear systems $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u} + w$, the Extended Kalman Filter (EKF) propagates the state estimate and covariance using the tangent-space (linearized) dynamics:

$$\dot{\hat{\boldsymbol{x}}} = f(\hat{\boldsymbol{x}}) + G(\hat{\boldsymbol{x}})\boldsymbol{u} + \mathbf{L}(t)\big(\boldsymbol{y} - h(\hat{\boldsymbol{x}})\big), \tag{7.96}$$

$$\dot{\mathbf{P}} = \mathbf{A}(t)\mathbf{P} + \mathbf{P}\mathbf{A}(t)^T + \mathbf{G}_w\mathbf{W}\mathbf{G}_w^T - \mathbf{P}\mathbf{C}(t)^T\mathbf{V}^{-1}\mathbf{C}(t)\mathbf{P}, \tag{7.97}$$

where $\mathbf{A}(t) = \frac{\partial f}{\partial \boldsymbol{x}}\big|_{\hat{\boldsymbol{x}}(t)}$ and $\mathbf{C}(t) = \frac{\partial h}{\partial \boldsymbol{x}}\big|_{\hat{\boldsymbol{x}}(t)}$.

Geometric Intuition

The EKF is the estimation analogue of DDP: both linearize the system along a trajectory (the estimated trajectory for EKF, the nominal trajectory for DDP) and solve a Riccati equation in the tangent space. The key difference is the direction of information flow: DDP propagates *cost* backward (from the terminal condition); the EKF propagates *uncertainty* forward (from the initial condition).

7.12.3 Contraction-Based Observer Design

The EKF has no convergence guarantees for nonlinear systems—the estimate may diverge if the linearization is poor. Contraction theory provides a rigorous alternative.

An observer $\dot{\hat{\boldsymbol{x}}} = f(\hat{\boldsymbol{x}}) + G(\hat{\boldsymbol{x}})\boldsymbol{u} + L(\hat{\boldsymbol{x}})(\boldsymbol{y} - h(\hat{\boldsymbol{x}}))$ is a *contracting observer* if the observer error dynamics $\dot{e} = \dot{\boldsymbol{x}} - \dot{\hat{\boldsymbol{x}}}$ are contracting in some metric $\mathbf{M}(\boldsymbol{x})$:

$$\operatorname{sym}\left(\mathbf{M}\left[\frac{\partial f}{\partial \boldsymbol{x}} - L(\boldsymbol{x})\frac{\partial h}{\partial \boldsymbol{x}}\right]\right) + \dot{\mathbf{M}} \preceq -2\lambda\mathbf{M}. \tag{7.98}$$

This is precisely the contraction condition of Chapter 5, applied to the observer error dynamics. The observer gain $L(\boldsymbol{x})$ plays the same role as the feedback gain $\mathbf{K}$ in control contraction metrics: it shapes the closed-loop dynamics to achieve contraction.

Theorem 7.9 (Observer-Controller Duality)**.** *The observer design problem — find $L(\boldsymbol{x})$ such that (7.98) holds — is the* dual *of the controller design problem via the substitutions $\mathbf{A} \leftrightarrow \mathbf{A}^T$, $\mathbf{B} \leftrightarrow \mathbf{C}^T$, $\mathbf{K} \leftrightarrow L^T$. A contraction metric for the observer is a contraction metric for the dual controller, and vice versa.*

7.12.4 The Observability Gramian

The observability Gramian $\mathcal{W}_o(t_0, t_1)$ quantifies how much information the output $\boldsymbol{y}$ reveals about the initial state $\boldsymbol{x}(t_0)$:

$$\mathcal{W}_o(t_0, t_1) = \int_{t_0}^{t_1} \Phi(s, t_0)^T \mathbf{C}(s)^T \mathbf{C}(s) \Phi(s, t_0)\, \mathrm{d}s, \tag{7.99}$$

where Φ is the state transition matrix from Chapter 2.

The system is *observable* on $[t_0, t_1]$ if $\mathcal{W}_o \succ 0$. The eigenvectors of $\mathcal{W}_o$ reveal the most and least observable state directions; the condition number $\kappa(\mathcal{W}_o)$ measures the anisotropy of observability.

Common Misconception

For the golf swing application (Chapter 9), the observability Gramian reveals which state variables can be reliably estimated from sensor measurements. Joint angles near the shoulder are typically well-observed (multiple muscle attachments create redundant signals); wrist angular velocity and shaft bending are poorly observed (few sensors, high velocity). The condition number of $\mathcal{W}_o$ quantifies this anisotropy.

7.12.5 The Separation Principle

When both observer and controller are available, the celebrated *separation principle* allows independent design:

Theorem 7.10 (Separation Principle)**.** *For linear time-invariant systems, the optimal controller using estimated states $\boldsymbol{u} = -\mathbf{K}\hat{\boldsymbol{x}}$ achieves the same closed-loop eigenvalues as the full-state feedback $\boldsymbol{u} = -\mathbf{K}\boldsymbol{x}$, with additional eigenvalues from the observer error dynamics. The controller gain $\mathbf{K}$ and observer gain $\mathbf{L}$ can be designed independently.*

For nonlinear systems, the separation principle does not hold in general. However, when both the controller and observer are designed via contraction metrics, the overall system (controller + observer) contracts at a rate determined by the *minimum* of the controller and observer contraction rates: $\lambda_{\text{overall}} \geq \min(\lambda_{\text{control}}, \lambda_{\text{observer}})$. This is a practical nonlinear extension of the separation principle.

7.13 Chapter Summary

1. **Historical foundation**: Kalman (1960) discovered that the LQR cost-to-go matrix is a Lyapunov function for the closed-loop system. Willems (1972) formalized this via dissipation inequalities. Anderson and Moore highlighted the robustness margins of LQR.

2. **Discrete duality**: The value decrease identity $\Delta V = -\delta \boldsymbol{x}^\mathsf{T}(\mathbf{Q}+\mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K})\delta \boldsymbol{x}$ is both a proof of optimality and a contraction certificate. The decay rate $\rho = 1 - \alpha/\overline{s}$ characterizes convergence speed.

3. **Continuous duality**: The time-derivative $\dot{V} = -\delta \boldsymbol{x}^\mathsf{T}(\mathbf{Q} + \mathbf{K}^\mathsf{T}\mathbf{R}\mathbf{K})\delta \boldsymbol{x}$ follows directly from the DRE and the closed-loop dynamics. Exponential decay follows with rate $\lambda = \alpha/(2\overline{s})$.

4. **Robustness margins**: LQR guarantees gain margin $[1/2, \infty)$, phase margin $\geq 60°$, and disk margin—all consequences of the Riccati structure and the positive-definiteness of $\mathbf{S}$.

5. **Condition number**: The condition number $\kappa(\mathbf{S}) = \overline{s}/\underline{s}$ quantifies the difficulty of controlling the system. Large κ indicates poor controllability and slow convergence.

6. **H∞ extension**: The duality extends to robust control: the H∞ Riccati solution is simultaneously an optimal performance certificate and a contraction metric under worst-case disturbances.

7. **Time-varying vs. steady-state**: The DRE provides time-adapted metrics for finite-horizon problems; the ARE provides a constant metric for infinite-horizon, time-invariant systems.

8. **Geometric interpretation**: The Riccati matrix defines a Riemannian metric on the tangent bundle. Optimal trajectories are geodesics, and the contraction property ensures neighboring geodesics converge exponentially.

9. **Practical workflow**: Design with duality integrated: solve LQR once, verify contraction margins and condition number, and adjust weights if needed. The single computation certifies both optimality and robustness.

10. **Limitations**: Duality requires bounded, well-conditioned Riccati matrices; meaningful weights; and linearity of perturbations. It breaks down near singular arcs, under large perturbations, or with hidden states.

This chapter has shown that optimality and stability are not independent concerns but two facets of a single geometric object. The Riccati matrix, solved once, provides both the controller and the robustness certificate. This is the power of duality: one computation, two major objectives accomplished.

Exercises

1. **Value decrease identity.** For the discrete pendulum in the worked example, verify the value decrease identity numerically: compute $V_{k+1} - V_k$ and $-\delta \boldsymbol{x}_k^T(\mathbf{Q} + \mathbf{K}^T\mathbf{R}\mathbf{K})\delta \boldsymbol{x}_k$ and confirm equality.

2. **Robustness margins.** For the continuous-time system $\dot{x} = ax + bu$ with LQR weights $Q = 1$, $R = 1$, compute the gain margin and phase margin as functions of a and b. Verify the theoretical bounds $[1/2, \infty)$ and $\geq$ 60ř.

3. **Dual Riccati.** For the system $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -0.1 \end{bmatrix}$, $\mathbf{C} = [1, 0]$, $W = 0.1$, $V = 1$, solve the estimation ARE for $\mathbf{P}$. Compare with the control ARE for $(\mathbf{A}^T, \mathbf{C}^T, W, V)$.

4. **Condition number and convergence.** For the 2D system in the worked example, vary $\mathbf{Q}$ from $0.1\mathbf{I}$ to $100\mathbf{I}$ while keeping $\mathbf{R} = \mathbf{I}$. Plot $\kappa(\mathbf{S})$ and the contraction rate λ as functions of $\|\mathbf{Q}\|$. Explain the observed relationship.

5. **EKF implementation.** Implement the EKF for the nonlinear pendulum $\ddot{\theta} + \sin\theta = u + w$ with measurement $y = \theta + v$. Compare the estimated trajectory with the true trajectory for different noise levels.

6. **Observability Gramian.** For the linearized pendulum at the upright equilibrium, compute the observability Gramian $\mathcal{W}_o(0, T)$ for $T = 1, 5, 10$. How does its condition number change? What does this imply about the observability of the system?

7. **H∞ vs LQR.** Solve both the LQR and H∞ Riccati equations for the double integrator with disturbance input. Compare the resulting contraction rates and gain magnitudes.

8. **Separation principle verification.** For the discrete pendulum, implement the separated controller (LQR gain) and observer (Kalman gain). Verify that the closed-loop eigenvalues are the union of the controller and observer eigenvalues.

Chapter 8

Forward Dynamics, Counterfactuals, and Drift

In Plain Language 8.1. Physics vs. The Driver

In a golf swing, a robot arm movement, or a spacecraft maneuver, how much of the motion is "physics doing its thing" and how much is the controller actively steering? This chapter develops tools to answer that question rigorously: forward dynamics models that separate passive drift from active control, and thought experiments (counterfactuals) that ask "What if the input were removed?"

8.1 The Causal Structure of Control-Affine Systems

The control-affine structure has been extensively studied in modern control theory [Kha02, Sas99, MLS94]. It provides a natural framework for understanding the interplay between passive and active dynamics.

The control-affine decomposition from Chapter 3,

$$\dot{\boldsymbol{x}} = \underbrace{f(\boldsymbol{x})}_{\text{Drift}} + \underbrace{G(\boldsymbol{x})\boldsymbol{u}}_{\text{Input}}, \tag{8.1}$$

is not merely a mathematical convenience. It encodes a *causal structure*:

Definition 8.1 (Drift)**.** The **drift vector field** $f(\boldsymbol{x})$ is the instantaneous state derivative when all active inputs are zero: $\dot{\boldsymbol{x}}\big|_{\boldsymbol{u}=0} = f(\boldsymbol{x})$. It captures the passive dynamics arising from accumulated state: inertia (velocity-dependent forces), gravity (configuration-dependent), elastic storage (deformation-dependent), and dissipation.

Definition 8.2 (Input)**.** The **input vector field** $G(\boldsymbol{x})\boldsymbol{u}$ is the additional contribution from active control. Because this enters linearly in $\boldsymbol{u}$ at fixed $\boldsymbol{x}$, its effect is instantaneous and separable from the drift.

The crucial structural property is **drift invariance**:

$$\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{u}} = 0. \tag{8.2}$$

The drift does not depend on the current input. This is the differential version of the superposition principle: the passive dynamics are a fixed "background" on which the input is superposed.

8.2 The Zero-Torque Counterfactual (ZTCF) Family

Definition 8.3 (Zero Torque Counterfactual)**.** Given a system in state $\boldsymbol{x}_0$ at time t_0, the **forward Zero-Torque Counterfactual** (ZTCF) trajectory is $\boldsymbol{x}^{(0)}(t)$ obtained by setting the declared applied control $\boldsymbol{u} \equiv 0$ for all $t \geq t_0$:

$$\dot{\boldsymbol{x}}^{(0)} = f\left(\boldsymbol{x}^{(0)}\right), \quad \boldsymbol{x}^{(0)}(t_0) = \boldsymbol{x}_0. \tag{8.3}$$

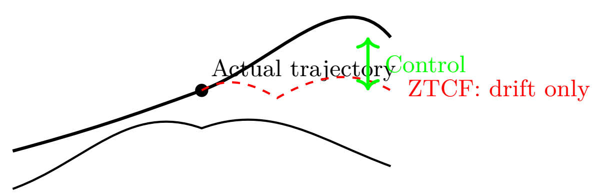
This is the integral curve of the drift vector field starting from the current state.

The forward ZTCF trajectory asks what the declared effective plant would do if its declared applied-control channel were removed. This does not imply zero muscle activation or removal of retained passive impedance.

For a mechanical system with state $\boldsymbol{x} = (\boldsymbol{q}, \dot{\boldsymbol{q}})$, the ZTCF represents the "passive shadow"—the motion that emerges purely from current velocities, gravity, elastic forces, and inertial coupling, with zero active muscular or motor effort.

Geometric Intuition

The ZTCF is like releasing the steering wheel of a car on a hilly road. The car continues to move—driven by its momentum and the terrain—but without any active guidance. Comparing the actual trajectory to the ZTCF reveals exactly how much the "driver" (controller) contributed at each instant.



Difference = active control contribution

8.2.1 Mathematical Foundations: Existence, Uniqueness, and Smoothness

The well-posedness of the ZTCF rests on classical existence and uniqueness theorems for ordinary differential equations. We formally state the conditions here.

Theorem 8.4 (Picard–Lindelöf Existence and Uniqueness)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a vector field that is continuously differentiable (i.e., $f \in C^1$). Then for any initial condition $\boldsymbol{x}_0 \in \mathbb{R}^n$, there exists a unique, continuously differentiable trajectory $\boldsymbol{x}^{(0)}(t)$ satisfying the initial value problem*

$$\dot{\boldsymbol{x}}^{(0)} = f(\boldsymbol{x}^{(0)}), \quad \boldsymbol{x}^{(0)}(t_0) = \boldsymbol{x}_0 \tag{8.4}$$

defined on some open interval $(t_0 - \epsilon, t_0 + \epsilon)$ for some $\epsilon > 0$.

Proof sketch: The Picard–Lindelöf theorem follows from the contraction mapping principle. Define the sequence of successive approximations

$$\boldsymbol{x}^{(k+1)}(t) = \boldsymbol{x}_0 + \int_{t_0}^{t} f(\boldsymbol{x}^{(k)}(s))\,\mathrm{d}s, \tag{8.5}$$

starting with $\boldsymbol{x}^{(0)}(t) = \boldsymbol{x}_0$. If f is Lipschitz continuous (which holds under C^1), this sequence converges uniformly to the unique solution on a sufficiently small interval.

Corollary 8.5 (Smoothness of ZTCF)**.** *If $f \in C^k$ (i.e., f is k times continuously differentiable), then the ZTCF trajectory $\boldsymbol{x}^{(0)}(t)$ is also C^k in both t and the initial condition $\boldsymbol{x}_0$.*

This means that if the drift is smooth (which it is for mechanical systems with smooth mass matrices and smooth potential energy functions), then the ZTCF is smooth. This justifies our use of ZTCF trajectories in subsequent numerical and analytical work.

8.2.2 The Flow Map and Properties

For systems where f is defined on all of $\mathbb{R}^n$ or an open domain $\mathcal{X}$, we define the **flow map** (or **solution map**):

$$\phi_t^f(\boldsymbol{x}_0) := \boldsymbol{x}^{(0)}(t; \boldsymbol{x}_0), \tag{8.6}$$

where $\boldsymbol{x}^{(0)}(t;\boldsymbol{x}_0)$ is the solution of the IVP (8.4) at time t, starting from $\boldsymbol{x}_0$ at $t = t_0$ (assuming $t_0 = 0$ for notational simplicity).

The flow map has the following properties:

Proposition 8.6 (Properties of the Flow Map)**.** *Suppose $f \in C^1(\mathbb{R}^n)$ and the solutions of $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ exist for all time. Then:*

1. ***Initial condition:*** $\phi_0^f(\boldsymbol{x}_0) = \boldsymbol{x}_0$.

2. ***Composition (group property):*** $\phi_{t_1+t_2}^f(\boldsymbol{x}_0) = \phi_{t_1}^f(\phi_{t_2}^f(\boldsymbol{x}_0))$.

3. ***Inverse:*** $\phi_{-t}^f(\phi_t^f(\boldsymbol{x}_0)) = \boldsymbol{x}_0$, *so* ϕ_t^f *is a diffeomorphism (invertible and smooth both ways).*

4. ***Differentiability with respect to initial condition:*** $\frac{\partial \phi_t^f(\boldsymbol{x}_0)}{\partial \boldsymbol{x}_0}$ *is the solution of the* ***variational equation***
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \phi_t^f}{\partial \boldsymbol{x}_0}\right) = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\phi_t^f(\boldsymbol{x}_0)} \cdot \frac{\partial \phi_t^f}{\partial \boldsymbol{x}_0}, \quad \frac{\partial \phi_0^f}{\partial \boldsymbol{x}_0} = I. \tag{8.7}$$

The variational equation (8.7) is crucial: it shows how perturbations to the initial state propagate along the ZTCF trajectory. The Jacobian matrix $\frac{\partial \phi_t^f}{\partial \boldsymbol{x}_0}$ grows or shrinks depending on the eigenvalues of the drift Jacobian $\frac{\partial f}{\partial \boldsymbol{x}}$ along the trajectory. We will return to this when analyzing contraction.

8.2.3 Mathematical Justification

The ZTCF is well-defined precisely because of drift invariance (8.2) and the input superposition principle (Theorem 3.1). Because $f(\boldsymbol{x})$ does not depend on $\boldsymbol{u}$, setting $\boldsymbol{u} = 0$ does not alter the drift field itself—it only removes the input contribution. The subtraction
$$G(\boldsymbol{x})\boldsymbol{u} = \dot{\boldsymbol{x}} - f(\boldsymbol{x}) \tag{8.8}$$
exactly recovers the active input contribution at each instant.

8.3 The Zero Velocity Counterfactual (ZVCF)

Definition 8.7 (Zero Velocity Counterfactual)**.** The **Zero Velocity Counterfactual** (ZVCF) evaluates the drift at the current configuration $\boldsymbol{q}$ and declared internal state but with velocity and declared control set to zero:
$$a_{\mathrm{ZVCF}} := -M(\boldsymbol{q})^{-1}h(\boldsymbol{q}, 0, z_0; p), \qquad \boldsymbol{u} = 0. \tag{8.9}$$
This reveals the static baseline: the acceleration due to gravity (and elastic preloads) without any velocity-dependent effects.

The difference between the full drift and the ZVCF isolates velocity-dependent forces:
$$a_{\mathrm{drift}} - a_{\mathrm{ZVCF}} = -M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}, \tag{8.10}$$
which captures Coriolis, centrifugal, and gyroscopic effects. These are the forces that arise purely from the system's inertial memory—the history encoded in $\dot{\boldsymbol{q}}$.

8.4 Decomposition of the Drift for Mechanical Systems

For mechanical systems without damping, the drift acceleration has a clean decomposition:
$$a_{\mathrm{drift}} = -M(\boldsymbol{q})^{-1}\left[\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q})\right]. \tag{8.11}$$

We can write this as a sum of two components:

$$a_{\text{grav}} := -M(\boldsymbol{q})^{-1}\boldsymbol{g}(\boldsymbol{q}), \tag{8.12a}$$

$$a_{\text{cor}} := -M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}, \tag{8.12b}$$

so that $a_{\text{drift}} = a_{\text{grav}} + a_{\text{cor}}$.

Common Misconception

This form assumes the system is unconstrained. Equation (8.11) and the input field $G(\boldsymbol{x}) = M(\boldsymbol{q})^{-1}B$ both change the moment a kinematic constraint is present — ground contact, a closed chain, a fixed clubhead path.

The reason is worth stating, because it is easy to get wrong in a way that leaves the formulas looking fine. Under a constraint $J_c\ddot{\boldsymbol{q}} + \dot{J}_c\dot{\boldsymbol{q}} = 0$, the multiplier $\boldsymbol{\lambda}$ is itself *affine in* $\boldsymbol{u}$. Carrying $J_c^{\top}\boldsymbol{\lambda}$ into the drift therefore makes the drift depend on the input, and $\partial f/\partial\boldsymbol{u} = 0$ — the property this entire construction rests on — silently fails. The ZTCF is defined by setting the input to zero and integrating the drift; that is only meaningful if the drift does not contain the input.

Chapter 10 eliminates the multiplier properly and gives the constrained drift and input fields, which satisfy $\partial f/\partial\boldsymbol{u} = 0$ by construction. **Use those whenever the system has a constraint**; the decomposition above applies only when it does not. The golf models in this book are unconstrained during the downswing, so the form here is the right one for them — but a golfer analysed with ground reaction forces is not.

8.4.1 Superposition of Drift Components

This decomposition is valid because $M(\boldsymbol{q})^{-1}$ is a linear operator (in the sense that it acts linearly on vectors). Therefore:

$$M(\boldsymbol{q})^{-1}\left[\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q})\right] = M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + M(\boldsymbol{q})^{-1}\boldsymbol{g}(\boldsymbol{q}), \tag{8.13}$$

and the two contributions add exactly, with no interaction terms. This is a mechanical manifestation of superposition at the drift level.

Geometric Intuition

Gravitational acceleration a_{grav} represents the component of drift that pulls the system toward lower potential energy (e.g., a pendulum hanging downward). It depends only on the configuration $\boldsymbol{q}$, not on velocities.

Coriolis/centrifugal acceleration a_{cor} represents inertial effects that arise from the system's motion. For example, in a rotating coordinate frame, a moving object experiences fictitious forces. Here, these emerge from the kinetic energy structure encoded in $\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$.

8.4.2 Magnitude Scaling

Because a_{grav} is independent of velocity while $a_{\text{cor}} \propto \dot{\boldsymbol{q}}$, we have:

$$\|a_{\text{grav}}\| \approx \text{const} \sim g\,(\text{gravitational acceleration scale}), \tag{8.14}$$

$$\|a_{\text{cor}}\| \sim \dot{\boldsymbol{q}}^2/L \quad (\text{for a system with length scale } L). \tag{8.15}$$

At low speeds, gravity dominates the drift. At high speeds, Coriolis dominates. This explains why a fast-swinging pendulum experiences primarily centrifugal effects, while a slowly tilting pendulum experiences primarily gravity.

8.4.3 Worked Decomposition Table: Double Pendulum Example

We will compute these components explicitly for the double pendulum in Section 8.12. For now, we show a schematic table of how the decomposition looks at various states:

q_1	q_2	$\dot{q}_1$	$\dot{q}_2$	$a_{\text{grav},1}$	$a_{\text{cor},1}$	$a_{\text{drift},1}$
0	0	0	0	(grav)	0	(grav)
0	0	2	0	(grav)	(small)	(grav+cor)
$\pi/4$	$\pi/6$	2	-1	(smaller)	(significant)	(sum)

Table 8.1: Schematic table showing the decomposition of drift acceleration into gravitational and Coriolis components. The exact values depend on the system parameters (masses, lengths) computed in Section 8.12. The key insight is that a_{cor} grows with velocity, while a_{grav} remains constant.

8.5 The Drift–Control Ratio

Definition 8.8 (Drift–Control Ratio)**.** At state $\boldsymbol{x}$, in a declared acceleration or task-projected space, the **Drift-Control Ratio (DCR)** is

$$\mathrm{DCR}_{W,\mathcal{U}}(\boldsymbol{x}) := \frac{\|W a_{\text{drift}}(\boldsymbol{x})\|}{\sup_{u \in \mathcal{U}(\boldsymbol{x})} \|W B_a(\boldsymbol{x}) u\| + \varepsilon}. \tag{8.16}$$

The drift–control ratio quantifies the *relative dominance* of passive versus active dynamics:

- $\rho \ll 1$: The system is "control-dominated." The controller can dictate the motion. Linearization and feedback design are straightforward.
- $\rho \approx 1$: Drift and control are comparable. The controller must work with the passive dynamics, not against them.
- $\rho \gg 1$: The system is "drift-dominated." Passive dynamics overwhelm available control authority. The controller can steer but cannot fundamentally alter the motion. This is the regime of high-speed underactuated systems.

Common Misconception

In drift-dominated regimes ($\rho \gg 1$), the mathematical fact that inputs enter linearly can be misleading. The *mechanism* is linear, but the *magnitude* of influence is tiny relative to the drift. Knowing that superposition holds exactly does not mean the controller has significant authority—it means we can precisely *quantify* how little authority remains.

8.5.1 Velocity Scaling

For mechanical systems, the drift scales quadratically with velocity (due to $\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$ terms), while maximum torque is roughly constant. Therefore:

$$\rho \sim \frac{\dot{\boldsymbol{q}}^2}{\text{const}} \to \infty \quad \text{as } \|\dot{\boldsymbol{q}}\| \to \infty. \tag{8.17}$$

At high speeds, every mechanical system becomes drift-dominated. This is why elite athletes "ride the physics" in the fast phases of movement: their skill lies in *setting up* the state so that the passive drift naturally produces the desired outcome.

8.6 Forward Dynamics as a Thought Experiment Engine

Forward dynamics simulation—integrating (8.1) given $\boldsymbol{x}_0$ and $\boldsymbol{u}(t)$—becomes a powerful analytical tool when combined with counterfactuals:

8.6.1 Attribution Analysis

Given an observed trajectory $\boldsymbol{x}(t)$ with known input $\boldsymbol{u}(t)$:

1. Compute the ZTCF from each time instant: "What would happen without input?"
2. Compute the input contribution: $\Delta\dot{\boldsymbol{x}} = G(\boldsymbol{x})\boldsymbol{u}$ at each instant.
3. Decompose the drift into gravitational, Coriolis, and elastic components.
4. Track the drift–control ratio $\rho(t)$ along the trajectory.

8.6.2 Sensitivity Counterfactuals

- **"What if actuator j failed?"**: Set $u_j = 0$ and re-simulate. Because input enters linearly, the effect of removing one channel is exactly $-g_j(\boldsymbol{x})u_j$—no need to re-solve the full dynamics.
- **"What if gravity changed?"**: Modify $\boldsymbol{g}(\boldsymbol{q})$ in the drift and re-simulate. This is a drift modification, not an input modification.
- **"What if the mass distribution changed?"**: Modify $M(\boldsymbol{q})$ and recompute both drift and input matrices. This changes the superposition structure itself.

8.7 Attribution Methodology: A Formal Procedure

To systematically attribute motion to drift versus control, we propose a five-step procedure that can be implemented as an algorithm or protocol.

Input: Observed state trajectory $\boldsymbol{x}(t)$ for $t \in [0, T]$, input signal $\boldsymbol{u}(t)$, system matrices $f(\boldsymbol{x})$ and $G(\boldsymbol{x})$, maximum input magnitude $\|\boldsymbol{u}_{\max}\|$.
Output: Drift–control ratio history $\rho(t)$, input contribution $G(\boldsymbol{x})\boldsymbol{u}(t)$, drift contribution $f(\boldsymbol{x}(t))$, ZTCF trajectories at key time points.

for each time step t_i in the trajectory **do**:

1. **Record** state $\boldsymbol{x}(t_i)$ and input $\boldsymbol{u}(t_i)$.
2. **Evaluate** drift $f(\boldsymbol{x}(t_i))$ and input contribution $G(\boldsymbol{x}(t_i))\boldsymbol{u}(t_i)$.
3. **Compute ZTCF**: integrate $\dot{\boldsymbol{x}}^{(0)} = f(\boldsymbol{x}^{(0)})$ from $\boldsymbol{x}(t_i)$ for a short horizon (e.g., 0.2 sec).
4. **Compute drift–control ratio**:
$$\rho(t_i) = \frac{\|f(\boldsymbol{x}(t_i))\|}{\|G(\boldsymbol{x}(t_i))\boldsymbol{u}_{\max}\|}.$$
5. **Classify phase**: If $\rho(t_i) < 0.5$, mark as *control-dominated*; if $0.5 \le \rho(t_i) \le 2$, mark as *transition*; if $\rho(t_i) > 2$, mark as *drift-dominated*.

end for

Algorithm 8.1: Drift–Control Attribution Protocol

This protocol produces a complete picture of when the controller is actively steering versus when it is passively managing drift. In practice, the protocol is implemented as a post-hoc analysis tool: you have a recorded trajectory and input, and you run the protocol offline to understand the causal roles of drift and control.

8.8 Sensitivity Counterfactuals: Detailed Mathematics

8.8.1 Single Actuator Failure

Suppose actuator j fails, so u_j drops from its nominal value to zero. The change in state derivative is:

$$\Delta\dot{\boldsymbol{x}} := \dot{\boldsymbol{x}}|_{u_j=0} - \dot{\boldsymbol{x}}|_{\text{nominal}} = -g_j(\boldsymbol{x})u_j, \tag{8.18}$$

where $g_j(\boldsymbol{x})$ is the j-th column of $G(\boldsymbol{x})$. This is exact because of the linearity of $G(\boldsymbol{x})\boldsymbol{u}$ in $\boldsymbol{u}$. No re-solving of the dynamics is needed; the effect is instantaneous and quantifiable.

8.8.2 Parameter Perturbation

When a system parameter p (e.g., a mass, length, or damping coefficient) changes, the effect propagates via the sensitivity equation. From Chapter 2, we have:

$$\frac{\partial\dot{\boldsymbol{x}}}{\partial p} = \frac{\partial f}{\partial p} + \frac{\partial G}{\partial p}\boldsymbol{u}. \tag{8.19}$$

This equation shows that the sensitivity is a linear combination of the drift sensitivity and the input matrix sensitivity. Over time, these sensitivities compound, so a small parameter error leads to a growing state error. The ZTCF at perturbed parameters can be computed by integrating the modified drift, providing a sensitivity counterfactual.

8.8.3 Gravity Modification

Gravity affects *only* the drift, not the input matrix $G(\boldsymbol{x})$. If we change the gravitational acceleration from g to g', then:

$$a'_{\text{drift}} = -M(\boldsymbol{q})^{-1}[\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + g'(\boldsymbol{q})], \quad G(\boldsymbol{x})' = G(\boldsymbol{x}) \quad \text{(unchanged)}. \tag{8.20}$$

The input contribution $G(\boldsymbol{x})\boldsymbol{u}$ is unaffected, so the change in behavior comes entirely from the ZTCF:

$$\text{Motion under gravity } g' = \text{Motion under } g + \text{Effect of } (g' - g) \text{ via ZTCF}. \tag{8.21}$$

This decomposition is notably clean because gravity is a pure drift effect.

8.8.4 Mass Redistribution

If the mass distribution changes (e.g., a robot carrying an object), then *both* $M(\boldsymbol{q})$ and therefore both $f(\boldsymbol{x})$ and $G(\boldsymbol{x})$ change. We have:

$$f(\boldsymbol{x})' = -M'(\boldsymbol{q})^{-1}[C'(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + g(\boldsymbol{q})], \quad G(\boldsymbol{x})' = \begin{bmatrix} 0_n \\ M'(\boldsymbol{q})^{-1} \end{bmatrix}. \tag{8.22}$$

Now the superposition structure itself changes: the input matrix G is no longer the same, and the drift cannot be simply subtracted. The effect is multiplicative, not additive. This breaks the simple linear decomposition and shows why mechanical changes (like adding inertia) are more subtle than external force changes (like changing gravity).

8.9 Data-Driven Drift Estimation

In many practical scenarios, the analytical model is unavailable or too complex to derive by hand. We must estimate $f(\boldsymbol{x})$ and $G(\boldsymbol{x})$ from data.

8.9.1 The Regression Approach

Suppose we have collected N triples of data: $\{(\boldsymbol{x}_i, \boldsymbol{u}_i, \dot{\boldsymbol{x}}_i)\}_{i=1}^N$, where each $\dot{\boldsymbol{x}}_i$ is the measured state derivative (e.g., from finite differencing of a recorded trajectory). We want to identify the system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$.

One approach is to assume a parametric form for f and G (e.g., polynomial or neural network basis functions), and then solve a regression problem:

$$\min_{f,G} \sum_{i=1}^{N} \left\| \dot{\boldsymbol{x}}_i - f(\boldsymbol{x}_i) - G(\boldsymbol{x}_i)\boldsymbol{u}_i \right\|^2 . \tag{8.23}$$

However, this is fundamentally ill-posed unless we have sufficient variation in the input $\boldsymbol{u}$.

8.9.2 Identification Condition: Input Variation

To separate f from G, we need the input to vary *independently* across the data. Specifically, if we write the regression in matrix form:

$$\dot{\boldsymbol{x}} = \Theta(\boldsymbol{x}) \cdot \theta, \tag{8.24}$$

where $\Theta(\boldsymbol{x})$ is a regressor matrix and θ is a parameter vector, then Θ must have full column rank. For the drift–input problem, this requires:

Persistent Excitation Condition: To identify both drift and input, the input signal $\boldsymbol{u}(t)$ must explore a region of the input space that is “large enough” relative to the state space. Formally, the data should include states with multiple different input values so that the regression can disentangle their effects.

Without this condition, the regression confounds drift and input: large drift values might be “explained” by small imaginary input contributions, or vice versa.

8.9.3 Practical Considerations

1. **Noise:** Measured state derivatives $\dot{\boldsymbol{x}}_i$ are corrupted by measurement noise. Regularization (e.g., L_2 penalty on parameters) is essential.

2. **Sampling Rate:** Finite differencing of discrete measurements introduces errors. Higher sampling rates (faster sensors) improve the estimate of $\dot{\boldsymbol{x}}$.

3. **Excitation Requirements:** The input must visit different regions of the state space with different control actions. A slowly varying or monotonic input signal will not provide sufficient variation.

4. **Model Class:** The choice of basis functions (polynomial, spline, neural network) affects identifiability. More flexible models can fit noise; more rigid models may miss nonlinearities.

8.9.4 Connection to System Identification

The problem of estimating f and G from data is a standard problem in **system identification**. The literature provides well-developed techniques such as:

- Subspace identification methods (Ho–Kalman, Balanced Model Reduction)
- Autoregressive models (ARX, ARMAX)
- Nonlinear regression (SINDy, Dynamic Mode Decomposition)
- Neural network-based identification

For control-affine systems specifically, the key is to ensure that the input variation is rich enough to separately identify the drift and the input gain.

8.10 Contraction Analysis of the Drift

A natural question is: does the drift itself have any stability or contraction properties? If the ZTCF is a contracting system, then the controller's job is simplified—the passive dynamics already provide a restoring force.

8.10.1 Drift Jacobian and Eigenvalue Analysis

Definition 8.9 (Drift Jacobian)**.** The drift Jacobian at state $\boldsymbol{x}$ is

$$J_f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}}, \tag{8.25}$$

an $n \times n$ matrix whose eigenvalues and singular values determine how perturbations propagate along the ZTCF.

Theorem 8.10 (Drift Contraction via Eigenvalues)**.** *Suppose the drift Jacobian $J_f(\boldsymbol{x})$ is symmetric and negative definite (all eigenvalues have negative real part) along a trajectory $\boldsymbol{x}(t)$. Then the ZTCF trajectory is locally contracting: small perturbations $\delta\boldsymbol{x}$ perpendicular to the flow direction decay exponentially.*

Intuition: If the drift Jacobian has negative eigenvalues, then the drift vector field is "pulling inward"—a hallmark of stability. Examples include:

- A damped pendulum where the velocity term $-c\dot{\boldsymbol{q}}$ (friction) dominates
- A spring-mass system $\ddot{\boldsymbol{q}} = -k\boldsymbol{q} - c\dot{\boldsymbol{q}}$ with positive damping $c > 0$
- A gene regulatory network with negative feedback

8.10.2 Contraction Implies Low Control Authority

When the drift is contracting, the optimal control strategy is often *minimal intervention*:

Proposition 8.11 (Minimal Intervention in Contracting Drift)**.** *If the ZTCF is contracting (drift Jacobian has negative eigenvalues), then the optimal control in the long term approaches zero: $\boldsymbol{u}^*(t) \to 0$ as $t \to \infty$.*

Rationale: If the passive dynamics already move the state toward a stable attractor, actively fighting these dynamics is wasteful. The optimal controller should allow the drift to do the work and apply control only to correct deviations from the desired attractor.

8.10.3 Proof via Optimal Control

Consider the cost function:

$$J = \int_0^\infty \left(\|\boldsymbol{x}(t)\|^2 + \lambda \|\boldsymbol{u}(t)\|^2 \right) \mathrm{d}t, \tag{8.26}$$

where $\lambda > 0$ is a regularization weight. The Bellman equation for the value function is:

$$0 = \min_{\boldsymbol{u}} \left[\|\boldsymbol{x}\|^2 + \lambda \|\boldsymbol{u}\|^2 + \nabla V \cdot (f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}) \right]. \tag{8.27}$$

The optimal input minimizes over $\boldsymbol{u}$:

$$\boldsymbol{u}^* = -\frac{1}{2\lambda} G(\boldsymbol{x})^\mathsf{T} \nabla V. \tag{8.28}$$

If the drift is contracting, the value function $V(\boldsymbol{x})$ has a very small gradient (the state is close to equilibrium), so $\boldsymbol{u}^*$ is small. This is the mechanism of minimal intervention.

8.11 Energy Interpretation of Counterfactuals

8.11.1 Energy of the ZTCF

For a mechanical system with kinetic energy $T(\boldsymbol{q}, \dot{\boldsymbol{q}})$ and potential energy $V(\boldsymbol{q})$, the total mechanical energy is:

$$E(\boldsymbol{q}, \dot{\boldsymbol{q}}) = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) + V(\boldsymbol{q}). \tag{8.29}$$

Along the ZTCF trajectory $(\boldsymbol{q}^{(0)}(t), \dot{\boldsymbol{q}}^{(0)}(t))$, the energy is:

$$E_{\mathrm{ZTCF}}(t) = T(\boldsymbol{q}^{(0)}(t), \dot{\boldsymbol{q}}^{(0)}(t)) + V(\boldsymbol{q}^{(0)}(t)). \tag{8.30}$$

8.11.2 Conservative Drift: Energy Conservation

If the drift is conservative (arising from a potential and frictionless kinematics), then $E_{\mathrm{ZTCF}}(t)$ is constant:

$$\frac{\mathrm{d}E_{\mathrm{ZTCF}}}{\mathrm{d}t} = 0. \tag{8.31}$$

This means that the ZTCF does not lose or gain energy; it oscillates at constant energy.

Example 8.12. An unpowered pendulum released from angle θ_0 will swing back and forth, with the energy divided between kinetic and potential as it swings. The ZTCF energy is constant: as it swings down, potential energy converts to kinetic; as it swings up, kinetic converts back to potential.

8.11.3 Dissipative Drift: Energy Decay

If the drift includes damping (e.g., friction or aerodynamic drag), then:

$$\frac{\mathrm{d}E_{\mathrm{ZTCF}}}{\mathrm{d}t} = -\mathcal{D}(t) < 0, \tag{8.32}$$

where $\mathcal{D}(t) \geq 0$ is the dissipation rate (power lost to friction).

For a linear damping model $\boldsymbol{f}_{\mathrm{damp}} = -C\dot{\boldsymbol{q}}$, the dissipation rate is:

$$\mathcal{D}(t) = \dot{\boldsymbol{q}}^{(0)}(t)^{\mathsf{T}} C \dot{\boldsymbol{q}}^{(0)}(t). \tag{8.33}$$

8.11.4 Control Energy Input

Along the actual trajectory (with control), the energy change due to control input is:

$$\left.\frac{\mathrm{d}E}{\mathrm{d}t}\right|_{\mathrm{control}} = \boldsymbol{u}(t)^{\mathsf{T}} \dot{\boldsymbol{q}}(t) = P_{\mathrm{control}}(t), \tag{8.34}$$

the power supplied by the controller (work per unit time).

The energy added by control over the time interval $[0, T]$ is:

$$\Delta E_{\mathrm{control}} := \int_0^T \boldsymbol{u}(t)^{\mathsf{T}} \dot{\boldsymbol{q}}(t)\,\mathrm{d}t. \tag{8.35}$$

If $\Delta E_{\mathrm{control}} > 0$, the controller added energy (e.g., by accelerating the system). If $\Delta E_{\mathrm{control}} < 0$, the controller removed energy (e.g., by braking).

8.11.5 Energy Comparison Counterfactual

We can directly compare the energy of the actual trajectory to the ZTCF:

$$\Delta E_{\text{actual}} := E(t) - E_{\text{ZTCF}}(t). \tag{8.36}$$

This difference quantifies how much the control has altered the energy budget. For example:

- If $\Delta E_{\text{actual}} > 0$, the controller added energy (likely accelerating).
- If $\Delta E_{\text{actual}} < 0$, the controller removed energy (likely braking).
- If $\Delta E_{\text{actual}} \approx 0$, the controller is purely steering without changing energy (a skillful balance).

Design Implication

Energy-optimal control often seeks to minimize the total energy input $\Delta E_{\text{control}}$ while achieving a goal. The ZTCF provides a baseline for comparison: in the best case, the controller can guide the motion without adding energy, exploiting gravity and momentum. This is the principle behind energy-efficient movement in biomechanics and efficient trajectory planning in robotics.

8.12 Worked Example: Underactuated Double Pendulum

We now provide a complete worked example with explicit numerical computation. Consider a double pendulum with a motor at joint 1 only.

8.12.1 System Parameters and Equations

$$m_1 = 1 \text{ kg}, \quad m_2 = 1 \text{ kg}, \quad l_1 = 1 \text{ m}, \quad l_2 = 1 \text{ m}, \quad g = 9.81 \text{ m/s}^2. \tag{8.37}$$

The state is $\boldsymbol{x} = (q_1, q_2, \dot{q}_1, \dot{q}_2)$, where q_i is the angle of joint i measured from the upright vertical position (positive downward).

8.12.2 Mass Matrix

The kinetic energy is:

$$T = \frac{1}{2} m_1 l_1^2 \dot{q}_1^2 + \frac{1}{2} m_2 [l_1^2 \dot{q}_1^2 + l_2^2 \dot{q}_2^2 + 2 l_1 l_2 \dot{q}_1 \dot{q}_2 \cos(q_1 - q_2)] \tag{8.38}$$

$$= \frac{1}{2} \begin{bmatrix} \dot{q}_1 & \dot{q}_2 \end{bmatrix} \begin{bmatrix} m_1 l_1^2 + m_2 l_1^2 & m_2 l_1 l_2 \cos(q_1 - q_2) \\ m_2 l_1 l_2 \cos(q_1 - q_2) & m_2 l_2^2 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}. \tag{8.39}$$

Thus the mass matrix is:

$$M(\boldsymbol{q}) = \begin{bmatrix} (m_1 + m_2) l_1^2 & m_2 l_1 l_2 \cos(q_1 - q_2) \\ m_2 l_1 l_2 \cos(q_1 - q_2) & m_2 l_2^2 \end{bmatrix}. \tag{8.40}$$

With the given parameters:

$$M(\boldsymbol{q}) = \begin{bmatrix} 2 & \cos(q_1 - q_2) \\ \cos(q_1 - q_2) & 1 \end{bmatrix}. \tag{8.41}$$

The determinant is:

$$\det M = 2 - \cos^2(q_1 - q_2) = 1 + \sin^2(q_1 - q_2) \geq 1, \tag{8.42}$$

so M is always invertible. The inverse is:

$$M(\boldsymbol{q})^{-1} = \frac{1}{1 + \sin^2(q_1 - q_2)} \begin{bmatrix} 1 & -\cos(q_1 - q_2) \\ -\cos(q_1 - q_2) & 2 \end{bmatrix}. \tag{8.43}$$

8.12.3 Coriolis Matrix

The Coriolis matrix can be computed from the kinetic energy. For this system:

$$\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & m_2 l_1 l_2 \sin(q_1 - q_2)\dot{q}_2 \\ -m_2 l_1 l_2 \sin(q_1 - q_2)\dot{q}_1 & 0 \end{bmatrix}. \tag{8.44}$$

With parameters:

$$\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & \sin(q_1 - q_2)\dot{q}_2 \\ -\sin(q_1 - q_2)\dot{q}_1 & 0 \end{bmatrix}. \tag{8.45}$$

The off-diagonal signs are opposite, and that asymmetry is the point: it is what makes $\dot{M} - 2\mathbf{C}$ skew-symmetric, and hence what makes the unforced system conserve energy. Writing both entries with the same sign — an easy slip, since the two terms look symmetric — injects energy at a rate comparable to the total energy of the swing over a single second, so it is worth checking numerically rather than by inspection.

8.12.4 Gravitational Vector

The potential energy is:

$$V(\boldsymbol{q}) = -m_1 g l_1 \cos q_1 - m_2 g(l_1 \cos q_1 + l_2 \cos q_2). \tag{8.46}$$

The gravitational vector is:

$$\boldsymbol{g}(\boldsymbol{q}) = -\frac{\partial V}{\partial \boldsymbol{q}} = \begin{bmatrix} -(m_1 + m_2) g l_1 \sin q_1 \\ -m_2 g l_2 \sin q_2 \end{bmatrix}. \tag{8.47}$$

With parameters:

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} -19.62 \sin q_1 \\ -9.81 \sin q_2 \end{bmatrix}. \tag{8.48}$$

8.12.5 Drift Components

The full drift acceleration is:

$$a_{\text{drift}} = -M(\boldsymbol{q})^{-1}[\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q})]. \tag{8.49}$$

The gravitational component is:

$$a_{\text{grav}} = -M(\boldsymbol{q})^{-1}\boldsymbol{g}(\boldsymbol{q}). \tag{8.50}$$

The Coriolis component is:

$$a_{\text{cor}} = -M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}. \tag{8.51}$$

8.12.6 Input Matrix

The input (torque τ_1 at joint 1) produces acceleration via:

$$G(\boldsymbol{x}) = \begin{bmatrix} 0 \\ 0 \\ [M(\boldsymbol{q})^{-1}]_{1\cdot} \\ [M(\boldsymbol{q})^{-1}]_{2\cdot} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{1+\sin^2(q_1-q_2)} \\ \frac{-\cos(q_1-q_2)}{1+\sin^2(q_1-q_2)} \end{bmatrix}. \tag{8.52}$$

8.12.7 Numerical Simulation

Initial condition:

$$q_1(0) = \pi/4 \approx 0.785 \text{ rad}, \quad q_2(0) = \pi/6 \approx 0.524 \text{ rad}, \tag{8.53}$$

$$\dot{q}_1(0) = 2 \text{ rad/s}, \quad \dot{q}_2(0) = -1 \text{ rad/s}. \tag{8.54}$$

Applied input:

$$\tau_1(t) = 5 \text{ N} \cdot \text{m} \quad \text{(constant torque)}. \tag{8.55}$$

We integrate both:

1. The actual trajectory with $\tau_1 = 5$ N $\cdot$ m,
2. The ZTCF trajectory with $\tau_1 = 0$.

over $T = 2$ seconds.

8.12.8 Results: ZTCF vs. Actual Trajectory

Both trajectories are integrated with RK4 at 5,000 steps, using the Coriolis matrix derived from $M(\boldsymbol{q})$ by its Christoffel symbols rather than a separately maintained expression. Table 8.2 gives the actual trajectory under $\tau_1 = 5$ N·m.

t (s)	q_1	q_2	$\dot{q}_1$	$\dot{q}_2$	$a_{\text{drift},1}$	$\tau_1[M^{-1}]_{11}$	$\rho(t)$
0.0	0.7854	0.5236	2.0000	-1.0000	-9.742	4.686	2.079
0.5	0.9514	0.3060	-1.4024	0.1886	-10.707	3.672	2.916
1.0	-0.0516	0.2870	-0.8654	-2.2171	4.953	4.503	1.100
1.5	-0.0242	-1.0481	-0.1023	-1.0901	-2.871	2.891	0.993
2.0	-0.0517	-0.5357	-0.3469	3.1249	-6.586	4.110	1.602

Table 8.2: Actual trajectory, integrated. $a_{\text{drift},1}$ and $\tau_1[M^{-1}]_{11}$ are evaluated at each sampled state, and $\rho(t)$ is their ratio — the drift–control ratio at joint 1.

t (s)	q_1	q_2	$\dot{q}_1$	$\dot{q}_2$	$\lVert\Delta\boldsymbol{q}\rVert$
0.0	0.7854	0.5236	2.0000	-1.0000	0.0000
0.5	0.5263	0.6641	-2.7217	1.2024	0.5558
1.0	-0.2760	-0.2033	0.4717	-4.6111	0.5393
1.5	-0.4161	-1.3513	-0.5783	0.2386	0.4955
2.0	-0.5652	-0.2543	0.6854	2.8132	0.5855

Table 8.3: The ZTCF, integrated from the same initial state with $\tau_1 = 0$. The final column is the configuration-space separation $\lVert\boldsymbol{q}_{\text{actual}} - \boldsymbol{q}_{\text{ZTCF}}\rVert$ between the two trajectories at that instant.

The two trajectories part quickly and then stay apart: the separation reaches 0.5558 rad within the first half-second and is 0.5855 rad at $t = 2$ s, having oscillated around that value in between rather than growing steadily. That is worth noting, because the intuitive picture — a counterfactual that drifts further and further from the actual motion — is not what a bounded oscillatory system does. Both trajectories are confined to the same energy shell, so their separation is bounded too. What the ZTCF isolates is not a widening gap but the *portion of the motion attributable to the torque*, which here is substantial from the outset and does not accumulate.

Common Misconception

An earlier revision of this table filled the state columns by hand and left $a_{\text{drift},1}$, $\tau_1[M^{-1}]_{11}$ and $\rho(t)$ as the literal placeholder "(eval)", with a caption saying the values "would be computed via ODE integration".
The hand-filled states were not a solution of the system stated above: integrating it gives values differing by up to 1.74 rad. They also disagreed internally — q_1 moved 0.202 rad over the first half-second, implying an average rate near 0.40 rad/s, while the $\dot{q}_1$ column read 2.0 to 2.34, about five times larger. And the velocity increments were constant to three decimals (0.341, 0.346, 0.345, 0.339), which is the signature of a straight-line extrapolation rather than an integration of a pendulum that actually swings.

8.12.9 Decomposition: Gravity vs. Coriolis

At $t = 1.0$ s the actual trajectory is at $q_1 = -0.0516$ rad, $q_2 = 0.2870$ rad, $\dot{q}_1 = -0.8654$ rad/s, $\dot{q}_2 = -2.2171$ rad/s. Evaluating the two drift components there:

1. Gravitational acceleration, from (8.50):

$$a_{\text{grav}} = -M^{-1}\boldsymbol{g} = (3.271,\ -5.862)^{\mathsf{T}}\ \text{rad/s}^2. \tag{8.56}$$

2. Coriolis acceleration, from (8.51):

$$a_{\text{cor}} = -M^{-1}\mathbf{C}\dot{\boldsymbol{q}} = (1.682,\ -1.835)^{\mathsf{T}}\ \text{rad/s}^2. \tag{8.57}$$

3. Total drift:

$$a_{\text{drift}} = a_{\text{grav}} + a_{\text{cor}} = (4.953,\ -7.698)^{\mathsf{T}}\ \text{rad/s}^2. \tag{8.58}$$

Both components matter here, and neither dominates the other by an order of magnitude — which is the point of decomposing them rather than quoting a single drift figure.

8.12.10 Drift–Control Ratio Evolution

As the system evolves:

- Initially ($t = 0$): ρ depends on the velocity magnitude and gravitational loading.
- Mid-trajectory ($t = 1$): As $\dot{\boldsymbol{q}}$ increases, Coriolis terms grow, and ρ increases.
- Late trajectory ($t = 2$): High velocities make $\rho \gg 1$ (drift-dominated regime).

This demonstrates the principle: at high speeds, the motor at joint 1 has less and less ability to steer the passively driven joint 2.

8.12.11 ZTCF Trajectory

We also compute the ZTCF by integrating $\dot{\boldsymbol{x}}^{(0)} = f(\boldsymbol{x}^{(0)})$ from the same initial condition with $\tau_1 = 0$. The ZTCF represents the passive shadow: how the system would evolve if we released the control at $t = 0$.

Qualitatively:

- The ZTCF will swing down due to gravity, accelerating faster than the controlled case (since gravity pulls harder than the control torque in the initial phase).
- The Coriolis coupling will cause joint 2 to accelerate and decelerate in a complex manner.
- The ZTCF energy is conserved (for this frictionless system), but the energy is distributed differently across the joints.

8.13 Connection to Contraction and Optimal Control (Detailed Proofs)

8.13.1 Contraction of Drift via Lyapunov Methods

We now provide a formal proof of the condition under which the drift itself is contracting.

Theorem 8.13 (Drift Contraction via Negative Eigenvalues)**.** *Let $f \in C^1(\mathcal{X})$ be the drift vector field on a domain $\mathcal{X}$. Suppose the drift Jacobian $J_f(\boldsymbol{x}) := \frac{\partial f}{\partial \boldsymbol{x}}$ is symmetric and has all eigenvalues in the left half-plane (i.e., $\lambda_i(J_f) < 0$ for all i). Then the ZTCF trajectory is exponentially stable: for any two initial conditions $\boldsymbol{x}_1, \boldsymbol{x}_2$ close to a trajectory, the distance between the solutions satisfies*

$$\|\boldsymbol{x}_1(t) - \boldsymbol{x}_2(t)\| \le e^{-\alpha t}\,\|\boldsymbol{x}_1(0) - \boldsymbol{x}_2(0)\| \tag{8.59}$$

for some $\alpha > 0$.

Proof: Consider the Lyapunov function $V(\delta\boldsymbol{x}) = \frac{1}{2}\|\delta\boldsymbol{x}\|^2$, where $\delta\boldsymbol{x} = \boldsymbol{x}_1 - \boldsymbol{x}_2$ is the perturbation. Along the ZTCF, the time derivative is:

$$\dot{V} = \delta\boldsymbol{x}^\mathsf{T} \frac{\partial f}{\partial \boldsymbol{x}}\bigg|_{\boldsymbol{x}_2(t)} \delta\boldsymbol{x} + O(\|\delta\boldsymbol{x}\|^2). \tag{8.60}$$

If J_f is symmetric with all eigenvalues negative, then:

$$\dot{V} \leq -\alpha \|\delta\boldsymbol{x}\|^2 = -2\alpha V, \tag{8.61}$$

where $\alpha = -\max_i \lambda_i(J_f) > 0$. Solving the differential inequality yields:

$$V(t) \leq e^{-2\alpha t} V(0), \tag{8.62}$$

which implies equation (8.59). □

8.13.2 Minimal Intervention in Drift-Dominated Regimes

We now prove that when drift dominates, the optimal control approaches zero.

Theorem 8.14 (Minimal Intervention Control Law)**.** *Consider the cost functional*

$$J = \int_0^\infty \left[\|\boldsymbol{x}(t) - \boldsymbol{x}^*\|_Q^2 + \|\boldsymbol{u}(t)\|_R^2\right] \mathrm{d}t, \tag{8.63}$$

where $\boldsymbol{x}^*$ is a desired attractor, Q and R are positive definite weights, and $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$.

If $\|f(\boldsymbol{x})\| / \|G(\boldsymbol{x})u_{\max}\|$ is very large (drift-dominated regime), then the optimal control is approximately

$$\boldsymbol{u}^* \approx -K(\boldsymbol{x})[\boldsymbol{x} - \boldsymbol{x}^*], \tag{8.64}$$

where the gain matrix $K(\boldsymbol{x})$ is small: $\|K(\boldsymbol{x})\| \to 0$ as the drift–control ratio $\to \infty$.

Sketch of proof: The Hamilton–Jacobi–Bellman equation is:

$$0 = \min_{\boldsymbol{u}} \left[\|\boldsymbol{x} - \boldsymbol{x}^*\|_Q^2 + \|\boldsymbol{u}\|_R^2 + \nabla V \cdot (f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u})\right]. \tag{8.65}$$

The minimizer over $\boldsymbol{u}$ is:

$$\boldsymbol{u}^* = -\frac{1}{2} R^{-1} G(\boldsymbol{x})^\mathsf{T} \nabla V. \tag{8.66}$$

In the drift-dominated regime, the term $\nabla V \cdot f(\boldsymbol{x})$ dominates, and ∇V is small (the state is already close to the attractor). Therefore $\boldsymbol{u}^*$ is small, and the control law reduces to a small correction term. The full solution requires solving the HJB equation numerically, but the insight is clear: when drift dominates, the optimal control is minimal.

8.13.3 Stability Without Control

Proposition 8.15 (ZTCF Stability Implies Easy Stabilization)**.** *If the ZTCF is exponentially stable (all drift Jacobian eigenvalues in the left half-plane), then any feedback law $\boldsymbol{u} = K[\boldsymbol{x} - \boldsymbol{x}^*]$ with K sufficiently small will stabilize the closed-loop system.*

Rationale: If the drift already moves toward $\boldsymbol{x}^*$, then a small feedback perturbation will not destabilize the system; the drift does the work, and the feedback just fine-tunes the convergence.

8.14 Summary Table: Key Concepts and Relationships

8.15 Example: Counterfactual Analysis of a Multisegment Swing

The golf swing provides a compelling example of the counterfactual framework because it exhibits all three drift–control regimes within a single motion lasting roughly one second. We sketch how the ZTCF, ZVCF, and drift–control ratio apply to a simplified kinematic chain (e.g., shoulder, elbow, wrist), noting that concrete numerical predictions require a calibrated biomechanical model with experimentally validated segment parameters.

Concept	Equation	Interpretation	Regime
Control-affine form	$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$	Superposition of drift and input	All
ZTCF	$\dot{\boldsymbol{x}}^{(0)} = f(\boldsymbol{x}^{(0)})$	Passive shadow (no control)	All
ZVCF	$a_{\text{ZVCF}} = -M^{-1}g(\boldsymbol{q})$	Static equilibrium acceleration	All
Drift decomposition	$a_{\text{drift}} = a_{\text{grav}} + a_{\text{cor}}$	Gravity + inertial coupling	Mechanical
Drift–control ratio	$\rho = \\|f\\| \,/\, \\|Gu_{\max}\\|$	Relative dominance	All
Drift Jacobian	$J_f = \partial f/\partial \boldsymbol{x}$	Local stability of ZTCF	All
Minimal intervention	$\boldsymbol{u}^* \approx -K[\boldsymbol{x} - \boldsymbol{x}^*]$, $K \to 0$	Optimal control when $\rho \gg 1$	Drift-dominated
ZTCF energy (conservative)	$E_{\text{ZTCF}} = \text{const}$	Energy conservation	No damping
ZTCF energy (dissipative)	$\dot{E}_{\text{ZTCF}} = -\mathcal{D} < 0$	Energy decay due to friction	With damping
Control power input	$P = \boldsymbol{u}^\intercal \dot{\boldsymbol{q}}$	Energy added by actuators	All

Table 8.4: A reference table of the major concepts introduced in this chapter, their defining equations, physical interpretations, and the regimes in which they apply.

8.15.1 Qualitative ZTCF Structure

At each instant during the swing, the ZTCF (Definition 8.3) answers: *what would happen if the golfer released all muscular effort?* The qualitative behavior follows directly from the equations of this chapter:

1. **Early motion (low velocity):** The drift field $f(\boldsymbol{x})$ is dominated by $a_{\text{grav}} = -M(\boldsymbol{q})^{-1}\boldsymbol{g}(\boldsymbol{q})$, which is small when the segments are near their resting configuration. The ZTCF trajectory barely moves, so $\|\boldsymbol{x}(t) - \boldsymbol{x}^{(0)}(t)\|$ (the gap between actual and passive motion) is large. All motion is attributable to active control: $\rho \ll 1$.

2. **Transition (direction reversal):** Near the top of the backswing, the system passes through a low-velocity state. The ZTCF from this point is dominated by gravitational acceleration pulling the arms back down—closely resembling the actual trajectory. The gap $\|\boldsymbol{x} - \boldsymbol{x}^{(0)}\|$ is small.

3. **Fast phase (high velocity):** As velocities grow, the Coriolis term $a_{\text{cor}} = -M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$ dominates the drift because it scales as $\dot{\boldsymbol{q}}^2$. The ZTCF closely tracks the actual trajectory: removing control barely changes the motion because the available muscle torque $\|G(\boldsymbol{x})\boldsymbol{u}_{\max}\|$ is small compared to $\|f(\boldsymbol{x})\|$. The drift–control ratio satisfies $\rho \gg 1$.

This qualitative pattern—large ZTCF divergence early, small divergence late—follows from the velocity scaling of the drift–control ratio (Eq. 8.17) and holds for any multisegment mechanical system with bounded actuator torques.

8.15.2 Structural Argument: Why Active Torque Is Negligible at Impact

The argument is structural and does not depend on specific parameter values. For a mechanical system with state $\boldsymbol{x} = (\boldsymbol{q}, \dot{\boldsymbol{q}})$ and input torque τ, the control contribution to acceleration is:

$$a_{\text{control}} = M(\boldsymbol{q})^{-1}B(\boldsymbol{q})\,\tau, \tag{8.67}$$

which is bounded by

$$\|a_{\text{control}}\| \leq \|M(\boldsymbol{q})^{-1}B(\boldsymbol{q})\| \cdot \|\tau_{\max}\|. \tag{8.68}$$

Meanwhile, the centrifugal acceleration from the drift scales as:

$$\|a_{\text{cor}}\| \sim \|M(\boldsymbol{q})^{-1}\mathbf{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}\| \sim O(\|\dot{\boldsymbol{q}}\|^2). \tag{8.69}$$

Since $\|\tau_{\max}\|$ is bounded (physiological or actuator limits) while $\|\dot{\boldsymbol{q}}\|$ grows during acceleration phases, there necessarily exists a velocity threshold $\dot{q}^*$ beyond which $\|a_{\text{cor}}\| \gg \|a_{\text{control}}\|$, i.e., $\rho(\boldsymbol{x}) \gg 1$. For fast athletic motions such as throwing, striking, or swinging, this threshold is typically reached well before the moment of peak velocity.

Geometric Intuition

The ZTCF analysis captures a principle recognized in biomechanics: in fast ballistic movements, active muscular contribution is concentrated early, while the late phase is governed by passive inertial dynamics. The counterfactual framework gives this observation a precise mathematical formulation via the drift–control ratio $\rho(\boldsymbol{x})$ and the ZTCF divergence $\|\boldsymbol{x}(t) - \boldsymbol{x}^{(0)}(t)\|$.

Common Misconception

Any numerical predictions of ZTCF divergence, phase timings, or drift–control ratios depend on the specific segment masses, lengths, inertias, and torque capacities of the system under study. These must be obtained from experimental measurements (e.g., motion capture combined with inverse dynamics). The structural argument above holds generically, but the quantitative details are model-dependent.

8.16 Chapter Summary

1. The drift–input decomposition $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$ is a causal structure: drift is the passive "memory" of the state; input is instantaneous active intervention.

2. The Zero Torque Counterfactual (ZTCF) is the trajectory under zero input, obtained by integrating the drift vector field. It is well-defined and smooth by the Picard–Lindelöf theorem and reveals the passive shadow—what happens without control.

3. The Zero Velocity Counterfactual (ZVCF) isolates static forces from velocity-dependent forces, decomposing the drift into gravitational and Coriolis components.

4. For mechanical systems, the drift decomposes linearly into gravitational and Coriolis contributions because the mass matrix inverse is a linear operator. This decomposition is clean and holds exactly.

5. The drift–control ratio ρ quantifies how much of the motion is determined by physics versus steered by the controller. At high speeds, $\rho \to \infty$ for all mechanical systems, shifting the system into a drift-dominated regime.

6. Forward dynamics simulation with counterfactuals provides a "thought experiment engine" for attribution analysis (Step 1–5 protocol), sensitivity studies (actuator failures, gravity changes, mass modifications), and controller design.

7. Data-driven drift estimation allows us to identify $f(\boldsymbol{x})$ and $G(\boldsymbol{x})$ from data when analytical models are unavailable. The key requirement is persistent excitation: the input must vary independently to separate drift from control.

8. Contraction analysis of the drift reveals whether the passive dynamics are stable. If the drift Jacobian has all negative eigenvalues, the system naturally contracts, and the controller's job is to apply minimal intervention.

9. The optimal control law in drift-dominated regimes ($\rho \gg 1$) approaches zero: the controller should allow physics to do the work. This is the principle of energy-efficient control and explains skillful movement in biomechanics and athletics.

10. Energy interpretation of counterfactuals: the ZTCF energy is conserved (for frictionless systems) or decays (for damped systems). Comparing ZTCF energy to actual energy reveals how much the controller is adding or removing energy.

11. A complete worked example of the underactuated double pendulum demonstrates the decomposition of drift into gravity and Coriolis, computation of the drift–control ratio, and the ZTCF trajectory. At high speeds, the single actuator at joint 1 can barely steer the passive joint 2.

12. The deepest design principle: the best controllers exploit the drift rather than oppose it, shaping the state so that passive dynamics produce the desired behavior. This is "effort-to-ease transition" in biomechanics and energy-efficient trajectory design in robotics.

This chapter has established that the drift–input decomposition is not merely algebraic bookkeeping but a causal framework for understanding, predicting, and attributing motion. The counterfactual machinery—ZTCF, ZVCF, and their energy interpretations—transforms abstract dynamics into actionable engineering insight: what would happen without control, and therefore what the controller must actually accomplish.

Exercises

1. **ZTCF of a simple system.** For $\ddot{x} + x^3 = u$ (Duffing oscillator), compute the ZTCF starting from $(x_0, \dot{x}_0) = (1, 0)$. Plot the actual trajectory under $u = -2x$ and the ZTCF side by side. Compute $\rho(t)$.

2. **ZVCF computation.** For the same system, compute the ZVCF at $t = 0$. Compare the ZVCF acceleration with the full acceleration. In what regime does gravity dominate?

3. **Drift-control ratio scaling.** For the double pendulum of the worked example, show analytically that $\rho \propto \|\dot{\boldsymbol{q}}\|^2 / \|\boldsymbol{u}\|$ when Coriolis forces dominate. What happens when $\|\dot{\boldsymbol{q}}\| \to 0$?

4. **Energy decomposition.** For a 2-DOF system with known mass matrix, decompose the total kinetic energy change $\Delta T = \int \dot{\boldsymbol{q}}^T \tau \, dt$ into drift and control contributions. Verify the identity $\Delta T = W_{\text{drift}} + W_{\text{control}}$.

5. **Data-driven drift estimation.** Generate simulated data from a known pendulum system under random inputs. Estimate the drift acceleration using the SINDy-type procedure described earlier in this chapter. Compare with the analytical drift.

6. **Contraction of the drift.** Compute the drift Jacobian for the undamped double pendulum at several configurations. Where is the drift contracting? Where is it expanding? Relate to the physical behavior of the system.

7. **Sensitivity counterfactual.** For the pendulum, compute $\frac{\partial \boldsymbol{x}(t_f)}{\partial \boldsymbol{u}(t_0)}$ using the sensitivity STM. Which initial torque direction has the largest effect on the terminal state?

8. **Attribution in a 3-link chain.** For a 3-DOF chain (torso–arm–forearm), compute the off-diagonal mass matrix terms M_{ij} and the inertial torques $\tau_{\text{inertial},j}$ during a prescribed deceleration of the torso. Which link receives the most inertial energy?

Chapter 9

Applications: Biomechanics, Robotics, and Beyond

9.1 What a Shared Mathematical Language Can Explain

A golf swing combines an actively driven body, a deforming club, changing contact forces, and a brief collision. A useful explanation must connect those parts without replacing them with a single slogan. Geometry determines how joint motion becomes club motion. Inertia determines how forces produce acceleration. Muscle activation, contact, and equipment determine which forces are available. Perturbation dynamics describe how errors evolve. An impact model translates the arriving club state into a ball outcome. Measurements then test this chain of predictions.

Control-affine equations and tangent maps help connect these questions, but they answer different questions. An instantaneous acceleration decomposition is not an energy budget. A local optimal controller is not automatically a contraction certificate throughout a swing. A contracting state error need not minimize impact error. A flexible shaft can change both the motion of the head relative to the grip and the motion of the grip itself. Those distinctions make the combined explanation stronger: each proposed mechanism has an equation, a stated intervention, and an observable consequence.

The golf example below is a manufactured teaching model. Its numerical values are reproducible, but they are not fitted to a golfer or offered as measured anatomical limits. The robot, spacecraft, and vehicle examples establish additional boundaries: task redundancy, orbital reference dynamics, and nonholonomic motion. They demonstrate transferable methods, rather than a universal theory of skilled human movement.

9.2 A Mechanically Consistent Golf Teaching Model

9.2.1 Coordinates, States, and Rigid-Body Inertia

Consider a planar serial chain with three revolute coordinates $q = (q_1, q_2, q_3)$. The first angle is measured from the horizontal; the second and third are relative joint angles. The absolute link orientations are $\theta_i = \sum_{j=1}^{i} q_j$. Three configuration degrees of freedom require six first-order states $(q, \dot{q})$. Adding two bending amplitudes $\eta = (\eta_1, \eta_2)$ requires ten states $(q, \eta, \dot{q}, \dot{\eta})$, not a six-state system with a different mass matrix.

Let l_i be link lengths, r_i the distances from each proximal joint to its center of mass, m_i the masses, and I_i the moments of inertia about those centers. With translational and angular Jacobians $J_{v,i}$ and $J_{\omega,i}$,

$$T_r = \frac{1}{2} \sum_{i=1}^{3} \left(m_i \|J_{v,i}\dot{q}\|^2 + I_i (J_{\omega,i}\dot{q})^2 \right), \qquad M_r = \sum_{i=1}^{3} \left(m_i J_{v,i}^{\mathsf{T}} J_{v,i} + I_i J_{\omega,i}^{\mathsf{T}} J_{\omega,i} \right). \tag{9.1}$$

This expression includes both translation and rotation. Replacing each link by a point mass changes the model and must be stated. For independent coordinates and positive link inertias, nonzero joint velocity produces positive kinetic energy; a singular end-point position Jacobian does not make this joint-space inertia indefinite.

An explicit formula provides a useful independent check. Define

$$
\begin{aligned}
&a = I_1 + I_2 + I_3 + m_1 r_1^2 + m_2(l_1^2 + r_2^2) + m_3(l_1^2 + l_2^2 + r_3^2),\\
&b = I_2 + I_3 + m_2 r_2^2 + m_3(l_2^2 + r_3^2), \qquad d = I_3 + m_3 r_3^2,\\
&h = l_1(m_2 r_2 + m_3 l_2)\cos q_2,\\
&j = m_3 l_1 r_3 \cos(q_2 + q_3), \qquad k = m_3 l_2 r_3 \cos q_3.
\end{aligned}
\tag{9.2}
$$

Then

$$
M_r(q) = \begin{bmatrix} a + 2(h + j + k) & b + h + j + 2k & d + j + k \\ b + h + j + 2k & b + 2k & d + k \\ d + j + k & d + k & d \end{bmatrix}. \tag{9.3}
$$

The cross terms are not optional corrections: they express the velocity of a distal mass produced by several joints simultaneously. In particular, the coefficient of k in the first two entries includes contributions from both participating velocities. Energy computed from this matrix must agree with energy computed directly from the moving centers of mass.

Use $m = (10, 5, 0.81)$ kg, $l = (0.30, 0.35, 1.15)$ m, $r_i = l_i/2$, and $I = (0.25, 0.08, 0.089)$ kg m^2. These represent a rigid carrier for the calculation; labeling its coordinates as particular anatomical joints does not validate that identification. At $q = (60, -30, -20)$ degrees, the reproducible result is

$$
M_r = \begin{bmatrix} 2.7750 & 1.3863 & 0.5998 \\ 1.3863 & 0.9955 & 0.5100 \\ 0.5998 & 0.5100 & 0.3568 \end{bmatrix} \ \mathrm{kg\,m^2}. \tag{9.4}
$$

Its determinant is approximately 0.0682 in the corresponding cubed inertia units; its smallest eigenvalue is approximately 0.0501 kg m^2. These are checks on this numerical model, not clinical or performance measures.

9.2.2 A Flexible Appendage Must Contribute All Its Kinetic Energy

For an illustrative flexible extension, add a beam of mass $m_s = 0.30$ kg along the third link. This mass is additional to the specified rigid carrier. It must not also be counted inside m_3 when interpreting the combined model. No separate clubhead mass moving with the flexible tip is included here. A physical club would require that mass and its rotational inertia, along with an appropriate shaft model and grip boundary condition.

Let $s \in [0, l_3]$, $t = (\cos\theta_3, \sin\theta_3)$, and $n = (-\sin\theta_3, \cos\theta_3)$. Write a beam material point as

$$
p(s, q, \eta) = p_h(q) + s\,t(q) + n(q)\sum_{a=1}^{2} \phi_a(s)\eta_a. \tag{9.5}
$$

At the undeformed configuration $\eta = 0$, its velocity has the form $J_q(s)\dot{q} + J_\eta(s)\dot{\eta}$, with $J_\eta = n[\phi_1\ \phi_2]$. The same material-point integral must generate every block:

$$
\begin{aligned}
&A = M_r + \int_0^{l_3} \rho_l J_q^{\mathsf{T}} J_q \, ds,\\
&C = \int_0^{l_3} \rho_l J_q^{\mathsf{T}} J_\eta \, ds, \qquad D = \int_0^{l_3} \rho_l J_\eta^{\mathsf{T}} J_\eta \, ds,\\
&M_f = \begin{bmatrix} A & C \\ C^{\mathsf{T}} & D \end{bmatrix}, \qquad \rho_l = m_s/l_3.
\end{aligned}
\tag{9.6}
$$

Including C and D while omitting the beam contribution to A does not describe that kinetic energy. It can even produce an indefinite matrix for otherwise positive physical parameters. Such a failure is an assembly error, not evidence of a kinematic singularity.

With dimensionless mode shapes and amplitudes in meters, A has units kg m^2, C has units kg m, and D has units kg. Positive definiteness is preserved by invertible coordinate changes, but raw eigenvalues and condition numbers of the full mixed-unit matrix depend on the amplitude scales. Compare them only after declaring those scales.

For the displayed posture, numerical integration gives

$$A = \begin{bmatrix} 3.2056 & 1.7293 & 0.8220 \\ 1.7293 & 1.2780 & 0.6990 \\ 0.8220 & 0.6990 & 0.4891 \end{bmatrix}, \qquad C = \begin{bmatrix} 0.1594 & -0.0496 \\ 0.1368 & -0.0371 \\ 0.0981 & -0.0157 \end{bmatrix}, \qquad D \simeq \begin{bmatrix} 0.0750 & 0.0000 \\ 0.0000 & 0.0750 \end{bmatrix}. \tag{9.7}$$

The units follow the preceding definitions. Rounding makes D appear exactly diagonal; the underlying quadrature has a small off-diagonal residual.

The shape functions are the first two clamped-free, uniform Euler–Bernoulli beam shapes without a tip mass. They can be derived from $EI\phi'''' = \rho_l \omega^2 \phi$ with $\phi(0) = \phi'(0) = \phi''(l_3) = \phi'''(l_3) = 0$. With $z = \beta_a s/l_3$,

$$\begin{aligned} \widetilde{\phi}_a(s) &= \cosh z - \cos z - \sigma_a(\sinh z - \sin z), \\ \sigma_a &= \frac{\cosh \beta_a + \cos \beta_a}{\sinh \beta_a + \sin \beta_a}, \qquad \cosh \beta_a \cos \beta_a = -1, \\ \phi_a(s) &= \widetilde{\phi}_a(s)/\widetilde{\phi}_a(l_3). \end{aligned} \tag{9.8}$$

The first two positive roots are approximately 1.875104 and 4.694091. Dividing by the signed tip value makes both $\phi_a(l_3) = 1$; dividing by a common positive constant leaves the second mode with the opposite sign. Either sign convention is physically acceptable if the amplitude and every coupling term transform consistently. It is inconsistent to claim a unit positive tip while using the other convention.

Here the reduced spring matrix is defined separately as $K_\eta = \mathrm{diag}(D_{11}\, 40^2, D_{22}\, 120^2)$, approximately $\mathrm{diag}(120, 1080)$ N/m. It is symmetric and positive. These are prescribed reduced-model frequencies in rad/s. They do not correspond to one uniform beam stiffness: that beam would have $\omega_2/\omega_1 = (\beta_2/\beta_1)^2 \simeq 6.267$, rather than 3. A measured nonuniform shaft, head inertia, axial loading, and rotation may require different modes and additional terms. The displayed inertia and springs alone are not a complete flexible forward-dynamics simulator.

9.2.3 Eliminate Accelerations Without Losing the Bias Terms

For a fully specified flexible model, collect gravity, velocity-dependent forces, elastic forces, and damping in h_q and h_η, using a consistent sign convention:

$$A\ddot{q} + C\ddot{\eta} + h_q = Bu, \qquad C^\mathsf{T}\ddot{q} + D\ddot{\eta} + h_\eta = 0. \tag{9.9}$$

Here C denotes an inertia cross block. It is distinct from the conventional Coriolis matrix in the product $\mathbf{C}\dot{\boldsymbol{q}}_{\mathrm{sys}}$, where $q_{\mathrm{sys}} = (q, \eta)$. A notation change cannot remove that velocity-dependent force from the model.

Define $S_q = A - CD^{-1}C^\mathsf{T}$, $H = S_q^{-1}$, and $\Gamma = -D^{-1}C^\mathsf{T}$. Direct substitution gives

$$\begin{aligned} \ddot{q} &= H(Bu - h_q + CD^{-1}h_\eta), \\ \ddot{\eta} &= -D^{-1}h_\eta + \Gamma\ddot{q}. \end{aligned} \tag{9.10}$$

Thus the acceleration input blocks are HB and ΓHB. The definition of Γ contains no H; including H there and multiplying by it again counts the mobility twice. The drift also contains the modal bias h_η. Removing it would remove elastic restoring forces and any other unactuated modal forces assigned to that term.

For the ten-state ordering $x = (q, \eta, \dot{q}, \dot{\eta})$, define $a_0 = H(-h_q + CD^{-1}h_\eta)$. The first-order affine model is

$$f(x) = \begin{bmatrix} \dot{q} \\ \dot{\eta} \\ a_0 \\ -D^{-1}h_\eta + \Gamma a_0 \end{bmatrix}, \qquad G(x) = \begin{bmatrix} 0 \\ 0 \\ HB \\ \Gamma HB \end{bmatrix}. \tag{9.11}$$

Each row has the appropriate derivative units. In particular, angles cannot be subtracted from gravity torques inside a state derivative.

At the example posture,

$$S_q = \begin{bmatrix} 2.8340 & 1.4142 & 0.6032 \\ 1.4142 & 1.0103 & 0.5123 \\ 0.6032 & 0.5123 & 0.3574 \end{bmatrix}, \qquad H = \begin{bmatrix} 1.3860 & -2.7604 & 1.6179 \\ -2.7604 & 9.1216 & -8.4168 \\ 1.6179 & -8.4168 & 12.1323 \end{bmatrix}. \tag{9.12}$$

Positive definiteness follows structurally: for any nonzero v, substitute the velocity pair $(v, -D^{-1}C^\top v)$ into the positive kinetic-energy quadratic form to obtain $v^\top S_q v > 0$. This proof requires the full, consistently assembled $M_f \succ 0$. Numerical inversion cannot rescue an inconsistent kinetic energy.

9.2.4 A Rigid Counterfactual Is a Different Model

The supplied zero-torque trajectory uses the six-state rigid chain, with

$$M_r(q)\ddot{q} + c_r(q,\dot{q}) + g_r(q) = u, \qquad a_{\text{drift}} = -M_r^{-1}(c_r + g_r). \tag{9.13}$$

It uses M_r, not the flexible Schur complement. Mixing the latter with rigid-only bias terms would generally break the energy balance. With conservative gravity and $u = 0$, the rigid model conserves $E = \frac{1}{2}\dot{q}^\top M_r \dot{q} + V_r(q)$. Its fixed base can exert reaction forces even when the generalized joint torques are zero.

Initialize $q = (60, -30, -20)$ degrees and $\dot{q} = (12, -18, -25)$ rad/s. The absolute angular velocities are $(12, -6, -31)$ rad/s; interpreting the three relative rates as independent measured segment speeds would be incorrect. At this state,

$$\begin{aligned} g_r &= (30.25,\ 14.34,\ 4.50)^\top \ \mathrm{N\,m}, \\ c_r &= (120.25,\ 11.14,\ -17.42)^\top \ \mathrm{N\,m}, \\ a_{\text{drift}} &= (-118.58,\ 70.47,\ 134.83)^\top \ \mathrm{rad/s^2}. \end{aligned} \tag{9.14}$$

The velocity-force contribution exceeds gravity in the first component at this state. That observation neither establishes dominance in every coordinate nor proves a universal transition during human downswing.

A fourth-order Runge–Kutta integration over 0.10 s uses 400 steps of 0.00025 s. Selected outputs are shown below. "Rigid tip" is the endpoint of this chain, not a separately modeled flexible clubhead.

t (s)	q_3 (rad)	$\dot{q}_3$ (rad/s)	Rigid-Tip Speed (m/s)	$\lVert a_{\text{drift}} \rVert_2$ (rad/s^2)
0.000	-0.3491	-25.00	35.37	192.9
0.050	-1.2301	-8.29	33.26	473.8
0.100	-0.9462	24.21	21.66	900.9

The tip speed decreases from 35.37 to 21.66 m/s in this particular calculation while the acceleration norm increases. Those two facts can coexist because acceleration changes direction as well as speed, and because energy redistributes between kinetic and potential terms. This table supplies no controlled comparison and no evidence that a golfer should relax their muscles. Numerical credibility requires step refinement and an energy check; physiological credibility would additionally require validated masses, contacts, passive tissues, muscle forces, and initial conditions.

9.3 From Drift Decomposition to a Testable Golf Argument

9.3.1 Declare the Intervention and the Quantity Being Compared

At a fixed state of a torque-driven rigid model, acceleration separates exactly as $\ddot{q} = a_0(q,\dot{q}) + M_r^{-1}u$. This is additivity in the input at one state. It does not imply that the trajectories produced by two different input histories add, because their evolving states change both terms.

A dimensionless drift-to-control ratio can be defined for a chosen acceleration output:

$$\rho_a = \frac{\lVert W_a a_0 \rVert_2}{\lVert W_a M_r^{-1} u \rVert_2}. \tag{9.15}$$

Here W_a specifies component scales or a declared projection and normalization. If the denominator is zero and the numerator is nonzero, the ratio is infinite; if both vanish, it is undefined. A numerical epsilon changes the definition and must be reported. The unforced trajectory above therefore cannot be assigned a finite drift-to-control ratio from its acceleration norm alone.

The full first-order drift $f = (\dot{q}, a_0)$ combines rates and accelerations. Its Euclidean norm is not $\|a_0\|$, and its comparison with $(0, M_r^{-1}u)$ depends on arbitrary units unless a state metric is specified. A metric-based ratio of full vector fields is a different quantity. Even acceleration itself changes nontrivially under nonlinear coordinates: for $z = \psi(q)$,

$$\ddot{z} = D\psi(q)\ddot{q} + D^2\psi(q)[\dot{q}, \dot{q}]. \tag{9.16}$$

The second term belongs to the transformed zero-input acceleration. Consequently, a large coordinate ratio is not an invariant amount of physiological effort or mechanical energy.

Velocity-dependent forces are often quadratic in velocity at fixed posture. Under $\dot{q} \mapsto s\dot{q}$, the rigid c_r scales as s^2, but it can vanish or cancel gravity in particular components. Meanwhile the applied torque and posture can change with speed. No theorem therefore makes this ratio increase through every swing or makes a particular numerical threshold a universal phase boundary.

A finite counterfactual should instead specify the same initial state, model parameters, contact rules, and future input intervention, then compare a stated output. "Set generalized active torques to zero" is one mathematical intervention. Removing neural feedback, reducing activation, removing all muscle force, and removing contact forces are different interventions. Already developed muscle force and an ongoing motor command do not disappear merely because a new sensory correction cannot arrive before impact. A meaningful claim about late control must include the actuator and delay model.

9.3.2 Acceleration Sequences Are Not Energy-Transfer Measurements

For the rigid chain, actuator power is $u^\mathsf{T}\dot{q}$ and the total energy balance is $\dot{E} = u^\mathsf{T}\dot{q}$ under the conservative assumptions above. For a single segment, power also enters through joint forces acting at moving points and moments acting with that segment's absolute angular velocity. At an ideal joint with a shared point velocity, equal and opposite joint-force powers cancel in the combined energy balance while transferring energy between the individual segments. If the application-point velocities differ, as in a sliding contact, their sum need not vanish. Joint moments have distinct proximal and distal powers whose sum corresponds to the relative-coordinate actuator power.

A proximal angular-speed decrease can accompany distal acceleration, but it does not, by itself, identify where the distal kinetic energy came from. Translational energy, potential energy, active work, and energy exchange with other segments can all matter. To establish transfer, define the bodies, reference points, sign conventions, and interval; calculate the force and moment powers; integrate them; and check each segment's energy balance. The temporal order of speed peaks is useful descriptive information, not a substitute for that accounting.

This distinction also resolves an apparent paradox in the teaching trajectory. A large acceleration norm can coexist with decreasing endpoint speed, while mechanical energy stays constant. A velocity component, an endpoint speed, a segment kinetic energy, and the total system energy are four different observables. An argument must identify which one it explains.

9.3.3 Shaft Recovery Depends on Phase and on the Moving Grip

For a frozen-coefficient linear bending model, write

$$D\ddot{\eta} + D_d\dot{\eta} + K_\eta\eta = -C^\mathsf{T}\ddot{q}. \tag{9.17}$$

With constant symmetric $D, K_\eta \succ 0$ and $D_d \succeq 0$, the modal energy obeys

$$E_\eta = \tfrac{1}{2}\dot{\eta}^\mathsf{T}D\dot{\eta} + \tfrac{1}{2}\eta^\mathsf{T}K_\eta\eta, \qquad \dot{E}_\eta = -\dot{\eta}^\mathsf{T}C^\mathsf{T}\ddot{q} - \dot{\eta}^\mathsf{T}D_d\dot{\eta}. \tag{9.18}$$

The sign of the base acceleration alone does not determine the direction of energy exchange; modal velocity and coupling orientation also enter. In a rotating, deforming model, coefficient derivatives and additional inertial terms must be included before using an energy balance. The frozen equation cannot silently replace the full flexible equations during a large swing.

For one mode driven harmonically at frequency Ω, the displacement response to force has transfer function $(k - m\Omega^2 + ic\Omega)^{-1}$. Stiffness changes amplitude and phase; damping and excitation timing matter as well. The velocity contribution at impact depends on the response at that particular instant. A greater stored elastic energy earlier in the swing does not guarantee a greater final clubhead speed or a preferred face orientation.

MacKenzie and Boucher's 2016 experiment tested two shafts with similar inertial properties in 33 male golfers, using 14 drives per shaft. The more flexible shaft produced a larger deflection-related speed contribution, accompanied by lower grip angular velocity at impact; the group difference in total clubhead speed was not statistically significant ($p = 0.14$). Shaft bending and grip orientation also contributed differently to delivered club orientation. The authors considered both mechanical reactions and changed motor patterns as possible explanations. This study supports analyzing the coupled golfer–club system. It does not establish equivalence between shafts, identify one neural mechanism, or guarantee an individual fitting response. Individual significance tests from one session also require attention to multiple comparisons and repeatability. See MacKenzie and Boucher, The Influence of Golf Shaft Stiffness on Grip and Clubhead Kinematics.

The important modeling question is what is held fixed. Prescribing identical grip motion for two shafts estimates an equipment response under a kinematic constraint. Prescribing identical applied forces allows the grip motion to change. Allowing the golfer's controller to adapt defines a third experiment. Differences among those results are scientifically informative; they should not be collapsed into a single claim that a shaft either adds or does not add speed.

9.3.4 Carry Uncertainty Through the Impact Event

Let $\Phi(t, t_0)$ propagate infinitesimal state perturbations along a specified smooth trajectory. At a fixed time T, an output $y = h(x(T), T)$ has initial-state sensitivity $h_x\Phi(T, t_0)$. Ball contact, however, generally occurs at a state-dependent time defined by $g(x, t) = 0$. Put $n = \nabla_x g$, let F^- be the pre-impact vector field, and evaluate quantities at the nominal event. If $d = g_t + n^\top F^- \neq 0$, then

$$\delta t_e = -\frac{n^\top \Phi \delta x_0}{d}, \qquad \delta x_e = \left(I - \frac{F^- n^\top}{d}\right)\Phi \delta x_0. \tag{9.19}$$

For an explicitly time-dependent impact output, $\delta y = h_x \delta x_e + h_t \delta t_e$. These are sensitivities of the state and output evaluated at each perturbed trajectory's own event. Comparing post-impact trajectories at a common later time instead requires the reset derivative and saltation correction. Near grazing contact, d can be small and the linear event sensitivity can become large or cease to apply.

An error direction invisible in today's club-position Jacobian can affect tomorrow's impact through Φ. Conversely, reducing every joint error equally can waste effort on directions that have little influence on the chosen outcome. To compare sensitivities, specify scales: if $\delta x = D_x \delta\bar{x}$ and $\delta\bar{y} = D_y^{-1}\delta y$, use the singular values of $D_y^{-1} J D_x$, not an unscaled matrix mixing meters, radians, and rates. A singular value is a local worst-case directional gain under those scales, not the expected error of a population of swings.

Parameter uncertainty enters alongside state uncertainty. For $\delta y = J_x \delta x + J_p \delta p$, propagate both covariance blocks and their cross covariance:

$$\Sigma_y \simeq J_x \Sigma_x J_x^\top + J_p \Sigma_p J_p^\top + J_x \Sigma_{xp} J_p^\top + J_p \Sigma_{xp}^\top J_x^\top. \tag{9.20}$$

Mass properties, shaft stiffness, contact location, face orientation, ball properties, and collision assumptions can alter the final prediction. This first-order expression requires small uncertainties and a differentiable local model; it does not cover unmodeled contact changes or large distribution tails. A high-quality golf model should report which uncertainty dominates the output it seeks to explain.

9.4 Robotics: Reachability, Redundancy, and a Verifiable Controller

9.4.1 Check the Reference Before Designing Its Controller

For a planar three-link robot with $l_i = 0.30$ m, the end position is

$$p(q) = \sum_{i=1}^{3} l_i \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}, \qquad \theta_i = \sum_{j=1}^{i} q_j. \tag{9.21}$$

A circle centered at $(0.60, 0.60)$ m with radius 0.20 m is not entirely reachable: its farthest point is about 1.049 m from the base, beyond the 0.90 m total link length. No feedback gain makes that reference feasible.

Instead choose the manufactured two-second reference

$$p_d(t) = \begin{bmatrix} 0.45 + 0.05\cos(\pi t) \\ 0.15 + 0.05\sin(\pi t) \end{bmatrix} \text{ m}, \qquad \psi_d = \pi/4, \qquad 0 \le t \le 2 \text{ s}. \tag{9.22}$$

At a quarter period, $t = 0.50$ s, the position is $(0.45, 0.20)$ m. Subtract the last link to obtain the wrist target $w = p_d - 0.30(\cos\psi_d, \sin\psi_d)^\mathsf{T}$. Its distance from the base stays between approximately 0.196 and 0.296 m, strictly within the two-link workspace and away from its folded and extended singularities.

A continuous elbow branch is

$$\begin{aligned} q_{2,d} &= \arccos\left(\frac{w_x^2 + w_y^2 - 0.18}{0.18}\right), \\ q_{1,d} &= \operatorname{atan2}(w_y, w_x) - \operatorname{atan2}(0.30\sin q_{2,d}, 0.30 + 0.30\cos q_{2,d}), \\ q_{3,d} &= \psi_d - q_{1,d} - q_{2,d}. \end{aligned} \tag{9.23}$$

Angles must be unwrapped consistently when differentiating the reference. Substituting these angles into forward kinematics verifies the pose; differentiating the resulting smooth branch supplies reference velocities and accelerations. Workspace feasibility alone does not establish torque feasibility, joint-limit clearance, or collision avoidance.

9.4.2 A Position Pullback Has a Nullspace

The position Jacobian has columns

$$J_{p,:,j} = \sum_{i=j}^{3} l_i \begin{bmatrix} -\sin\theta_i \\ \cos\theta_i \end{bmatrix}. \tag{9.24}$$

It is a 2×3 matrix. At a regular configuration it has rank two, so the position-error pullback $Q_q = J_p^\mathsf{T} W_p J_p$, for $W_p \succ 0$, has rank two and a one-dimensional nullspace. It cannot have three positive eigenvalues merely because the robot is away from a position singularity. A determinant of J_p itself is not defined; manipulability can instead use $\det(J_p J_p^\mathsf{T})$ with declared units and scales.

Adding orientation yields the square pose Jacobian with third row $(1, 1, 1)$. It is nonsingular when the first two links are not collinear. These are different tasks with different nullspaces. In a humanoid swing, the same distinction appears when a cost constrains club position while leaving face orientation or redundant body motion weakly penalized.

Adding ϵI to Q_q makes that particular coordinate quadratic form positive definite for $\epsilon > 0$. It changes the cost in every direction, not only the nullspace. It does not create a physical barrier against singularities, guarantee a feasible trajectory, or prove contraction. A state metric for a torque-driven three-joint robot also needs six-state treatment; a 3×3 posture penalty alone does not measure velocity error.

9.4.3 An Exact Local Example With Its Assumptions Exposed

For an ideal fully actuated robot with known dynamics $M(q)\ddot{q} + h(q, \dot{q}) = u$, computed torque can impose a desired acceleration:

$$u = M(q)(\ddot{q}_d + v) + h(q, \dot{q}), \qquad \ddot{e} = v, \qquad e = q - q_d. \tag{9.25}$$

Use normalized angle and time coordinates to consider one error channel $z = (e, e')$, where the prime denotes differentiation with respect to dimensionless time $\tau = t/t_s$. The physical virtual acceleration is $(q_s/t_s^2)v$ when $e = (q - q_d)/q_s$. In these normalized variables,

$$z' = A_0 z + B_0 v, \qquad A_0 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{9.26}$$

For the infinite-horizon objective $\int_0^\infty (z^\top z + v^2)\, d\tau$, direct substitution into the continuous Riccati equation gives

$$S = \begin{bmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{bmatrix}, \qquad K = B_0^\top S = \begin{bmatrix} 1 & \sqrt{3} \end{bmatrix}, \qquad v = -Kz. \tag{9.27}$$

This standard double-integrator example is also derived in MIT's Linear Quadratic Regulators notes. The cost is on normalized error and virtual acceleration; it is not a claim about actual motor electrical energy or a human's physiological objective.

Let $A_c = A_0 - B_0 K$ and $W = I + K^\top K$. The Riccati identity is

$$A_c^\top S + S A_c = -W. \tag{9.28}$$

It follows that the metric length of an error difference contracts at any rate $\lambda \le \frac{1}{2}\lambda_{\min}(W, S)$ in normalized time, where the generalized eigenvalue uses $Wv = \mu S v$. Here the smallest generalized eigenvalue is $\sqrt{3} - 1/\sqrt{2}$, so $\lambda = (\sqrt{3} - 1/\sqrt{2})/2$. Physical-time rate is λ/t_s. The other common quantity, the real part of a closed-loop eigenvalue, is not interchangeable with this particular metric's instantaneous bound.

For three independent ideal error channels, repeat this block in a six-state metric after fixing the state ordering. The exact error equation makes the certificate valid in the chosen unwrapped coordinate chart as long as computed torque remains realizable and the model is exact. Saturation, contact changes, uncertain inertia, actuator dynamics, and branch changes invalidate that exact cancellation and require a new residual analysis. The certificate should not be advertised as a verified global property of an arbitrary robot or human.

9.4.4 Finite-Horizon and Sampled Certificates Need Their Own Equations

For a time-varying linear error system with the finite-horizon Riccati solution $S(t)$ and $K(t) = R^{-1}B^\top S$, the closed-loop identity is

$$\dot{S} + A_c^\top S + S A_c = -Q - K^\top R K. \tag{9.29}$$

Uniform positive definiteness of S and a uniform generalized lower bound on the right-hand quadratic form are needed for a uniform contraction statement. A zero terminal cost can make S degenerate at the horizon. Plotting a decreasing Riccati trace is neither that test nor a substitute for it.

For sampled error dynamics $z_{k+1} = A_{c,k} z_k$, the corresponding requirement is $A_{c,k}^\top S_{k+1} A_{c,k} \preceq \rho^2 S_k$ with $\rho < 1$. In the fixed-metric case, compute ρ^2 as the largest generalized eigenvalue of $(A_c^\top S A_c, S)$. It is not generally the squared spectral radius of A_c. These distinctions matter whenever a controller designed in continuous time is executed on sampled measurements.

9.5 Spacecraft: Respect the Orbit and the Meaning of the Cost

9.5.1 Relative Motion About a Circular Chief Orbit

Let x be radial displacement, y along-track displacement, and z normal displacement in the rotating frame of a circular chief orbit. For small relative separation, the Clohessy–Wiltshire linearization is

$$\ddot{x} = 3n^2 x + 2n\dot{y} + u_x, \qquad \ddot{y} = -2n\dot{x} + u_y, \qquad \ddot{z} = -n^2 z + u_z. \tag{9.30}$$

The inputs are accelerations. The circular-orbit and small-separation assumptions are part of the model; a different axis convention changes the displayed signs. See MIT Astrodynamics, Lecture 26 for the relative-motion derivation.

For radius $r_0 = 6.7 \times 10^6$ m and $\mu = 3.98600435507 \times 10^{14}$ $\mathrm{m}^3/\mathrm{s}^2$, $n = \sqrt{\mu/r_0^3} \simeq 0.001151$ rad/s and $T = 2\pi/n \simeq 5458$ s, about 90.96 minutes. This is a low Earth orbit, not a geosynchronous orbit. The Earth gravitational parameter is from JPL's DE440 astrodynamic parameters.

For $X = (x, y, z, \dot{x}, \dot{y}, \dot{z})$, write $\dot{X} = AX + Bu$ with

$$A = \begin{bmatrix} 0 & I_3 \\ N & C_o \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ I_3 \end{bmatrix}, \quad N = \operatorname{diag}(3n^2, 0, -n^2), \quad C_o = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{9.31}$$

The blocks act on position and velocity with different units; it is misleading to attach a single dimensional unit to every entry of this unscaled state matrix.

A reproducible reference can be manufactured rather than claimed as flight validation. Over $t_f = 900$ s, set $s = t/t_f$, $p(s) = 10s^3 - 15s^4 + 6s^5$, and $y_d = 1000 - 990p(s)$ m, with $x_d = z_d = 0$. Then $u_{x,d} = -2n\dot{y}_d$, $u_{y,d} = \ddot{y}_d$, and $u_{z,d} = 0$ satisfy the linear equations. The reference starts and ends at rest, with zero endpoint acceleration, and ends at a 10 m along-track standoff. It is not a completed docking maneuver. Thruster limits, approach corridors, collision geometry, navigation error, and nonlinear orbital-model error still require evaluation.

9.5.2 Quadratic Effort Is Not Propellant Consumption

An LQR objective $\int (e^\mathsf{T} Q e + \delta u^\mathsf{T} R \delta u)\, dt$ balances weighted tracking error and squared control effort around a feasible reference. Position and velocity weights require scales; Q here is a cost weight, not a process-noise covariance. A terminal penalty is a soft cost, not an exact terminal docking constraint.

For an ideal thruster with constant specific impulse, $\dot{m} = -\|F\|/(I_{sp} g_0)$ describes propellant use. The actual thruster allocation and operating model can require modifications. With $F = mu$, mass changes can enter the dynamics. The time integral of squared acceleration is therefore not a direct fuel measurement. A fuel-optimal or fuel-limited claim requires the appropriate objective, thrust model, mass evolution, and constraints. Neither a quadratic regulator nor an unreported simulation establishes a numerical percentage of fuel savings.

To implement a sampled controller under zero-order-held acceleration, compute

$$\exp\left(\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix} \Delta t \right) = \begin{bmatrix} A_d & B_d \\ 0 & I_3 \end{bmatrix}. \tag{9.32}$$

Design a discrete regulator for (A_d, B_d) and evaluate $A_d - B_d K_d$. Subtracting a discrete gain from the continuous matrix and interpreting the result as a sampled stability certificate mixes two different systems. A covariance propagation or Monte Carlo study can then test navigation uncertainty, but it must report its noise model and cannot validate unmodeled thruster or contact physics.

9.6 Vehicles: Input Coordinates and Speed Change the Argument

9.6.1 An Exact Affine Bicycle Model

For the kinematic bicycle with rear-axle position (x, y), heading ψ, speed v, wheelbase L, steering angle δ, and longitudinal acceleration a,

$$\dot{x} = v\cos\psi, \quad \dot{y} = v\sin\psi, \quad \dot{\psi} = \frac{v}{L}\tan\delta, \quad \dot{v} = a. \tag{9.33}$$

This model is nonlinear in the steering-angle input. With curvature $\kappa = \tan\delta/L$, it is exactly control affine:

$$\dot{X} = \begin{bmatrix} v\cos\psi \\ v\sin\psi \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} a + \begin{bmatrix} 0 \\ 0 \\ v \\ 0 \end{bmatrix} \kappa. \tag{9.34}$$

Alternatively, append steering angle to the state and use steering rate as an input. A Taylor expansion in δ around a reference gives a local input approximation, not the exact original affine model. Steering limits transform to curvature limits, and steering-rate constraints do not disappear under this reparameterization.

The no-side-slip condition restricts velocities. It does not by itself reduce the dimension of the configuration space: sequences of admissible motions can produce displacements not available instantaneously. Lie brackets describe this local mechanism, as illustrated in Modern Robotics, Controllability of Wheeled Mobile Robots. A rank calculation alone should not be turned into an unrestricted global parking guarantee; reversibility, admissible controls, obstacles, and the particular controllability theorem's assumptions matter.

9.6.2 A Speed Schedule Requires Its Own Dynamics

For a straight reference and small heading error, with fixed $v > 0$, the lateral error equations are $\dot{e}_y = v e_\psi$ and $\dot{e}_\psi = v\kappa$. The curvature feedback

$$\kappa = -\frac{\omega_n^2}{v^2} e_y - \frac{2\zeta\omega_n}{v} e_\psi \tag{9.35}$$

gives $\ddot{e}_y + 2\zeta\omega_n \dot{e}_y + \omega_n^2 e_y = 0$. For positive ω_n and ζ, this linearized constant-speed model is stable. The position and heading gains decrease as speed increases for this fixed target polynomial. There is no general rule that all gains should increase with speed.

If v varies, differentiation introduces $(\dot{v}/v)\dot{e}_y$, and simply inserting the current speed into the same law no longer gives the stated constant-coefficient equation. The gain formula is singular at zero speed. Reverse motion, curved references, steering dynamics, and saturation also need separate treatment. Frozen-speed eigenvalues do not certify stability under arbitrary switching or rapid scheduling.

At fixed steering angle, yaw rate scales as v and the no-slip lateral acceleration as $v^2 \tan\delta/L$. For $L = 3$ m, $\delta = 0.10$ rad, and $v = 25$ m/s, the latter is about 20.9 m/s^2. That requested acceleration signals that the simple kinematic assumptions need scrutiny; the equation does not establish that the tires can supply it. A dynamic tire model and friction constraints determine feasibility. Comparing unscaled norms that mix position rates, heading rate, and longitudinal acceleration cannot establish control dominance at high speed.

The connection to golf is methodological. A parameter change can alter the dynamics, input map, feasible controls, and error metric simultaneously. A gain schedule or equipment change must be evaluated within that coupled system; conclusions from a fixed-parameter calculation do not automatically survive the change.

9.7 Reproducible Computation and Evidence

9.7.1 Check an Equation Before Interpreting Its Output

The following complete Python example checks the normalized double-integrator result. It requires NumPy and SciPy. The residual verifies the equation solved; the generalized eigenvalue verifies the stated metric inequality. Neither check validates an unmodeled actuator.

```python
import numpy as np
from scipy.linalg import eigvalsh, solve_continuous_are

A = np.array([[0.0, 1.0], [0.0, 0.0]])
B = np.array([[0.0], [1.0]])
Q = np.eye(2)
R = np.eye(1)
S = solve_continuous_are(A, B, Q, R)
K = np.linalg.solve(R, B.T @ S)
Ac = A - B @ K
W = Q + K.T @ R @ K
residual = Ac.T @ S + S @ Ac + W
assert np.linalg.norm(residual) < 1e-10
rate = 0.5 * eigvalsh(W, S)[0]
assert np.isclose(rate, (np.sqrt(3.0) - 1.0 / np.sqrt(2.0)) / 2.0)
```

For an equilibrium regulator, the physical command includes any nonzero equilibrium input. For a trajectory regulator, it includes a dynamically feasible reference input and the tracking correction. Omitting that feedforward term changes the error dynamics. Apply limits and actuator dynamics before interpreting simulated performance; clipping an ideal command invalidates an unconstrained certificate unless the resulting residual is bounded in a new analysis.

In MATLAB, `dlqr(A,B,Q,R,N)` solves an infinite-horizon discrete LQR problem; the fifth argument is the state–input cross-weight matrix N, not a horizon length. A finite-horizon problem requires a backward recursion with a declared terminal cost and time-indexed gains. The MathWorks dlqr documentation specifies the cross term and discrete Riccati equation. Likewise, an outline containing undefined model objects or placeholder matrices is an implementation sketch, not an executable validation.

9.7.2 Separate Verification, Calibration, and Validation

Verification asks whether the equations were implemented and solved correctly. For the golf calculation this includes Cartesian-versus-joint kinetic energy, mass symmetry and definiteness under admissible parameters, correct signed mode normalization, correct Schur elimination, unforced energy conservation, and step refinement. A reproducible table is valuable only after these model identities are satisfied.

Calibration estimates model parameters from data. Validation compares held-out predictions with observations for a declared use. Fitting the same swing used to assess accuracy is not an independent prediction test. A claimed error percentage needs the measured quantity, normalization, sample, uncertainty, and protocol; it cannot be inferred from the apparent plausibility of a simulated trajectory.

For perturbation experiments, align the model and measurement definitions. Separate fixed-time errors from event-time errors, and choose whether the comparison is between two closed-loop trajectories or between a trajectory and its nominal reference. Measurement noise can dominate a ratio with a small initial-error denominator. A worst-case contraction bound need not be attained by typical perturbations, and a confidence interval containing a predicted bound is not a validation criterion by itself. Test the inequality with appropriate uncertainty, check the assumed region and disturbances, and report violations as well as agreement.

For the golf swing, an informative progression is to verify rigid-body energy accounting, add and identify shaft dynamics, verify contact and impact behavior, then assess the coupled model on held-out swings and equipment conditions. Measure grip and head motion together when studying shaft effects. Include force or moment information when testing an energy-transfer explanation. Include event timing and output uncertainty when testing an accuracy explanation. Each additional model component should resolve a stated discrepancy or prediction need rather than merely increase the number of states.

9.8 Exercises That Test the Argument

1. Derive the six independent entries of M_r from Cartesian center-of-mass velocities. Check at random postures and velocities that both kinetic-energy calculations agree. Explain why an endpoint singularity need not make M_r singular.

2. Assemble the beam's three inertia blocks from the same material-point velocity. Repeat with twice the beam mass. Show which rigid-coordinate diagonal term gains $m_s l_3^2/3$, and test the complete kinetic energy rather than only a matrix determinant.

3. Change the sign of the second mode and transform its amplitude and coupling consistently. Show that physical displacement and kinetic energy are unchanged. Explain why assigning two arbitrary frequencies does not identify a uniform beam's EI.

4. Verify the two block acceleration equations by substituting the Schur solution. Include a nonzero modal bias so an omitted elastic term cannot pass the check. Distinguish this test from integrating the rigid six-state counterfactual.

5. Refine the rigid zero-torque integration step and report both endpoint convergence and mechanical-energy error. Explain why the displayed acceleration norm is not a finite drift-to-control ratio or an estimate of muscular effort.

6. Construct a motion with decreasing proximal angular speed but no positive distal energy transfer across the chosen joint. State the external powers and show the segment energy balance. Identify the extra measurements required to interpret a human speed-peak sequence.

7. For two shaft stiffnesses, compare identical prescribed grip motion, identical applied force histories, and an adapting controller. State which causal question each comparison addresses and which output would distinguish the hypotheses.

8. Reconstruct the robot circle using the given inverse kinematics. Compute the nullspace of J_p and compare position and pose pullbacks. Test whether adding ϵI imposes any obstacle or joint-limit constraint.

9. Verify the exact Riccati solution and its generalized metric rate. Add a torque limit to the computed-torque model and identify where the ideal proof ceases to apply rather than extending its rate by assertion.

10. Substitute the quintic spacecraft reference into the relative-motion equations. Compare integrated squared acceleration with a declared propellant model. Report the assumptions needed before calling the final standoff a docking solution.

11. Derive the additional lateral-error term for variable vehicle speed. Test a rapid speed change and a steering limit. Explain why frozen-speed eigenvalues and kinematic lateral acceleration do not establish dynamic feasibility.

12. For a transverse impact event, compare finite-difference sensitivities with the event-time formula and a fixed-time formula. Repeat near grazing. Propagate parameter as well as state uncertainty, using declared input and output scales.

Chapter 10

Parallel Mechanisms, Constrained Dynamics, and Geometric Control

10.1 What the Geometry Explains

Holding a club with both hands closes a mechanical loop. Ground contact closes further loops through the legs and pelvis. These connections restrict relative motion and transmit loads, but they do not uniquely prescribe muscle forces, determine a neural control policy, or supply energy merely by exerting a force. The useful geometric question is how configuration, inertia, constraints, and actuation jointly determine feasible motion and load transmission.

We develop that question in a smooth, fixed contact mode. Let q have n local configuration coordinates, $v = \dot{q}$, and $u \in \mathbb{R}^m$ denote specified applied inputs. There are c independent, stationary, ideal holonomic constraints $\Phi(q) = 0$. Their Jacobian is $J_c = D\Phi \in \mathbb{R}^{c\times n}$, and $M(q) \succ 0$ is the mass matrix. These assumptions permit exact identities. Grip compliance, sliding, lift-off, tissue activation, and ball impact require additional constitutive states or contact modes; they cannot be inferred from the identities alone.

The mechanics below follows the vector/covector and virtual-work treatment of constrained systems [BL04, MLS94]. Its worked examples are mathematical checks, not measurements of a golfer. The final sections explain what measurements would be needed to turn the framework into an empirical account of a swing.

10.2 Configuration, Constraints, and Regularity

10.2.1 Configuration Is Different From State

For N unconstrained spatial rigid bodies, $Q_{\text{free}} = SE(3)^N$ has dimension $6N$; planar bodies contribute three configuration dimensions each. Joint coordinates reduce this description before contact constraints are imposed. Angles have periodic topology: a pair of revolute angles lives on $S^1 \times S^1$, with intervals used only as coordinate charts. The mechanical state is in the tangent bundle TQ, with local coordinates (q, v) and twice the configuration dimension. Activation and flexible coordinates may enlarge it further.

If J_c has rank c on the level set, the regular level-set theorem gives

$$Q_c = \{q : \Phi(q) = 0\}, \qquad \dim Q_c = n - c =: d, \qquad T_q Q_c = \ker J_c(q). \tag{10.1}$$

The admissible state satisfies both $\Phi(q) = 0$ and $J_c v = 0$. A configuration constraint does not imply $J_c \dot{v} = 0$: differentiating again gives

$$J_c \dot{v} + \dot{J}_c v = 0. \tag{10.2}$$

Thus admissible velocities form a linear tangent space at fixed q, whereas admissible coordinate accelerations form an affine set at fixed (q, v).

Rank loss invalidates the regular-level-set argument; it does not by itself prove that the physical set is singular or unstabilizable. The equation $x^2 = 0$ has zero derivative at its solution, although its solution set is the smooth zero-dimensional set $\{0\}$. Redundant constraint equations can also lose rank without a change in physical mobility. At a true mechanism singularity, inspect independent coordinates, higher-order compatibility, actuator directions, and contact inequalities before drawing a control conclusion.

10.2.2 A Movable Planar Closed Chain

Take a two-link arm based at the origin and a third link based at (a, b). Let all three angles be absolute angles from the same horizontal axis. Loop closure is

$$\Phi_1 = \ell_1 \cos\theta_1 + \ell_2 \cos\theta_2 - \ell_3 \cos\theta_3 - a, \tag{10.3}$$

$$\Phi_2 = \ell_1 \sin\theta_1 + \ell_2 \sin\theta_2 - \ell_3 \sin\theta_3 - b. \tag{10.4}$$

The Jacobian has columns $\ell_1(-\sin\theta_1, \cos\theta_1)^\top$, $\ell_2(-\sin\theta_2, \cos\theta_2)^\top$, and $\ell_3(\sin\theta_3, -\cos\theta_3)^\top$. At rank two, three free angles minus two independent constraints leave one local degree of freedom. If instead there were only two grounded links meeting at their tips, two angles minus two regular closure constraints would leave isolated configurations. Equal link lengths alone would not establish a movable chain; coincident bases and compatible geometry would matter too.

For the two-grounded-link Jacobian with columns $\ell_1(-\sin\theta_1, \cos\theta_1)^\top$ and $\ell_2(\sin\theta_2, -\cos\theta_2)^\top$, its determinant is $\ell_1\ell_2 \sin(\theta_1 - \theta_2)$. This detects first-order rank loss when the links align, but finite mobility still requires examination of the closure equations.

10.2.3 A Revolute Joint and Its Frames

Choose frames at the joint pivot, with their local z axes along the joint axis. The allowed relative transform has $p_{\text{rel}} = 0$ and $R_{\text{rel}} = R_z(\theta)$. Locally near aligned axes, five independent constraints are the three components of p_{rel}, $R_{13} = 0$, and $R_{23} = 0$, on the branch $R_{33} > 0$. In contrast, setting $R_{12} = R_{21} = 0$ would prohibit ordinary rotation about z, because those entries are $-\sin\theta$ and $\sin\theta$. The relative motion has one degree of freedom; two free bodies connected by this joint have seven total configuration degrees of freedom, including the common spatial pose.

10.2.4 Integrability Needs More Than a Derivative Test

For a nonvanishing one-form $\alpha = \sum_i a_i(q)dq_i$, the velocity constraint $\alpha(v) = 0$ defines a distribution. Locally, its codimension-one integrability condition is $\alpha \wedge d\alpha = 0$. When integrable, it can be written $\alpha = \mu\, d\phi$ with a nonzero integrating factor μ; α itself need not be exact. For example, $e^x dy = 0$ means $dy = 0$ even though $d(e^x dy) = e^x dx \wedge dy \neq 0$. Closed forms are locally exact on a contractible neighborhood, but global exactness requires topological conditions. For several independent forms, use the Frobenius involutivity condition on their common kernel.

A wheel with planar heading θ has the lateral no-slip constraint $-\sin\theta\, \dot{x} + \cos\theta\, \dot{y} = 0$. Its distribution is nonintegrable. Rolling on a fixed straight track can instead give $x - r\varphi = \text{constant}$. Rolling is therefore not universally nonholonomic. Sliding friction, unilateral joint limits, and contact opening are different phenomena again; they are not all smooth equality constraints.

10.3 Mass Metric, Forces, and Inertial Terms

10.3.1 Vectors and Covectors Have Different Roles

Velocity $v \in T_qQ$ is a vector. A generalized load $F \in T_q^*Q$ is a covector: $F^\top v$ is power. The mass matrix maps velocity to momentum and defines $\langle a, b\rangle_M = a^\top M b$. Ideal stationary constraint reactions are covectors in range $J_c^\top$, because

$$(J_c^\top \lambda)^\top v = \lambda^\top J_c v = 0. \tag{10.5}$$

Their associated mass-metric normal vectors are in a different space:

$$(\ker J_c)^{\perp_M} = \text{range}(M^{-1} J_c^\top). \tag{10.6}$$

Indeed $v^\top M(M^{-1}J_c^\top \lambda) = 0$ for every tangent v. With $M = \text{diag}(2, 1)$ and $J_c = (1, 1)$, the vector $(1, -1)$ is tangent. Its mass inner product with $(1, 1)$ is one, but its mass inner product with $(1/2, 1)$ is zero. Omitting M^{-1} changes the orthogonality claim.

10.3.2 Constructing the Mass Matrix

For rigid link k, let $J_{v,k}$ map generalized rates to center-of-mass velocity in a fixed spatial frame, and let $J_{\omega,k}$ map them to angular velocity in that same frame. If R_k maps body coordinates to that frame and $I_{b,k}$ is inertia about the center of mass in body coordinates, then

$$M(q) = \sum_k \left(m_k J_{v,k}^\top J_{v,k} + J_{\omega,k}^\top R_k I_{b,k} R_k^\top J_{\omega,k} \right). \tag{10.7}$$

Every summand is an $n \times n$ matrix. Positive definiteness requires a nondegenerate model with independent coordinates and positive inertias in its physical directions. Redundant orientation parameters require their own constraints; one cannot assume an unconstrained positive-definite mass matrix in arbitrary redundant coordinates.

With upward height z_k and downward gravity of magnitude $g > 0$, use $V = \sum_k m_k g z_k$. Kinetic energy is $T = \frac{1}{2} v^\top M v$. It measures energy at a velocity, not an unspecified physiological cost of acceleration.

10.3.3 Coriolis Loads and the Energy Identity

Write the forced equations as

$$M\dot{v} + h(q,v) = Bu + J_c^\top \lambda, \qquad h = c(q,v) + \nabla V, \tag{10.8}$$

with any additional known nonconservative loads added explicitly on the right. The inertial velocity load is

$$c_i = \sum_{j,k} \Gamma_{ijk} v_j v_k, \tag{10.9}$$

$$\Gamma_{ijk} = \tfrac{1}{2} \left(\partial_k M_{ij} + \partial_j M_{ik} - \partial_i M_{jk} \right). \tag{10.10}$$

One valid matrix representation is $C_{ij} = \sum_k \Gamma_{ijk} v_k$, so $c = Cv$. It satisfies $\dot{M} - 2C$ skew-symmetric and

$$v^\top c = \tfrac{1}{2} v^\top \dot{M} v. \tag{10.11}$$

In general $C \neq \dot{M}$. The matrix representation is not unique, while the load c for the specified coordinates is fixed. The second-kind connection coefficients are $\Gamma^i_{jk} = \sum_\ell (M^{-1})_{i\ell} \Gamma_{\ell jk}$, giving $M^{-1}c$ component $\sum_{j,k} \Gamma^i_{jk} v_j v_k$. There is no extra minus sign in this identity; the minus appears when solving the force-free equation for acceleration.

For a simple abstract two-coordinate metric, introduce a length scale ℓ so the entries have angular inertia units:

$$M = \ell^2 \begin{bmatrix} m_1 + m_2 & m_2 \cos q_2 \\ m_2 \cos q_2 & m_2 \end{bmatrix}, \qquad m_1, m_2 > 0. \tag{10.12}$$

Its determinant is $\ell^4 (m_1 m_2 + m_2^2 \sin^2 q_2) > 0$. Direct differentiation gives $c = (-\ell^2 m_2 \sin q_2\, v_2^2, 0)^\top$. Thus $\Gamma^1_{22} = -m_2^2 \sin q_2 / (m_1 m_2 + m_2^2 \sin^2 q_2)$ and $\Gamma^1_{12} = 0$. Substitution verifies the energy identity. This is a metric example, not a derived anthropometric two-link arm.

Connection coefficients can be nonzero in curvilinear coordinates on a flat space. Their presence does not establish nonzero intrinsic curvature. Coordinate-free equations retain covariant acceleration; they do not eliminate the physical inertial coupling. Curvature concerns how covariant differentiation fails to commute, a separate question from whether a coordinate expression contains centrifugal terms.

10.4 Eliminating Reactions Without Losing Their Input Dependence

Define $K_c = J_c M^{-1} J_c^\top \succ 0$ and $\gamma = \dot{J}_c v$. The acceleration constraint and force balance give

$$\lambda = -K_c^{-1} \left[J_c M^{-1} (Bu - h) + \gamma \right]. \tag{10.13}$$

The reaction changes when the input changes, even at the same (q, v). It cannot be held at its measured value and called an input-independent drift.

Define a vector projector and its dual force projector:

$$P_v = I - M^{-1} J_c^\top K_c^{-1} J_c, \qquad P_F = P_v^\top. \tag{10.14}$$

These projectors act on different spaces. They satisfy $P_v^2 = P_v$, $J_c P_v = 0$, $P_v^\top M = M P_v$, and $P_F J_c^\top = 0$. A load decomposes as $F = P_F F + (I - P_F)F$; the second part belongs to the reaction covector space. This algebraic partition is not a label for useful versus wasted physiological effort.

The reduced acceleration equation is

$$\dot{v} = f(q, v) + G(q)u, \tag{10.15}$$

$$f = -P_v M^{-1} h - M^{-1} J_c^\top K_c^{-1} \gamma, \tag{10.16}$$

$$G = P_v M^{-1} B. \tag{10.17}$$

Here f is an acceleration function in a coordinate chart. The first-order vector fields on TQ_c are $F_0(q, v) = (v, f)$ and $F_i(q, v) = (0, G_i)$. This distinction matters for Lie brackets and state dimension. All columns of G are tangent, while $J_c f = -\gamma$ supplies the normal acceleration required by the changing tangent plane.

For $\Phi(q) = \frac{1}{2}(q^\top q - R^2)$ on a circle with $M = I$, $J_c = q^\top$ and $\gamma = v^\top v$. The admissible acceleration for an unconstrained candidate a_0 is

$$a = P_v a_0 - \frac{q}{R^2} \|v\|^2. \tag{10.18}$$

At $q = (1, 0)$ and $v = (0, 2)$, its radial component must be -4. Projecting a_0 onto the tangent line alone gives radial component zero and violates the acceleration constraint. Acceleration-level satisfaction also does not guarantee that an ordinary numerical integrator preserves position and velocity constraints.

A zero-input counterfactual evaluates $u = 0$ with the chosen constitutive model and contact mode, then evolves its own state. The drift includes the zero-input reaction computed above. If a computed reaction violates a friction cone or requires tensile ground contact, the assumed mode is infeasible and must change. A prescribed measured base motion is an external boundary intervention, not an isolated passive swing.

10.5 Three Different Forms of Redundancy

10.5.1 Constraint Freedom, Task Freedom, and Input Freedom

Constraint-compatible velocity freedom is $\ker J_c$. For a task map $y = \psi(q)$ with Jacobian J_t, task-preserving velocity freedom is $\ker J_c \cap \ker J_t$. Neither is automatically an unactuated direction. Input redundancy is the kernel of the effective acceleration map G; its elements change input without changing instantaneous acceleration in this ideal model, though reactions can change.

For desired acceleration a_d, let $b = a_d - f$. Exact instantaneous tracking requires $b \in \operatorname{range} G$ and some feasible $u \in \mathcal{U}$ with $Gu = b$. At rank r, the unconstrained algebraic solution family is

$$u = G^+ b + (I - G^+ G)z, \qquad z \in \mathbb{R}^m, \tag{10.19}$$

provided the range condition holds. Its input redundancy dimension is $m - r$. If $m < d$ and G has full column rank, the system is instantaneously underactuated and each feasible b has a unique input. Fewer inputs do not create an infinite family of solutions. Bounds, force signs, actuator rates, and friction restrictions can eliminate an otherwise algebraically feasible solution.

If reduced coordinates give a full-row-rank acceleration map $\bar{G} \in \mathbb{R}^{d \times m}$ and the objective is $\frac{1}{2} u^\top W u$, $W \succ 0$, Lagrange multipliers give

$$u_W = W^{-1} \bar{G}^\top (\bar{G} W^{-1} \bar{G}^\top)^{-1} \bar{b}. \tag{10.20}$$

For $\bar{G} = (1, 1)$, $\bar{b} = 1$, and $W = \operatorname{diag}(1, 4)$, this is $(0.8, 0.2)$, not the Euclidean minimum-norm allocation $(0.5, 0.5)$. Differentiating W with respect to posture is not a substitute for solving this fixed-state allocation problem. State planning and instantaneous input allocation are distinct optimizations.

10.5.2 Minimum-Energy Velocity and Muscle Allocation

To find a velocity with task rate $\dot{y}_d$ while preserving the constraint, stack $A = (J_c^\top, J_t^\top)^\top$ and $b_y = (0, \dot{y}_d)^\top$. With independent rows, the minimizer of $\frac{1}{2}v^\top M v$ subject to $Av = b_y$ is

$$v_* = M^{-1}A^\top(AM^{-1}A^\top)^{-1}b_y. \tag{10.21}$$

The Euclidean pseudoinverse minimizes Euclidean norm; it minimizes kinetic energy only under the corresponding metric choice. Position-level feasibility and finite-time task execution still require integration and input checks.

Muscle allocation is another inverse problem. If a signed moment-arm matrix $R(q)$ maps muscle forces s to required joint load τ, one may solve $Rs = \tau$ with $0 \le s \le s_{\max}$ and a chosen objective such as $\sum_i (s_i/A_i)^p$. For $p = 2$ and fixed positive areas this is a convex quadratic objective with linear constraints. A nonzero-load solution is not in $\ker R$; differences between unconstrained solutions lie in $\ker R$. Capacity limits and activation dynamics can remove that freedom. Selecting one solution with an objective supplies a modeling assumption, not a measurement of the nervous system's objective [Zaj89].

10.6 Twists, Wrenches, and a Two-Hand Grip

10.6.1 Consistent Frame Transformations

Let $g_{AB} = (R, p)$ map coordinates in frame B into frame A. Use twist ordering $\xi = (\omega, v)$, where v is the linear component associated with the chosen frame origin. Then

$$\xi_A = \mathrm{Ad}_{g_{AB}}\,\xi_B, \qquad \mathrm{Ad}_{g_{AB}} = \begin{bmatrix} R & 0 \\ [p]_\times R & R \end{bmatrix}. \tag{10.22}$$

In particular $v_A = Rv_B + p \times R\omega_B$. The equivalent matrix identity is $\widehat{\xi}_A = g_{AB}\widehat{\xi}_B g_{AB}^{-1}$. The translation term must be retained. A six-dimensional twist cannot be transformed by a four-dimensional homogeneous matrix acting directly on its column of components.

For wrench ordering $W = (\tau, F)$, invariance of $W^\top\xi$ requires

$$W_A = \mathrm{Ad}_{g_{AB}}^{-\top}\,W_B = \begin{bmatrix} R\tau_B + p \times RF_B \\ RF_B \end{bmatrix}. \tag{10.23}$$

The reverse transformation is $W_B = \mathrm{Ad}_{g_{AB}}^\top\,W_A$. Stating the direction of the frame map avoids ambiguous uses of the word coadjoint. Screw reciprocity between a wrench and an admissible twist means $W^\top\xi = 0$, not $\xi^\top M\eta = 0$ between two velocities.

For integrated twist coordinates $\zeta = (\omega, v)$ and $\theta = \|\omega\|$, the exponential is

$$\exp\widehat{\zeta} = \begin{bmatrix} e^{[\omega]_\times} & J(\omega)v \\ 0 & 1 \end{bmatrix}, \tag{10.24}$$

$$J(\omega) = I + \frac{1-\cos\theta}{\theta^2}[\omega]_\times + \frac{\theta - \sin\theta}{\theta^3}[\omega]_\times^2. \tag{10.25}$$

Use the continuous limit $J(0) = I$, with series evaluation for small θ. The last coefficient has denominator θ^3 because ω already contains the rotation magnitude; mixing unit-axis and integrated-coordinate formulas changes the translation. A time-varying twist generally requires an ordered integration, not the exponential of its simple time integral [MLS94].

10.6.2 Different Contact Points Have Different Velocities

At hand contact i, let r_i be the vector from the club reference point O to that contact, expressed in one spatial frame. No translational slip imposes

$$J_{h,i}(q)\dot{q} = v_O + \omega_c \times r_i. \tag{10.26}$$

The left and right hand point velocities are generally unequal when the club rotates. Full rigid attachment also constrains relative angular velocity; point-contact and soft-finger models constrain fewer components. If limb Jacobians are converted to the same club reference point and frame, equality of their full twists can be used. Equating unshifted velocities at different grip points is incorrect.

For two point forces on the club, the resultant wrench at O is

$$F = f_L + f_R, \qquad \tau_O = r_L \times f_L + r_R \times f_R. \tag{10.27}$$

Adding $\delta f_L = a(r_L - r_R)$ and $\delta f_R = -\delta f_L$ changes neither resultant component. An internal squeeze along the hand separation can therefore be invisible to a measured resultant wrench. Distributed contact moments, compliance, and friction alter the admissible internal-load set, but they do not make bilateral force inference unique without further evidence. Internal grip loading may affect stiffness and future contact behavior even when its instantaneous net wrench is zero.

10.6.3 Singular Values Need Physical Scaling

For a task Jacobian, $\sqrt{\det(J_t J_t^\top)}$ measures the volume of its Euclidean velocity ellipsoid when rows are independent. It is not a complete condition-number measure: $\operatorname{diag}(100, 0.01)$ has determinant one but condition number 10^4. Translational and rotational components need declared scales before such measures can be compared. Report singular values and physical input limits, not just a determinant.

A rank-k screw span defines a point of $\mathrm{Gr}(k, 6)$. When the spanning matrix loses rank, it ceases to represent a k-plane; the rank-deficient set is not a collection of points inside that same Grassmannian with reduced dimension. Rank-loss codimension depends on the matrix family. Neither arm alignment nor a small manipulability determinant by itself proves high geometric curvature, loss of feedback stabilizability, or beneficial energy transfer.

10.7 Ground Contact, Momentum, and Power

10.7.1 Count the Floating Base Before Subtracting Constraints

As an explicitly idealized model, give each leg seven joint coordinates and the pelvis a six-dimensional floating pose. There are then $6 + 7 + 7 = 20$ free configuration dimensions. One rigid stationary foot supplies up to six independent constraints, leaving 14; two such feet supply up to 12, leaving eight at regular configurations. Starting with 14 joint coordinates and subtracting foot constraints while also allowing a floating pelvis omits the base. Anatomical joint models may use different joint counts. Point contacts, rolling contacts, deformable shoes, and dependent constraints also change the count.

A rigid bilateral contact model remains admissible only while its calculated wrench respects nonpenetration, compressive normal force, friction, and the support region. A formal equality solution that pulls a foot into the ground or requires excessive tangential traction is not a feasible physical prediction.

10.7.2 Center of Pressure and Reference-Point Conventions

Express the force exerted by the plate on the body in a right-handed frame with z upward, and shift its measured moment to an origin in the contact plane. For $F_z \neq 0$, a resultant applied at $(x_c, y_c, 0)$ with a remaining vertical free moment gives

$$x_c = -\frac{M_y}{F_z}, \qquad y_c = \frac{M_x}{F_z}, \qquad M_{z,\mathrm{free}} = M_z - x_c F_y + y_c F_x. \tag{10.28}$$

These signs follow from $r_c \times F = (y_c F_z, -x_c F_z, x_c F_y - y_c F_x)$. A plate origin below the surface requires a moment shift first; a manufacturer may report the opposite action-reaction convention. Near zero normal load the ratios become ill-conditioned. A center of pressure does not replace the complete wrench, especially its free moment.

10.7.3 Inverse Dynamics With All Loads Declared

For a body-fixed offset r from a moving reference point to the center of mass, expressed spatially,

$$a_C = a_O + \alpha \times r + \omega \times (\omega \times r). \tag{10.29}$$

The last term has a plus sign; its double cross product already points inward for circular motion. Sum all applied forces and all moments about the center of mass:

$$\sum F = m a_C, \qquad \sum \tau_C = I_C \alpha + \omega \times (I_C \omega), \tag{10.30}$$

with I_C represented in the same frame. Gravity, measured contact loads, distal and proximal joint loads, and their moment arms belong in these sums. A backward Newton–Euler recursion is a systematic application of these balances with explicit action-reaction signs and frame shifts. Kinematic accelerations come from the motion model or measurements; measured ground force is a load boundary condition, not an acceleration measurement.

Inverse dynamics gives the required net generalized load after known loads are accounted for. Unknown bilateral load sharing and muscle redundancy remain inverse problems. Recursive inverse dynamics is linear in the number of links for a tree of fixed-size joints; recursive forward-dynamics algorithms can also avoid assembling and inverting a dense mass matrix. Closed loops add constraint solves and cannot be assigned the same computational cost without specifying the solver.

10.7.4 Support Can Redirect Motion Without Supplying Work

With matching reference point and frame, contact power is $P_c = F_c^\top v_c + \tau_c^\top \omega_c$. A stationary rigid foot has zero twist, so its ideal ground wrench delivers zero mechanical power, even if its forces and moments are large. A moment does not supply rotational energy when angular velocity is zero. A foot pivoting or deforming needs the actual contact motion and traction distribution; no-slip at one point does not mean the entire foot is stationary.

Nevertheless, the ground can change total linear and angular momentum. About a fixed inertial origin, $\dot{H}_O$ equals the total external moment, including ground reactions and gravity where applicable. Zero work does not imply zero impulse or zero angular impulse. This is central to the golf swing: support reactions can redirect motion and enable muscles to do work elsewhere without being an independent source of mechanical energy.

For stationary ideal constraints, stored mechanical energy $E = T + V$ obeys

$$\dot{E} = u^\top y + P_{\text{other}} - D, \qquad y = B^\top v, \qquad D \geq 0, \tag{10.31}$$

when dissipation is modeled by D and all other power sources are declared. Passivity is a storage inequality, $E(t) - E(0) \leq \int u^\top y \, dt$ for the chosen port when other sources are absent and storage is bounded below. Equality is the lossless special case. Torque dotted with torque is not power. Muscle activation can maintain force with nonzero metabolic cost even at zero joint power. Mechanical work, squared torque, metabolic expenditure, and fatigue therefore define different optimization problems.

10.8 Reachability and Optimal Control

10.8.1 Instantaneous Rank Does Not Decide Finite-Time Motion

For smooth first-order fields, $[X, Y] = DY\,X - DX\,Y$. If $X = (x_2, -x_1)$ and $Y = (0, 1)$, then $[X, Y] = (-1, 0)$. This is a second state direction relative to the single input Y, not a third independent direction in a two-dimensional space.

The Chow–Rashevskii bracket-generating condition concerns driftless systems with reversible admissible controls on an appropriate connected region. Mechanical dynamics have drift, so distinguish accessibility, small-time local controllability, and reachability over a prescribed forward-time horizon. For $\dot{x} = 1$, $\dot{y} = u$, the drift and input span the plane but no forward trajectory can decrease x. For second-order mechanics, input fields $(0, G_i(q))$ have zero mutual brackets when they depend only on q; brackets with the drift (v, f) are essential. Checking the acceleration columns alone against the full state dimension is invalid.

As a driftless example, car kinematics have $g_1 = (\cos\theta, \sin\theta, \tan\varphi/L, 0)$ and $g_2 = (0, 0, 0, 1)$. Their bracket $[g_2, g_1] = (0, 0, \sec^2\varphi/L, 0)$ first gives a heading direction; further brackets produce lateral displacement directions. Steering while stationary does not translate the car. Maneuvers realizing bracket effects use sequences of controls and, for the usual reversible argument, both forward and reverse travel.

10.8.2 Necessary Conditions in Independent State Coordinates

Use a local chart of TQ_c with independent state $z \in \mathbb{R}^{2d}$ and dynamics $\dot z = \mathcal{F}(z, u)$. This avoids silently treating dependent ambient position and velocity coordinates as independent. For fixed final time, terminal cost $\phi(z(T))$, and running cost ℓ, define the normal minimization Hamiltonian

$$H = \ell + p^\top \mathcal{F}, \qquad \dot p = -H_z, \qquad u_* \in \arg\min_{u\in\mathcal{U}} H. \tag{10.32}$$

For a free terminal state, $p(T) = \nabla\phi$; endpoint constraints change transversality. These are necessary conditions under the usual regularity assumptions, not a general sufficient certificate. Abnormal extremals need the cost multiplier formulation. Ambient formulations must handle both position and velocity consistency and cannot obtain all costate conditions by adding one unexplained position multiplier.

For a control-linear Hamiltonian $H = H_0 + \sigma^\top u$ on a box, minimization selects $u_i = u_{i,\min}$ when $\sigma_i > 0$ and $u_i = u_{i,\max}$ when $\sigma_i < 0$. A switching function that vanishes on an interval requires higher-order conditions and may admit a singular arc. With $\ell = \ell_0 + u^\top R u$, $R \succ 0$, the unconstrained minimizer is $-\frac{1}{2}R^{-1}\mathcal{G}^\top p$ for $\mathcal{F} = \mathcal{F}_0 + \mathcal{G}u$. Under bounds, solve the resulting convex quadratic program. Componentwise clipping is exact for a separable diagonal R and independent box bounds, not for an arbitrary coupled R.

Torque input, muscle force input, excitation input, and activation state are different model choices. Smooth activation does not refute a bang-bang excitation solution when activation dynamics filter that excitation. A torque optimum does not establish that muscle activation follows the same profile or that the nervous system optimizes the chosen objective.

10.8.3 Value Functions and Local Numerical Methods

Where differentiable, the finite-horizon value $\mathcal{V}(z, t)$ satisfies

$$-\mathcal{V}_t = \min_{u\in\mathcal{U}} \left(\ell + \nabla\mathcal{V}^\top \mathcal{F}\right), \qquad \mathcal{V}(z, T) = \phi(z). \tag{10.33}$$

Nonsmooth value functions need an appropriate generalized, typically viscosity, solution. Autonomous dynamics do not remove finite-horizon time dependence. A stationary HJB equation belongs to a separately specified infinite-horizon or stationary problem. A grid with ten values in each of 28 independent state dimensions has 10^{28} nodes, illustrating why direct grids become impractical; the dimension must come from the chosen constrained model.

For a discrete full-state transition $z_{k+1} = F_k(z_k, u_k)$, iLQR uses $\delta z_{k+1} = A_k\delta z_k + B_k\delta u_k$ and second-order cost terms. Its backward pass produces both a feedforward step and feedback gain, and its forward pass rolls out $u_k^{\text{new}} = u_k + \alpha k_k + K_k(z_k^{\text{new}} - z_k)$. DDP additionally retains value-gradient-weighted second derivatives of the dynamics. Both need regularization, step acceptance, feasibility handling, and convergence checks. Projecting a cost gradient alone does not enforce a nonlinear contact constraint. Chapter 6 derives the Riccati and local quadratic calculations; neither algorithm automatically proves a global nonlinear optimum.

10.9 What Measurements Can Identify

With known parameters and an explicit zero-input contact model, $f(q, v)$ is computable from an estimated state. That is model evaluation, not a theorem that kinematics alone identify the physical drift. The acceleration constraint supplies c equations for n accelerations and generally leaves d tangent directions undetermined. Actual reactions additionally depend on applied inputs. Unknown inertias, tendon state, compliance, and loads cannot be made known by writing them as functions of q.

Observability asks whether different initial states can yield distinguishable output histories under specified inputs. For an autonomous smooth system, full rank of differentials of sufficiently many output Lie derivatives is a sufficient local condition under appropriate hypotheses. Controlled systems require the relevant input fields or a specified known-input trajectory. A local rank condition does not prove global uniqueness, numerical conditioning, or parameter identifiability. Parameters may be augmented as constant states, but adequate excitation and sensitivity remain necessary.

Nonlinear indistinguishable sets need not be affine subspaces. For $\dot{x} = 0$ and output $y = x_1^2 + x_2^2$, a nonzero output level is a circle. The kernel of Dy gives its tangent line, not the entire set of indistinguishable states. Linearized null directions describe first-order sensitivity; they must not be promoted to exact finite-state equivalence.

Motion capture and force plates can constrain inverse dynamics when synchronized, calibrated, and accompanied by inertial estimates. Surface EMG adds electrical observations with muscle-specific processing and activation dynamics; integrating its signal does not directly yield force or activation. A universal fixed delay is not an adequate constitutive law. Optimization can select among feasible muscle patterns, but selection by a prior or cost function does not resolve ambiguity empirically [Zaj89].

For the two-hand grip, report which sensors constrain the net club wrench and which constrain the internal hand-load family. For the feet, report separate contact wrenches rather than assuming a unique bilateral distribution from a combined resultant. Compare alternative parameter sets and input allocations that fit the same observations; test whether the proposed conclusion survives that ambiguity.

10.10 Symmetry, Mechanical Connections, and Geometric Phase

10.10.1 Momentum Conservation Has a Boundary Condition

A cyclic coordinate has conserved conjugate momentum only when the Lagrangian is independent of that coordinate and its generalized applied force vanishes. Constraints and external loads must respect the symmetry. A grounded golfer generally receives external angular impulse, so total angular momentum of the golfer-plus-club subsystem is not automatically conserved. Conservation in a freely floating model cannot be asserted for the grounded experiment.

For a local planar symmetry coordinate θ and shape s, consider

$$T = \tfrac{1}{2} I(s)\dot{\theta}^2 + \dot{\theta}\, a(s)^\top \dot{s} + \tfrac{1}{2}\dot{s}^\top H(s)\dot{s}. \tag{10.34}$$

The conjugate momentum is $p_\theta = I\dot{\theta} + a^\top \dot{s}$. If it is conserved, reconstruction gives

$$\dot{\theta} = I^{-1} p_\theta - A(s)\dot{s}, \qquad A = I^{-1} a^\top. \tag{10.35}$$

The locked inertia is $I = \partial^2 T / \partial \dot{\theta}^2$; there is no additional factor of one half. The connection needs the group–shape cross block a, not the derivative of I alone. At fixed momentum, a conventional Routhian is $L - p_\theta \dot{\theta}$ with $\dot{\theta}$ eliminated; forced or noncyclic systems require a modified reduction [BL04].

10.10.2 Separate Dynamic and Geometric Contributions

For a closed shape loop in this planar model,

$$\Delta\theta = \int_0^T I(s(t))^{-1} p_\theta \, dt - \oint A(s)\, ds. \tag{10.36}$$

The first term is a dynamic phase and the second a geometric contribution with the sign fixed by the reconstruction equation. At zero momentum, only the latter remains. On a suitable planar shape patch, Stokes' theorem converts $\oint A$ to an integral of dA. General noncommuting spatial rotations require a group reconstruction and ordered transport; a scalar area formula does not directly apply.

For a checkable example, choose $I = 1$, $a = (0, ks_1)^\top$, and $H = I_2 + aa^\top$. The full kinetic metric is positive definite because its Schur complement is I_2. At zero momentum, $\dot{\theta} = -ks_1\dot{s}_2$. A counterclockwise rectangular shape loop of side lengths a_0, b_0 gives $\Delta\theta = -ka_0 b_0$. This proves a geometric effect in the specified model. It does not identify an analogous contribution in a golf swing until the inertia blocks, external moment, and measured shape trajectory have been established.

10.11 Numerical Methods That Preserve the Stated Problem

The position-constrained first-order mechanical DAE is ordinarily differentiation index three under regularity assumptions. Differentiating position once yields velocity consistency and twice yields an algebraic equation for λ; differentiating that relation yields a differential equation for the multiplier. Position to velocity to acceleration is two differentiations, not three. This terminology does not make a twice-differentiated constraint automatically stable under time stepping.

Baumgarte stabilization replaces the acceleration equation by $J_c\dot{v} + \gamma + 2\alpha J_c v + \beta^2\Phi = 0$. In continuous arithmetic the violation obeys $\ddot{\Phi} + 2\alpha\dot{\Phi} + \beta^2\Phi = 0$. Positive gains damp errors, but α and β have inverse-time units, and large gains can make integration stiff. Constraint scaling and time-step studies are necessary. Stabilization introduces corrective forces and does not guarantee exact energy conservation.

A local Newton correction for position may use

$$q^{j+1} = q^j - M(q^j)^{-1}J_c(q^j)^\top \left[J_c(q^j)M(q^j)^{-1}J_c(q^j)^\top\right]^{-1}\Phi(q^j). \tag{10.37}$$

It requires a suitable initial guess and stopping criterion. After correcting position, velocity projection is $v = P_v(q)v_{\text{pred}}$. These are projection operations; by themselves they do not specify SHAKE or RATTLE. Those methods combine particular constrained position/momentum updates with an underlying integration scheme. Their symplectic properties require their actual assumptions and solve tolerances; arbitrary after-step projection is not equivalent [HLW06].

For constant mass, a RATTLE-style velocity-Verlet step first computes a half-step momentum with a constraint impulse along $J_c(q_n)^\top$, chooses its multiplier so the updated position satisfies $\Phi(q_{n+1}) = 0$, then updates momentum with the new applied force and a second multiplier so $J_c(q_{n+1})M^{-1}p_{n+1} = 0$. The half-step factors, old/new Jacobians, and both multipliers are part of the method. Configuration-dependent inertia and velocity-dependent forces require an appropriate generalized scheme, not a direct substitution into that separable formula.

SVD can reveal the numerical rank of a scaled J_c and supply a tangent basis N. The Euclidean projector $NN^\top$ assumes Euclidean-orthonormal columns; the kinetic-metric projector is $N(N^\top MN)^{-1}N^\top M$. Neither tangent projector alone includes the acceleration offset $-M^{-1}J_c^\top K_c^{-1}\gamma$. Solve linear systems instead of explicitly forming inverses in numerical code.

Validation should report position, velocity, and acceleration residuals separately; input/reaction feasibility; changes under tighter solve tolerances and smaller time steps; and the energy balance including applied work and dissipation. Energy constancy is a test for an isolated lossless model, not for every passive, driven, or damped system. Good energy behavior is not proof of correct contact geometry or scientifically valid parameters.

10.12 Connecting the Pieces in a Golf Investigation

Begin with a measurable outcome, such as delivered clubhead speed and face orientation at the model's own impact event. Specify the body/club model, coordinate frames, grip model, foot contact modes, and inputs. Estimate state and parameter uncertainty before decomposing effects. A useful same-state calculation compares the drift acceleration and individual projected input contributions; a forward intervention branches the state and recomputes reactions and contact events.

Then ask distinct questions. How does posture change the effective input map? Which internal hand loads leave the net wrench unchanged while altering grip compliance? How much mechanical work has already entered the system, and where is it stored? How does ground impulse redirect momentum while doing little or no work in an ideal stationary-contact model? Which feedback corrections remain feasible under torque, activation, and contact limits? Impact depends on the delivered state, so differences earlier in the swing matter through their effects on that state and its uncertainty.

These questions are intertwined without being interchangeable. A favorable configuration may improve load transmission while increasing sensitivity to parameter error. Co-contraction can change impedance without changing the instantaneous net joint torque, yet it can have metabolic cost and change later behavior. A zero-torque continuation can retain substantial speed because of stored momentum and energy, while still being an unrealistic intervention on activation or grip. A force plate can measure a large ground moment without showing that the ground supplied rotational work. An optimizer can expose a feasible strategy without establishing that a golfer uses it.

For a defensible argument, report the model identity, the empirical observation, and the proposed causal mechanism separately. Publish the equations and reference-point conventions, reproducible numerical counterexamples, uncertainty ranges, and interventions that would falsify the proposed explanation. A prediction should survive reasonable alternative contact models and muscle allocations before being presented as a robust swing mechanism. Coaching conclusions require their own outcome evidence; geometric feasibility alone does not establish a beneficial instruction for a person.

10.12.1 Exercises and Reproducible Checks

1. Verify $P_v^2 = P_v$ and $P_v^\top M = M P_v$ for the unequal-mass example. Compare vector and force projections and explain their different domains.
2. Derive the circle's normal acceleration and show why a tangent-only acceleration update fails at nonzero speed.
3. Differentiate the two-coordinate metric to recover c and verify $v^\top c = \frac{1}{2} v^\top \dot{M} v$ for arbitrary rates.
4. Derive the weighted allocation $(0.8, 0.2)$ and identify input bounds that make the requested acceleration infeasible.
5. Transform an arbitrary twist and wrench between translated, rotated frames and verify power invariance. Then construct a nonzero internal hand-force pair with zero resultant wrench.
6. Derive the force-plate center-of-pressure signs from $r \times F$ and explain why the free moment must be retained.
7. Integrate the rectangular geometric-phase example and reverse the loop orientation. Add a known external torque and identify the extra momentum-dependent contribution.
8. Simulate a constrained system at successively smaller time steps. Report constraint residuals and the work-inclusive energy balance rather than certifying accuracy from one plot.

Glossary of Important Concepts

These definitions distinguish geometric objects, physical models, and the guarantees obtained from a calculation. A definition does not establish that a particular golf model satisfies its assumptions.

Articulated Body Algorithm
A recursive forward-dynamics algorithm for a rigid-body tree. With bounded joint dimension, its arithmetic cost scales linearly with the number of bodies. Closed-loop constraints and contact require additional treatment; the tree algorithm alone does not solve every constrained contact problem.

Configuration Space
The set of configurations allowed by a model, including positions, orientations, and declared geometric constraints. It is a smooth manifold when suitable regularity conditions hold; redundant coordinates and constraint singularities require care. Velocities are additional state information.

Contraction Analysis
Analysis of convergence between trajectories under stated common inputs or closed-loop dynamics, in a specified metric and domain. A differential contraction inequality needs regularity, metric bounds, and domain/existence conditions to imply the corresponding finite-distance result. Contraction does not by itself prove input feasibility or a desired contact outcome.

Control Contraction Metric (CCM)
A positive-definite differential metric satisfying control-dependent contraction conditions that support feedback construction for feasible reference trajectories. The result depends on the theorem's domain, regularity, metric bounds, and controller construction. Actuator saturation and state constraints require explicit treatment.

Degrees of Freedom (DOF)
The number of locally independent coordinates needed to specify a configuration. Independent constraints can reduce it; redundant parameters can exceed it. The number of actuators and the dimension of the full state space are different quantities.

Drift
The evolution of the declared effective plant when the declared control channel is zero. It includes the configuration-velocity kinematics and all retained uncontrolled dynamics, such as gravity, inertial coupling, passive elasticity, and damping. Its meaning depends on the chosen state, control channel, and environment; zero control does not mean zero acceleration or zero velocity.

Drift-Control Ratio (DCR)
A model-based comparison of drift and bounded control authority in a declared acceleration or task-projected space:

$$\mathrm{DCR}_{W,\mathcal{U}}(x) = \frac{\|W a_{\mathrm{drift}}(x)\|_2}{\sup_{u\in\mathcal{U}(x)} \|W B_a(x) u\|_2 + \varepsilon}.$$

State the projection/scaling W, admissible input set $\mathcal{U}$, acceleration input map B_a, and regularizer ε, with compatible units. This scalar magnitude ratio omits directional alignment and does not by itself establish controllability, cancellation feasibility, or a universal coaching threshold.

Euler Angles
Three parameters describing an orientation using a declared sequence of rotations. Different sequences have different singular sets. Angle rates are related to angular velocity by a configuration-dependent map and are not generally its Cartesian components.

Exponential Map
For a Lie group, the map from a Lie-algebra element to the group element reached at parameter one along

its generated one-parameter subgroup; for a matrix Lie group it is the matrix exponential. It need not be globally one-to-one. It is not general integration of an arbitrary time-varying velocity. The Riemannian exponential defined by geodesics is a related but distinct construction requiring a connection or metric.

Flow
The solution map of an ordinary differential equation where solutions exist uniquely. For an autonomous smooth vector field it forms a local one-parameter family with the composition property. Time-dependent systems use a two-time solution map; hybrid resets need not belong to a single smooth flow.

Gimbal Lock
A singularity of an Euler-angle rate-to-angular-velocity map. The parameterization loses local rank; the rigid body's physical rotational freedom does not disappear. Quaternions or overlapping coordinate charts handle orientation differently and have their own representation conventions.

Jacobian
The derivative of a map expressed in coordinates. For a task map $y = h(q)$, $\dot{y} = J(q)\dot{q}$. Acceleration also contains $\dot{J}\dot{q}$. Spatial and body Jacobians use different frame conventions, which must match the represented velocities and forces.

Kinematics
Description of position, orientation, velocity, and acceleration relationships without using forces to determine the motion. A kinematically allowed path is not automatically dynamically feasible under available actuation.

Lagrangian Mechanics
A formulation using a Lagrangian, often $L = T - U$, together with applied generalized forces and any constraints. For unconstrained coordinates, $d(\partial L/\partial \dot{q})/dt - \partial L/\partial q = Q$. Dissipation, actuation, nonconservative loads, and contact must be specified; kinetic and potential energy alone do not determine a forced system's motion.

Lie Algebra
A vector space equipped with a bilinear antisymmetric bracket satisfying the Jacobi identity. For a Lie group, it is the tangent space at the identity with the induced bracket. Brackets of smooth vector fields are another important example; accessibility arguments require the appropriate system and theorem assumptions.

Lie Group
A smooth manifold that is also a group, with smooth multiplication and inversion. Examples include the rotation group $SO(3)$ and rigid-transformation group $SE(3)$. A choice of coordinates represents these objects without changing the group itself.

Lyapunov Stability
For an equilibrium, the property that sufficiently small initial perturbations remain small for all future times of the stated system. It does not require return to equilibrium. Asymptotic stability additionally requires attraction; orbital and finite-horizon stability statements refer to different targets and must be defined separately.

Manifold
A space locally described by Euclidean coordinates, with the usual topological regularity conditions. A smooth manifold also has smoothly compatible charts. Its dimension is local and does not depend on how many coordinates are used to embed or redundantly represent it.

Screw Axis
For a rigid-body twist with nonzero angular velocity, an instantaneous axis and pitch describing rotation together with translation along that axis. Pure translation is a limiting case, often represented as an axis at infinity; a finite unique rotational axis is then unavailable.

State Space
The set of variables sufficient to determine future evolution under specified inputs and a model. Mechanical

models often include configuration and velocity or momentum, and may additionally include actuator states, flexible modes, estimators, or discrete contact modes.

Tangent Bundle
The union of tangent spaces T_qQ over configurations $q \in Q$, with its bundle and smooth-manifold structure. For an n-dimensional smooth configuration manifold, TQ has dimension $2n$. It represents configuration-velocity states for a basic mechanical model; momentum states instead lie in the cotangent bundle T^*Q.

Tangent Space
The vector space of instantaneous derivatives of smooth curves through a point on a manifold. Its vectors are local velocities, not finite displacements between arbitrary configurations. Applied differential constraints can restrict the admissible velocities further.

Twist
A six-component representation of an instantaneous rigid-body velocity field, combining angular and linear terms with a declared reference point and frame. In one frame, $v(r) = v_0 + \omega \times (r - r_0)$. Spatial and body representations are related by the appropriate adjoint transformation; the linear term must not be confused with the velocity of an unspecified point.

Wrench
The dual force representation consisting of a moment about a declared reference point and a resultant force in a declared frame. With matching twist conventions, mechanical power is $\tau \cdot \omega + F \cdot v_0$. Changing point or frame requires the corresponding moment shift and dual transformation so that power is unchanged.

Zero-Torque Counterfactual (ZTCF) Family
Model interventions setting the declared applied generalized-control channel to zero while preserving the declared effective plant. Pointwise samples, stitched traces, forward trajectories, and branched trajectories answer different questions. A stitched pointwise acceleration trace is not automatically a feasible unforced trajectory.

Zero-Velocity Counterfactual (ZVCF)
The instantaneous acceleration after the declared velocity and control intervention:

$$a_{\mathrm{ZVCF}}(q, z) = -M(q)^{-1} h(q, 0, z_0; p).$$

Specify the auxiliary-state reset z_0, retained loads, parameters, and contact mode. This separates the retained configuration-dependent loads under that intervention; it is not necessarily gravity alone, and it is not a forward simulation of a stationary body.

Further Reading and References

The following books develop complementary parts of the framework used here. Choose a source for the particular claim or calculation being investigated, and consult its assumptions, notation, edition, and errata. A robotics or control theorem does not become evidence about human golf performance merely because it can be applied to a simplified swing model.

Geometry, Kinematics, and Mechanical Models

- **A Mathematical Introduction to Robotic Manipulation**, Murray, Li, and Sastry [MLS94], treats rigid-body motion, manipulator and hand kinematics, dynamics, and nonholonomic motion planning. The authors' companion site provides chapter information and corrections. Use it to connect Lie groups and Jacobians to explicitly defined frames and mechanical systems.
- **Geometric Control of Mechanical Systems**, Bullo and Lewis [BL04], develops differential-geometric modeling and control of mechanical systems. Its companion page includes contents, selected material, examples, and errata. This is a suitable route into connections, mechanical control structures, and the hypotheses behind geometric analysis.
- **Robot Modeling and Control**, Spong, Hutchinson, and Vidyasagar [SHV06], covers kinematics, Jacobians, trajectory planning, dynamics, joint and multivariable control, force control, and vision-based methods. The publisher's chapter listing identifies the scope of the cited edition. Its robot models still require adaptation and validation before supporting a biomechanical interpretation.

Computational Rigid-Body Dynamics

- **Rigid Body Dynamics Algorithms**, Featherstone [Fea08], develops spatial-vector conventions and recursive dynamics algorithms. The author's teaching materials and software resources support implementation. Distinguish the linear-cost articulated-body calculation for a tree from the additional work needed for closed loops, constraints, and contacts.

Nonlinear Stability and Feedback

- **Applied Nonlinear Control**, Slotine and Li [SL91], develops nonlinear stability and feedback methods. Slotine's accompanying course materials identify the relevant book sections and separate the later contraction lectures and papers. Check the domain and model conditions of a stability argument before extending it to saturated or hybrid systems.
- **Contraction Theory for Dynamical Systems**, Bullo [Bul24], develops contractivity and incremental stability using specified norms and metrics. The author's book page provides version information and materials. The bibliography cites the 2024 edition; later revisions may reorganize or extend the treatment. Contraction claims must retain their metric, input, domain, and existence assumptions.

The chapter-level primary references supply more specialized results for transverse dynamics, phase constraints, nonlinear funnels, stochastic models, and learning. Keep mathematical derivations, numerical verification, model calibration, and experimental validation identifiable when combining those sources.

Index

Notation Reference

Symbol	Meaning
$\boldsymbol{x} = (\boldsymbol{q}, \dot{\boldsymbol{q}})$	State vector (generalized coordinates and velocities)
$\boldsymbol{u} \in \mathbb{R}^m$	Control input vector
$f(\boldsymbol{x})$	Drift vector field (passive dynamics)
$G(\boldsymbol{x})$ or $g_i(\boldsymbol{x})$	Input (control) vector fields
$\mathbf{A}(t)$	State Jacobian $\partial f/\partial x$ along nominal trajectory
$\mathbf{B}(t)$	Input Jacobian $\partial(Gu)/\partial u$ along nominal trajectory
$\mathbf{M}(\boldsymbol{x}, t) \succ 0$	Contraction metric (positive definite)
$\mathbf{S}_t$	Riccati (cost-to-go curvature) matrix
$\mathbf{K}_t$	Feedback gain matrix
$\mathbf{Q}_t \succeq 0$	State cost weight
$\mathbf{R}_t \succ 0$	Input cost weight
$\Phi(t_1, t_0)$	State transition matrix
$T_{\bar{\boldsymbol{x}}}\mathcal{M}$	Tangent space at $\bar{\boldsymbol{x}}$ on manifold $\mathcal{M}$
$\delta\boldsymbol{x}, \delta\boldsymbol{u}$	Infinitesimal perturbations in state and input
$\lambda > 0$	Contraction rate
$H(\boldsymbol{q})$ or $M(\boldsymbol{q})$	Mass/inertia matrix of mechanical system
$C(\boldsymbol{q}, \dot{\boldsymbol{q}})$	Coriolis/centrifugal matrix
$\boldsymbol{g}(\boldsymbol{q})$	Generalized gravity vector
$J(\boldsymbol{q})$	Jacobian (task or constraint)
$\preceq, \succeq$	Matrix inequality (Loewner order)
$[f, g]$	Lie bracket of vector fields f and g
$L_f h$	Lie derivative of h along f
$\mathbf{P}(t)$	Estimation error covariance matrix
$\mathbf{L}(t)$	Observer (Kalman) gain
$\mathcal{W}_o$	Observability Gramian
$H(\boldsymbol{x})$	Hamiltonian (total energy)
$J(\boldsymbol{x}), R(\boldsymbol{x})$	Port-Hamiltonian interconnection and dissipation matrices
$\rho(t)$	Drift–control ratio
$\varphi(t)$	Funnel boundary

Differential Geometry Primer

This appendix collects the key definitions from differential geometry used throughout the text. The presentation is deliberately concrete; for full generality, see do Carmo [dC92] or Abraham and Marsden [AM78].

.1 Smooth Manifolds

A *smooth manifold* $\mathcal{M}$ of dimension n is a topological space that locally looks like $\mathbb{R}^n$. Formally, it is equipped with an atlas of charts $\{(U_\alpha, \phi_\alpha)\}$ where each $U_\alpha \subset \mathcal{M}$ is open and $\phi_\alpha : U_\alpha \to \mathbb{R}^n$ is a homeomorphism, with smooth transition maps $\phi_\beta \circ \phi_\alpha^{-1}$ on overlaps.

Examples used in this book:

- $\mathbb{R}^n$ (state space of most control systems)
- S^1 (circle; configuration space of a revolute joint)
- $\mathbb{T}^n = (S^1)^n$ (torus; configuration space of an n-joint robot)
- SO(3) (rotation group; attitude of a rigid body)
- SE(3) (rigid-body pose: rotation + translation)

.2 Tangent Spaces and Tangent Bundles

The *tangent space* $T_p\mathcal{M}$ at a point $p \in \mathcal{M}$ is the vector space of all tangent vectors at p. It can be defined as the set of equivalence classes of curves through p, or as the set of derivations on smooth functions at p. In either case, $T_p\mathcal{M} \cong \mathbb{R}^n$.

The *tangent bundle* $T\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p\mathcal{M}$ is itself a smooth manifold of dimension $2n$. A point in $T\mathcal{M}$ is a pair (p, v) with $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$. For mechanical systems, $T\mathcal{M}$ is the *state space* $(\boldsymbol{q}, \dot{\boldsymbol{q}})$.

.3 Vector Fields and Flows

A *vector field* f on $\mathcal{M}$ assigns a tangent vector $f(p) \in T_p\mathcal{M}$ to each point p. In coordinates, $f = (f_1(\boldsymbol{x}), \ldots, f_n(\boldsymbol{x}))^T$. The ODE $\dot{\boldsymbol{x}} = f(\boldsymbol{x})$ defines the *flow* $\varphi_t : \mathcal{M} \to \mathcal{M}$, which maps initial conditions to their time-t images.

.4 Riemannian Metrics

A *Riemannian metric* on $\mathcal{M}$ is a smoothly varying inner product $\langle \cdot, \cdot \rangle_p$ on each tangent space $T_p\mathcal{M}$. In coordinates, it is represented by a positive-definite matrix $\mathbf{G}(p)$:

$$\langle v, w \rangle_p = v^T \mathbf{G}(p) w.$$

For mechanical systems, the mass matrix $M(\boldsymbol{q})$ is a Riemannian metric on configuration space. The contraction metric $\mathbf{M}(\boldsymbol{x})$ of Chapter 5 is a Riemannian metric on state space.

.5 Lie Groups and Lie Algebras

The most important manifolds in rigid-body control are matrix Lie groups. A Lie group is both a smooth manifold and a group, with smooth multiplication and inversion.

- $\mathrm{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3\times 3} : \mathbf{R}^T\mathbf{R} = \mathbf{I},\, \det(\mathbf{R}) = 1\}$ (rotations)
- $\mathrm{SE}(3) = \left\{ \begin{bmatrix} \mathbf{R} & \boldsymbol{p} \\ \mathbf{0}^T & 1 \end{bmatrix} : \mathbf{R} \in \mathrm{SO}(3),\, \boldsymbol{p} \in \mathbb{R}^3 \right\}$ (poses)

Their Lie algebras are tangent spaces at identity:

$$\mathfrak{so}(3) = \{\mathbf{\Omega} \in \mathbb{R}^{3\times 3} : \mathbf{\Omega}^T = -\mathbf{\Omega}\}, \quad \mathfrak{se}(3) = \left\{ \begin{bmatrix} \mathbf{\Omega} & \boldsymbol{v} \\ \mathbf{0}^T & 0 \end{bmatrix} \right\}.$$

Using the hat map $\widehat{\omega}$, angular velocity vectors become algebra elements. This is the algebraic backbone of variational dynamics in Chapter 2 and motion composition in Chapter 9.

.6 Exponential and Logarithmic Maps

The exponential map converts tangent directions into finite motions. For SO(3):

$$\exp(\widehat{\omega}\theta) = \mathbf{I} + \frac{\sin\theta}{\theta}\widehat{\omega}\theta + \frac{1-\cos\theta}{\theta^2}(\widehat{\omega}\theta)^2.$$

Given $\mathbf{R} \in \mathrm{SO}(3)$ with $\theta = \cos^{-1}\left(\frac{\mathrm{tr}(\mathbf{R})-1}{2}\right)$, the log map recovers axis-angle:

$$\log(\mathbf{R}) = \frac{\theta}{2\sin\theta}(\mathbf{R} - \mathbf{R}^T).$$

For SE(3), the exponential combines rotation with translated screw motion and is used directly in product-of-exponentials kinematics and geodesic interpolation.

Example .1. Python Example: SO(3) Exp/Logapp-so3-exp-log

```python
import numpy as np
from scipy.linalg import expm, logm

def hat(w: np.ndarray) -> np.ndarray:
    return np.array([[0, -w[2], w[1]],
                     [w[2], 0, -w[0]],
                     [-w[1], w[0], 0]], dtype=float)

w = np.array([0.3, -0.2, 0.5])
R = expm(hat(w))              # exp map to SO(3)
w_hat = logm(R).real          # log map back to so(3)
```

.7 Adjoint Representation

Adjoint operators transport twists and wrenches across frames:

$$\mathrm{Ad}_{(\mathbf{R},\boldsymbol{p})} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\boldsymbol{p}]_\times \mathbf{R} & \mathbf{R} \end{bmatrix}, \quad \mathrm{ad}_{(\omega,v)} = \begin{bmatrix} [\omega]_\times & \mathbf{0} \\ [v]_\times & [\omega]_\times \end{bmatrix}.$$

In Chapter 9, this map explains why equivalent perturbations look different between body-fixed and world-fixed coordinates.

.8 Fiber Bundles

Tangent and cotangent bundles are canonical fiber bundles:

$$\pi_T : T\mathcal{M} \to \mathcal{M}, \quad \pi_T(p, v) = p, \qquad \pi_{T^*} : T^*\mathcal{M} \to \mathcal{M}.$$

The base point is configuration; the fiber stores admissible velocities ($T\mathcal{M}$) or covectors/momenta ($T^*\mathcal{M}$). This viewpoint clarifies why state and costate evolve on different but coupled geometric objects in Chapters 6 and 7.

.9 Connections and Parallel Transport

A connection specifies how to compare tangent vectors at different points. In local coordinates, the Levi-Civita connection of metric $\mathbf{G}$ has Christoffel symbols Γ^k_{ij}. Covariant acceleration is

$$\frac{D\dot{q}^k}{dt} = \ddot{q}^k + \Gamma^k_{ij}\dot{q}^i\dot{q}^j.$$

Parallel transport ($Dv/dt = 0$ along a curve) is the right way to carry perturbations when curvature is nonzero. This geometric correction underlies stable trajectory tracking on curved configuration manifolds.

.10 Curvature and Trajectory Sensitivity

Curvature quantifies how geodesics separate. Sectional curvature $K(u, v)$ for a 2-plane $\mathrm{span}\{u, v\} \subset T_p\mathcal{M}$ predicts local sensitivity:

- $K > 0$: geodesics tend to reconverge (spherical-like geometry)
- $K < 0$: geodesics diverge faster (hyperbolic-like sensitivity)

Contraction analysis in Chapter 5 can be interpreted as enforcing metric dynamics that dominate this geometric divergence.

Linear Algebra for Control

.11 Matrix Inequalities and the Loewner Order

For symmetric matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$, we write:

- $\mathbf{A} \succ 0$ ($\mathbf{A}$ is *positive definite*): $v^T \mathbf{A} v > 0$ for all $v \neq 0$.
- $\mathbf{A} \succeq 0$ ($\mathbf{A}$ is *positive semidefinite*): $v^T \mathbf{A} v \geq 0$ for all v.
- $\mathbf{A} \succeq \mathbf{B}$: $\mathbf{A} - \mathbf{B} \succeq 0$.

Key property: The set of positive semidefinite matrices forms a *convex cone*. This is why LMI-based searches for contraction metrics (Chapter 5) are convex optimization problems.

.12 Schur Complement

For a block matrix $\mathbf{M} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix}$ with $\mathbf{C} \succ 0$:

$$\mathbf{M} \succ 0 \quad \Leftrightarrow \quad \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^T \succ 0. \tag{38}$$

The matrix $\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^T$ is the *Schur complement* of $\mathbf{C}$ in $\mathbf{M}$. This identity is used extensively in Riccati equation derivations (Chapters 6 and 7) and in the mass matrix decomposition for flexible systems (Chapter 9).

.13 Singular Value Decomposition

Every matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ has an SVD: $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal and $\boldsymbol{\Sigma}$ is diagonal with non-negative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. The SVD is used in this book for:

- Analyzing the STM $\Phi(t_1, t_0)$: singular values quantify amplification of perturbations (Chapter 2).
- Impact sensitivity analysis: singular values of $J_{\text{impact}} \cdot C \cdot \Phi$ reveal the most sensitive perturbation directions (Chapter 9).
- Condition number $\kappa(\mathbf{A}) = \sigma_1/\sigma_n$: measures sensitivity to perturbations.

.14 Matrix Exponential

For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the matrix exponential is:

$$e^{\mathbf{A}t} = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!}.$$

It satisfies $\frac{\mathrm{d}}{\mathrm{d}t} e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t}$ and is the state transition matrix for LTI systems: $\Phi(t, t_0) = e^{\mathbf{A}(t-t_0)}$.

Caution: For time-varying $\mathbf{A}(t)$, $\Phi(t, t_0) \neq e^{\int_{t_0}^{t} \mathbf{A}(s)\,\mathrm{d}s}$ in general. Equality holds only when $[\mathbf{A}(t_1), \mathbf{A}(t_2)] = 0$ for all t_1, t_2 (the commutativity condition for matrix exponentials).

.15 Kronecker Products

The Kronecker product $\mathbf{A} \otimes \mathbf{B}$ expands block structure and is essential for vectorized Lyapunov/Riccati equations:

$$\mathrm{vec}(\mathbf{AXB}) = (\mathbf{B}^T \otimes \mathbf{A})\mathrm{vec}(\mathbf{X}).$$

For $\mathbf{A}^T\mathbf{P} + \mathbf{PA} + \mathbf{Q} = 0$, vectorization yields

$$\big(\mathbf{I} \otimes \mathbf{A}^T + \mathbf{A}^T \otimes \mathbf{I}\big)\mathrm{vec}(\mathbf{P}) = -\mathrm{vec}(\mathbf{Q}),$$

which is directly useful in Chapter 7.

.16 Matrix Calculus

Control derivations repeatedly require gradients with respect to vectors and matrices:

$$\frac{\partial}{\partial x}(x^T\mathbf{A}x) = (\mathbf{A} + \mathbf{A}^T)x, \quad \frac{\partial}{\partial \mathbf{X}}\,\mathrm{tr}(\mathbf{A}^T\mathbf{X}) = \mathbf{A}.$$

These identities support second-order expansions in DDP and local quadratic models in Chapter 6.

.17 Generalized Eigenvalue Problems

Many stability questions appear as matrix pencils:

$$\mathbf{A}v = \lambda\mathbf{B}v, \quad \det(\mathbf{A} - \lambda\mathbf{B}) = 0.$$

Generalized eigenanalysis is required when constraints, descriptor dynamics, or mass matrices lead to non-identity state weighting.

.18 Perturbation Theory and Pseudospectra

Eigenvalues can be highly sensitive in non-normal systems. The ε-pseudospectrum is

$$\Lambda_\varepsilon(\mathbf{A}) = \{z \in \mathbb{C} : \|(z\mathbf{I} - \mathbf{A})^{-1}\| > 1/\varepsilon\}.$$

This captures transient growth that eigenvalues alone miss, which is directly relevant to “small perturbation, large excursion” behavior in Chapter 2.

.19 Sparse Matrix Methods

Large-scale discretizations produce sparse operators. Storing sparse matrices in CSR/CSC formats reduces memory and enables scalable iterative solvers (CG, GMRES, MINRES). This matters for trajectory optimization and model predictive control where linear solves dominate runtime.

.20 LMI Fundamentals

Linear Matrix Inequalities (LMIs) take the form $\mathbf{A}_0 + \sum_i x_i\mathbf{A}_i \prec 0$. These are convex constraints. Tools like cvxpy and SciPy can solve LMI optimization problems efficiently.

Example .2. Python Example: Sparse Solve with SciPyapp-sparse-solve

```
import nu
```

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